

MGT 510 TOTAL QUALITY MANAGEMENT GRAND QUIZ AND QUIZ FILE

1. Goods, healthy individuals and services come under which of the following sections of the following sections of the organizational system. **PPT 2**
 - a. Input
 - b. Process
 - c. Output**
 - d. None of the given options
2. Which of the following is not the ISO quality management principle?
(Lecture no 21 Page no 82)
 - a. Involvement of People
 - b. Process Orientation
 - c. Management**
 - d. Continual Improvement
3. Which one of the following options focuses on long term profits and continual improvement?
 - a. Quality control
 - b. Total quality (Lecture no 3 Page no 11 Handouts)**
 - c. Traditional quality
 - d. Cultural model
4. Which one of the following options is not the task facilitating behavior?
 - a. Encouraging (Lecture no 6 Page no 27,28 Handouts)**
 - b. Seeking information
 - c. Initiating
 - d. Coordinating
5. Which of the following is a unit for measuring process performance according to traditional view of quality? **Page 11**
 - a. Effective parts per thousand produced
 - b. Defective parts per thousand produced
 - c. Effective parts per hundred produced
 - d. Defective parts per hundred produced**
6. Which of the following is typically viewed to lie between fully vertically integrated firms? **Google**
 - a. Supply chain management**
 - b. Production chain management
 - c. Value chain management
 - d. Demand chain management
7. Which one of the following group maintenance behaviors refers to reconciling differences among group members?
 - a. Compromising
 - b. Encouraging
 - c. Harmonizing (Lecture no 6 Page no 28 Handouts)**
 - d. Blocking

22. Which of the following refers to bridge the gaps in the dispersion of quality and to attain the expected value?
- a. Quality control
 - b. Quality assurance (Lecture no 1 Page no 2 Handouts)**
 - c. Quality planning
 - d. Quality managing
23. Quality assurance is usually associated with which of the following activities?
- a. Measurement and inspection (Lecture no 1 Page no 1 Handouts)**
 - b. Specialization activities
 - c. Support services
 - d. None of the given options
24. Which of the following quality dimensions tells about the ability to repair a product quickly and easily?
- a. Serviceability (Lecture no 4 Page no 13 Handouts)**
 - b. Aesthetic
 - c. Conformance
 - d. Feature
25. From the total quality perspective, there is an emphasis on continuous improvement of which of the following?
- a. Products and people only
 - b. Processes and people only
 - c. Products, processes, and purpose
 - d. Products, processes, and people (Lecture no 3 Page no 11 Handouts)**
26. Total organization excellence (TOE) can be done for which of the following? PPT 2
- a. Production people
 - b. Marketing people
 - c. Finance people
 - d. All staff members**
27. Which of the following shows that how does product feel, look and sound?
- a. Reliable
 - b. Durable
 - c. Feature
 - d. Aesthetics (Lecture no 4 Page no 13 Handouts)**

28. Customer value strategy includes all of the following offers, EXCEPT:

- a. Product characteristics
- b. Mode of deliver
- c. Flat hierarchy (Lecture no 2 Page no 5 Handouts)**
- d. Support services

29. Which one of the following depicts conformance – a dimension of quality? **Page 13**

- a. Exterior finish
- b. **Meeting specifications**
- c. Brand name
- d. Ease of repair

30. All of the following are the grounds to give birth to total quality approach EXCEPT:

- a. Obsession with quantity
- b. Confusion and poor leadership
- c. One employee in charge
- d. Political disorder**

31. Total quality view is that 80 percent of quality problems are due to which of the following options?

- a. Employee's fault
- b. Production's fault
- c. Customer's fault
- d. Management's fault (Lecture no 3 Page no 11 Handouts)**

32. Which of the following options can be used to assess internal customer needs and to communicate internal supplier capabilities?

- a. Retailing concepts and tools
- b. Marketing concepts and tools (Lecture no 5 Page no 21 Handouts)**
- c. Management concepts and tools
- d. None of the given options

33. Which of the following results in low costs?

- a. High productivity and high capacity utilization (Lecture no 5 Page no 24 Handouts)**
- b. Low productivity and low capacity utilization
- c. Low productivity and high capacity utilization
- d. High productivity and low capacity utilization

34. All of the following are the group maintenance behaviors, EXCEPT:

- a. Encouraging
- b. Harmonizing
- c. Compromising
- d. Coordinating (Lecture no 6 Page no 28 Handouts)**

35. Manufacturing based view of quality focuses on which one of the following options? **Page 33**

- a. Conformance to specified requirements**
- b. Set of product characteristics
- c. Translating requirements into physical dimensions
- d. Measurement of capacity

36. Gauging and measurement were the methods of measurement in which of the following eras?

- a. Inspection (Lecture no 10 Page no 41 Handouts)**
- b. Statistical quality control
- c. Quality assurance
- d. Strategic total quality management

37. The traditional view is that employees are _____ for quality.

- a. Rewarded
- b. Blamed (Lecture no 3 Page no 11 Handouts)**
- c. Powered
- d. Motivated

38. In traditional management, performance appraisal, recognition, and reward system place people in which one of the following environments?

- a. Social environment
- b. Political environment
- c. External competitive environment
- d. Internal competitive environment (Lecture no 5 Page no 22 Handouts)**

39. Which of the following is/are the cause(s) of disparity in products?
Page 2

- a. Raw material
- b. Weather
- c. Machinery condition
- d. All of the given options**

40. Which of the following forms the basis of TQM as an organization wide process? **(Page 26)**
- a. Not using the resources of the total organization properly
 - b. Organizational flexibility and response to change**
 - c. Not bound in business relationships
 - d. Not measuring continuous improvement
41. Which one of the following depicts durability – a dimension of quality? **(Page 13)**
- a. Exterior finish
 - b. Quality of work
 - c. Useful life**
 - d. Ease of repair
42. Which of the following models has the viewpoint that change and learning are valued in themselves? **Page 40**
- a. Organism model
 - b. Mechanistic model
 - c. Cultural model**
 - d. Total Quality model
43. Taguchi methods emphasize consistency of performance and reduced _____. **PPT 8**
- a. Price
 - b. Cost
 - c. Variation**
 - d. Competition
44. Which one the following is not included in the concept of quality? **PPT 12**
- a. Quality by definition
 - b. Quality by philosophy
 - c. Quality by technology**
 - d. Quality by attitude
45. Which of the following concepts and tools can be used to assess internal customer needs and to communicate internal supplier capabilities? **(Page 21)**
- a. Financial
 - b. Management
 - c. Marketing**
 - d. Production
46. Quality can be derived as which of the following? **PPT 8**
- a. $Q = \text{Total Give} / \text{Total Get}$
 - b. $Q = \text{Total Get} / \text{Total Give}$**
 - c. $Q = \text{Total Get} + \text{Total Give}$
 - d. $Q = \text{Total Give} + \text{Total Get}$

47. “Retiring from the group, being silent or refusing to deal with a particular aspect of the group’s work”, depicts which one of the following self-oriented behaviors?
- a. Controlling
 - b. Excluding
 - c. Belittling
 - d. Withdrawing (Lecture no 6 Page no 28 Handouts)**
48. Which one of the following depicts feature, a dimension of quality? **Page 13**
- a. Product looks and sounds
 - b. Brand name and price
 - c. Bells and whistles of the product**
 - d. Ability to repair a product
49. Which of the following are called the staff members of the organization?
- a. Investor Customers
 - b. Internal Customers (Lecture no 11 Page no 50 Handouts)**
 - c. External Customers
 - d. Social Customers
50. Asking for examples, pictures, diagrams, and data in meetings, refers to which of the following? **(Page 26)**
- a. Acting as gatekeepers
 - b. Closing the discussion
 - c. Asking for clarification
 - d. Testing for consensus
51. With increasing variability. Taguchi suggests an increasing loss for which of the following?
- a. The customer
 - b. The producer
 - c. The society
 - d. All of the given options (Lesson no 8 Page no 36 Handouts)**

52. Taguchi suggested that loss in a process is increased with increase in which of the following?
- a. Specifications
 - b. Standards
 - c. Competition
 - d. **Variability (Lecture no 8 Page no 36 Handouts)**
53. All of the following are the styles of writing minutes of meeting, EXCEPT: **Page 30**
- a. Narrative
 - b. Resolution
 - c. **Withdrawing**
 - d. Action
54. Which one of the following quality leaders said that quality is “non faulty system”? **PPT 7**
- a. **W. Edwards Deming**
 - b. Joseph M. Juran
 - c. Philip B. Crosby
 - d. Ishikawa
55. What was the role of quality professionals during 1950s? **Page 41**
- a. Troubleshooting and the application of statistical methods
 - b. Inspection, sorting, counting and grading
 - c. **Quality measurement, quality planning, and program design**
 - d. Goal setting, education and training, and program design
56. Which of the following options was the view of quality during inspection era? **Page 41**
- a. A problem to be detect
 - b. A competitive opportunity
 - c. **A problem to be solved (Lecture no 10 Page no 41 Handouts)**
 - d. A threat to organization

57. Which one of the following types of managers spend their almost 40 percent time in meetings? Page 27
- a. Average professional/managers
 - b. Senior executives
 - c. **Middle level managers**
 - d. Lower managers
58. Which of the following equations represents the correct definition of quality? PPT 8
- a. $Q = \text{Level of give} / \text{level of get}$
 - b. $Q = \text{Total give} / \text{total get}$
 - c. **$Q = \text{Level of sustainable and emotional benefits}$**
 - d. $Q = \text{Total profit} / \text{total get}$
59. During 1990s, the following institution(s) began to pay increase attention to quality. Page 12
- a. Education and health care only
 - b. **Education, health care, and government**
 - c. Health care and government only
 - d. Government and financial institutions only
60. What is the relationship between Ishikawa and Deming? Page 37
- a. Son and father
 - b. Nephew and uncle
 - c. **Student and teacher**
 - d. Grandson and grandfather
61. Which of the following options was the emphasis of quality assurance era? (Lecture no 10 Page no 41 Handouts)
- a. Programs and systems
 - b. The market and customer needs
 - c. Product uniformity
 - d. **The entire production chain**
62. Total quality views people as the enterprise's true? Page 22
- a. Positive asset
 - b. **Competitive edge**
 - c. Value addition
 - d. None of the given options

63. Who developed Quality Function Deployment (QFD)? **Page 46**

- a. Frederick Taylor
- b. Joseph M. Juran
- c. Y. Akao**
- d. Armand Feigenbaum

64. All of the following are the discussion skills of quality meeting, EXCEPT: **(Lecture no 6 Page no 26 Handouts)**

- a. Ask for clarification
- b. Findings cost advantage**
- c. Contain digressions
- d. Test for agreement

65. Whose concepts are referred to as statistical quality control (SQC)?

- a. Shewhart's (Lecture no 7 Page no 32 Handouts)**
- b. Deming's
- c. Juran's
- d. Crosby's

66. What is the objective of a TQM system? **(Page 82)**

- a. Continuous improvement
- b. Continual improvement**
- c. Business improvement
- d. Process improvement

67. Which of the following quality leaders defines the quality as degree of match between true and substitute? **PPT 8**

- a. Juran
- b. Ishikawa**
- c. Crosby
- d. Deming

68. In the time of high competition, competitors often played according to which of the following strategies?

- a. Economies of scale
- b. Mass production
- c. Live and let live (Lecture no 2 Page no 5 Handouts)**
- d. None of the given options

70. Corporate culture is shifting its focus from upgrading technology to upgrading _____.

- a. Process
- b. People (Lecture no 9 Page no 39 Handouts)**
- c. Place
- d. Product

71. "The car seats maintain their shape after years of use" represents an example of which one of the following options?

- a. Quality of performance (Lecture no 7 Page no 35 Handouts)**
- b. Quality of design
- c. Quality of conformance
- d. None of the given options

72. Who suggested, "A quality product is one that provides performance or conformance at an acceptable price or cost"?

- a. Crosby
- b. Ishikawa
- c. Garvin (Lecture no 7 Page no 34 Handouts)**
- d. Shewhart

73. What was the orientation and approach during 1930s?

- a. Inspects in quality
- b. Controls in quality (Lecture no 10 Page no 41 Handouts)**
- c. Builds in quality
- d. Manages in quality

74. Which one of the following shows that how correct is your system? Page 124

- a. Consistency
- b. **Accuracy**
- c. Completeness
- d. Responsiveness

75. Which of the following options has the responsibility of quality in strategic total quality management era?

- a. The inspection department
- b. Everyone in the organization (Lecture no 10 Page no 41 Handouts)**
- c. All departments
- d. The manufacturing department

76. Which one of the following quality leaders said that quality is "conformance for requirement"? **PPT 7**

- a. W. Edwards Deming
- b. Joseph M. Juran
- c. Philip B. Crosby**
- d. Ishikawa

77. Learn how to tell when nothing can be gained from further discussion. This refers to which of the following discussion skills in meetings? **(Page 26)**

- a. Acting as gatekeepers
- b. Closing the discussion**
- c. Asking for clarification
- d. Testing for consensus

78. Which of the following options refers to the view of quality as the satisfaction of customer needs?
- a. Fitness for product
 - b. Fitness for customer
 - c. Fitness for use (Lecture no 11 Page no 47 Handouts)**
 - d. Fitness for quality
79. Which one of the following techniques is used to link strategy and operations through the hierarchy?
- a. Management by practice
 - b. Management by objective (Lecture no 2 Page no 6 Handouts)**
 - c. Management by organizing
 - d. Management by task
80. For successful quality implementation it should start and end with the: **PPT 5**
- a. Supplier
 - b. Customer**
 - c. Technology
 - d. Process
81. Which one of the following quality leaders said that quality and profitability are associated together? PPT 11
- a. W. Edwards Deming**
 - b. Kaoru Ishikawa
 - c. Joseph M. Juran
 - d. Genichi Taguchi
82. How many steps are involved in Deming Chain Reaction?
- a. Four
 - b. Five
 - c. Six (Lecture no 11 Page no 48 Handouts)**
 - d. Seven
83. All of the following are the themes of new paradigm for managing organizations, EXCEPT: **Page 5**
- a. Customer value strategy
 - b. Mass production**
 - c. Cross functional systems
 - d. Continuous improvement
84. In the new paradigm, managers make strategic positioning decisions with a focus on _____.
- a. Warfare model on competition
 - b. Market segmentation (Lecture no 2 Page no 6 Handouts)**
 - c. Internal measures
 - d. None of the given options

85. Which of the following are the internal customers of the organization? **Page 50**
- a. Customers
 - b. Stakeholders
 - c. Employees**
 - d. Suppliers
86. Which of the following is an outcome considered by the managers in the old paradigm for making tradeoffs?
- a. Profit
 - b. Scheduling
 - c. Totality
 - d. Efficiency**
87. Which of the following is **NOT** an example of organization? **Page 1**
- a. Mosque
 - b. Colleague**
 - c. Hospital
 - d. None of the given options
 - e.

Ref: Hospitals, **Colleges**, Factories, Farms and Government offices. Mosque/Church is also an example of an organization.

88. Making openings for less aggressive members by directly asking their opinion or making a general request for input. This refers to which of the following discussion skills in meetings?
- a. Closing the discussion
 - b. Acting as gatekeepers**
 - c. Testing for consensus
 - d. Containing digression
89. In which of the following models, management coordinates and provides visible control?
- a. Organism model
 - b. Mechanistic model pg 40**
 - c. Cultural model
 - d. Total Quality model
90. Which of the following is a process for focusing on customer wants or needs in the design or redesign of a

product or service?

- a. Quality Function Deployment (Lecture no 11 Page no 46 Handouts)**
- b. New Product Development
- c. Supplier-Input-Process-Output-Customer
- d. Total Quality Management

91. Quality is a race without having any _____.

- a. Finish line PPT#4**
- b. Start line
- c. Boundary line
- d. Score line

92. All of the following are the tasks of senior executive leaders and leadership committee in the organization, EXCEPT:

- a. Participation in strategic projects
- b. Regular interaction with customers
- c. Manufacturing of products (Lecture no 5 Page no 20 Handouts)**
- d. Reinforce values and expectations

93. What is quality for any organization? **(Doubtful)**

- a. A solution**
- b. A problem
- c. A specification
- d. A project

94. In which century, the US broke further from the European tradition and adopted a new management approach developed by Frederick W. Taylor? **Page 42**

- a. 17th century
- b. 18th century
- c. 19th century**
- d. 20th century

95. A manager must perform all the management functions with which one of the following options? **Page 1**

- a. Managing
- b. Assurance**
- c. Condition
- d. Specialization

96. For becoming the International Standard Organization, company should meet the _____ requirement. **(Doubtful)**
- a. **Standard production**
 - b. Government/Market
 - c. Technical
 - d. Geographical
97. Which of the following models has a vertical chain of command? **Page 40**
- a. Organism model
 - b. **Mechanistic model**
 - c. Cultural model
 - d. Total Quality model
98. Which of the following models considers the goal of meeting individual needs?
- a. Organism model
 - b. Mechanistic model
 - c. **Cultural model pg 40**
 - d. Total Quality model
99. Which one of the following quality leaders said that quality is “fitness for use”?
- a. W. Edwards Deming
 - b. **Joseph M. Juran (Lecture no 11 Page no 49 Handouts)**
 - c. Philip B. Crosby
 - d. Ishikawa
100. Which one of the following depicts aesthetics, a dimension of quality?
- a. **Exterior finish**
 - b. Quality of work
 - c. Ranking first
 - d. Ease of repair
101. Which of the following is a benefit of ISO 9000? PPT 20
- a. **Reduction in on-site customer (second party) audits**
 - b. Inadequate Funding for establishing the quality system
 - c. Process that takes many months to complete
 - d. Increase in probability for external failure
102. Which of the following options is included in quality management principles?
Page no 82
- a. **Leadership**
 - b. Owner satisfaction
 - c. Prevention cost
 - d. Competitor satisfaction
103. What is the other name used for Deming Cycle? **Page 58**
- a. **Shewhart cycle**
 - b. Product life cycle
 - c. Data life cycle management
 - d. Information life cycle management

104. Which of the following statements is TRUE about Deming's philosophy? **Page 51**

- a. It is based on improving products and services by reducing uncertainty and variability in the design and manufacturing processes**
- b. Quality is either or not present in the whole organization; that quality is the responsibility of everyone in the organization
- c. Increasing conformance to specifications through elimination of defects, supported extensively by statistical tools for analysis
- d. Increasing loss, for the producer, the customer, and society, associated with increasing variability from a target value

105. Special causes result in _____ that disrupt the random pattern of common causes. **Page 53**

- a. Natural variation
- b. Common variation
- c. Unique variation
- d. Unnatural variation**

106. Theory establishes a _____ relationship that can be used for prediction. **Page 54**

- a. Common and special
- b. Cause and effect**
- c. Direct and indirect
- d. None of the given options

107. In which one of the following views, productivity and quality are always in conflict? **Page 11**

- a. Cultural view
- b. Traditional view**
- c. Quality control view
- d. Total quality view

108. Which of the following models has a goal of organizational survival? **Page 40**

- a. Organism model**
- b. Mechanistic model
- c. Cultural model
- d. Total Quality model

109. Which of the following is not included in benefits of the ISO 9000?

- a. Increased productivity
- b. Improved performance
- c. Provide confidence to internal customers**
- d. Improved customer satisfaction

110. Identify the example of internal failure costs. **Page 60**

- a. Quality planning
- b. Re-inspection**
- c. Cost to plan for quality
- d. Customer returns

111. Product based definition of quality best suits to which of the following options?

- a. Doctors
- b. Engineers Page 33**
- c. Teachers
- d. Students

112. Which one of the following is not in external failure costs?

- a. Customer returns
- b. Material review**
- c. Product recalls
- d. Warranty Claims

113. Which of the following is the key benefit of customer focus principle?

- a. Clear vision
- b. **Increased revenue**
- c. Setting challenging goals
- d. Motivated employees

114. Which of the following options is included in leadership triad of the criteria for performance excellence? **Page 77**

- a. Process management
- b. **Customer and market focus**
- c. Human resource focus
- d. None of the given options

115. According to Deming, which one of the following statements is false in case if people don't understand psychology? **Page 55**

- a. They don't understand motivation
- b. They are less likely to distinguish between fact and opinion
- c. They don't understand the process of change
- d. **The revert to coercive and paternalistic approaches**

116. All of the following options tell about the quality of product, EXCEPT: **PPT 7**

- a. Conformance to requirement
- b. Fast service
- c. **High price**
- d. Long life of product

117. Manufacturing based view of quality focuses on all of the following, EXCEPT **Page 33**

- a. Statistical analysis to measure quality
- b. **Measurement of capacity**
- c. Conformance to specified requirements
- d. Engineering and manufacturing practices

118. Quality loss function is developed by which one of the following quality gurus? **Page 36**

- a. W. Edwards Deming
- b. Kaoru Ishikawa
- c. Joseph M. Juran
- d. **Genichi Taguchi**

119. All of the following steps are required for quality planning EXCEPT: **Page 65**

- a. Determines who the customers are
- b. Deploy the plans to operational levels
- c. Identify customers needs
- d. **Assess actual quality performance**

120. Cause and effect diagram or fish bone diagram is developed by which one of the following quality gurus? Page 168

- a. W. Edwards Deming
- b. Kaoru Ishikawa
- c. Joseph M. Juran
- d. Philip B. Crosby

121. Which of the following models reflects horizontal processes beginning with suppliers and ending with customers? Page 40

- a. Organism model
- b. Mechanistic model
- c. Cultural model
- d. Total Quality model

122. Quality assurance is _____ system and quality control is _____ system.

- a. Corrective, Preventive
- b. Preventive, Corrective
- c. Structural, Unstructured
- d. Unstructured, Structural

123. Which one of the following was the primary concern during statistical quality control era? Page 41

- a. Detection
- b. Control
- c. Coordination
- d. Strategic impact

124. According to Deming, when people don't understand the system. Page 54

- a. They see events as individual
- b. They see symptoms but not the root causes of problems
- c. They blame individuals for problems
- d. All of the given

125. From system perspective, Deming said that most organizational processes are _____ PPT 13

- a. Functional
- b. Cross Functional
- c. Development
- d. Systematic

126. Which of the following factors has negligible impact on the product performance and can be ignored? Page 38

- a. Heat
- b. Temperature
- c. Cost
- d. Noise

- a. Inspection
b. Statistical quality control
c. **Quality assurance**
d. Strategic total quality management

134. Which of the following statements is TRUE for ISO 9004? **Doubtful**

- a. Describes the principles of a quality management system and defines the terminology
- b. Describes the requirements relative to a quality management system either for internal use or for contractual or certification purposes**
- c. Intended for internal use and not for contractual purposes, focuses particularly on continually improving performance
- d. Contains the guidelines for auditing quality management and/or environmental management systems

135. All of the following are the parts of system thinking, EXCEPT: **PPT 2**

- a. Input
- b. Process
- c. Customer**
- d. Output

136. ISO 9000 is a series of international standard and it deals with _____

- a. Product Standard
- b. Service Standard
- c. Quality System Standard
- d. Food Safety Standard

137. Which of the following are the dimensions of quality in service sector?
Page 15

- a. Time, Timeliness & Aesthetics
- b. Accessibility, Accuracy & Responsiveness**
- c. Completeness, Consistency & Durable
- d. Time, Consistency & Features

138. Based in Geneva, Switzerland, International Organization for Standardization is a consortium of the national standard bodies, which has how many countries? **PPT 19**

- a. More than 100**
- b. Less than 100
- c. Less than 90
- d. Less than 80

139. Quality of design focuses on which of the following options? **Page 64**

- a. Market research
- b. Product concept
- c. Design specification
- d. All of the given options**

140. Customer complaints and product recalls are the examples of which of the following costs? **Page 69**

- a. Internal failure cost
- b. Prevention cost
- c. External failure cost**
- d. Appraisal cost

d. Customer improvement

149. In EFQM model, which of the following are caused by 'Enablers'? **PPT 5**

- a. Profits b. Losses **c. Results** d. Strategies

150. The concept of quality has often been defined from a transcendent view as _____. **Page 32**

- a. **Innate excellence** b. Meet customer need
c. Exceed customer need d. Conformance to requirements

151. In the new paradigm, managers view error as which one of the following?

- a. A threat b. A substitute
c. An opportunity d. An end

152. Total quality management concept is applied in which of the following departments of the organization?

- a. Finance b. Marketing
c. Operation **d. Every/All**

153. Quality must be measured by the interaction between which of the following participants'? **Page 34**

- a. The product itself b. The user and how he uses the product
c. Instructions for use **d. All of the given options**

154. In EFQM model, which of the following are improved using feedback from 'Results'? **Google**

- a. Societies **b. Enablers** c. Cultures d. Ethics

155. The quality initiatives focused initially on reducing defects and errors in products and services with the help of which of the following? **Page 12**

- a. **Measurement, statistics, and other problem solving tools**
b. Measurement, statistics, and cause and effect analysis
c. Statistics and cause and effect analysis only
d. Failure modes and cause and effect analysis only

156. Which of the following is true for subsystems? **PPT 2**

- a. These are not linked together as internal customers and suppliers
b. These are linked together as external customers and suppliers
c. These are linked together as external and internal customers and suppliers
d. These are linked together as internal customers and suppliers

157. In old paradigm of quality, organization structure was based on _____

Page 7

- a. Flat hierarchy
- b. Fewer levels of managers
- c. Specialization of task**
- d. None of the given options

158. Which one of the following models values stability? **Page 40**

- a. Organism model
- b. **Mechanistic model**
- c. Cultural model
- d. Total Quality model

159. Guidelines for quality management in projects are discussed in which one of the following ISO standards? **PPT 19**

- a. ISO-9003 (2004)
- b. ISO-2000
- c. ISO-10006 (2003)**
- d. ISO-22000

160. Juran is best known for which of the following contributions to quality philosophy? **Page 62**

- a. Three basic steps to progress
- b. Ten steps to quality improvements
- c. The Juran trilogy
- d. All of the given options**

161. Process audit and test of purchased material belong to which of the following types of quality costs? **Page 60**

- a. Internal failure cost
- b. Prevention cost
- c. External failure cost
- d. Appraisal cost**

162. Which one of the following themes refers to the combination of benefits derived from using a product or services? **Page 5**

- a. Customer buying power
- b. Continuous improvement
- c. Customer value strategy**
- d. Cross functional system

163. "Trying to focus the group discussion on topics of personal interests rather than the group", depicts which of the following self-oriented behaviors? **Page 28**

- a. Diverting**
- b. Excluding
- c. Belittling
- d. Blocking

164. Which of the following is not included in internal failure? **Page 60**

- a. Cost of rework
- b. Downtime
- c. Retesting Cost
- d. Product Recalls**

165. Which of the following is a requirement of ISO 9000 2000? **PPT 21**

- a. **Processes to measure and analyze customer satisfaction**
- b. An Environmental Policy to be in existence within the organization
- c. Procedures for registration of font-related identifiers
- d. Irregular improvement of internal process and customer satisfaction

166. Juran's prescriptions focus on three major aspects of quality that is called: **Page 63**

- a. System of profound knowledge
- b. Quality of design
- c. **Quality trilogy**
- d. Quality process

167. Total quality views the enterprise as a system of which of the following options? **Page 21**

- a. **Independent processes**
- b. Dependent processes
- c. Managing processes
- d. Strategic processes

168. How the requirements for documentation and records may not arise? **Page 82**

- a. Contractual requirements from the customer or other interested parties
- b. Acceptance of international, national, regional and industry sector standards
- c. **Irrelevant statutory and regulatory requirements**
- d. Decisions by the organization

169. People, technology and capital come under which of the following sections of the organizational system **PPT 2**

- a. **Input**
- b. Process
- c. Output
- d. None of the given options

170. All of the following are the examples of failure costs EXCEPT: **Page 60**

- a. Processing customer complaints
- b. Customer returns
- c. Warranty claims
- d. **Quality planning**

171. Which of the following is NOT the evaluation dimension for the Baldrige Criteria Framework? **Page 79**

- a. Approach
- b. **Unemployment**
- c. Deployment
- d. Results

172. Who is known as a father of modern-day statistical quality control? **Page 32**

- a. **Shewhart**
- b. Deming
- c. Juran
- d. Crosby

173. Which countries are encouraging the publication of quality books in their native language to make them more accessible? **Page 12**

- a. **Spain and Brazil**
- b. China and Korea
- c. India and China
- d. Canada and America

174. Organizations need to do which of the following options in order to beat the competition? **Page 13**

- a. **Exceed customer expectation**
- b. Satisfy customer need
- c. Satisfy customer demands
- d. Meet customer expectation

175. All of the following are included in quality control, EXCEPT: **Page 65**

- a. Establishing standards
- b. **Identify need of the customers**
- c. Taking action on the difference
- d. Measuring actual performance

176. According to Ishikawa, which of the following options is not the broader interpretation of quality? **Page 71**

- a. Quality of process
- b. **Quality of product**
- c. Quality of service
- d. Quality of information

177. All of the following are the examples of appraisal costs EXCEPT: **Page 60**

- a. Quality measurement criteria
- b. **Quality improvement projects**
- c. Laboratory acceptance testing
- d. Inspection and testing

178. Who emphasized that there is no knowledge without theory and that experience alone does not establish a theory? **Page 54**

- a. Shewhart
- b. Deming
- c. Juran
- d. Crosby

179. Which of the following is not an example of assignable causes in the production process? **Page 53**

- a. Bad batch raw material
- b. Excessive tool wear
- c. **Machine vibration**
- d. Poorly trained operator

180. Which of the following helps to identify the process outputs and the customers of those outputs so that the voice of the customer can be captured? **Page 48**

- a. Fishbone diagram
- b. **SIPOC diagram**
- c. Matrix diagram
- d. Relations diagram

181. Whose success is due primarily to continuous, small, incremental improvements in design and production as noted by Deming?

- a. Americans manufacturers
- b. Chinese manufacturers
- c. Russian manufacturers
- d. Japanese manufacturers**

182. During which century, the manufacturers began to include quality processes in quality practices?

- a. 17th century
- b. 18th century
- c. 19th century
- d. 20th century pg 43**

183. "Conformance to specification" refers to which of the following definitions of quality?

- a. User based
- b. Product based
- c. Manufacturing based pg13**
- d. Value based

184. Which one of the following shows that product is staying with you for a lifetime?

- a. Reliability
- b. **Durability pg 13**
- c. Serviceability
- d. Repair ability

185. In which of the following fields, Deming got his training?

- a. Mathematics
- b. **Statistics pg 37**
- c. Philosophy
- d. Physics

186. Some organizations emphasizes the use of quality tools but failed to do which of the following?

- a. Focus on what is truly important to the distributors
- b. Incorporate continuous improvement efforts pg 36**
- c. Make fundamental changes in their processes and culture
- d. Focus on what is truly important to the customer

187. Which of the following is a unit in which two or more persons are working together for some purpose?

- a. Organization pg 1**
- b. Industry
- c. Process
- d. Commerce

188. Total quality organizations expect to make at least which of the following improvements?

- a. 5 or more improvements per employee per year
- b. 10 or more improvements per employee per year pg 11**
- c. 40 or more improvements per employee per year
- d. 60 or more improvements per employee per year

189. Which of the following theories states that improvements in quality lead to lower cost because of less rework and better use of material?

- a. Theory X
- b. **Deming chain reaction pg 47**
- c. Theory Y
- d. None of the given options

190. Quality applies to which aspect(s) of the organization?

- a. Products only
- b. Products and services
- c. People and product
- d. **Products, services, and people**

191. In traditional management, motivation is often analogous to which of the following models?

- a. **Theory X pg 22**
- b. Theory Y
- c. Hierarchy of needs
- d. Statistical theory

192. All of the following are the main pillars of management, EXCEPT: **(POLCA)**

- a. Planning
- b. **Organization**
- c. Controlling
- d. Leading

193. Which of the following options refers to the transcendent definition of quality?

- a. Conformance to specifications
- b. **Level of excellence pg 32**
- c. Quality vs. price
- d. Fitness for intended use

194. Total quality approach focuses on long term profits and _____.

- a. Incremental improvement
- b. **Continuous improvement pg 11**
- c. Competitor improvement
- d. Customer improvement

195. Which one of the following factors can't change the personal attitude?

- a. Actions
- b. **Abilities PPT#8**
- c. Awards
- d. Awareness

196. It is compulsory for everyone to _____ ones knowledge otherwise knowledge will be

- a. Remind
- b. **Update**
- c. Remember
- d. Memorize

197. The majority of advertisers appeal the public on the basis of which of the following?

- a. **Quality of product pg 32**
- b. Quality of staff
- c. Inferiority of product
- d. Inferiority of service

198. Which of the following refers to SIPOC (Supplier-Input- Process-Output-Customer) diagram?

- a. **A high-level process map pg 48**
- b. A low-level process map
- c. A high-level performer map
- d. A low-level performer map

199. If organization wants to see better performance in the organization, it will be on the basis of _____

- a. Technology **b. People** c. Capital d. Material

220. Which of the following is not the step of Deming Chain Reaction?

- a. Improve quality b. Stay in business
c. Job promotion 48 d. Productivity improves

221. In RATER, which of the following is the measuring of responsiveness?

- a. All items are placed in order
b. Employees greet each customer with politeness
c. Service personnel react quickly and resolve problems pg 15
d. Service is easy to obtain

222. The quality loss function allows financial decisions to be made at which of the following stages?

- a. Design stage pg 37** b. Product stage
c. Manufacturing stage d. Service stage

223. Efficiency and short term probability are the goals of which one of the following models?

- a. Organic Model **b. Mechanistic Model pg 40**
c. Structural Model d. Unstructured Model

224. Which of the following refers to the gathering of two or more people who interact to achieve an expected outcome?

- a. Agreement **b. Meeting pg 27**
c. Digression d. None of the given options

225. Which of the following options reinforce values and expectations of company through performance reviews?

- a. Senior executive leaders b. Strategic owners
c. Leadership committee **d. All of the given options**

PRINCIPLES OF TOTAL QUALITY – Page 16

There probably are as many **different approaches to TQ** as there are businesses. However, most share some basic elements: (1) customer focus, (2) a process orientation, (3) continuous improvement and learning, (4) empowerment and teamwork, (5) management by fact, and (6) leadership and strategic planning.

Page no 82

ISO 9000:2000 made a giant leap in comparison, especially in the area of continual improvement, which has gone from receiving just cursory treatment to becoming a firm requirement. In addition, the standard now incorporates eight quality management principles that come directly from TQM. They are:

1. **Customer focus** – understanding customer's needs, striving to exceed their expectations.
2. **Leadership** – establishing direction, unity of purpose, and a supportive work environment.
3. **Involvement of people** – ensuring that all employees at all levels are able to fully use their abilities for the organization's benefit.
4. **Process approach** – recognizing that all work is done through processes, and managing them accordingly.
5. **System approach to management** – expands on the previous principle in that achieving any objective requires a system of interrelated processes.
6. **Continual improvement** – as a permanent organizational objective, recognizing and acting on the fact that no process is so good that further improvement is impossible.
7. **Factual approach to decision making** – acknowledging that sound decisions must be based on analysis of factual data and information.
8. **Mutually beneficial supplier relationships** – to take advantage of the synergy that can be found in such relationships

How is Total Quality Different? Page 3

What **distinguishes the total quality approach from traditional ways** of doing business can be found in how it is achieved. The distinctive characteristics of total quality are these:

1. customer focus (internal and external),
2. obsession with quality,
3. use of the scientific approach in decision making and problem solving,
4. long-term commitment,
5. teamwork,
6. employee involvement and empowerment,
7. continual process improvement and product improvement
8. bottom-up education and training, Education and training intensive
9. freedom through control,
10. unity of purpose,

all deliberately aimed at supporting the organizational strategy.

MANUFACTURED products have several quality dimensions including the following: **Page 13**

1. **Performance:** a product's primary operating characteristics
2. **Feature:** the bells and whistles of a product.
3. **Reliability:** the probability of a product's surviving over a specified period of time under stated conditions of use.
4. **Conformance:** the degree to which physical and performance characteristics of a product match pre-established standards.
5. **Durability:** the amount of use one gets from a product before it physically deteriorates or until replacement is preferable.
6. **Serviceability:** the ability to repair a product quickly and easily.
7. **Aesthetics:** how a product looks, feels, sounds, tastes, or smells.
8. **Perceived quality:** subjective assessment resulting from image, advertising, or brand names.

The most important dimensions of **SERVICE QUALITY** include the following; you may remember the most important ones by **RATER: Page 15**

- 1 **Reliability:** How much reliable is the service provider?
- 2 **Accessibility and convenience:** Is the service easy to obtain?
- 3 **Timeliness:** Will a service be performed when promised?
- 4 **Completeness:** Are all items in the order included?
- 5 **Consistency:** Are services delivered in the same fashion for every customer, and every time for the same customer?
- 6 **Tangibility:** After the service is over, is there anything to take home to remind the service experience?
- 7 **Empathy or Courtesy:** Do frontline employees greet each customer cheerfully?
- 8 **Responsiveness:** Can service personnel react quickly and resolve unexpected problems?

Theme 1: Customer Value Strategy Old Paradigm and New Paradigm Page 6

Customer value is defined as the combination of benefits derived from using a product (or service) and the sacrifices required of the customer.

Topics	Old Paradigm	New (Emerging) Paradigm
Quality	Meeting specifications, inspected into product, make tradeoffs among quality, cost, schedule	One component of customer value, managed into process, seek synergies among quality, cost, schedule.
Measurement	Internal measures of efficiency, productivity, costs, and profitability, not necessarily linked to customers	All measures linked to customer value
Positioning	Competition	Customer segments
Key stakeholder	Stockholder, boss	Customer (other stakeholders are beneficiaries)
Product design	Internal, sell what we can build	External, build what customers need

Theme 2: Organizational Systems Old Paradigm and New Paradigm page 7

Organizational systems are the means that provide customer value.

Topics	Old Paradigm	New (Emerging) Paradigm
Cross-functional approach	Negotiation across functional interfaces to obtain cooperation	Cross-functional systems defined, owned, and optimized
Technology	To deal with complexity, to eliminate people problems	To reduce complexity, source of optimization for customer value
Employee involvement	Focused on hygiene factors	Focused on strategic factors
Human resource management	Regarded as a staff responsibility, administration of personnel hiring, firing, handling complaints	Regarded as a critical resource, managed as system input
Role definition	Task and job descriptions set limits	Vision inspires flexibility
Culture	Social and emotional issues are suppressed, politics and power dominate	Connect with individual sense of purpose, emotions, and social meaning
Structure	Specialization, tall hierarchy with functional emphasis	Integration, flat hierarchy with team emphasis

Theme 3: Continuous Improvement **Old Paradigm and New Paradigm page 8**

To keep pace with the changes in the external environment, managers have to change the organization.

Topics	Old Paradigm	New (Emerging) Paradigm
Occasion	Focused new product development, episodic, reactive to problems, big breakthroughs only	Focused on broader systems, unending, proactive to opportunities, big breakthroughs and small steps
Approach	Trial and error	Scientific method
Response to error	Punish, fear, cover-up, seek people fix, employees are responsible	Learning, openness, seek process/system fix, management is responsible
Decision-making perspective	Individual political expediency, short term	Strategic, long-term, purposeful for organization
Managerial roles	Administer and maintain status quo, control other	Challenge status quo, prompt strategic improvement
Authority	Top-driven via rules and policies	Customer-driven through vision, enablement, and empowerment
Focus	Business results through quotas and targets	Business results through capable systems, means tied to results
Control	Scoring, reporting evaluating	Statistical study of variation to understand causes
Means	Delegated by managers to staff and subordinates	Owned by managers who lead staff and subordinates

The **Deming Chain Reaction** page 48

1. Improve Quality
2. Costs decrease because of less rework, fewer mistakes, fewer delays and snags, and better use of time and materials,
3. Productivity improves
4. Capture the market with better quality and lower price
5. Stay in business
6. Provide jobs and more jobs

Organizational Models for Management Page 40

Dimension	Mechanistic Model	Organism Model	TQ Model	Cultural Model
Goal	Organizational efficiency and performance	Organizational survival	Long-term survival	Meet individual needs; human development
Definition of quality	Conformance to standards	Customer satisfaction	Satisfying or delighting the customer	Constituent satisfaction
Role/nature of environment	Objective; outside boundary	Objective; inside boundary	Blurred organization and environmental boundaries	Enacted/boundaries defined through relationships
Role of management	Coordinate and provide visible control	Coordinate and provide invisible control by creating vision	Focus on improvement and creating a system that can produce quality outcomes	Coordinate and mediate negotiations
Structural rationality	Chain of command (vertical) Technical rationality	Process flow (horizontal and vertical) Organizational rationality	Horizontal processes beginning with suppliers and ending with customers and supported by teams	Mutual adjustment in any direction Political rationality
Philosophy toward change	Stability is valued; learning arises from specialization	Change and learning assist adaptation	Change, continuous improvement, and learning are encouraged	Change and learning are valued in themselves
Dimension	Mechanistic Model	Organism Model	TQ Model	Cultural Model

The Four Major Quality Eras Page 41

Identifying Characteristics/ Date of inception	Inspection (1800s)	Statistical Quality Control (1930s)	Quality Assurance (1950s)	Strategic Total Quality Management (1980s)
Primary concern	Detection	Control	Coordination	Strategic impact
View of quality	A problem to be solved	A problem to be solved	A problem to be solved, but one that is attacked proactively	A competitive opportunity
Emphasis	Product uniformity	Product uniformity with reduced inspection	The entire production chain, from design to market, and the contribution of all functional groups, especially designers, to preventing quality failures	The market and consumer needs
Methods	Gauging and measurement	Statistical tools and techniques	Programs and systems	Strategic planning, goal-setting, and mobilizing the organization
Role of quality professionals	Inspection, sorting, counting, and grading	Troubleshooting and the application of statistical methods	Quality measurement, quality planning, and program design	Goal-setting, education and training, consultative work with other departments, and program design
Who has responsibility for quality	The inspection department	The manufacturing and engineering departments	All department, although top management is only peripherally involved in designing, planning, and executing quality policies	Everyone in the organization, with top management exercising strong leadership
Orientation and approach	Inspects in quality	Controls in quality	Builds in quality	Manages in quality
Identifying Characteristics/ Date of inception	Inspection (1800s)	Statistical Quality Control (1930s)	Quality Assurance (1950s)	Strategic Total Quality Management (1980s)

WTO (in 20XX) The Shifting Focus of Corporate Culture

From	To
Thinking Global	Being Global
Focus on Local Market	Focus on Global Market
MNC in manufacturing	MNC in Services
Antagonistic towards Rivals	Collaboration with Rivals
Work with Hands	Work with 3H & H3
Upgrade Technology	Upgrade People
Focus on Process Quality	Focus on Design Quality
Mass Mindset	Lean and Me Mindset
Manage by Control	Manage by Commitment
Individual Values	Shared Values

OVERVIEW OF QUALITY MANAGEMENT

PROFESSIONAL MANAGERIAL ERA (1950)

In our present age of market driven capitalism and futuristic knowledge driven economic markets, the decision are made and the trends are set by the professional managers. Unlike their predecessors, the captains of today's business do not own their own companies. They must know the whole business but have control over only one small part. They must be product oriented, process conscious, financially responsible, and public spirited. They must be all things to all people, yet still function as only one cog in the wheel.

If the history of management tells us anything, it is that, no matter what happens; peace or war, prosperity or famine, this world will always be in need of good managers . . . the kind who can get society from “where it is” to “where it wants to be.” Can you be one?

WHAT IS AN ORGANIZATION?

“An entity where two or more persons work together to achieve a goal or a common purpose is called Organization.”

There are so many organizations around us. Daily we visit and see many organizations. Hospitals, Colleges, Factories, Farms and Government offices. Mosque/Church is also an example of an organization. People go there and say prayers. Activities of praying are to achieve a certain goal. Similarly, any unit in which two or more persons are working together for some purpose is called an organization.

Unit of Organization:

- **People**
- **Purpose**
- **Process**
- **POLCA**

If there is an organization, then there must be some people. They work as whole for a common purpose, so there must be a defined purpose. If an organization doesn't have any purpose, it will not survive for long run. To achieve the purposes by using people, the processes are needed. Without any process, you cannot achieve any type of purpose or goal. If we see in our daily life, we have some goals. For achieving these goals, we use some processes. So that process is also obvious and important for an organization. The last important thing for any organization is that it requires main pillars of management i.e. **POLCA:**

- **Planning**
- **Organizing**
- **Leading**
- **Controlling**

A manager must perform all theses management functions with **Assurance!**

QUALITY FOCUS APPROACH TO MANAGEMENT

“There are really only three types of people: those who make things happen, those who watch things happen, and those who say, ‘What happened?’”

The total quality concept as an approach to doing business began to gain wide acceptance in the west in the late 1980s and early 1990s. However, individual elements of the concept – such as the use of statistical data, teamwork, continual improvement, customer satisfaction, and employee involvement – have been used by visionary organizations for years. It is the pulling together and coordinated use of these and other previously disparate elements that gave birth to the comprehensive concept known as total quality.

Why Focus on Quality?

To understand total quality, one must first understand quality. Customers of businesses will define quality very clearly using specifications, standards, and other measures. This makes the point that quality can be defined and measured. Although few consumers could define quality if asked, all know it when they see it. This makes the critical point that quality is in the eye of the beholder. With the total quality approach, customers ultimately define quality.

People deal with the issue of quality continually in their daily lives. We concern ourselves with quality when grocery shopping, eating in a restaurant, and making a major purchase such as an automobile, a home, a television, or a personal computer. Perceived quality is a major factor by which people make distinctions in the market place. Whether we articulate them openly or keep them in the back of our minds. We all apply a number of criteria when making a purchase. The extent to which a purchase meets these criteria determines its quality in our eyes.

One way to understand quality as a consumer-driven concept is to consider the example of eating at a restaurant. How will you judge the quality of the restaurant? Most people apply such criteria as the following:

- ◆ Service
- ◆ Response time
- ◆ Food preparation
- ◆ Environment/atmosphere
- ◆ Price
- ◆ Selection

The example gets at one aspect of quality the results aspect. Does the product or service meet or exceed customer expectations? This is a critical aspect of quality, but it is not the only one. Total quality is a much broader concept that encompasses not just the results aspect but also the quality of people and the quality of processes.

How Is Total Quality Different?

What distinguishes the total quality approach from traditional ways of doing business can be found in how it is achieved. The distinctive characteristics of total quality are these: customer focus (internal and external), obsession with quality, use of the scientific approach in decision making and problem solving, long-term commitment, teamwork, employee involvement and empowerment, continual process improvement, bottom-up education and training, freedom through control, and unity of purpose, all

deliberately aimed at supporting the organizational strategy. Each of these characteristics is explained later in this chapter.

The Historic Development of Total Quality

The total quality movement had its roots in the time and motion studies conducted by Frederick Taylor in the 1920s. Taylor is now known as “the father of scientific management.”

The most fundamental aspect of scientific management was the separation of planning and execution. Although the division of labor spawned tremendous leaps forward in productivity, it virtually eliminated the old concept of craftsmanship in which one highly skilled individual performed all the tasks required to produce a quality product. In a sense, a craftsman was CEO, production worker, and quality controller all rolled into one person. Taylor’s scientific management did away with this by making planning the job of management and production the job of labor. To keep quality from falling through the cracks, it was necessary to create a separate quality department. Such departments had shaky beginnings, and just who was responsible for quality became a clouded issue.

As the volume and complexity of manufacturing grew, quality became an increasingly difficult issue. Volume and complexity together gave birth to quality engineering in the 1920s and reliability engineering in the 1950s. Quality engineering, in turn, resulted in the use of statistical methods in the control of quality, which eventually led to the concepts of control charts and statistical process control, which are now fundamental aspects of the total quality approach.

Lesson # 02

TOTAL QUALITY MANAGEMENT AND TOTAL ORGANIZATION EXCELLENCE

TRADITIONAL MANAGEMENT AND QUALITY MANAGEMENT:

Twentieth-century management has been strongly influenced by Taylor's scientific management and Weber's theory of bureaucracy. These approaches have led managers to work within functional hierarchies, with their responsibilities divided according to specialized activities, such as accounting, marketing, engineering, and manufacturing. Economic principles for competing in well-defined markets emphasized economies of scale, efficiencies, mass production, and technological innovation.

While there had been competition, competitors often played according to a "live and let live" strategy. Because monopolies were precluded by law, companies had little incentive to completely drive competitors from the marketplace.

Even when new product technologies created new markets, such as plastics in the 1950s, management practices changed very little. Managers set goals for productivity, efficiency, and profitability, using management by objectives (MBO) to link strategy and operations through the hierarchy. Managers motivated employees to fulfill those goals by inducements such as profit sharing, stock options, and bonuses, or other rewards such as job enrichment or participative management. However, the job of management remained much the same: set goals, define roles, provide technologies, and motivate employees. Accounting, marketing, engineering, and manufacturing practices also did not change. Occasionally, new techniques were introduced within the traditional functions, such as quality control in manufacturing. But such changes went largely unnoticed by the rest of the organization.

No one challenged this approach to management as long as it served society well. While managers in Japan were rewriting the rules of business practice and management and planning to win the world markets by focusing on quality management, U.S. managers continued in the stage of normality. The US, and western society in general, was focused on another agenda: the cold war in the 1950s and 1960s.

THE STAGE OF REPLACEMENT: A NEW PARADIGM

The stage of replacement means shifting to a new paradigm. Managers must shift to a new paradigm for managing organizations because of the anomalies that threaten their survival and prosperity. To make this shift, however, they must understand the new paradigm and how it differs from the old paradigm. To initiate this understanding, we will contrast the new (but still emerging) paradigm with the old paradigm.

Themes of the New Paradigm

The differences between the new and the old paradigm are organized around three themes: customer value strategy, cross-functional systems, and continuous improvement.

Theme 1: Customer Value Strategy

Customer value is defined as the combination of benefits derived from using a product (or service) and the sacrifices required of the customer. The customer value strategy is the business plan for offering value to customers, including product characteristics, attributes, mode of delivery, support services, and so on.

The theme of customer value strategy may be addressed in many topics, including quality, measurement, positioning, key stakeholder, and product design.

Topics	Old Paradigm	New (emerging) paradigm
Quality	Meeting specifications, inspected into product, make tradeoffs among quality, cost, schedule	One component of customer value, managed into process, seek synergies among quality, cost, schedule.
Measurement	Internal measures of efficiency, productivity, costs, and profitability, not necessarily linked to customers	All measures linked to customer value
Positioning	Competition	Customer segments
Key stakeholder	Stockholder, boss	Customer segments Customer (other stakeholders are beneficiaries)
Product design	Internal, sell what we can build	External, build what customers need

Quality

In the old paradigm, managers define quality in terms of meeting specifications. Quality is assured by weeding out the “bad” products before they are shipped to customers. Managers make tradeoffs among quality, cost, and scheduling under the assumption that relationships among these outcomes are fixed. By contrast, in the new paradigm, managers recognize that product quality is only one component of customer value, and managers seek synergies among quality, cost, and schedule, not just tradeoffs. For example, improving quality by reducing variation in outputs reduces defects, reduces costs, and makes performance to schedule more predictable. Further, quality is more broadly defined than just product quality. Quality applies to every aspect of the organization. It must be managed into processes and systems, and not just inspected into products. **System thinking included to think of all interdependent parts of the system into one whole.**

Measurement

In the old paradigm, measurement systems are focused on internal measures of efficiency, productivity, costs, and profitability. This is the tradition of management by objectives. Managers do not necessarily understand how these internally focused measures are related to customer value. In the new paradigm, managers may use internally focused measures, but they are linked to customer value in a broader measurement system. Managers interpret measures in terms of the impact on customer value in the long term and short term.

Positioning

In the old paradigm, managers make strategic positioning decisions based primarily on warfare models on the competition. In the new paradigm, managers make **strategic positioning decisions with a focus on market segmentation and customer needs, wants and demands.**

Key Stakeholder

In the old paradigm, the key external stakeholder is the stockholder, and the key internal stakeholder is one's boss. All other stakeholders, such as customers, employees, suppliers, and business partners, are pawns to serve the goals of the key stakeholders. In the new paradigm, the key stakeholders are

customers, both internal and external customers. Providing value to customers is the key to serving all other stakeholders over the long term.

Product Design

In the old paradigm, the product design process is internally driven, based on the assumption that “we know what is best for the customer.” Managers enact a “push” strategy that aims to “sell what we can build.” In the new paradigm, managers develop products after first determining what customers need. Managers both react to improve products in existing markets and actively seek to create new markets with new products.

Theme 2: Organizational Systems

Organizational systems are the means that provide customer value. These systems broadly include material and human inputs, process technology, operating methods and work practices, streams of work activity, information flows, and decision making. The approaches to managing these systems in the old and new paradigms are discussed below.

Topics	Old Paradigm	New (emerging) paradigm
Cross-functional approach	Negotiation across functional interfaces to obtain cooperation	Cross-functional systems defined, owned, and optimized
Technology	To deal with complexity, to eliminate people problems	To reduce complexity, source of optimization for customer value
Employee involvement	Focused on hygiene factors	Focused on strategic factors
Human resource management	Regarded as a staff responsibility, administration of personnel hiring, firing, handling complaints	Regarded as a critical resource, managed as system input
Role definition	Task and job descriptions set limits	Vision inspires flexibility
Culture	Social and emotional issues are suppressed, politics and power dominate	Connect with individual sense of purpose, emotions, and social meaning
Structure	Specialization, tall hierarchy with functional emphasis	Integration, flat hierarchy with team emphasis

Cross-Functional Approach

The old paradigm does not acknowledge systems that cut across functional or unit boundaries. Managers simply negotiate across functional interfaces to obtain minimal cooperation. In the new paradigm, managers define, own, and optimize cross-functional systems for customer value.

Technology

In the old paradigm, managers use technology to help them deal with the overly complex systems that have grown up in the organization. Also, they use technology to eliminate people problems (robots don't talk back). In the new paradigm, managers prefer to eliminate complexity rather than automate it or computerize it. Managers use technology only to optimize systems for customer value.

Employee Involvement

In the old paradigm, employee involvement programs are implemented without a focus to contribute to systems. Employee involvement in improvement programs tends to focus on quality of work life issues and some limited operational changes. In the new paradigm, employee involvement is strategically focused and contributes to system purposes.

Human Resource Management

In the old paradigm, managers regard human resource management (HRM) as a staff responsibility. HR specialists process paperwork to hire and fire, and handle personnel complaints. In the new paradigm, line managers regard human resources as critical resources and strategically manage them as inputs to systems.

Role Definition

In the old paradigm, managers use task and job descriptions to prescribe and set limits to personal responsibilities. In the new paradigm, managers convey a vision to lead and inspire flexibility. Employees participate in any activities required to provide superior value to customers.

Culture

In the old paradigm, managers suppress social and emotional issues that are regarded as irrational and sources of distraction away from goals and objectives. Power and politics dominate the culture, with individuals jockeying for personal gain. In the new paradigm, managers connect organizational mission and purpose with each individual's sense of purpose, emotions, and social meaning. Individuals channel their needs for pride in workmanship toward strategic purposes.

Structure

In the old paradigm, organizational structure is based on specialization of tasks. The hierarchy is tall, with many levels of managers, and it emphasizes functional lines of authority. In the new paradigm, the hierarchy is flat, with fewer levels of managers, and it emphasizes teamwork to serve super ordinate objectives.

Theme 3: Continuous Improvement

To keep pace with the changes in the external environment, managers have to change the organization. Managers have always made improvements. However, with rates of change increasing in the external environment, managers must improve differently and more frequently than in the past. They must pursue continuous improvement, which is a constant striving to change and make things better.

Topics	Old Paradigm	New (emerging paradigm)
Occasion	Focused new product development, episodic, reactive to problems, big breakthroughs only	Focused on broader systems, unending, proactive to opportunities, big breakthroughs and small steps
Approach	Trial and error	Scientific method
Response to error	Punish, fear, cover-up, seek people fix, employees are responsible	Learning, openness, seek process/system fix, management is responsible
Decision-making perspective	Individual political expediency, short term	Strategic, long-term, purposeful for organization
Managerial roles	Administer and maintain status quo, control other	Challenge status quo, prompt strategic improvement
Authority	Top-driven via rules and policies	Customer-driven through vision, enablement, and empowerment
Focus	Business results through quotas and targets	Business results through capable systems, means tied to results
Control	Scoring, reporting evaluating	Statistical study of variation to understand causes
Means	Delegated by managers to staff and subordinates	Owned by managers who lead staff and subordinates

Occasion

In the old paradigm, the occasions for improvement were primarily new product development and reaction to salient problems. Managers mostly looked for big breakthroughs to get improvement. In the new paradigm, the occasions for improvement are everywhere, every day. Managers improve proactively at every opportunity, even in the absence of salient problems. Manager improve all aspects of the organization's systems through both big breakthroughs and small steps.

Approach

In the old paradigm, managers accomplish improvements through trial and error. In the new paradigm, managers use the scientific method to study proposed changes and their effects.

Response to Error

In the old paradigm, if they care at all, managers are intolerant of error. They regard error as a personal failure, and they respond with punishment to instill fear in those blamed. The result is fear and cover-up in the future. In the new paradigm, error is not desired; however, managers view error as an opportunity for learning. People openly acknowledge error because managers do not assign personal blame, but seek to fix a process or system.

Decision-Making Perspective

In the old paradigm, managers make decisions that are politically expedient or that serve short-term personal objectives. In the new paradigm, managers make decisions that serve long-term strategic purposes.

Managerial Roles

In the old paradigm, managers primarily administer existing systems and maintain the status quo. In the new paradigm, managers challenge the status quo for strategic improvement to meet future demands. At the same time, they consistently execute existing systems to meet current demands.

Authority

In the old paradigm, managers impose authority from the top down via rules and policies. In the new paradigm, top managers still hold authority but they impose it by communicating a vision, enabling people with systems, and empowering them to make the vision real.

Focus

In the old paradigm, managers focus on improving business results through the imposition of quotas and targets. They delegate responsibility, often without giving real authority to change broader systems that constrain results. In the new paradigm, managers focus on improving business results through improving the capabilities of systems. They focus on the means as well as the results, because they have retained responsibility for improving systems.

Control

In the old paradigm, managers control the organization through scoring individual performance, reviewing regular reports, and evaluating performance as either good or bad. In the new paradigm, managers statistically study variation to understand the causes of poor performance and make changes in systems to improve performance.

Means

In the old paradigm, managers delegate the means of improvement to staff and subordinates who must figure out how to meet established targets. In the new paradigm, managers assume responsibility for the means of improvement. They lead improvement by staff and subordinates.

The Relationship between Quality and Competitiveness

At each successive level of competition the quality of the competitors increased. A similar phenomenon happens to businesses in the marketplace. Companies that used to compete only on a local, regional, or national level now find themselves competing against companies from throughout the world. Some of these companies find the competition to be more intense than any they have ever encountered. Only those who are able to produce world-class quality can compete at this level. In practical terms, it is extremely important for a country's businesses to be able to compete globally. When they can't, jobs are lost and the quality of life in that country declines correspondingly.

Lesson # 03

INTEGRATING PEOPLE AND PERFORMANCE THROUGH QUALITY MANAGEMENT

VIEWS ON QUALITY

The total quality philosophy introduced a whole new way of looking at performance of a company through people and managing the quality of products and processes. The traditional view of quality measured process performance in defective parts per hundred produced. With total quality the same measurement is thought of 1 part per million. The traditional view focused on after-the-fact inspections of products. With total quality the emphasis is on continuous improvement of products, processes, and people in order to prevent problems before they occur. The traditional view of quality saw employees as passive workers who followed orders given by supervisors and managers. It was their labor, not their brains that was wanted. With total quality, employees are empowered to think and make recommendations for continual improvement. They are also shown the control boundaries within which they must work and are given freedom to make decisions within those boundaries.

The traditional view of quality expected one improvement per year per employee. Total quality organizations expect to make at least 10 or more improvements per employee per year. Organizations that think traditionally focus on short-term profits. The total quality approach focuses on long-term profits and continual improvement.

The following statements summarize some of the major differences between the traditional view of quality and the total quality perspective:

- ◆ **Productivity versus quality.** The traditional view is that productivity and quality are always in conflict. You cannot have both. The total quality view is that lasting productivity gains are made only as a result of quality improvement.
- ◆ **How quality is defined.** The traditional view is that quality is defined solely as meeting customer specifications. The total quality view is that quality means satisfying customer needs and exceeding customer expectations.
- ◆ **How quality is measured.** The traditional view is that quality is measured by establishing an acceptable level of nonconformance and measuring against that benchmark. The total quality view is that quality is measured by establishing high-performance benchmarks for customer satisfaction and then continually improving performance.
- ◆ **How quality is achieved.** The traditional view is that quality is inspected into the product. The total quality view is that quality is determined by product design and achieved by effective control techniques.
- ◆ **Attitude toward defects.** The traditional view is that defects are an expected part of producing a product. Measuring defects per hundred is an acceptable standard. The total quality view is that defects are to be prevented using effective control systems and should be measured in defects per million (Six Sigma)

- ◆ **Quality as a function.** The traditional view is that quality is a separate function. The total quality view is that quality should be fully integrated throughout the organization – it should be everybody's responsibility.
- ◆ **Responsibility for quality.** The traditional view is that employees are blamed for quality. The total quality view is that 80 percent of quality problems are management's fault.
- ◆ **Supplier relationships.** The traditional view is that supplier relationships are short term and cost driven. The total quality view is that supplier relationships are long term and quality oriented.

Lesson #04

FUNDAMENTALS OF TOTAL QUALITY AND RATERS VIEW

Total quality—a comprehensive, organization-wide effort to improve the quality of products and service—applies not only to large manufacturers, but to small companies alike. All organization—large and small, manufacturing and service, profit and not-for-profit—can benefit from applying the principles of total quality.

During the 1990s, health care, government, and education began to pay increased attention to quality. As more public and government attention focuses on the nation's health care system, its providers turn toward quality as a means of achieving better performance and lower costs. One hospital, for example, lowered its rate of post-surgical infections to less than one-fifth of the acceptable national norms through the use of quality tools.

Although quality initiatives focused initially on reducing defects and errors in products and services through the use of measurement, statistics, and other problem-solving tools, organizations began to recognize that lasting improvement could not be accomplished without significant attention to the quality of the management practices used on a daily basis. Managers began to realize that the approaches they use to listen to customers and develop long-term relationships, develop strategy, measure performance and analyze data, reward and train employees, design and deliver products and services, and act as leaders in their organizations are the true enablers of quality, customer satisfaction, and business results. In other words, they recognized that the “quality of management” is as important as the “management of quality.” As organizations began to integrate quality principles into their management systems, the notion of total quality management, or TQM, became popular. Quality took on a new meaning of organization-wide performance excellence rather than an engineering-based technical discipline.

As quality principles have matured in organizations, attention to quality as “something new” has faded, and the term “total quality management (TQM),” which was popular throughout the 1980s and early 1990s has all but fallen out of the business vernacular. Critics suggested that “TQM is as dead as a pet rock” (Business Week, June 23, 1997, p. 47). Perhaps it is unfortunate that a three-letter acronym was chosen to represent such a powerful management concept. It is equally unfortunate that people point to the demise of faddish terminology as a generalization of the concepts themselves. Reasons for failure of quality initiatives are rooted in organizational approaches and systems. As the editor of Quality Digest put it: “No, TQM isn't dead. TQM failures just prove that bad management is still alive and kicking.” The most successful organizations have found that the fundamental principles of total quality are essential to effective management practice, and continue to represent a sound approach for achieving business success.

The real challenge today is to ensure that managers do not lose sight of the basic principles on which quality management and performance excellence are based. The global marketplace and domestic and international competition has made organizations around the world realize that their survival depends on high quality. Many countries, such as Korea and China, are mounting

national efforts to increase quality awareness, including conferences, seminars, radio shows, school essay contests, and pamphlet distribution. Spain and Brazil are encouraging the publication of quality books in their native language to make them more accessible. These trends will only increase the level of competition in the future. Even the tools used to achieve quality a decade ago are no longer sufficient to achieve the performance levels necessary to compete in today's world. Many organizations are embracing highly sophisticated, statistically based tools as part of popular "Six Sigma" initiatives. These require increased levels of training and education for managers and frontline employees alike, as well as the development of technical staff. As Tom Engibous, president and chief executive officer of Texas Instruments commented on the present and future importance of quality in 1997: Quality will have to be everywhere, integrated into all aspects of a winning organization.

THE CONCEPT OF QUALITY

People define quality in many ways. Some think of quality as superiority or excellence, others view it as a lack of manufacturing or service defects, still others think of quality as related to product features or price. Followings are some of many ways to look at quality.

1. perfection
2. consistency
3. eliminating waste
4. speed of delivery
5. compliance with policies and procedures
6. providing a good, usable product
7. doing it right the first time
8. delighting or pleasing customers
9. total customers service and satisfaction⁶

Today most managers agree that the main reason to pursue quality is to satisfy customers. The American National Standards Institute (ANSI) and the American Society for Quality (ASQ) define quality as "the totality of features and characteristics of a product or service that bears on its ability to satisfy given needs." The view of quality as the satisfaction of customer needs is often called fitness for use. In highly competitive markets, merely satisfying customer needs will not achieve success. To beat the competition, organizations often must exceed customer expectations. Thus, one of the most popular definitions of quality is meeting or exceeding customer expectations.

QUALITY IN MANUFACTURING

Well-developed quality systems have existed in manufacturing for some time. However, these systems focused primarily on technical issues such as equipment reliability, inspections, defect measurement, and process control. The transition to a customer-driven organization has caused fundamental changes in manufacturing practices, changes that are particularly evident in areas such as product design, human resource management, and supplier relations. Product design activities, for example, now closely integrate marketing, engineering, and manufacturing operations. Human resource practices concentrate on empowering workers to collect and analyze

data, make critical operations decisions, and take responsibility for continuous improvements, thereby moving the responsibility for quality from the quality control department onto the factory floor. Suppliers have become partners in product design and manufacturing efforts. Many of these efforts were stimulated by the automobile industry, which forced their network of suppliers to improve quality.

Manufactured products have several quality dimensions including the following:

1. **Performance:** a product's primary operating characteristics.
2. **Feature:** the "bells and whistles" of a product.
3. **Reliability:** the probability of a product's surviving over a specified period of time under stated conditions of use.
4. **Conformance:** the degree to which physical and performance characteristics of a product match pre-established standards.
5. **Durability:** the amount of use one gets from a product before it physically deteriorates or until replacement is preferable.
6. **Serviceability:** the ability to repair a product quickly and easily.
7. **Aesthetics:** how a product looks, feels, sounds, tastes, or smells.
8. **Perceived quality:** subjective assessment resulting from image, advertising, or brand names.

Most of these dimensions revolve around the design of the product.

Quality control in manufacturing is usually based on conformance, specifically conformance to specifications. Specifications are targets and tolerances determined by designers of products and services. Targets are the ideal values for which production strives; tolerances are acceptable deviations from these ideal values. For example, a computer chip manufacturer might specify that the distance between pins on a computer chip should be 0.095 ± 0.005 inches. The value 0.090 and 0.100 would be acceptable.

QUALITY IN SERVICES

Service can be defined as "any primary or complementary activity that does not directly produce a physical product—that is, the non goods part of the transaction between buyer (customer) and seller (provider)." A service might be as simple as handling a complaint or as complex as approving a home mortgage. Service organizations include hotels; health, legal, engineering, and other professional services; educational institutions; financial services; retailers; transportation; and public utilities.

Today services account for nearly 80 percent of the U.S., Singapore and Sweden workforce. The importance of quality in services cannot be underestimated, as statistics from a variety of studies reveals:

- The average company never hears from more than 90 percent of its unhappy customers. For every complaint it receives, the company has at least 25 customers with problems, about one-fourth of which are serious.

- Of the customers who make a complaint, more than half will do business again with that organization if their complaint is resolved. If the customer feels that the complaint was resolved quickly, this figure jumps to about 95 percent.
- The average customer who has had a problem will tell nine or ten others about it. Customers who have had complaints resolved satisfactorily will only tell about five others.
- It costs six times more to get a new customer than to keep a current customer.

So why do many companies treat customers as commodities? In Japan the notion of customer is equated with “honored guest.” Service clearly should be at the forefront of a firm’s priorities.

The service sector began to recognize the importance of quality several years after manufacturing had done so. This can be attributed to the fact that service industries had not confronted the same aggressive foreign competition that faced manufacturing. Another factor is the high turnover rate in service industry jobs, which typically pay less than manufacturing jobs. Constantly changing personnel makes establishing a culture for continuous improvement more difficult.

The production of services differs from manufacturing in many ways, and these differences have important implications for managing quality. The most critical differences are:

1. Customer needs and performance standards are often difficult to identify and measure, primarily because the customers define what they are, and each customer is different.
2. The production of services typically requires a higher degree of customization than does manufacturing. Doctors, lawyers, insurance salespeople, and food-service employees must tailor their services to individual customers. In manufacturing, the goal is uniformity.
3. The output of many service systems is intangible, whereas manufacturing produces tangible, visible products. Manufacturing quality can be assessed against firm design specifications, but service quality can only be assessed against customers’ subjective, nebulous expectations and past experiences. Manufactured goods can be recalled or replaced by the manufacturer, but poor service can only be followed up by apologies and reparations.
4. Services are produced and consumed simultaneously, whereas manufactured goods are produced prior to consumption. In addition, many services must be performed at the convenience of the customer. Therefore, services cannot be stored, inventoried, or inspected prior to delivery as manufactured goods are. Much more attention must therefore be paid to training and building quality into the service as a means of quality assurance.
5. Customers often are involved in the service process and present while it is being performed, whereas manufacturing is performed away from the customer. For example, customers of a quick-service restaurant place their own orders, carry their food to the table, and are expected to clear the table when they have finished eating.
6. Services are generally labor intensive, whereas manufacturing is more capital intensive. The quality of human interaction is a vital factor for services that involve human contact. For example, the quality of hospital care depends heavily on interactions among the

patients, nurses, doctors, and other medical staff. Hence, the behavior and morale of service employees is critical in delivering a quality service experience.

7. Many service organizations must handle very large numbers of customer transactions. For example, on a given business day, the National Bank of Pakistan might process more than 5.5 million transactions for 7.5 million customer through 1,600 branches and more than 3,500 banking machines, and TCS or Fed Ex might handle more than 1.5 million shipments across the globe. Such large volumes increase the opportunity for error.

These differences have made it difficult for many service organizations to apply total quality principles.

Many service organization have well-developed quality assurance systems. Most of them, however, are based on manufacturing analogies and tend to be more product-oriented than service-oriented. Many of the key dimensions of product quality apply to services. For instance, “on time arrival” for an airline is a measure of service performance; frequent flyer awards and “business class” sections represent features. A typical hotel’s quality assurance systems focus on technical specifications such as properly made-up rooms. However, service organizations have special requirements that manufacturing systems cannot fulfill. The most important dimensions of service quality include the following; you may remember the most important ones by **RATER**:

- **Reliability**: How much reliable is the service provider?
- **Accessibility and convenience**: Is the service easy to obtain?
- **Timeliness**: Will a service be performed when promised?
- **Completeness**: Are all items in the order included?
- **Consistency**: Are services delivered in the same fashion for every customer, and every time for the same customer?
- **Tangibility**: after the service is over, is there any thing to take home to remind the service experience?
- **Empathy** or **Courtesy**: Do frontline employees greet each customer cheerfully?
- **Responsiveness**: Can service personnel react quickly and resolve unexpected problems?

Service organizations must look beyond product orientation and pay significant attention to customer transactions and employee behavior. Several points that service organizations should consider are as follows:

- The quality characteristics that a firm should control may not be the obvious ones. Customer perceptions are critical although it may be difficult to define what the customer wants. For example, speed of service is an important quality characteristic, yet perceptions of speed may differ significantly among different service organization and customers. Marketing and consumer research can play a significant role.
- Behavior is a quality characteristic. The quality of human interaction is vital in every transaction that involves human contact. For example, banks have found that the friendliness of tellers is a principal factor in retaining depositors.
- Image is a major factor in shaping customer expectations of a service and in setting standards by which customers evaluate that service. A breakdown in image can be as harmful as a

breakdown in delivery of the service itself. Top management is responsible for shaping and guiding the image that the firm projects.

- Establishing and measuring service levels may be difficult. Service standards, particularly those relating to human behavior, are often set judgmentally and are hard to measure. In manufacturing, it is easy to quantify output, scrap, and rework. Customer attitudes and employee competence are not as easily measured.
- Quality control activity may be required at times or in places where supervision and control personnel are not present. Often work must be performed at the convenience of the customer. This calls for more training of employees and self-management.

These issues suggest that the approach to managing quality in services differs from that used in manufacturing. However, manufacturing can be seen as a set of interrelated services, not only between the company and the ultimate consumer, but within the organization. Manufacturing is a customer of product design; assembly is a customer of manufacturing; sales are a customer of packaging and distribution. If quality is meeting and exceeding customer expectations, then manufacturing takes on a new meaning, far beyond product orientation. Total quality provides the umbrella under which everyone in the organization can strive to create customer satisfaction.

Quality in ICT Sector

Quality in IT and IS was taken seriously only after SW Engineering principles were established. SWE Institute at CM University developed the CMM levels to indicate the maturity levels of an organization taking care of RATER along with issues of Configuration Management, Verification and Validation issues along with Scalability and Reusability issues.

PRINCIPLES OF TOTAL QUALITY

A definition of total quality was endorsed in 1992 by the chairs and CEOs of nine major U.S. corporations in cooperation with deans of business and engineering departments of major universities, and recognized consultants:

Total Quality (TQ) is a people-focused management system that aims at continual increase in customer satisfaction at continually lower real cost. TQ is a total system approach (not a separate area or program) and an integral part of high-level strategy; it works horizontally across functions and departments, involves all employees, top to bottom, and extends backward and forward to include the supply chain and the customer chain. TQ stresses learning and adaptation to continual change as keys to organizational success.

The foundation of total quality is philosophical: TQ includes systems, methods, and tools. The systems permit change; the philosophy stays the same. TQ is anchored in values that stress the dignity of the individual and the power of community action.

There probably are as many different approaches to TQ as there are businesses. However, most share some basic elements: (1) customer focus, (2) a process orientation, (3) continuous

improvement and learning, (4) empowerment and teamwork, (5) management by fact, and (6) leadership and strategic planning.

Customer Focus

The customer is the judge of quality. Understanding customer needs, both current and future, and keeping pace with changing markets requires effective strategies for listening to and learning from customers, measuring their satisfaction relative to competitors, and building relationships. Customer needs—particularly differences among key customer groups – must be linked closely to an organization’s strategic planning, product design, process improvement, and workforce training activities. Satisfaction and dissatisfaction information are important because understanding them leads to the right improvements that can create satisfied customers who reward the company with loyalty, repeat business, and positive referrals. Creating satisfied customers includes prompt and effective response and solutions to their needs and desires as well as building and maintaining good relationships. A business can achieve success only by understanding and fulfilling the needs of customers. From a total quality perspective, all strategic decisions a company makes are “customer-driven.” In other words, the company shows constant sensitivity to emerging customer and market requirements. This requires an awareness of developments in technology and rapid and flexible response to customer and market needs.

Customer-driven firms measure the factors that drive customer satisfaction. A company close to its customer knows what the customer wants, how the customer uses its products, and anticipates the needs that the customer may not even be able to express. It also continually develops new techniques to obtain customer feedback. Customer opinion surveys and focus groups can help companies understand customer requirements and values. Some companies require their sales and marketing executives to meet with random group of key customers on a regular basis. Other companies bring customers and suppliers into internal product design and development meetings.

A firm also must recognize that internal customers—the recipients of any work output, such as the next department in a manufacturing process or the order-picker who receives instructions from an order entry clerk – are as important in assuring quality as are external customers who purchase the product. Failure to meet the needs of internal customers will likely affect external customers. Employees must view themselves as customers of some employees and suppliers to others. Employees who view themselves as both customers of and suppliers to other employees understand how their work links to the final product. After all, the responsibility of any supplier is to understand and meet customer requirements in the most efficient and effective manner possible.

Customer focus extends beyond the consumer and internal relationships, however. Society represents an important customer of business. A world-class company, by definition, is an exemplary corporate citizen. Business ethics, public health and safety measures, concern for the environment, and sharing quality-related information in the

company” business and geographic communities are required. In addition, company support—within reasonable limits of its resources—of national, industry, trade, and community activities and the sharing of nonproprietary quality-related information demonstrate far-reaching benefits.

Customers may be of following types:

1. External Customer
2. Internal Customer
3. Investor Customer
4. Social or Society Customer

Process Orientation

The traditional way of viewing an organization is by surveying the vertical dimension – by keeping an eye on an organization chart. However, work gets done (or fails to get done) horizontally or cross-functionally, not hierarchically. A process is a sequence of activities that is intended to achieve some result. According to AT&T, a process is how work creates value for customers. We typically think of processes in the context of production: the collection of activities and operations involved in transforming inputs—physical facilities, materials, capital, equipment, people, and energy—into **outputs—products and services**. Common types of production processes include machining, mixing, assembly, filling orders, or approving loans. However, nearly every major activity within an organization involves a process that crosses traditional organizational boundaries. For example, an order fulfillment process might involve a salesperson placing the order; a marketing representative entering it on the company’ computer system; a credit check by finance; picking, packaging, and shipping by distribution and logistics personnel; invoicing by finance; and installation by field service engineers. A process perspective links all necessary activities together and increases one’s understanding of the entire system, rather than focusing on only a small part. Many of the greatest opportunities for improving organizational performance lie in the organizational interfaces – those spaces between the boxes on an organization chart.

CONTINUOUS IMPROVEMENT AND LEARNING

Continuous improvement is part of the management of all systems and processes. Achieving the highest levels of performance requires a well-defined and well-executed approach to continuous improvement and learning. “Continuous improvement” refers to both incremental and “breakthrough” improvement. Improvement and learning need to be embedded in the way an organization operates. This means they should be a regular part of daily work, seek to eliminate problems at their source, and be driven by opportunities to do better as well as by problems that need to be corrected. Improvements may be of several types:

- Enhancing value to the customer through new and improved products and services;
- Improving productivity and operational performance through better work processes and reductions in errors, defects, and waste; and
- Improving flexibility, responsiveness, and cycle time performance.

MANAGEMENT BY FACT

Organizations need performance measures for three reasons:

- To lead the entire organization in a particular direction; that is, to drive strategies and organizational change,
- To manage the resources needed to travel in this direction by evaluating the effectiveness of action plans, and
- To operate the processes that make the organization work and continuously improve.²¹

Data and information support analysis at all organizational levels. The types of information and how it is disseminated and aligned with organizational levels are equally vital to success. At the work level, data provide real-time information to identify assignable reasons for variation, determine root causes, and take corrective action as needed. This might require lean communication a channel consisting of bulletins, computerized quality reports, and digital readouts of part dimensions to provide immediate information on what is happening and how things are progressing. At the process level, operational performance data such as yields, cycle times, and productivity measures help manager determine whether they are doing the right job, whether they are using resources effectively, and whether they are improving. Information at this level generally is aggregated; for example daily or weekly scrap reports, customer complaint data obtained from customer service representatives or monthly sales and cost figures faxed in from field offices. At the organization level, quality and operational performance data from all areas of the firm, along with relevant financial, market, human resources, and supplier data, form the basis for strategic planning and decision making. Such information is highly aggregated and obtained from many different sources throughout the organization.

A company should select performance measures and indicator that best represent the factor that lead to improved customer, operational, and financial performance. These typically include

- Customer satisfaction,
- Product and service performance,
- Market assessments,
- Competitive comparisons,
- Supplier performance,
- Employee performance, and
- Cost and financial performance.

A comprehensive set of measures and indicators tied to customer and company performance requirements provides a clear basis for aligning all activities of the company with its goals.

LEADERSHIP AND STRATEGIC PLANNING

Leadership for quality is the responsibility of top management. Senior leadership must set directions; create a customer orientation, clear quality values, and high expectations that address the needs of all stakeholders; and build them into the way the company operates. Senior leaders need to commit to the development of the entire workforce and should encourage participation, learning, innovation, and creativity throughout the organization. Reinforcement of the values and expectations requires the substantial personal commitment and involvement of senior management. Through their personal roles in planning, reviewing company quality performance, and recognizing employees for quality achievement, the senior leaders serve as role models, reinforcing the values and encouraging leadership throughout the organization.

If commitment to quality is not a priority, any initiative is doomed to failure. Lip service to quality improvement is the kiss of death. Many companies have a corporate quality council made up of top executives and managers, which sets quality policy and reviews performance goals within the company. Quality should be a major factor in strategic planning and competitive analysis processes.

Many of the management principles and practices required in a TQ environment may be contrary to long-standing practice. Top managers, ideally starting with the CEO, must be the organization's TQ leaders. The CEO should be the focal point providing broad perspectives and vision, encouragement, and recognition. The leader must be determined to establish TQ initiatives and committed to sustain TQ activities through daily actions in order to overcome employees' inevitable resistance to change.

Unfortunately, many organizations do not have the commitment and leadership of their top managers. This does not mean that these organizations cannot develop a quality focus. Improved quality can be fostered through the strong leadership of middle managers and the involvement of the workforce. In many cases, this is where quality begins. In the long run, however, an organization cannot sustain quality initiatives without strong top management leadership.

Achieving quality and market leadership requires a strong future orientation and a willingness to make long-term commitments to key stakeholders-customers, employees, suppliers, stockholders, the public, and the community. Strategic business planning should be the driver for quality excellence throughout the organization and needs to anticipate many changes, such as customer's expectations, new business and partnering opportunities, the global and electronic marketplaces, technological developments, new customer segments, evolving regulatory requirements, community / societal expectations, and strategic changes by competitors. Plans, strategies, and resource allocations need to reflect these influences. Improvements do not happen overnight. The success of market penetration by Japanese manufacturers evolved over several decades.

Lesson# 05

TOTAL QUALITY MANAGEMENT AND GLOBAL COMPETITIVE ADVANTAGE

Bringing TQ to Life at ABC Engineering Company:

ABC is a contract manufacturer of precision sheet metal and machined components for telecommunications semi-conductor, and medical equipment industries. Some of the ways it exemplifies the principles of TQ are described below.

Customer Focus. ABC made a strategic decision to carefully select customers that support its values—particularly a systematic approach to business and performance management, desire for long-term partnerships, and global leadership. Management and Team Leaders work with each customer to establish current requirements and future needs, and each customer is assigned a three-person Customer Service team that is on call 24 hours a day for day-to-day production issues.

Process Orientation. Processes such as prototype development, scheduling, production setup, fabrication, assembly, and delivery have process owners responsible for maintaining the process to customer requirements. A Quality Assurance team member works with manufacturing teams to create process documentation.

Continuous Improvement and Learning. Teams use a structured approach to evaluate and improve their processes, documenting them, and presenting a status report of improvements to senior leaders and the ABC Steering Committee. Teams benchmark competitors, “best practice” companies, and customers to learn from others.

Empowerment and Teamwork. Production and delivery processes are designed around cell manufacturing. Teams are responsible for knowing their customer’s requirements and producing according to those requirements. Teams are empowered to change targets recommended during strategic planning if they believe it will help them achieve higher performance, as well as to schedule work, manage inventory, and design the layout of their work areas.

Management by Fact. Team analyzes defect data, customer-reported problems, and control charts generated during production to identify problems and opportunities for improvement. Every business goal and project has defined methods for measurement, and senior leaders meet weekly to review company performance and ensure alignment with directions and plans.

Leadership and Strategic Planning. Senior Executive Leaders (SELs) and the Leadership Committee (LC) set the strategic direction of the company, and communicate and reinforce values and expectations through performance reviews, participation in improvement or strategic projects, regular interactions with customers and team members, and recognition of team member achievements.

All this has contributed to an annual average increase in sales growth of 35 percent from 1995 to 2000, and high levels of customer and employee satisfaction, and quality and operational performance.

TQM AND STRATEGIC FOCUS

The nature of TQ differs from common management practices in many respects.

1. Strategic Planning and Management

In traditional management, financial and marketing issues such as profitability, return on investments, and market share drive strategic planning. Quality planning activities are delegated to the “quality control” department. Long-term quality initiatives are viewed as being costly and not contributing to the ultimate performance measure – profit. Quality planning and strategic business planning are indistinguishable in TQ. Quality goals are the cornerstone of the business plan. Measures such as customer satisfaction, defect rates, and process cycle times receive as much attention in the strategic plan as financial and marketing objectives.

2. Changing Relationship with Customers and Supplier

In traditional management, quality is defined as adherence to internal specifications and standards. Quality is defined as adherence to internal specifications and standards. Quality is measured only by the absence of defects. Inspection of people’s work by others is necessary to control defects. In TQ, quality is defined as products and services beyond present needs and expectations of customers. Innovation is required to meet and exceed customers’ needs.

Traditional management places customers outside of the enterprise and within the domain of marketing and sales. TQ views everyone inside the enterprise as a customer of an internal or external supplier, and a supplier of an external or internal customer. Marketing concepts and tools can be used to assess internal customer needs and to communicate internal supplier capabilities.

3. Organizational Structure

Traditional management views an enterprise as a collection of separate, highly specialized individual performers and units, loosely linked by a functional hierarchy. Lateral connections are made by intermediaries close to the top of the organization. TQ views the enterprise as a system of interdependent processes, linked laterally over time through a network of collaborating (internal and external) suppliers and customers. Each process is connected to the enterprise’s mission and purpose through a hierarchy of micro-and macro processes. Every process contains sub processes and is also contained within a higher process. This structure of processes is repeated throughout the hierarchy.

4. Organizational Change

Once a traditional organization has found a formula for success, it keeps following it. Management's job is to prevent change, to maintain the status quo. In TQ the environment in which the enterprise interacts is changing constantly. If the enterprise continues to do what it has done in the past, its future performance relative to the competition will deteriorate. Management's job, therefore, is to provide the leadership for continual improvement and innovation in processes and systems, products, and services. External change is inevitable, but a favorable future can be shaped.

5. Team work

In traditional management, individuals and departments work for themselves. Individuals are driven by short-term performance measures, have narrowly defined jobs, and rarely see how they fit into the whole process or system. Little communication and cooperation exists between design and manufacturing, manufacturing and marketing, and sales / service and design. In TQ individuals cooperate in team structures such as quality circles, steering committees, and self-directed work teams. Departments work together toward system optimization through cross-functional teamwork.

The adversarial relationship between union and management is inevitable in traditional management. The only room for negotiation is in areas such as wages, health, and safety. In TQ the union is a partner and a stakeholder in the success of the enterprise. The areas for partnership and collaboration are broad, particularly in education, training, and meaningful involvement of employees in the improvement of processes that they affect and that affect their work.

6. Motivation and Job Design

Motivation in traditional management is often akin to McGregor's Theory X model of motivation: workers dislike work and require close supervision and control. TQ organizations support the premise of Theory Y: workers are self-motivated, seek responsibility, and exhibit a high degree of imagination and creativity at work. TQ managers provide leadership rather than overt intervention in the processes of their subordinates, who are viewed as process managers rather than functional specialists. People are motivated to make meaningful contributions to what they believe is an important and noble cause, of value to the enterprise and society.

In traditional management, competition is inevitable and inherent in human nature. Performance appraisal, recognition, and reward systems place people in an internally competitive environment. Individualism is reinforced to the detriment of teamwork. Competitive behavior – one person against another or one group against another – is not a natural state in TQ. TQ reward systems recognize individual as well as team contributions and reinforce cooperation.

7. Management and Leadership

Traditional management views people as interchangeable commodities, developed to meet the perceived needs of the enterprise. People are passive contributors with little autonomy-doing what they are told and nothing more. TQ views people as the enterprise's true competitive edge. Leadership provides people with opportunities for personal growth and development. People are able to take pride and joy in learning and accomplishment, and the ability of the enterprise to succeed is enhanced. People are active contributors, valued for their creativity and intelligence. Every person is a process manager presiding over the transformation of inputs to outputs of greater value to the enterprise and to the ultimate customer.

Competitive Advantage on basis of Quality Strategy

- A firm has many options in defining its long-term goals and objectives, the customers it wants to serve, the products and services it produces and delivers, and the design of the production and service system to meet these objectives. Strategic planning is the process by which the members of an organization envision its future and develop the necessary procedures and operations to carry out that vision. Strategy – the result of strategic planning – is the pattern of decisions that determines and reveals a company's goals, policies, and plans to meet the needs of its stakeholders. An effective strategy allows a business to create a sustainable competitive advantage.

Quality as a Strategy

The concept of strategy has different meanings to different people. James Brian Quinn characterizes strategy as follows:

A strategy is a pattern or plan that integrates an organization's major goals, policies, and action sequences into a cohesive whole. A well formulated strategy helps to marshal and allocate an organization's resources into a unique and viable posture based on its relative internal competencies and shortcomings, anticipated changes in the environment, and contingent moves by intelligent opponents.

Formal strategies contain three elements:

1. Goals to be achieved,
2. Policies that guide or limit action, and
3. Action sequences, or programs, that accomplish the goals.

Effective strategies develop around a few key concepts and thrusts that provide focus. The essence of strategy is to build a posture that is so strong in selective ways that the organization can achieve its goals despite unforeseeable external forces that may arise.

The traditional focus of business strategies has been finance and marketing. These parallel two of the principal sources of competitive advantage i.e. cost and differentiation. Total quality – with a

focus on people – leads to improvements in both areas. Therefore, quality can be viewed as a strategy in itself.

The role of quality in business strategy has taken two significant steps since 1980. First, many firms have recognized that a strategy driven by quality can lead to significant market advantages. Second, the lines between quality strategy and generic business strategies have become blurred to the point where TQ principles are integrated into most businesses' normal business planning; that is, TQ is a basic operating philosophy that provides the foundation for effective management

For most companies, integration of TQ into strategic business planning is the result of a natural evolution. For most new companies – or those that have enjoyed a reasonable measure of success – quality takes a back seat to increasing sales, expanding capacity, or boosting production. Strategic planning usually focuses on financial and marketing strategies.

As a company begins to face increasing competition and rising consumer expectations, cost-cutting objectives take precedence. Some departments or individuals may champion quality improvement efforts, but quality is not integrated in the company's strategic business plan. In the face of market crises, which many U.S. firms experienced in the 1970s and 1980s, top management begins to realize the importance of quality as a strategic operating policy. In many cases, however, quality is considered separate from financial and marketing plans. Companies that aspire to world-class status reach the highest level of evolution where quality becomes an integral part of the overall strategic plan and is viewed as a central operating strategy.

Competitive advantage denotes a firm's ability to achieve market superiority over its competitors. In the long run, a sustainable competitive advantage provides above-average performance. A strong competitive advantage has six characteristics:

1. It is driven by customer wants and needs. A company provides value to its customers that competitors do not.
2. It makes a significant contribution to the success of the business.
3. It matches the organization's unique resources with the opportunities in the environment. No two companies have the same resources; a good strategy uses them effectively.
4. It is durable and lasting and difficult for competitors to copy. A superior research and development department, for example, can consistently develop new products or processes to remain ahead of competitors.
5. It provides a basis for further improvement.
6. It provides direction and motivation to the entire organization.

As each of these characteristics relates to quality, quality can be an important means of gaining competitive advantage. Let us see how total quality contributes to competitive advantage.

- Discuss cost leadership, differentiation, and people as principal sources of competitive advantage, and their relationship to quality;
- Relate quality to the achievement of higher profitability;

- Describe the importance of quality in meeting customer expectations in product design, service, flexibility and variety, innovation, and rapid response; and
- Discuss empirical results showing the impact of quality on business results.

Sources of Competitive Advantage

The classic literature on competitive strategy suggests that a firm can possess two basic types of competitive advantage: low cost and differentiation.

Cost Leadership

Many firms gain competitive advantage by establishing themselves as the low-cost leader in an industry. These firms produce high volumes of mature products and achieve their competitive advantage through low prices. Such firms often enter markets that were established by other firms. They emphasize achieving economies of scale and finding cost advantages from all sources. Low cost can result from high productivity and high capacity utilization. More importantly, improvements in quality lead to improvements in productivity, which in turn lead to lower costs. Thus a strategy of continuous improvement is essential to achieve a low-cost competitive advantage.

To achieve cost leadership for high volume products, companies use a variety of approaches:

- Early manufacturing involvement in the design of the product both for make-versus-buy decisions and for assurance that the production processes can achieve required tolerances.
- Product design to take advantage of automated equipment by minimizing the number of parts, eliminating fasteners, making parts symmetric whenever possible, avoiding rigid and stiff parts and using one-sided assembly designs.
- Limited product models and customization in distribution centers rather than in the factory.
- A manufacturing system designed for a fixed sequence of operations. Every effort is made to ensure zero defects at the time of shipment. Work-in-process inventory is reduced as much as possible, and multi skilled, focused teams of employees are used.

A cost leader can achieve above-average performance if it can command prices at or near the industry average. However, it cannot do so with an inferior product. The product must be perceived as comparable with competitors or the firm will be forced to discount prices well below competitors' prices to gain sales. This can cancel any benefits that result from cost advantage.

Differentiation

To achieve differentiation, a firm must be unique in its industry along some dimensions that are widely valued by customers. It selects one or more attributes that customers perceive as important and positions itself uniquely to meet those needs. For instance, Dell's direct business

model was the first of its kind in the computer industry and continues to be a principal source of the company's success.

Often, a firm with a differentiation strategy can command premium prices and achieve higher profits. Juran cites an example of a power tool manufacturer that improved reliability well beyond that of competitors. Field data showing that the differences in reliability resulted in significantly lower operating cost were publicized, and the company was able to secure a premium price.

However, a firm that uses differentiation as its source of competitive advantage must make its products or systems difficult to copy. Often this involves culture, habits, and sunk costs. For example, why doesn't every company copy Dell's superior direct business model? Dell's approaches are hardly a secret; even Michael Dell has written a book about it. Competitors have copied its Web site with stunning precision, but they face far greater difficulty copying the supporting activities—purchasing, scheduling, and logistics—that Dell has built around its direct model over several decades. Competitors are burdened by long-standing relationships with suppliers and distributors and by a different culture.⁷

People

The competitive advantage resulting from an organization's people can drive low cost and differentiation. For example, over several decades, Southwest Airlines has been the most profitable U.S. carrier. It has fewer employees per aircraft and flies more passengers per employee. Much of its cost advantage comes from its very productive, motivated, and unionized workforce.⁹ Is its competitive advantage low cost, or is it the people? It would appear that the real driver of Southwest's competitive advantage is its people. Herb Kelleher, former CEO, once stated, "It's the intangibles that are the hardest things for competitors to imitate. You can get on an airplane. You can get ticket-counter space, you can get baggage conveyors. But it is our esprit de corp.—the culture, the spirit—that is truly our most valuable competitive asset." Providing a work environment that fosters cooperation, initiative, and innovation; educating and training the workforce; and enhancing the factors that affect well-being, satisfaction, and motivation are very difficult for competitors to copy. This is a significantly different philosophy from the work environment that came into being during the Industrial Revolution.

The Importance of Quality to Competitive Advantage

The role of quality in achieving competitive advantage was demonstrated by several research studies during the 1980s. PIMS Associates, Inc., a subsidiary of the Strategic Planning Institute, maintains a database of 1,200 companies and studies the impact of product quality on corporate performance.¹¹ PIMS researchers have found that

- Product quality is the most important determinant of business profitability.
- Business offering premium quality products and services usually have large market shares and were early entrants into their markets.

- Quality is positively and significantly related to a higher return on investment for almost all kinds of products and market situations. PIMS studies have shown that firms with products of superior quality can more than triple return on sales over products perceived as having inferior quality.
- A strategy of quality improvement usually leads to increased market share, but at a cost in terms of reduced short-run profitability.
- High quality producers can usually charge premium prices.

The value of product in the marketplace is influenced by the quality of its design. Improvements in performance, features, and reliability will differentiate the product from its competitors, improve a firm's quality reputation, and improve the perceived value of the product. This allows the company to command higher prices and achieve an increased market share. This, in turn, leads to increased revenues that offset the added costs of improved design and provides sustainable basis for the competitive advantage.

Lesson #0 6**Total quality management
And planning for quality at office**

Total Quality Management is an organization wide process based on:

- ✚ best use of the resources of the total organization
- ✚ organizational flexibility and response to change
- ✚ defined internal and external customer/supplier relationships embracing:
 - ✚ external customers
 - ✚ internal customers
 - ✚ external suppliers
 - ✚ internal suppliers

Bound together in long term business relationships

- ❖ Measurement of performance. The standard is the “agreed customer requirement “and the required performance is
 - absolute conformance to agreed customer requirements
 - customer satisfaction
 - process efficiency
 - anticipating customer needs and expectations
 - delivering products and services that delight customers
 - benchmarking - identifying and adopting world-wide best practice
 - measuring and monitoring continuous improvement

TQM PRINCIPLES PROVIDE A FRAMEWORK AND MODELS HOW TO SPEND THE TIME IN ORGANIZATIONS TO DO THE QUALITY WORK. CONDUCTING MEETINGS AND MAKING DECISION IS ALSO ONE THE IMPORTANT TASK OF MANAGERS.

Now we are going to discuss in detail the nine basics steps which professional managers can use in their meetings in order to improve the performance and productivity .After implementing these basic skills in any organization company can save their valuable and precious time and can come to conclusion easily.

Nine Discussion Skills

1. Open the discussion
2. Listen
3. Ask for clarification
4. Manage participation
5. Summarize
6. Manage time
7. Contain digressions
8. Test for agreement
9. Close the discussion

Lesson # 07

Leaders in quality revolution And defining for quality

THE CONCEPT AND DEFINATION OF QUALITY

While managers have shown interest in the concept of quality, many have been frustrated by its elusiveness. They find diverse and often conflicting definitions in professional books, journals, and news media. Despite common themes such as continuous improvement, customer focus, and excellence, different people emphasize different things. For example, in a 1991 public television special, “Quality or Else,” executives, managers, workers, academics, and others defined quality variously as follows:

- A pragmatics system of continual improvement, a way to successfully organize man and machines.
- The meaning of excellence
- The unyielding and continuing effort by everyone in an organization to understand, meet, and exceed the needs of its customers
- The best product that you can produce with the materials that you have to work with
- Continuous good product which a customer can trust.
- Producing a product or service that meets the needs or expectations of the customers.
- Not only satisfying customers, but delighting them, innovating, creating.

Your own sample of definitions would probably reveal similar variety. Different companies, and different people within the same company, often disagree on the definition of quality. Sometimes the disagreement is merely due to semantics. Sometimes they are the result of focusing on different dimensions of quality. Other times the differences are more profound, implying conflicting courses of action and approaches to management. Here we look at several views of quality and then offer a definition that should help to integrate managerial efforts to improve quality throughout an organization.

The Transcendent View of Quality

The concept of quality has often been defined, from a transcendent view, as “innate excellence”. This view implies that high quality is something timeless and enduring, an essence that transcends or rises above individual tastes or styles. It often regards quality as an un-analyzable property that people learn to recognize through experience, just as Plato argued that beauty can be understood only after exposure to a series of objects that display its characteristics.

A Critique of the Transcendent View of Quality

Walter A. Shewhart, the father of modern-day statistical quality control, offered the following criticism of the transcendent view of quality:

Dating at least from the time of Aristotle, there has been some tendency to conceive of quality as indicating the goodness of an object. The majority of advertisers appeal to the public upon the basis of the quality of product. In so doing, they implicitly assume that there is a measure of goodness which can be applied to all kinds of product, whether it be vacuum tubes, sewing machines automobiles.

The transcendent view of quality essentially tells a manager “you will know it when you see it” and does not inform managers how to pursue excellence. Certainly the notion of excellence is an important and inspirational component of quality. However, future managers must have a better understanding of the concept. The definition of quality must be more pragmatic, more objective, and more tangible. It must inform managers about how to make improvements.

To better understand how this total views of quality impacts managerial practices, it is useful to understand how managerial approaches to quality have evolved from a narrow view, focused on inspection and conformance to specified standards, toward a broader view, focused on organizational strategy for providing superior customer value. We now discuss the evolution of quality approaches.

Product-Based

Shewhart explains that the word “quality,” in Latin, *qualitas*, comes from quails, meaning “How constituted” and signifies a thing’s basic nature. Emphasizing a product-based view of quality, Shewhart argued that the quality of a manufactured product may be described in terms of a set of characteristics. Product-based definitions of quality suit engineers because they are concerned with translating product requirements into specific components and physical dimensions that can be produced. For example, measurement of capacity, inductance, and resistance may be used to define the quality of a relay. So, according to this view, quality is a precise and measurable variable: differences in quality reflect differences in the quantity of an attribute the product possess (Garvin, 1988). For example, high-quality rugs have a large number of knots per square inch. Quality in a rug can be seen as an inherent characteristic that can be assessed objectively. Since quality reflects the quantity of attributes contained by a product, and because attributes are costly to produce, high quality means higher cost. In this view, a Cadillac loaded with a number of amenities is a higher-quality car than a stripped-down Chevrolet.

The product-based view has some merit, but it does not accommodate differences in individual tastes and preferences.

Manufacturing-Based

Another meaning of the word “quality” is “the degree of excellence that a thing possess” once it is manufactured. The manufacturing-based view of quality focuses on manufacturing and engineering practices, emphasizes conformance to specified requirements, and relies on statistical analysis to measure quality. As you will see, it contradicts the notion that higher quality necessarily corresponds to higher cost. Returning to the example of a relay, Shewhart suggests the overall quality of a relay can be further expressed in terms of whether it meets

engineering specifications for product-based characteristics (qualities), such as capacity, inductance, and resistance. To simplify, let's just consider two dimensions, resistance and inductance.

The quality or degree of excellence of a product (represented by a point or a set of two measurements) falling within the rectangular region would be characterized as good or satisfactory. A product with quality outside the region, not conforming to satisfactions, would be characterized as bad or unsatisfactory. Of course, real manufacturing processes produce a stream of products. Ideally, each individual product has quality that conforms to specification. However, variation in the production process may produce some products that are outside the specifications.

Shewhart suggested that the fraction of nonconforming items produced by a manufacturing process can be studied statistically to assess quality. The knowledge gained from statistical studies can be used to improve the control of quality, thus ensuring that a larger fraction of the products conform to specification. By stabilizing and reducing variation in the process, managers can ensure that product quality is always within specification. Such improvement would mean fewer defects, less scrap, less rework, and consequently, less cost.

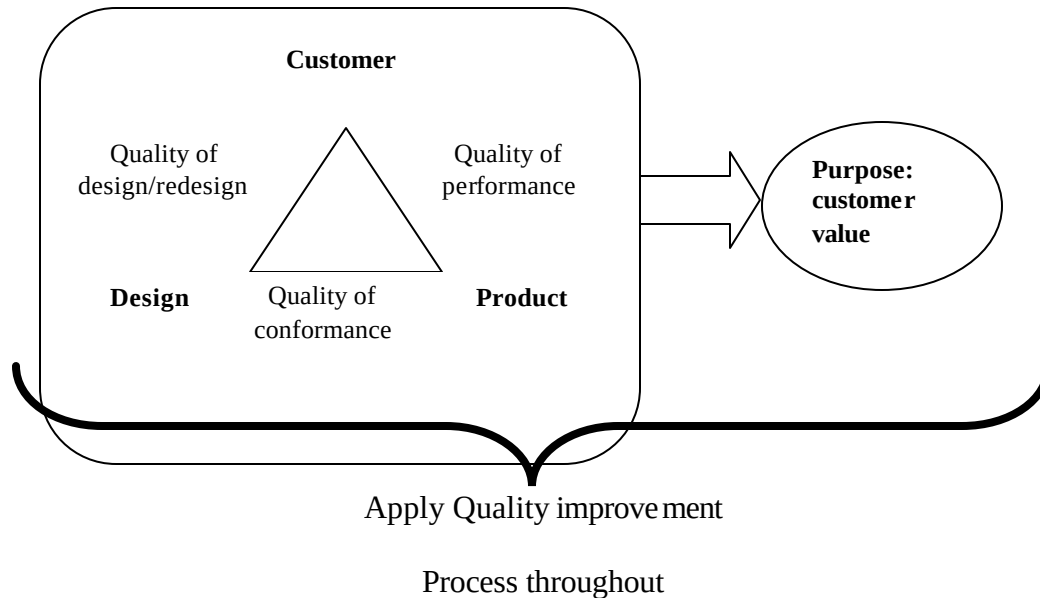
User-Based

The user-based perspective does not abandon manufacturing quality as a strategic objective, but provides a context for it. As Shewhart says, "The broader concept of economic control naturally includes the problem of continually shifting the standards expressed in terms of measurable physical properties to meet best the shifting economic value of these particular physical characteristics depending upon shifting human wants". The *user-based view of quality*, popular with people in marketing, presumes that quality rests in "the eye of the beholder," the user of the product, rather than an engineer's specified standards.

The Concept of Customer Value

Value-based approaches expand on the user-based view of quality by incorporating the notion of "price of costs." Regarding value-based approaches, Garvin (1988) suggests "a quality product is one that provides performance or conformance at an acceptable price or cost. By this reasoning, a \$ 500 running shoe, no matter how well-constructed, could not be a quality product, for it would find few buyers." In support of this view, researchers have demonstrated a positive relationship between market share and value-based measures of quality. Other examples contradict this value-based approach which assumes that lower cost always means higher value to the customer. Designer dresses that sell for \$5,000, or luxury cars that cost more than a home, suggest that there is another dimension of value. The definition of customer value offered below expands on this value-based definition of quality. In fact, the customer value concept encompasses all of the foregoing definitions of quality.

Quality improvement for customer value



We define customer value as a combination of benefits and sacrifices occurring when a customer uses a product or service to meet certain needs. Those consequences that contribute to meeting one's needs are benefits, while those consequences that detract from meeting one's needs are sacrifices.

For example, a person who purchases a large luxury car to satisfy a need for pleasurable travel might enjoy such benefits as comfort, restfulness, and audio entertainment. On the other hand, the person also has to make certain sacrifices, such as paying for the vehicle, difficulties in parking a large vehicle, and fuel and maintenance costs. These sacrifices detract from meeting customer needs. For example, if the car owner lives in a town that doesn't have a dealership for the luxury car, he or she will incur costs such as frustration, time, and the inconvenience of going to another town for service.

The concept of customer value encompasses the benefits and sacrifices associated with the customer's use process throughout the life cycle of product ownership. As Deming (1986) suggests: "Quality must be measured by the interaction between three participants: (1) the product itself; (2) the user and how he uses the product, how he installs it, how he takes care of it; and (3) instructions for use, training of customer and training of repairman, service provided for repairs, and availability of parts."

The concept of customer value also encompasses all the definitions of quality mentioned so far. To provide value to customers, managers must ensure the following:

1. *Quality of Design/Redesign*: Product designs conform to customer needs (product-based and user-based quality). For example, automobile producers design car seat to conform to the contours of the driver's back.
2. *Quality of Conformance*: Product manufactured conforms to product designs (manufacturing-based quality). For example, each car seat produced meets the targeted design specifications.
3. *Quality of Performance*: Products manufactured conform to customer needs by performing in the field (user-based quality). For example, the car seats maintain their shape after years of use.

All of these dimensions of quality should be managed through a quality improvement process that enhances customer value. Placing quality in the broader context of customer value counteracts the tendency of people with different functional orientations within an organization to take different views of quality (Garvin, 1988). For example, marketing people tend to have a user-based and product-based view that focuses on matching product characteristics with customer perceptions. Engineers, on the other hand, tend to take a product-based view that focuses on defining product characteristics. Manufacturing people tend to view quality as conformance to specification and targets.

Lesson # 08

Taguchi Loss Function And Quality Management

Total Quality Paradigms

Adopting a TQ philosophy requires significant changes in organization design, work processes, and culture. Organizations use a variety of approaches. Some emphasize the use of quality tools, such as statistical process control or Six Sigma (which we discuss in the next chapter), but have not made the necessary fundamental changes in their processes and culture. It is easy to focus on tools and techniques but very hard to understand and achieve the necessary changes in human attitudes and behavior. Others have adopted a behavioral focus in which the organization's people are indoctrinated in a customer-focused culture, or **emphasize error prevention and design quality**, but fail to incorporate continuous improvement efforts. Still other companies focus on problem-solving and continuous improvements, but fail to focus on what is truly important to the customer. Although these firms will realize limited improvements, the full potential of total quality is lost due to a lack of complete understanding by the entire organization.

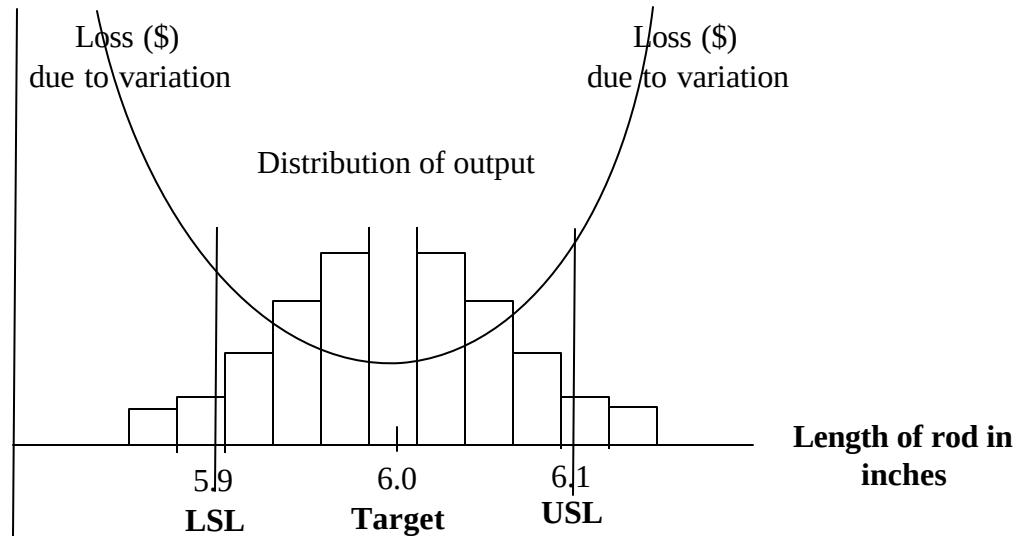
Single approaches, such as statistical tools, behavioral change, or problem solving can have some short-term success, but do not seem to work well over time. Total quality requires a comprehensive effort that encompasses all of these approaches. A total change in thinking, not a new collection of tools, is needed. Total quality requires a set of guiding principles. Such principles have been promoted by the many "quality gurus" – Deming, Juran, and Crosby, Ishikawa and Taguchi. Their insights on measuring, managing, and improving quality have had profound impacts on countless managers and entire corporations around the world.

DEFINING QUALITY AS LOSS FUNCTION

Taguchi (1986) suggests that there is increasing loss, for the producer, the customer, and society, associated with increasing variability, or deviation from a target value that reflects the “ideal state.” This relationship to variability can be expressed as a loss function, as shown for the distribution of rods from grinding operation C, in Figure 12. The greater the variability, deviation from target, the greater the loss.

Traditional specifications, used in the manufacturing-based approach to quality, define conformity in terms of upper and lower specification limits. For example steel rods should meet the engineering specification for length of six inches, plus or minus 10 one-hundredths of an inch ($6 \pm .10$). This approach tends to allow complacency concerning variation within that range. It assumes that a product just barely meeting specifications, just within the limit, is just as “good” as one right in the middle, but one just outside the limit is “bad.” Taguchi questions these assumptions, and suggests the degree of “badness” or “loss” increases gradually as the deviation from the target value increases. Although managers may choose to do the right thing (the target), in order to provide superior value to customers through superior “quality,” they must also continuously improve their systems and reduce variation to meet the target.²⁰ In the 1980s, Motorola committed to a campaign called Six Sigma, which is one way of saying reduce variation so much that the chance of producing defects is down to about 3.4 defects per million, or 99.99966 percent perfect.

Taguchi's loss function: loss increases as a function of variation



The Deming Management Philosophy

Deming was trained as a statistician and worked for Western Electric during its pioneering era of statistical quality control development in the 1920s and 1930s. During World War II he taught quality control courses as part of the national defense effort. Although Deming taught many engineers in the United States, he was not able to reach upper management. After the war, Deming was invited to Japan to teach statistical quality control concepts. Top managers there were eager to learn, and he addressed 21 to executives who collectively resented 80 percent of the country's capital. They embraced Deming's message and transformed their industries. By the mid-1970s, the quality of Japanese products exceeded that of Western manufacturers, and Japanese companies had made significant penetration into Western markets.

Deming taught quality to Japanese and Ishikawa was Deming's student. Americans did not listen to Deming as attentively as Japanese did and took his a prophet of quality.

Lesson # 09

WTO, SHIFTING FOCUS OF CORPORATE CULTURE AND ORGANIZATIONAL MODEL OF MANAGEMENT

WTO (in 20XX) The Shifting Focus of Corporate Culture

From	To
Thinking Global	Being Global
Focus on Local Market	Focus on Global Market
MNC in manufacturing	MNC in Services
Antagonistic towards Rivals	Collaboration with Rivals
Work with Hands	Work with 3H & H3
Upgrade Technology	Upgrade People
Focus on Process Quality	Focus on Design Quality
Mass Mindset	Lean and Me Mindset
Manage by Control	Manage by Commitment
Individual Values	Shared Values

Organizational Models for Management

Dimension	Mechanistic Model	Organism Model	TQ Model	Cultural Model
Goal	Organizational efficiency and performance	Organizational survival	Long-term survival	Meet individual needs; human development
Definition of quality	Conformance to standards	Customer satisfaction	Satisfying or delighting the customer	Constituent satisfaction
Role/nature of environment	Objective; outside boundary	Objective; inside boundary	Blurred organization and environmental boundaries	Enacted/boundaries defined through relationships
Role of management	Coordinate and provide visible control	Coordinate and provide invisible control by creating vision	Focus on improvement and creating a system that can produce quality outcomes	Coordinate and mediate negotiations
Structural rationality	Chain of command (vertical) Technical rationality	Process flow (horizontal and vertical) Organizational rationality	Horizontal processes beginning with suppliers and ending with customers and supported by teams	Mutual adjustment in any direction Political rationality
Philosophy toward change	Stability is valued; learning arises from specialization	Change and learning assist adaptation	Change, continuous improvement, and learning are encouraged	Change and learning are valued in themselves

HISTORY OF QUALITY MANAGEMENT PARADIGMS

THE EVOLUTION OF QUALITY APPROACHES

The shift to Total Quality Management may be revolutionary for many managers because the tenets of the new paradigm are so radically different from past managerial practices. It will require both a thought revolution and a behavioral revolution. Approaches to quality have evolved through a series of gradual refinements over the last century. The shift seems dramatic and revolutionary to many managers because they have not kept up with the evolving approaches over the years. However, they have not defined their managerial roles in terms of the latest advancements or they feel skeptical about its success.

The Four Major Quality Eras

Identifying Characteristics Date of inception	Inspection (1800s)	Statistical Quality Control (1930s)	Quality Assurance (1950s)	Strategic Total Quality Management (1980s)
Primary concern	Detection	Control	Coordination	Strategic impact
View of quality	A problem to be solved	A problem to be solved	A problem to be solved, but one that is attacked proactively	A competitive opportunity
Emphasis	Product uniformity	Product uniformity with reduced inspection	The entire production chain, from design to market, and the contribution of all functional groups, especially designers, to preventing quality failures	The market and consumer needs
Methods	Gauging and measurement	Statistical tools and techniques	Programs and systems	Strategic planning, goal-setting, and mobilizing the organization
Role of quality professionals	Inspection, sorting, counting, and grading	Troubleshooting and the application of statistical methods	Quality measurement, quality planning, and program design	Goal-setting, education and training, consultative work with other departments, and program design
Who has responsibility for quality	The inspection department	The manufacturing and engineering departments	All department, although top management is	Everyone in the organization, with top management

			only peripherally involved in designing, planning, and executing quality policies	exercising strong leadership
Orientation and approach	“Inspects in” quality	“Controls in” quality	“Builds in” quality	“Manages in” quality

Customer-Craft OR the Inspection Era

Until the nineteenth century, skilled craftsmen manufactured goods in small volume. They handcrafted and fit together parts to form a unique product that was only informally inspected. Population growth and industrialization brought about production in larger volume. Manufacturing in the industrialized world tended to follow this craftsmanship model till the factory system, with its emphasis on product inspection, started in Great Britain in the mid-1750s and grew into the Industrial Revolution in the early 1800s.

The factory system, a product of the Industrial Revolution in Europe, began to divide the craftsmen’s trades into specialized tasks. This forced craftsmen to become factory workers and forced shop owners to become production supervisors, and marked an initial decline in employees’ sense of empowerment and autonomy in the workplace.

Quality in the factory system was ensured through the skill of laborers supplemented by audits and/or inspections. Defective products were either reworked or scrapped.

Mass Production and Inspection

In the 1800s, increased specialization, division of labor, and mass production required more formal inspection. Parts had to be interchangeable. Inspectors examined products to detect flaws and separate the good from the bad. They used gauges to catch deviant parts and make sure parts fit together at final assembly. The gauging system made inspections more consistent than those conducted solely by eye, and gave inspection a new respectability.

Formalizing the Inspection Function

By the early 1900s, gauging had become more refined, and inspection was even more important. It was prominent in Henry Ford’s moving assembly line and Frederick W. Taylor’s system of shop floor management. In 1922, G.S. Radford formally linked inspection to quality control. For the first time, quality was regarded as an independent function and a distinct management responsibility. Radford

defined quality in term of conformance to “established requirements” and emphasized inspection. He also suggested some lasting quality principles, such as getting designers involved early, closely coordinating various departments, and achieving the quality improvement results of increased output and lower costs.

Late in the 19th century the United States broke further from European tradition and adopted a new management approach developed by Frederick W. Taylor. Taylor’s goal was to increase productivity without increasing the number of skilled craftsmen. He achieved this by assigning factory planning to specialized engineers and by using craftsmen and supervisors, who had been displaced by the growth of factories, as inspectors and managers who executed the engineers’ plans.

Taylor’s approach led to remarkable rises in productivity, but it had significant drawbacks: Workers were once again stripped of their dwindling power, and the new emphasis on productivity had a negative effect on quality.

To remedy the quality decline, factory managers created inspection departments to keep defective products from reaching customers. If defective product did reach the customer, it was more common for upper managers to ask the inspector, “Why did we let this get out?” than to ask the production manager, “Why did we make it this way to begin with?”

Through the 1920s, however, quality control was most often limited to inspection and focused on activities such as counting, grading, and rework, which is antithetical to Total Quality Management’s emphasis on prevention to avoid defects. Inspection departments and quality professionals were not required to troubleshoot, to understand and address the causes of poor quality, until the 1930s, with the creation of statistical quality control.

In the early 20th century, manufacturers began to include quality processes in quality practices. After the United States entered World War II, quality became a critical component of the war effort: Bullets manufactured in one place, for example, had to work consistently in rifles made in another. The armed forces initially inspected virtually every unit of product; then to simplify and speed up this process without compromising safety, the military began to use sampling techniques for inspection, aided by the publication of military-specification standards and training courses in Walter Shewhart’s statistical process control techniques

The Statistical Quality Control Era

In 1931, Walter A. Shewhart gave quality a scientific footing with the publication of his book *Economic Control of Quality of Manufactured Product*. Shewhart was one of a group of people at Bell Laboratories investigating problems of quality. The statistical quality control approach that Shewhart advocated is based on his views of quality. Statistical quality control requires that numbers derived from measures of processes or products be analyzed according to a theory of variation that links outcomes to uses.

Shewhart's Views of Quality

Shewhart offered a pragmatic concept of quality: "The measure of quality is a quantity which may take on different numerical values. In other words, the measure of quality, no matter what the definition of quality may be, is a variable". Shewhart's emphasis on measurement in his definition of quality obviously relates to his prescriptions for statistical quality control, which requires numbers.

Shewhart recognized that industrial processes yield data. For example, a process in which metal is cut into sheets yields certain measurements, such as each sheet's length, height and weight. Shewhart determined this data could be analyzed using statistical techniques to see whether a process is stable and in control, or if it is being affected by special causes that should be fixed. In doing so, Shewhart laid the foundation for control charts, a modern-day quality tool.

Shewhart's concepts are referred to as statistical quality control (SQC). They differ from product orientation in that they make quality relevant not only for the finished product but also for the process that created it.

The Quality Assurance Era

During the quality assurance era, the concept of quality in the United States evolved from a narrow, manufacturing-based discipline to one with implications for management throughout a firm. Statistics and manufacturing control remained important, but coordination with other areas, such as design, engineering, planning, and service activities, also became important to quality. While quality remained focused on defect prevention, the quality assurance era brought a more proactive approach and some new tools.

The quality assurance era significantly expanded the involvement of all other functions through total quality control, and inspired managers to pursue perfection actively. However, the approaches to achieving quality remained largely defensive. Controlling quality still meant acting on defects. Quality was something that could hurt a company if ignored, rather than a positive characteristic necessary in obtaining competitive advantage. This view started to change in the 1970s and 1980s, when managers started to recognize the strategic importance of quality.

Total Quality Control and Customer Driven Quality

The beginning of the 20th century marked the inclusion of “processes” in quality practices. A “process” is defined as a group of activities that takes an input, adds value to it and provides an output, such as when a chef transforms a pile of ingredients into a meal.

In 1956, Armand Feigenbaum extended this principle by suggesting that high-quality products are more likely to be produced through total quality control than when manufacturing works in isolation:

The underlying principle of this total quality view. . . . is that, to provide genuine effectiveness, control must start with the design of the product and end only when the product has been placed in the hands of a customer who remains satisfied the first principle to recognize is that quality is everybody’s job.

The birth of total quality in the United States came as a direct response to the quality revolution in Japan following World War II. The Japanese welcomed the input of Americans Joseph M. Juran and W. Edwards Deming and rather than concentrating on inspection, focused on improving all organizational processes through the people who used them.

Feigenbaum’s message reinforced Juran’s emphasis on managerial responsibility. To make total quality control work, many companies developed matrices or relationship charts. These charts list functions (departments or groups) across the top and required activities down the side, and shows responsibility relationships in each cell. The considerable overlap among functions means that cross-functional teams are needed to ensure required communication and collaboration, for example, in assessing the “manufacturability” of a design and debugging new manufacturing techniques through pilot runs.

Both Juran and Feigenbaum acknowledged that statistical methods and manufacturing control were still important. However, they also felt total quality control would require new management skills to deal with areas such as new product development and vendor selection. Managers also would be required to engage in activities such as quality planning and coordinating cross-functional teamwork. Despite the emphasis on teamwork, Feigenbaum’s TQC suggests that more than half of the primary responsibilities for quality belong to the quality control department, another practice that is antithetical to modern Total Quality Management.

W Edwards Deming, a statistician with the U.S. Department of Agriculture and Census Bureau, became a proponent of Shewhart's SQC methods and later became a leader of the quality movement in both Japan and the United States.

THE STRATEGIC QUALITY MANAGEMENT ERA

The present quality era, Strategic Quality Management, incorporates elements of each of the preceding eras, particularly the contributions of Shewhart, Deming, Juran, and Feigenbaum. So many elements of previous eras are incorporated into Strategic Quality Management that the last two decades may at first appear to be just a repackaging of old ideas. There are, however, dramatic differences from earlier eras. For the first time, top managers began to view quality positively as a competitive advantage, and to address it in their strategic planning processes, which are focused on customer value.

Because quality started to attract the attention of top managers, it impacted management throughout the organization. Quality was not just for the inspectors or people in the quality assurance department to worry about. This era marks the emergence of a new paradigm for management. A number of developments were brought together and reconfigured into a new approach to management in all departments and specialties.

A variety of external forces brought quality to the attention of top managers. They began to see a link between losses of profitability and poor quality. The forces that brought this connection to their attention included a rising tide of multimillion-dollar product liability suits for defective products and constant pressures from the government on several fronts, including closer policing of defects, product recalls. Perhaps the most salient external force was the growing market share incursions from foreign competitors, particularly the Japanese, in such diverse industries as semiconductors, automobiles, machine tools, radial tires, and consumer electronics.

Producing products with superior quality, lower cost, and more reliable delivery, Japanese firms gained market shares and achieved immense profitability. The onslaught of these events in the mid-1970s and 1980s seemed rather sudden. However, Japanese firm had been building their industrial capabilities for decades, developing and refining approaches to quality grounded in the principles taught to them by Americans after World War II. Manager and theorists have been captivated by "Japanese management" over the last two decades. Indeed, the Strategic Quality Management era borrows a number of its elements from the developments that quality took place in Japan at the same time as the quality assurance era in the United States.

Total Quality Management

Just as the definition of quality has been a source of confusion, so has the definition of Total Quality Management. There is no consensus on what constitutes TQM. Almost every organization defines it differently or calls it something other than TQM.

In the United States, Total Quality Management is often used to refer to the management approaches being developed in the current era of Strategic Quality Management while the new paradigm is emerging. Ideally, managers in the Strategic Quality Management era regard total Quality management as something more than a “program,” and take it beyond all the deficiencies mentioned earlier.

In this context, the word “Total” conveys the idea that all employees, throughout every function and level of an organization, pursue quality. The word “quality” suggests excellence in every aspect of the organization. “Management” refers to the pursuit of quality results through a quality management process. This begins with strategic management processes and extends through product design, manufacturing, marketing, finance, and so on. It encompasses, yet goes beyond, all of the earlier definitions of quality by pulling them together into a never-ending process of improvement.. Accordingly, TQM is as much about the quality process as it is about quality results or quality products. It began with people, particularly managers.

Lesson # 11**DEFINING QUALITY, QUALITY MANAGEMENT
AND LINKS WITH PROFITABILITY****BEYOND TOTAL QUALITY**

By the end of the 1990s Total Quality Management (TQM) was considered little more than a fad by many American business leaders (although it still retained its prominence in Europe).

As the 21st century begins, the quality movement has matured. New quality systems have evolved beyond the foundations laid by Deming, Juran and the early Japanese practitioners of quality.

Some examples of this maturation:

- In 2000 the ISO 9000 series of quality management standards was revised to increase emphasis on customer satisfaction.
- Beginning in 1995, the Malcolm Baldrige National Quality Award added a business results criterion to its measures of applicant success.
- Six Sigma, a methodology developed by Motorola to improve its business processes by minimizing defects, evolved into an organizational approach that achieved breakthroughs – and significant bottom-line results.
- Quality Function Deployment was developed by Y. Akao as a process for focusing on customer wants or needs in the design or redesign of a product or service.
- Sector-specific versions of the ISO 9000 series of quality management standards were developed for such industries as automotive (QS-9000), aerospace (AS9000) and telecommunications (TL 9000 and ISO/TS 16949) and for **environmental management (ISO 14000)** and also for IT Sector
- Quality has moved beyond the manufacturing sector into such areas as service, healthcare, education and government.
- The CMM levels 1-5 emphasize quality in IT sector from the maturity point of view of processes and people.
- Emphasis on learning by doing as a means of continual improvement has been accepted and has paved the way to consider Knowledge as a strategic resource of production in 21st century work place and in work organizations.
- Discussion of Knowledge Management and Learning Organizations are becoming common in all sorts of companies and all sectors of economy and is rather being taken as a source of national competitive advantage in global interconnected and interdependent infocom oriented world.

THE CONCEPT OF QUALITY

People define quality in many ways. Some think of quality as superiority or excellence, others view it as a lack of manufacturing or service defects, still others think of quality as related to product features or price. As study that asked managers of 86 firms to define quality produced several dozen different responses, including.

1. perfection
2. consistency
3. eliminating waste
4. speed of delivery
5. compliance with policies and procedures

6. providing a good, usable product
7. doing it right the first time
8. delighting or pleasing customers
9. total customers service and satisfaction⁶

Today most managers agree that the main reason to pursue quality is to satisfy customers. The American National Standards Institute (ANSI) and the American Society for Quality (ASQ) define quality as "the totality of features and characteristics of a product or service that bears on its ability to satisfy given needs." The view of quality as the satisfaction of customer needs is often called fitness for use. In highly competitive markets, merely satisfying customer needs will not achieve success. To beat the competition, organizations often must exceed customer expectations. Thus, one of the most popular definitions of quality is meeting or exceeding customer expectations.

What Is Quality?

To understand total quality, one must first understand quality. Customers that are business organizations will define quality very clearly using specifications, standards, and other measures. This makes the point that quality can be defined and measured. Although few consumers could define quality if asked, all know it when they see it. This makes the critical point that quality is in the eye of the beholder. With the total quality approach, customers ultimately define quality.

People deal with the issue of quality continually in their daily lives. We concern ourselves with quality when grocery shopping, eating in a restaurant, and making a major purchase such as an automobile, a home, a television, or a personal computer. Perceived quality is a major factor by which people make distinctions in the market place. Whether we articulate them openly or keep them in the back of our minds. We all apply a number of criteria when making a purchase. The extent to which a purchase meets these criteria determines its quality in our eyes.

One way to understand quality as a consumer-driven concept is to consider the example of eating at a restaurant. How will you judge the quality of the restaurant? Most people apply such criteria as the following:

- ◆ Service
- ◆ Response time
- ◆ Food preparation
- ◆ Environment/atmosphere
- ◆ Price
- ◆ Selection

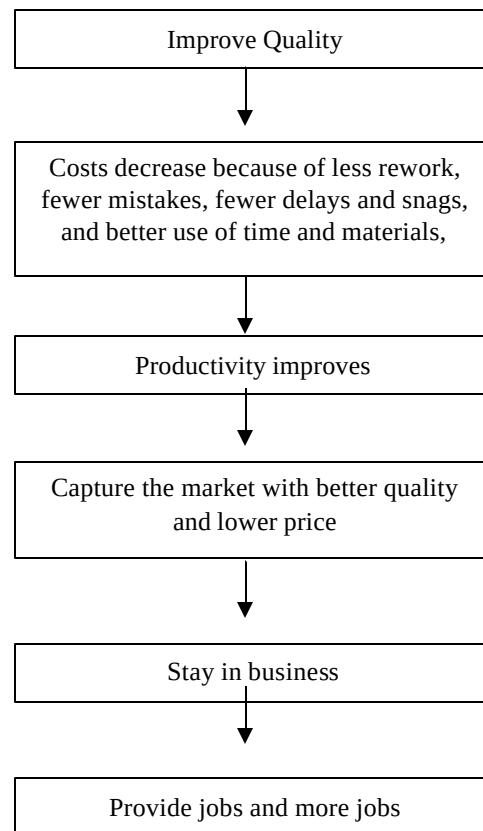
The example gets at one aspect of quality – the results aspect. Does the product or service meet or exceed customer expectations? This is a critical aspect of quality, but it is not the only one. Total quality is a much broader concept that encompasses not just the results aspect but also the quality of people and the quality of processes.

Chain Reaction and Profitability

Deming stresses that higher quality and improvement in quality leads to higher productivity, which in turn leads to long-term competitive strength. The Deming "chain reaction," shown in Figure below, summarizes this view. This theory states that improvements in quality lead to lower cost because of less

rework, fewer mistakes, fewer delays and snags, and better use of time and materials. Lower costs, in turn, lead to productivity improvements. With better quality and lower prices, the firm can achieve a higher market share and thus stay in business, providing more and more jobs. Deming states emphatically that top management has the overriding responsibility for quality improvement.

The Deming Chain Reaction



SIPOC Analysis:

A fundamental step in improving a process is to understand how it functions from a process management perspective. This can be seen through the analysis diagram of the process to identify the Supplier-Input-Process-Output-Customer (SIPOC) linkages.

It begins with defining the process of interest and listing the outputs that the process creates that go to the customers.

Quality Terminology:

Quality: is fitness for use (Juran)

Quality Control: It is a system where the qualities of products or services are inspected into to produce them economically to meet the requirement of the purchaser. It is the operational technique.

Quality Assurance: It means to assure quality in a product so that a customer can buy it with confidence and use it for a large period of time with satisfaction.

According to ISO, QA means, "all those planned and systematic actions necessary to provide adequate confidence that a product or service will satisfy given requirements for quality."

Quality Management system refers to the organization's structure for managing its processes - or activities - that transform inputs of resources into a product or service which meet the organization's objectives, such as satisfying the customer's quality requirements, complying to regulations, or meeting environmental objectives.

Total Quality Management: TQM is both philosophy and a set of guiding principles that represent the foundation of a continuously improving organization. It is the application of quantitative methods as well as human resources to improve the whole supply chain for customers and suppliers.

Customer: Any one who receives or is affected by the product, service, or process.

External Customer: The one outside the company walls or office/department walls or the next one in chain who receives your product, service or idea.

Internal Customers: Staff members, employees or any one who works for the interest of a company or office or boss and expects a reward or salary or benefit from the company or office or boss.

Investor Customer: The one who has invested his fortune and finance to build a company and expects a good return on his/her financial capital or fortune. Shareholders, Stockholders

Social Customer: The society at large.

Lesson # 12**LEARNING ABOUT QUALITY AND APPROACHES FROM
QUALITY PHILOSOPHIES**

Many individuals have made substantial contributions to the theory and practice of quality management. These include the well-known “gurus”: W. Edwards Deming, Joseph M. Juran, and Philip B. Crosby, as well as many other consultants, business executives, and academic researchers. Their philosophical writings and lectures have helped shape management thought as well as provide the foundation for practical management frameworks designed around quality.

Total quality requires a set of guiding principles. Such principles have been promoted by the three “quality gurus” – Deming, Juran, and Crosby. Their insights on measuring, managing, and improving quality have had profound impacts on countless managers and entire corporations around the world. Deming has generated the most interest and controversy. A discussion of his philosophy, which is actually more about management than quality, follows.

The Deming Management Philosophy

Deming was trained as statistician and worked for Western Electric during its pioneering era of statistical quality control development in the 1920s and 1930s. During World War II he taught quality control courses as part of the national defense effort. Although Deming taught many engineers in the United States, he was not able to reach upper management. After the war, Deming was invited to Japan to teach statistical quality control concepts. Top managers there were eager to learn, and he addressed 21 to executives who collectively resented 80 percent of the country’s capital. They embraced Deming’s message and transformed their industries. By the mid-1970s, the quality of Japanese products exceeded that of Western manufacturers, and Japanese companies had made significant penetration into Western markets.

Deming’s contributions were recognized early by the Japanese. The Deming Application Prize was instituted in 1951 by the Union of Japanese Scientists and Engineers in recognition and appreciation for his achievements in statistical quality control. Deming also received the nation’s highest honor, the Royal Order of the Sacred Treasure, from the emperor of Japan. The former chairman of NEC Electronics once said, “There is not a day I don’t think about what Dr. Deming meant to us.”

Deming was virtually unknown in the United States until 1980 where NBC aired a white paper entitled “If Japan Can . . . Why Can’t We?” This program made Deming a household name among corporate executives, and companies such as Ford invited him to assist them in revolutionizing their quality approaches. Deming worked with passion until his death in December 1993 at the age of 93, knowing he had little time left to make a difference in his home country. When asked how he would like to be remembered, Deming replied, “I probably won’t even be remembered in my home country USA” Then after a long pause, he added, “Well, maybe.... As someone who spent his life trying to keep America from committing suicide.”

Unlike other management gurus and consultants, Deming never defined or described quality precisely. In his last book, he stated, “A product or a service possesses quality if it helps somebody and enjoys a good and sustainable market.” Deming’s philosophy is based on improving products and services by reducing uncertainty and variability in the design and manufacturing processes. In Deming’s view, variation is the chief culprit of poor quality. In mechanical assemblies, for example, variations from specifications for part dimensions lead to inconsistent performance and premature wear and failure. Likewise,

inconsistencies in service frustrate customers and damage a firm's image. To achieve reduced variation, he advocates a never-ending cycle of product design, manufacture, test, and sales, followed by market surveys, then redesign, and so forth.

Deming has summarized his philosophy in what he calls "A system of Profound Knowledge."

System of Profound Knowledge

Profound knowledge consists of four parts: (1) appreciation for a system, (2) some knowledge of the theory of variation, (3) theory of knowledge, and (4) psychology.

APPRECIATION FOR A SYSTEM

A system is a set of functions or activities within an organization that work together to achieve organizational goals. For example, McDonald's restaurant can be viewed as a system. It consists of the order-taker/cashier subsystem, grill and food preparation subsystem, drive-through subsystem, and so on.

The components of any system must work together for the system to be effective. When parts of a system interact, the system as a whole cannot be understood or managed solely in terms of its parts. To run any system, manager must understand the interrelationships among all subsystems and the people that work in them. One example is performance appraisal. The following are some of the factors within a system that affect the individual performance of an employee:

- Training received
- Information and resources provided
- Leadership of supervisors and managers
- Disruptions on the job
- Management policies and practices

According to Deming, however, most performance appraisals do not recognize these factors.

Management must have an aim, a purpose to which the system continually strives. Deming believes that the aim of any system is for everybody – stockholders, employees, customers, community, the environment – to gain over the long term. Stockholders can realize financial benefits, employees can have opportunities for training and education, customers can receive products and services that meet their needs and create satisfaction, the community can benefit from business leadership, and the environment can benefit from socially responsible management.

Deming emphasizes that management's job is to optimize the system. By making decisions that are best for only a small part of the system (often encouraged by competition), we sub-optimize. Sub-optimization results in a loss to everybody in the system. For example, a common practice is to purchase materials or services at the lowest bid. Inexpensive material may be of such inferior quality that they will cause excessive costs in adjustment and repair during manufacture and assembly. Although the purchasing department's track record might look good, the overall system will suffer.

This theory applies to managing people also. Pitting individuals or departments against each other for resources is self-destructive. The individuals or departments will perform to maximize their expected gain, not that of the firm as a whole. Employees must cooperate with each other. Likewise, sales quotas or arbitrary cost reduction goals do not motivate people to improve the system and, ultimately, customer satisfaction; workers will perform only to meet the quotas and goals.

THEORY OF VARIATION

The second part of Profound Knowledge some understands of statistical theory, particularly as it applies to variation. Just as no two pizzas or Qeema Nans are exactly alike, no two outputs from any production process are exactly alike. A production process contains many sources of variation. Different lots of martial will vary in strength, thickness, or moisture content, for example. Cutting tools will have inherent variation in strength and composition. During manufacturing, tools will experience wear, machine vibrations will cause changes in settings, and electrical fluctuations will cause variations in power. Operators may not position parts on fixtures consistently.

The complex interaction of all these variations in materials, tools, machines, operators, and the environment cannot be understood. Variation due to any individual source appears random; however, their combined effect is stable and can usually be predicted statically. Factors that are present as a natural part of a process are called common causes of variation.

Common causes generally account for about 80 to 90 percent of the observed variation in a production process. The remaining 10 to 20 percent result from special causes of variation, often called **assignable causes**. Special causes arise from external sources that are not inherent in the process. A **bad batch or material purchased from a supplier, poorly trained operator, excessive tools wear, or mis-calibration of measuring instruments** are examples of special causes. **Special causes result in unnatural variations that disrupt the random pattern of common causes.** Hence they are generally easy to detect using statistical methods, and it is usually economical to remove them.

A system governed only by common causes is said to be stable. Understanding a stable system and the differences between special and common causes of variation is essential for managing nay system. Management can make two fundamental mistakes in attempting to improve a process:

1. To treat as special cause any fault, complaint, mistake, breakdown, accident, or shortage when it actually came from common causes.
2. To attribute to common causes any fault, complaint, mistake, breakdown, accident, or shortage when it actually came from a special cause.

In the first case, tampering with a stable system will actually increase the variation in the system. In the second case, we can miss the opportunity to eliminate unwanted variation by assuming that it is not controllable. Changing a system on the basis of a special cause can damage the system and add cost. Variation should be minimized. The producer and consumer both benefit from reduced variation. The producer benefits by having less need for inspection, less scrap and rework, and higher productivity. The consumer is assured that all products have similar quality characteristics; this especially important when the consumer is another firm using large quantities of the product in its own manufacturing or service operation.

Variation increases the cost of doing business. The only way to reduce variation due to common causes is to change the technology of the process —the machines, people, materials, methods, or measurement system. The process is under the control of management, not the production operators. Pressuring operators to perform at higher quality levels may not be possible and may be counterproductive. Variation due to special causes can be identified through the use of control charts, which shall be introduced latter.

Lesson # 13**Total quality management theories
Edward Deming's system of profound knowledge****Theory of Knowledge**

The third part of Profound Knowledge is the theory of knowledge – a branch of philosophy and management concerned with the nature and scope of knowledge, its presuppositions and bases, and the general reliability of claims to knowledge.

Deming emphasizes that there is no knowledge without theory and that experience alone does not establish a theory. To copy an example of success without understanding it with the aid of theory may lead to disaster. Experience only describes; it cannot be tested or validated. Theory establishes a cause-and-effect relationship that can be used for prediction. Theory leads to questioning and can be tested and validated – it explains why. Many consultant methods that have sustained success are grounded in theory. Managers have responsibility to learn and apply theory.

PSYCHOLOGY

Psychology helps us to understand people, interactions between people and circumstances, interactions between leaders and employees, and any system of management. People differ from one another. A leader must be aware of these differences and use them to optimize everybody's abilities and inclinations.

Many managers operate under the supposition that all people are alike and treat them as interchangeable components of a process. However, people learn in different ways and at different speeds and perform at different levels. Leaders have an obligation to make changes in the system of management that will bring improvement. People have an innate need for relationships with other people and for self-esteem and respect. Circumstances provide some people with dignity and self-esteem and deny them to other people. People inherit the right to enjoy work. Psychology helps us to nurture and preserve people's positive innate attributes.

PERSPECTIVES ON PROFOUND KNOWLEDGE

Little in Deming's system of Profound Knowledge is original. The concept of common and special causes of variation was developed by Walter Shewhart in the 1920s; behavioral theories to which Deming subscribes were developed in the 1960s; systems theory was refined by management scientists from the 1950s through the 1970s; and scientists in all fields have long understood the relationships among prediction, observation, and theory. Deming's contribution was in tying together some basic concepts. He recognized the synergy among these diverse subjects and developed them into a theory of management.

Peter Scholtes, a noted consultant, makes some salient observations about the failure to understand the components of Profound Knowledge:

1. When people don't understand systems;

- They see events as individuals incidents rather than the net result of many interactions and interdependent forces;
- They see the symptoms but not the deep causes of problems;
- They don't understand how an intervention in one part of [an organizational] can cause havoc in another place or at another time;

- They blame individuals for problems even when those individuals have little or no ability to control the events around them; and
2. **When people don't understand variation;**
- They don't see trends that are occurring;
 - They see trends where there are none;
 - They don't know when expectations are realistic;
 - They don't understand past performance so they can't predict future performance;
 - They don't know the difference between prediction, forecasting, and guesswork;
 - They give others credit or blame when those people are simply either lucky or unlucky. This usually occurs because people tend to attribute everything to human effort, heroics, frailty, error, or deliberate sabotage, no matter what the systemic cause; and
 - They are less likely to distinguish between fact and opinion.
3. **When people don't understand psychology;**
- They don't understand motivation or why people do what they do;
 - They resort to carrots and sticks and other forms of induced motivation that have no positive effect and impair the relationship between the motivator and the one being motivated;
 - They don't understand the process of change and the resistance to it;
 - They revert to coercive and paternalistic approaches when dealing with people; and
 - They create cynicism, demoralization, demotivation, guilt, resentment, burnout, craziness, and turnover.
4. **When people don't understand the theory of knowledge;**
- They don't know how to plan and accomplish learning and improvement;
 - They don't understand the difference between improvement and change; and
 - Problems will remain unsolved, despite their best efforts.

DEMING'S PHILOSOPHY & 14 POINTS FOR MANAGEMENT

The 14 Points for Management, listed below in table have been the subject of considerable controversy and debate. They have their basis in System of Profound Knowledge. Many companies have studied and applied them to their organizations with success.

Deming's 14 Points for Management

- | | |
|-----|--|
| 1. | Create and publish to all employees a statement of the aims and purposes of the company of other organization. The management must demonstrate constantly their commitment to this statement. |
| 2. | Learn the new philosophy, top management and everybody. |
| 3. | Understand the purpose of inspection, for improvement of processes and reduction of cost. |
| 4. | End the practice of awarding business on the basis of price tag alone. |
| 5. | Improve constantly and forever the system of production and service. |
| 6. | Institute training. |
| 7. | Teach and instituted leadership |
| 8. | Drive out fear. Create trust. Create a climate for innovation. |
| 9. | Optimize toward the aims and purposes of the company the efforts of teams, groups, staff areas. |
| 10. | Eliminate exhortations for the workforce. |
| 11. | (a) Eliminate numerical quotas for production. Instead, learn and institute methods for improvement.
(b) Eliminate MBO (Management by Objective). Instead, learn the capabilities of processes and how to improve them. |
| 12. | Remove barriers that rob people of pride of workmanship. |
| 13. | Encourage education and self-improvement for everyone. |
| 14. | Take action to accomplish the transformation. |

Detail Explanation of Deming's Fourteen Points of TQM

- Management Commitment** – Businesses should not exist simply for profit; their true purpose should be to serve their customers and employees. To do this, they must take a long-term view and invest in innovation, training, and research. Thus, an organization needs a clear mission and statement of purpose.
- Learn the New Philosophy** – Western management has been built on the Taylor system, which has led to numbers-driven production, quotas, and adversarial work relationships. Old methods of management create mistrust, fear, and anxiety with a focus on “Satisfying” rather than on “optimizing.” Eliminating defects is not good enough. Defect-free production is taken for granted in Japan. Achieving competitive success in today’s global economy requires a customer-driven approach based on mutual cooperation between labor and management and a never-ending cycle of improvement. Everyone, from the boardroom to the stockroom, must learn the new philosophy.
- Understand Inspection** – Routine inspection acknowledges defects but does not add value to the product. Instead, it encourages defects because “someone else” catches and fixes the problems. This procedure increases costs and decreases productivity. Workers must take responsibility for their own work and be able to take appropriate action to assure good quality. Manager need to understand how variation affects their processes and to take steps to reduce the causes of variation. Inspection should be used as information-gathering tools for improvement, not as an end in itself.

Lesson # 14

DEMING'S PHILOSOPHY AND 14 POINTS FOR MANAGEMENT

4. **End Price Tag Decisions** – Purchasing decisions traditionally have been driven by cost through competitive bidding, not by quality. Costs due to inferior materials and components increase costs in later stages of production and can far exceed the “savings” realized through competitive bidding. The purchasing department is a supplier to the production department and must understand its new role. Suppliers themselves are part of the whole system.
5. **Improve constantly** – Western management has typically thought of improvement in the context of large, expensive innovations such as robotics and computer-integrated manufacturing. The success of Japanese manufacturers, however, is due primarily to continuous, small, incremental improvements in design and production. Improved design results from understanding customer needs and from continual market surveys and other sources of feedback. Improved production is achieved by reducing the causes of variation in order to establish a stable, predictable production process. Statistical methods provide one means for doing this. Improvement should go beyond production, encompassing transportation, engineering, maintenance, sales, service, and administration – all areas of the organization.
6. **Institute Training** – Employees need the proper tools and knowledge to do a good job, and it is management's responsibility to provide these. In addition to specific job skill, all employees should be trained in statistical tools for quality problem solving and continuous improvement. Training not only improves quality and productivity, but also enhances workers' morale by showing them that the company is dedicated to helping them and is investing in their future. Deming notes that in Japan, entry-level managers spend 4 to 12 years on the factory floor and in other activities to learn the problems of production. At Honda of America in Maryville, Ohio, all employees start out on the production floor regardless of their job classification.
7. **Institute Leadership** – The job of management is leadership and guidance, not supervision and work direction. Supervisors should be coaches, not policemen, and supervision should provide the link between management and the workforce. Leadership can help to eliminate fear and encourage teamwork.
8. **Drive out Fear** – Fear in work manifests in many ways: fear of reprisal, fear of failure, fear of the unknown, fear of change. Many workers fear punishment or reprisals for not meeting quotas and for problems of the system that are beyond their control. Managers compete against each other to protect their own jobs or to receive higher performance ratings. Fear encourages short-term, selfish thinking, not long-term improvement for the benefit of all.
9. **Optimize Team Efforts** – Barriers between individuals and departments lead to poor quality, because “customers” do not receive what they need from their “suppliers.” This is often the result of internal competition for raises or performance ratings. Teamwork helps to break down barriers between internal customers and suppliers. The focus should be on meeting customer needs and improving processes. Teamwork is an important means of achieving a company's goals.
10. **Eliminate Exhortations** – Motivation can be better achieved through trust and leadership than slogans. Slogans calling for improved quality usually assume that poor quality results from a lack of motivation. Workers cannot improve solely through motivational methods when the system in which they work constrains their performance. On the contrary, they will become frustrated and their performance will decrease further.

11. **Eliminate Quotas and MBO** – Numerical quotas reflect short-term perspectives and do not encourage long-term improvement, particularly if rewards or performance appraisals are tied to meeting quotas. Workers may shortcut quality to reach the goal. If the quota is met, they have no incentive to continue production or to improve quality. Arbitrary management goals without a method for achieving them have no meaning. Further, variation in the system makes year-to-year or quarter-to-quarter comparisons meaningless. The typical American MBO system focuses on results, not process, and encourages short-term behavior. Management must understand the system and the variation within it and seek to improve it in the long term.
12. **Remove Barriers to Pride in Workmanship** – The Taylor system has promulgated the view of workers as a “commodity.” Factory workers are given monotonous tasks, provided with inferior machines, tools, or materials, told to run defective items to meet sales pressures, and report to supervisors who know nothing about the job.

Deming believed that one of the biggest barriers to pride in workmanship is performance appraisal. Performance appraisals

- Destroy teamwork by promoting competition among employees for limited resources;
- Foster mediocrity since objectives typically are driven by numbers and what the boss wants;
- Focus on short-term results and discourage risk taking; and
- Are not focused on serving the customer.

Deming suggest that there are three categories of performance: the majority who work within the system, those outside the system on the superior side, and those outside the system on the inferior side. Statistical methods provide the means of making this classification. Superior performers should be compensated specially; inferior performers need extra training or a different job.

13. **Institute Education** – “Training” in number 6 refers to job skills; education refers to self-development. Firms have a responsibility to develop the value and self-worth of the individual. Investing in people is a powerful motivation method.
14. **Take Action** – The TQM philosophy is a major cultural change and many firms find it difficult. Top management must institute the process and include everyone in it.

Continuous improvement is an ongoing effort to improve products, services or processes. These efforts can seek “incremental” improvement over time or “breakthrough” improvement all at once.

Among the most widely used tools for continuous improvement is a four-step quality model—the plan-do-check-act (PDCA) cycle, also known as Deming Cycle or Shewhart Cycle:

- **Plan:** Identify an opportunity and plan for change.
- **Do:** Implement the change on a small scale.
- **Check:** Use data to analyze the results of the change and determine whether it made a difference.
- **Act:** If the change was successful, implement it on a wider scale and continuously assess your results. If the change did not work, begin the cycle again.

Other widely used methods of continuous improvement — such as Six Sigma, Lean, and Total Quality Management — emphasize employee involvement and teamwork; measuring and systematizing processes; and reducing variation, defects and cycle times.

Continuous or Continual?

The terms *continuous improvement* and *continual improvement* are frequently used interchangeably. But some quality practitioners make the following distinction:

- *Continual improvement*: a broader term preferred by W. Edwards Deming to refer to general processes of improvement and encompassing “discontinuous” improvements—that is, many different approaches, covering different areas.
- *Continuous improvement*: a subset of continual improvement, with a more specific focus on linear, incremental improvement within an existing process. Some practitioners also associate *continuous improvement* more closely with techniques of statistical process control.

The cost of quality.”

It’s a term that’s widely used – and widely misunderstood.

The “cost of quality” isn’t the price of creating a quality product or service. It’s the cost of NOT creating a quality product or service.

Every time work is redone, the cost of quality increases. Obvious examples include:

- The reworking of a manufactured item.
- The retesting of an assembly.
- The rebuilding of a tool.
- The correction of a bank statement.
- The reworking of a service, such as the reprocessing of a loan operation or the replacement of a food order in a restaurant.

In short, any cost that would not have been expended if quality were perfect contributes to the cost of quality.

Total Quality Costs

As the figure below shows, quality costs are the total of the cost incurred by:

- Investing in the prevention of nonconformance to requirements.
- Appraising a product or service for conformance to requirements.
- Failing to meet requirements.

Quality Costs—general description

Prevention Costs

The costs of all activities specifically designed to prevent poor quality in products or services.

Examples are the costs of:

- New product review
- Quality planning
- Supplier capability surveys
- Process capability evaluations
- Quality improvement team meetings
- Quality improvement projects
- Quality education and training

Appraisal Costs

The costs associated with measuring, evaluating or auditing products or services to assure conformance to quality standards and performance requirements.

These include the costs of:

- Incoming and source inspection/test of purchased material
- In-process and final inspection/test
- Product, process or service audits
- Calibration of measuring and test equipment
- Associated supplies and materials

Total Quality Costs:

The sum of the above costs. This represents the difference between the actual cost of a product or service and what the reduced cost would be if there were no possibility of substandard service, failure of products or defects in their manufacture.

Failure Costs

The costs resulting from products or services not conforming to requirements or customer/user needs. Failure costs are divided into internal and external failure categories.

Internal Failure Costs

Failure costs occurring prior to delivery or shipment of the product, or the furnishing of a service, to the customer.

Examples are the costs of:

- Scrap
- Rework
- Re-inspection
- Re-testing
- Material review
- Downgrading

External Failure Costs

Failure costs occurring after delivery or shipment of the product -- and during or after furnishing of a service -- to the customer.

Examples are the costs of:

- Processing customer complaints
- Customer returns
- Warranty claims
- Product recalls

Lesson # 15**Deming Cycle and Quality Trilogy****DEMING CYCLE**

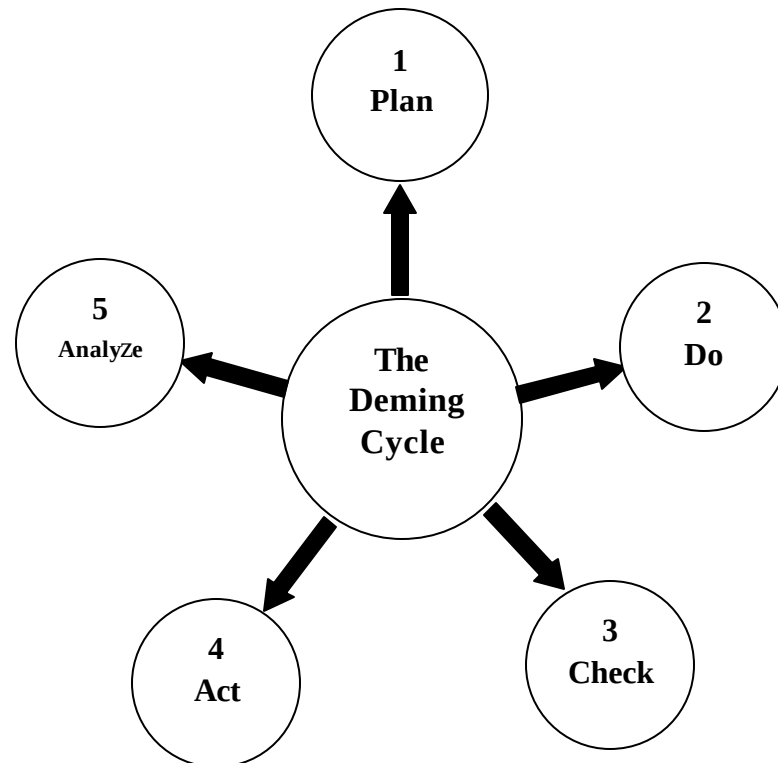
Deming cycle is a tool for continuous improvement and it is a tool for an ongoing effort to improve products, services or processes. These efforts can seek “incremental” improvement over time or “breakthrough” improvement all at once.

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- **Act:** If the change was successful, implement it on a wider scale and continuously assess your results. If the change did not work, begin the cycle again.

Deming cycle was developed to link the production of a product with consumer needs and focus the resources of all departments (research, design, production, and marketing) in a cooperative effort to meet those needs. The Deming Cycle proceeds as follows:

1. Conduct consumer research and use it in planning the product (PLAN).
2. Produce the product (DO).
3. Check the product to make sure it was produced in accordance with the plan (CHECK).
4. Market the product (ACT).
5. Analyze how the product is received in the marketplace in terms of quality, cost, and other criteria (ANALYZE).



DEMING CYCLE

Other widely used methods of continuous improvement — such as Six Sigma, Lean Production, and Kaizen — emphasize employee involvement and teamwork; measuring and systematizing processes; and reducing variation, defects and cycle times.

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associate *continuous improvement* more closely with techniques of statistical process control.

Juran's Contribution

Joseph M. Juran ranks near Deming in the contributions he has made to quality and the recognition he has received as a result. His Juran Institute. Inc., in Connecticut, USA is an international leader in conducting training, research, and consulting activities in the area of quality management. Quality materials produced by Juran have been translated into 14 different languages.

Juran holds degrees in both engineering and law. The emperor of Japan awarded him the Order of the Sacred Treasure medal, in recognition of his efforts to develop quality in Japan and to promote friendship between Japan and the United States. Juran is best known for the following contributions to the quality philosophy:

- ◆ Juran's Three Basic Steps to Progress
- ◆ Juran's Ten Steps to Quality Improvement
- ◆ The Juran Trilogy

Juran's Three Basic Steps to Progress

Juran's Three Basic Steps to Progress are broad steps that, in Juran's opinion, companies must take if they are to achieve world-class quality. He also believes there is a point of diminishing return that applies to quality and competitiveness. An example illustrates his observation:

Say that an automobile maker's research on its cars reveals that buyers drive them an average of 50,000 kms before trading them in. Applying Juran's theory, this automaker should invest the resources necessary to make this line of cars run trouble free for perhaps 60,000 kms. According to Juran, resources devoted to improving quality beyond this point will run the cost up higher than the typical buyer is willing to pay.

- | |
|--|
| <ol style="list-style-type: none">1. Achieve structured improvements on a continual basis combined with dedication and a sense of urgency.2. Establish an extensive training program.3. Establish commitment and leadership on the part of higher management |
|--|

Juran's Ten Steps to Quality Improvement

Examining Juran's Ten Steps to Quality Improvement, you will see some overlap between them and Deming's Fourteen Points. They also mesh well with the philosophy of other quality experts.

1. Build awareness of both the need for improvement and opportunities for improvement.
2. Set goals for improvement.
3. Organize to meet the goals that have been set.
4. Provide training.
5. Implement projects aimed at solving problems.
6. Report progress.
7. Give recognition.
8. Communicate results.
9. Keep score.
10. Maintain momentum by building improvement into the company's regular systems.

The Juran Trilogy

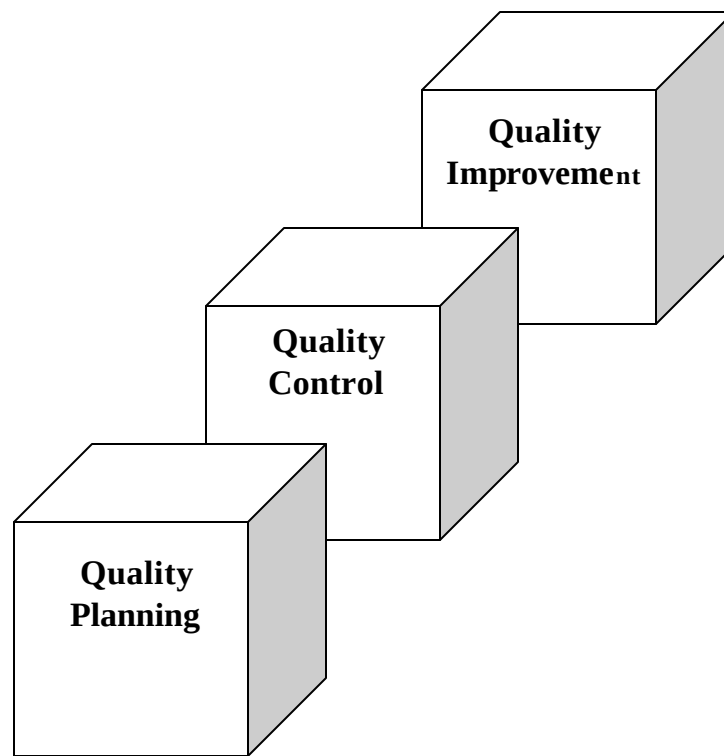
The Juran Trilogy summarizes the three primary managerial functions.

Quality Planning, Quality Control, and Quality Improvement

Lesson # 16

**JURAN AND CROSBY ON QUALITY AND
QUALITY IS FREE****The Juran Philosophy and Trilogy**

The Juran Trilogy® is a registered trademark of Juran Institute, Inc.



Juran taught quality principles to the Japanese in the 1950s just after Deming and was a principal force in their quality reorganization. Like Deming, he concludes that we face a major crisis due to the loss of sales to foreign competition and the huge costs of poor quality. To solve this crisis, new thinking about quality that includes all levels of the managerial hierarchy is required. Upper management in particular requires training and experience in managing for quality.

Juran's programs are designed to fit into a company's current strategic business planning with minimal risk of rejection. This is in contrast to Deming who proposes sweeping cultural change. Juran contends that employees at different levels of an organization speak in different "languages." (Deming believes statistics should be the common language.)

Top management speaks in the language of dollars, workers speak in the language of things, and middle management must be able to speak both languages and translate between dollars and things. Thus, to get top management's attention, quality issues must be cast in the language they understand – dollars.

Juran advocates the accounting and analysis of quality costs to focus attention on quality problems. At the operational level, Juran's focus is on increasing conformance to specifications through elimination of defects, supported extensively by statistical tools for analysis. Thus, his philosophy fits well into existing management systems.

Juran defines quality as "fitness for use." (Deming advocates no specific definition.) This is broken down into four categories: quality of design, quality of conformance, availability, and field service. Quality of design focuses on market research, the product concept, and design specifications. Quality of conformance includes technology, manpower, and management. Availability focuses on reliability, maintainability, and logistical support. Field service quality comprises promptness, competence, and integrity.

Juran views the pursuit of quality on two levels: (1) the mission of the firm as whole is to achieve high product quality, and (2) the mission of each individual department in the firm is to achieve high production quality. Like Deming, Juran advocates a never-ending spiral of activities that includes market research, product development, design, planning for manufacture, purchasing, production process control, and inspection and testing, followed by customer feedback. Because of the interdependence of these functions, the need for competent company-wide quality management is great. Senior management must play an active and enthusiastic leadership role in the quality management process.

Juran's prescriptions focus on three major aspects of quality called the Quality Trilogy.

Quality planning – the process for preparing to meet quality goals,

Quality control – the process for meeting quality goals during operations, and

Quality improvement – the process for breaking through to unprecedented levels of performance.

Quality planning begins with identifying customers, both external and internal, determining their needs, and developing product features that respond to customer needs.

Quality control involves determining what to control, establishing units of measurement so that data may be objectively evaluated, establishing standards of performance, measuring actual performance,

interpreting the difference between actual performance and the standard, and taking action on the difference.

Juran's program for quality improvement involves demonstrating the need for improvement, identifying specific projects for improvement, organizing to guide the projects, diagnosing the causes, providing remedies for the causes, proving that the remedies are effective under operating conditions, and providing control to hold improvements.

Quality Planning: Quality planning involves developing the products, systems, and process needed to meet or exceed customer expectations. The following steps are required:

1. Determine who the customers are.
2. Identify customers' needs.
3. Develop products with features that respond to customer needs.
4. Develop systems and processes that allow the organization to produce these features.
5. Deploy the plans to operational levels.

Quality Control: The control of quality involves the following processes:

1. Assess actual quality performance.
2. Compare performance with goals.
3. Act on differences between performance and goals.

Quality Improvement: The improvement of quality should be ongoing continual:

1. Develop the infrastructure necessary to make annual quality improvements.
2. Identify specific areas in need of improvement, and implement improvement projects.
3. Establish a project team with responsibility for completing each improvement project.
4. Provide teams with what they need to be able to diagnose problems to determine root causes develop solutions, and establish control that will maintain gains made.

Juran's assessment of most companies is that quality control is far and away the top priority among the trilogy and most companies feel they are strong in this category. Quality planning and quality improvement, however, are not important priorities and are significantly weaker in most organizations. He feels that more effort needs to be placed on quality planning and even more on quality improvement.

Japanese efforts at quality improvement were supported by massive training programs and top management leadership. Training in managerial quality-oriented concepts as well as training in the tools for quality improvement, cost reduction, data collection, and analysis is one of the most important components of Juran's philosophy. Juran maintains that the Japanese experience leaves little doubt as to the significance of the return on quality training in competitive advantage, reduced failure costs, higher productivity, smaller inventories, and better delivery performance.

Crosby's Philosophy and Contributions

Philip B. Crosby started his career in quality later than Deming and Juran. His corporate background includes 14 years as director of quality at ITT (1965-1979). He left ITT in 1979 to form Philip Crosby Associates, an international consulting firm on quality improvement, which he ran until 1992, when he retired as CEO to devote his time to lecturing on quality related issues.

Crosby, who defines quality simply as conformance to customer requirements, is best known for his advocacy of zero defects management and prevention as opposed to statistically acceptable levels of quality. He is also known for his work on Costs of Quality, Quality is Free and Crosby's Fourteen Steps to Quality Improvement.

The essence of Crosby's quality philosophy is embodied in what he calls the Absolutes of Quality Management and the Basic Elements of Improvement..

Crosby's Absolutes of Quality Management are as follows:

- ***Quality means conformance to requirements not elegance.*** Crosby dispels the myth that quality is simply a feeling of "excellence." Requirements must be clearly stated so that they cannot be misunderstood. Requirements are communication devices and are ironclad. Once a task is done, one can take measurements to determine conformance to requirements. The nonconformance detected is the absence of quality. Quality problems become nonconformance problems – that is, variation in output. Setting requirements is the responsibility of management.
- ***There is no such thing as a quality problem.*** Problems must be identified by the individuals or departments that cause them. There are accounting problems, manufacturing problems, design problems, front-desk problems, and so on. Quality originates in functional departments, not in the quality department, and the burden of responsibility for such problems lies with the functional departments. The quality department should measure conformance, report results, and lead the drive

the develop a positive attitude toward quality improvement. This is similar to number 3 of Deming's Points.

- ***There is no such thing as the economics of quality: it is always cheaper to do the job right the first time.*** Crosby supports the premise that “economics of quality” has no meaning. Quality is free. What costs money are all the actions that involve not doing jobs right the first time. The Deming Chain Reaction provides a similar message.

CROSBY'S CONCEPT OF COST OF QUALITY

- ***The only performance measurement is the cost of quality.*** The cost of quality is the expense of nonconformance. Crosby notes that most companies spend 15 to 20 percent of their sales dollars on quality costs. Accompany with a well-run quality management program can achieve a cost of quality that is less than 2.5 percent of sales, primarily in the prevention and appraisal categories. Crosby's program calls for measuring and publicizing the cost of poor quality. Quality cost data are useful in calling problems to management's attention, selecting opportunities for corrective action, and tracking quality improvement over time. Such data provide visible proof of improvement and recognition of achievement. Juran also supports this theme.
- ***The only performance standard is Zero Defects.*** Crosby feels that the Zero Defects (ZD) concept is widely misunderstood and resisted. Zero Defects is not a motivational program. It is as follows:

Zero Defects is a performance standard. It is the standard of the craftsman regardless of his or her assignment. The theme of ZD is doing it right the first time. That means concentrating on preventing defects rather than just finding and fixing them.

People are conditioned to believe that error is inevitable; thus they not only accept error, they anticipate it. It does not bother us to make a few errors in our work. . . . To err is human. We all have our own standards in business or academic life—our own points at which errors begin to bother us. It is good to get an A in school, but it may be OK to pass a course with a C.

We do not maintain these standards, however, when it comes to our personal life. If we did, we should expect to be shortchanged every now and then when we cash our paycheck; we should expect hospital nurses to drop a constant percentage of newborn babies. . . . We as individuals do not tolerate these things. We have a dual standard: one for ourselves and one for our work.

Most human error is caused by lack of attention rather than lack of knowledge. Lack of attention is created when we assume that error is inevitable. If we consider this condition carefully and pledge ourselves to make a constant conscious effort to do our jobs right the first time, we will take a giant step toward eliminating the waste of rework, scrap, and repair that increases cost and reduces individual opportunity.

Crosby summarized his approach to management in what he refers to as the Absolutes of Quality Management, which answer the following questions.

1. What is quality?
2. What system is needed to cause quality?
3. What performance standard should be used?
4. What measurement system is required?

Answers

1. The First Absolute: The Definition of Quality is conformance to Requirements.
2. The Second Absolute: The System of Quality Is Prevention.
3. The Third Absolute: the Performance Standard Is Zero Defects.
4. The Fourth Absolute: The Measurement of Quality Is the Price of Nonconformance.

Cost of Quality

Until the 1950s, manager assumed it was important to improve quality because defects were costly. But they had no idea just how costly defects were and consequently did not know how much they should improve. They had no yardstick for measuring the costs of quality. In addition to emphasizing management's role in quality, Joseph Juran and Philip Crosby gave managers a means of answering a critical question which remained in their minds: "How much quality is enough?"

The cost of achieving a given level of quality is divided into avoidable costs and unavoidable costs. Unavoidable costs are those related to preventing defects. These include inspection, sampling, sorting, and other quality control initiatives.

Avoidable costs are related to defects and product failures. These include scrapped materials, labor hours required for rework and repair, complaint processing, and financial losses resulting from unhappy customers. Juran and Crosby called avoidable costs "gold in the mine" because investment in quality improvement can sharply reduce them and lead to substantial savings. With Juran and Crosby's cost of quality concept, manager could calculate when additional expenditures on prevention were justified.

Unlike Juran and Deming, Crosby's program is primarily behavioral. He places more emphasis on management and organizational processes for changing corporate culture and attitudes than on the use of statistical techniques. Like Juran and unlike Deming, his approach fits well within existing organizational structures.

Cost of Quality Attitude:

In fact in daily life we pay the price for our not so good attitude in rupees, or in the form of lost respect or rapport or even anger and frustration which are the emotional costs. How to plan to prevent them is every body's responsibility. Best price or zero cost is the prerogative cost. It is the compliment or comment given to you by your customer, by your family or by they boss when he or she ask that has he or she paid you enough for your quality work and are you really happy with him on the given reward!

Quality Management Characteristics for the Future

To succeed in the global marketplace for now and in the future, organizations will have to operate according to the principles of quality management. Such companies will have the following characteristics:

- ◆ A total commitment to continually increasing value for customers, investors, and employees.
- ◆ A firm understanding that market driven means that quality is defined by customers, not the company
- ◆ A commitment to leading people with a bias for continuous improvement and communication.
- ◆ A recognition that sustained growth requires the simultaneous achievement of four objectives all the time, forever: (a) customer satisfaction, (b) cost leadership, (c) effective human resources, and (d) integration with the supplier base.
- ◆ A commitment to fundamental improvement through knowledge, skills, problem solving, and teamwork.

Companies that develop these characteristics will be those that fully institutionalize the principles of quality management. Consequently, quality management as both a practice and a profession has a bright future. In fact, in terms of succeeding in the global marketplace, quality management is the future.

Consequently, more and more companies are making quality management the way they do business, and more and more institutions of higher education are offering quality management courses and programs.

Lesson # 18

COSTS OF QUALITY AND RETURN ON QUALITY**The Cost of Quality:**

It's a term that's widely used – and widely misunderstood.

The “cost of quality” isn't the price of creating a quality product or service. It's the cost of NOT creating a quality product or service.

Every time work is redone, the cost of quality increases. Obvious examples include:

- The reworking of a manufactured item.
- The retesting of an assembly.
- The rebuilding of a tool.
- The correction of a bank statement.
- The reworking of a service, such as the reprocessing of a loan operation or the replacement of a food order in a restaurant.

In short, any cost that would not have been expended if quality were perfect contributes to the cost of quality.

Total Quality Costs

As the table below shows, quality costs are the total of the cost incurred by:

- Investing in the prevention of nonconformance to requirements.
- Appraising a product or service for conformance to requirements.
- Failing to meet requirements.

Quality Costs—general description**Prevention Costs**

The costs of all activities specifically designed to prevent poor quality in products or services.

Examples are the costs of:

- New product review
- Quality planning
- Supplier capability surveys
- Process capability evaluations
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- Quality improvement projects
- Quality education and training

Appraisal Costs

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The costs resulting from products or services not conforming to requirements or customer/user needs. Failure costs are divided into internal and external failure categories.

Internal Failure Costs

Failure costs occurring prior to delivery or shipment of the product, or the furnishing of a service, to the customer.

Examples are the costs of:

- Scrap
- Rework
- Re-inspection
- Re-testing
- Material review
- Downgrading

External Failure Costs

These include the costs of:

- Incoming and source inspection/test of purchased material
- In-process and final inspection/test
- Product, process or service audits
- Calibration of measuring and test equipment
- Associated supplies and materials

Failure costs occurring after delivery or shipment of the product -- and during or after furnishing of a service -- to the customer.

Examples are the costs of:

- Processing customer complaints
- Customer returns
- Warranty claims
- Product recalls

Total Non Quality Costs and Return on Quality:

The sum of the above costs to produce a product or to deliver a service. This represents the difference between the actual cost of a product or service and what the reduced cost would be if there were no possibility of substandard service, failure of products or defects in their manufacture. The measure of revenue saved by producing the right product the first time and every time will actually be the return one gets by implementing the system of quality.

In general if preventive system costs Rs. 1 like any QA system then QC overheads are RS 10 in comparison. The costs incurred on failure is Rs 100 and even sometime higher (1000 or more) as you may totally loose your customers, your rapport or your total market.

Hence we need to learn that quality system is an investment which ultimately reduces the total cost of production or working as we have to consider every effort of ours financially accountable to make quality a business case.

Quality is a continuous journey and hence not a destination.

Lesson # 19

OVERVIEW OF TOTAL QUALITY APPROACHES

“A company that claims that it cannot standardize and must rely on experience is a company without technology.”

Kaoru Ishikawa

The Total View of Quality

The concept of customer value represents a dramatic improvement over the traditional approach to quality, the “conformance to specified standards” approach. It extends the concept of quality to include user perceptions and use consequences. However, it still falls short of the concept of Total Quality, which stresses the importance of quality in every aspect of an organization.

Perhaps the Japanese best express this broader and more holistic view of quality, Ishikawa states: “Narrowly interpreted, quality means quality of product. Broadly interpreted, quality means quality of work, quality of service, quality of information, quality of process, quality of division, quality of people including workers, engineers, managers, and executives, quality of system, quality of company, quality of objective, etc.” This view of quality may at first seem to be too idealistic. However, managers who are committed to this view of quality have pragmatic solutions for translating the word “quality” into organizational realities.

As Imai suggests, it all begins with the “quality of people.” He states:

There is very little agreement on what constitutes quality. In its broadest sense, quality is anything that can be improved. When speaking of “quality” one tends to think first in terms of product quality. When discussed in the context of kaizen strategy nothing could be further off the mark. The foremost concern here is with the quality of people. The three building blocks of a business are hardware, software, and “human ware.” Only after human ware is squarely in place should the hardware and software aspects of a business be considered. Building quality into people means helping them become kaizen conscious.

This total view of quality includes all of the above quality themes we have already seen, integrating them into a comprehensive approach to continuous improvement.

The Future of Quality Management

In an article for Quality Digest, another quality pioneer Armand Feigenbaum explains several trends that will shape the future of quality management. Those trends are as follows:

- ◆ **Demanding global customers.** The provision of quality begets an ever-increasing demand for quality. Today’s customers share two common characteristics: (a) they are part of regional trade alliances such as the Americas, Europe, and Asia: and (b) they expect both high quality and added value.
- ◆ **Shifting customer expectations.** Increasingly, today’s global customer is interested not just in the quality of a product provided but also the quality of the organization that backs it up. Customers want an excellent product or service from an organization that also provides accurate billing, reliable delivery, and after-purchase support.

- ◆ **Opposing economic pressures.** The global marketplace exerts enormous, unrelenting pressure on organizations to continually improve quality while simultaneously reducing the prices they charge for goods and services. The key to achieving higher quality and lower prices for customers is the reduction of the expenses associated with satisfying unhappy customers – expenses that amount to as much as 25% of the cost of sales in many companies.
- ◆ **New approaches to management.** Companies that succeed in the global marketplace have learned that you manage budgets, but lead people. The old approach of providing an occasional seminar or motivational speech for employees without making any fundamental changes in the way the organization operates will no longer work.

The total in total quality indicates a concern for quality in the broadest sense – what has come to be known as the “Big Q.” Big Q refers to quality of products, services, people, processes, and environments. Correspondingly, “Little Q” refers to a narrower concern that focuses on the quality of one of these elements or individual quality criteria within an individual element.

Which Q Approach?

QMS implementation has to be a systemic approach but which system to choose depends where are you and how far the management is committed for quality. Each QMS requires companies to deploy various quality tools for the improvement of quality.

INTERNATIONAL QUALITY AWARDS

A focus on total quality has permeated organizations throughout the world. Numerous countries and regions of the world have established awards and award criteria.

The Malcolm Baldrige National Quality Award (MBNQA) has been one of the most powerful catalysts of total quality in the United States, and indeed, throughout the world. More importantly, the Award's Criteria for Performance excellence establishes a framework for integrating total quality principles and practices in any organization. Many other award programs are similar in nature to the Baldrige criteria.

THE DEMING PRIZE

The Deming Application Prize was instituted in 1951 by the Union of Japanese Scientists and Engineers (JUSE) in recognition and appreciation of W. Edwards Deming's achievements in statistical quality control and his friendship with the Japanese people. The Deming Prize has several categories, including prizes for individuals, factories, and small companies, and the Deming application prize, which is an annual award presented to a company or a division of a company that has achieved distinctive performance improvements through the application of Company-wide Quality Control (CWQC). As defined by JUSE, CWQC is

A system of activities to assure that quality products and services required by customers are economically designed produced and supplied while respecting the principle of customer-orientation and the overall public well-being. These quality assurance activities involve market research, research and development, design, purchasing, production, inspection and sales, as well as all other related activities inside and outside the company. Through everyone in the company understanding both statistical concepts and methods, through their application to all the aspects of quality assurance and through repeating the cycle of rational planning, implementation, evaluation and action, CWQC aims to accomplish business objectives.²³

The judging criteria consist of a checklist of 10 major categories: policies, the organization and its operations, education and dissemination, information gathering, communication and its utilization, analysis, standardization, control/management, quality assurance, effects, and future plans. Each major category is divided into subcategories, or “checking points.” For example, the policy category includes policies pursued for management, quality, and quality control; methods for establishing policies; appropriateness and consistency of policies; utilization of statistical methods; communication and dissemination of policies; checks of policies and the status of their achievement; and the relationship between policies and long- and short-term plans. Each category is weighted equally.

Hundreds of companies apply for the award each year. After an initial application accepted as eligible for the process, the company must submit a detailed description of its quality practices. Based on review of the written descriptions, only a few companies believed to be successful in CWQC are selected for a site visit. The site visit consists of a company presentation, in-depth questioning by examiners, and an executive session with top managers. Examiners visit plants and are free to ask any worker any question.

The Deming Prize is awarded to all companies that meet the prescribed standard. However, the small number of awards given each year is an indication of the difficulty of achieving the standard. The objectives are to ensure that a company has so thoroughly deployed a quality process that it will continue to improve long after a prize is awarded. The application process has no “losers.” For companies that do not qualify, the examination process is automatically extended up to two times over three years.

EUROPEAN QUALITY AWARD

In October 1991, the European Foundation for Quality Management (EFQM) in partnership with the European Commission and the European Organization for Quality announced the creation of the European Quality Award. The award was designed to increase awareness throughout the European Community, and businesses in particular, of the growing importance of quality to their competitiveness in the increasingly global market and to their standards of life. The European Quality Award consists of two parts: the European Quality Prize, given to companies that demonstrate excellence in quality management practice by meeting the award criteria, and the European Quality Award, awarded to the most successful applicant. In 1992, four prizes and one award were granted for the first time.

The award process is similar to the Deming Prize and Baldrige Award. The assessment is based on customer satisfaction, business results, processes, leadership, people satisfaction, resources, people management, policy and strategy, and impact on society. Like Baldrige, results – including customer satisfaction, people (employee) satisfaction, and impact on society – constitute a high percentage of the total score. These are driven by “enablers” – constitute a high percentage of the total score. These are driven by “enablers” – the means by which an organization approaches its business responsibilities. The categories are roughly equivalent to those in Baldrige. However, the results criteria of people satisfaction, customer satisfaction, impact on society, and business results are somewhat different. The impact on society results category focuses on the perceptions of the company by the community at large and the company’s approach to the quality of life, the environment, and the preservation of global resources. The European Quality Award criteria place greater emphasis on this category than is placed on the public responsibility item in the Baldrige Award criteria.

CANADIAN AWARDS FOR BUSINESS EXCELLENCE

Canada’s National Quality Institute (NQI) recognizes Canada’s foremost achievers of excellence through the prestigious Canada Awards for Excellence. NQI is a nonprofit organization designed to stimulate and support quality-driven innovation within all Canadian enterprises and institutions, including business,

government, education, and health care. The Canadian Awards for Business Excellence quality criteria are similar in structure to the Baldrige Award Criteria, with some key differences. The major categories and items within each category are:

1. Leadership: strategic direction, leadership involvement, and outcomes.
2. Customer focus: voice of the customer, management of customer relationships, measurement, and outcomes.
3. Planning for improvement: development and content of improvement plan, assessment, and outcomes.
4. People focus: human resource planning, participatory environment, continuous learning environment, employee satisfaction, and outcomes.
5. Process optimization: process definition, process control, process improvement, and outcomes.
6. Supplier focus: partnering and outcomes.

AUSTRALIAN BUSINESS EXCELLENCE AWARDS

The Australian Quality Awards (now called Business Excellence Awards) were developed independently from the Baldrige Awards in 1988. The awards are administered by the Australian Quality Awards Foundation, a subsidiary of the Australian Quality council. Four levels of awards are given.

1. The Business Improvement Level: encouragement recognition for “Progress Toward Business Excellence” or “Foundation in Business Excellence”;
2. The Award Level: representing Australian best practices; recognition as a winner or finalist;
3. The Award Gold Level: open only to former award winners; represents a revalidation and ongoing improvement;
4. The Australian Business Excellence Prize: open only to former award winners; represents international best practices evident throughout the organization.

As with Baldrige, the framework emphasizes the holistic and interconnected nature of the management process. The criteria are benchmarked with the Baldrige criteria and the European Business Excellence Model. One of the distinctive aspects of Australia’s program is solid union support.

ISO 9000: WHAT IT IS

The International Organization for Standardization (ISO) is a federation of the national standards bodies of nations from around the world. ISO 9000 is about standardizing the systemic approach organizations everywhere take to managing and improving the processes that ultimately result in their products and services. Specifically, ISO 9000 establishes the requirements for quality management systems (QMS) that must be employed by all organizations registered to the standard. Registered organizations should enjoy:

- Wider customer acceptance of products and services
- Improved effectiveness and reliability of its processes
- Improved quality of products and services
- Improved organizational performance and competitiveness

Since ISO 9000 was first released in 1987 it has evolved through two revisions, the first in 1994 and the most recent at the end of 2000. This evolution has aligned it more closely with the Total Quality Management philosophy. It seemed to many observers, including the authors, that the 1987 and 1994 versions shied away from association with TQM, or from acknowledging its existence. Even the 2000 version, which borrows heavily from TQM, scarcely acknowledges it. The fact is, of course, that with the

tutelage of W. Edwards Deming and Joseph Juran, the Japanese started the development of the management system we now know as TQM in 1950. Over the years several Japanese experts – Kaoru Ishikawa, Shigeo Shingo, Taiichi Ono, and others – emerged, and by the early 1970s TQM had been widely accepted in Japan. By 1980 the Western world began taking note. By the time ISO 9000: 1987 was released, TQM was a mature management system, well understood by many in the West. It is clear that ISO's Technical Committee 176 (TC 176), which was charged with ISO 9000's development, borrowed some TQM elements, most notably its documentation requirements. ISO 9000: 1994 moved a bit closer to TQM, at least mentioning (though not requiring) continual improvement. But any acknowledgement of TQM's influence or superiority seemed to be deliberately avoided. ISO 9000:2000 made a giant leap in comparison, especially in the area of continual improvement, which has gone from receiving just cursory treatment to becoming a firm requirement.

Lesson # 20**BUSINESS EXCELLENCE MODELS****Excellence starts with ‘Marketing’**

The ‘marketing’ function of an organization must take the lead in establishing the true requirements for the product or service. Having determined the need, marketing should define the market sector and demand. This will determine product or service features such as the grade, price, quality, timings, etc. For example, a major hotel chain, before opening a new hotel or refurbishing an old one, will need to consider its location and accessibility, before deciding whether it will be predominantly a budget, first class, business or family hotel.

Marketing will also need to establish customer requirements by reviewing the market needs, particularly in terms of unclear or unstated expectations or preconceived ideas held by customers. Marketing is responsible for determining the key characteristics which determine the suitability of the product or service in the eyes of the customer. This may, of course, involve the use of market research techniques, data gathering, and analysis of customer complaints.

Excellent communication between customers and suppliers is the key to total organizational excellence. This will eradicate the ‘demanding nuisance/idiot’ view of customers, which pervades some organizations. Poor communications often occur in the supply chain between organizations, when neither party realizes how poor they are. Feedback from both customers and suppliers needs to be improved, where dissatisfied customers and suppliers do not communicate their problems. In such cases non-conformance of purchased products or services is often due to the customer’s inability to communicate their requirements clearly. If these ideas are also used within an organization, then the internal supplier/customer interfaces will operate much more smoothly.

Marketing must also establish systems for feedback of customer information and reaction, which should be designed on a continuous monitoring basis. Any information pertinent to the product or service should be collected and collated, interpreted, analyzed, and communicated to improve the response to customer experience and expectations. These same principles must also be applied inside the organization for continuous improvement at every transformation process interface to be achieved. If one department has problems recruiting the correct sort of staff, and ‘HR’ have not established mechanisms for gathering, analyzing, and responding to information on new employees, then frustration and conflict will replace communication and co-operation.

Excellence in all functions

For an organization to be truly excellent, each part of it must work properly together. Each part, each activity, each person in the organization affects and is in turn affected by others. Errors have a way of multiplying and failure to meet the requirements in one part or area creates problems elsewhere, leading to yet more errors, yet more problems and so on. The benefits of getting it right first time everywhere are enormous.

Everyone experience – almost accepts – problems in working life. This causes people to spend a large part of their time on useless activities, correcting errors, looking for things, finding out why things are late, and checking suspect information, rectifying and reworking, apologizing to customers for mistakes, poor quality and lateness. The list is endless and it is estimated that about one-third of our efforts are wasted in this way. In the service sector it can be much higher.

Quality, the way we have defined it as meeting the customer requirements, gives people indifferent functions of an organization a common language for improvement. It enables all the people, with different abilities and priorities, to communicate readily with one another, in pursuit of a common goal. When business and industry was local, the craftsman could manage more or less on his own. Business is now so complex and employs so many different specialist skills that everyone has to rely on the activities of others in doing their jobs.

Frameworks for Quality and Performance Excellence

The philosophies of Deming, Juran, and Crosby provide fundamental principles on which total quality is based. Business firms tend to be highly individualized. As a result, it is difficult to apply one specific philosophy. Company leaders must understand the differences and commonalities in the three philosophies and tailor an approach that fits their unique culture. Some of the most successful firms, such as Texas Instruments and Dana Corporation, have done this. Aspects of implementation are addressed further in chapter 11.

None of these philosophies, however, provide a framework for how to implement total quality within an organization or a means of assessing total quality efforts relative to one's peers or world-class companies. Award criteria and certification procedures fill this important role. The two most prominent frameworks for quality that have had world-wide influence are ISO 9000 and the Malcolm Baldrige National Quality Award Criteria.

THE MALCOLM BALDRIGE NATIONAL QUALITY AWARD

In this section we present an overview of the Award, the Criteria, and the Award Process. The Baldrige Award recognizes U.S. companies that excel in quality management practice and performance. The Baldrige Award does not exist simply to practice and performance. The Baldrige Award does not exist simply to recognize product excellence, nor does it exist for the purposes of "winning." Its principal focus is on promoting high-performance management practices that lead to customer satisfaction and business results. Up to three companies can receive an award in each of the categories of manufacturing, small business, service, nonprofit health care, and nonprofit education. Health care and education award categories were established in 1999.

The purposes of the award are to

- Help stimulate American companies to improve quality and productivity for the pride of recognition while obtaining a competitive edge through increased profits;
- Recognize the achievements of those companies that improve the quality of their goods and services and provide an example to others;
- Establish guidelines and criteria that can be used by business, industrial, governmental, and other enterprises in evaluating their own quality improvement efforts; and
- Provide specific guidance for other American enterprises that wish to learn how to manage for high quality by making available detailed information on how winning enterprises were able to change their cultures and achieve eminence.

THE CRITERIA FOR PERFORMANCE EXCELLENCE

The award examination is based upon a rigorous set of criteria, called the Criteria for Performance Excellence, designed to encourage companies to enhance their competitiveness through an aligned approach to organizational performance management that result in:

1. Deliver of ever-improving value to customers, contributing to marketplace success.
2. Improvement of overall company performance and capabilities
3. Organizational and personal learning

The criteria consist of a hierarchical set of categories, items, and areas to address. The seven categories are

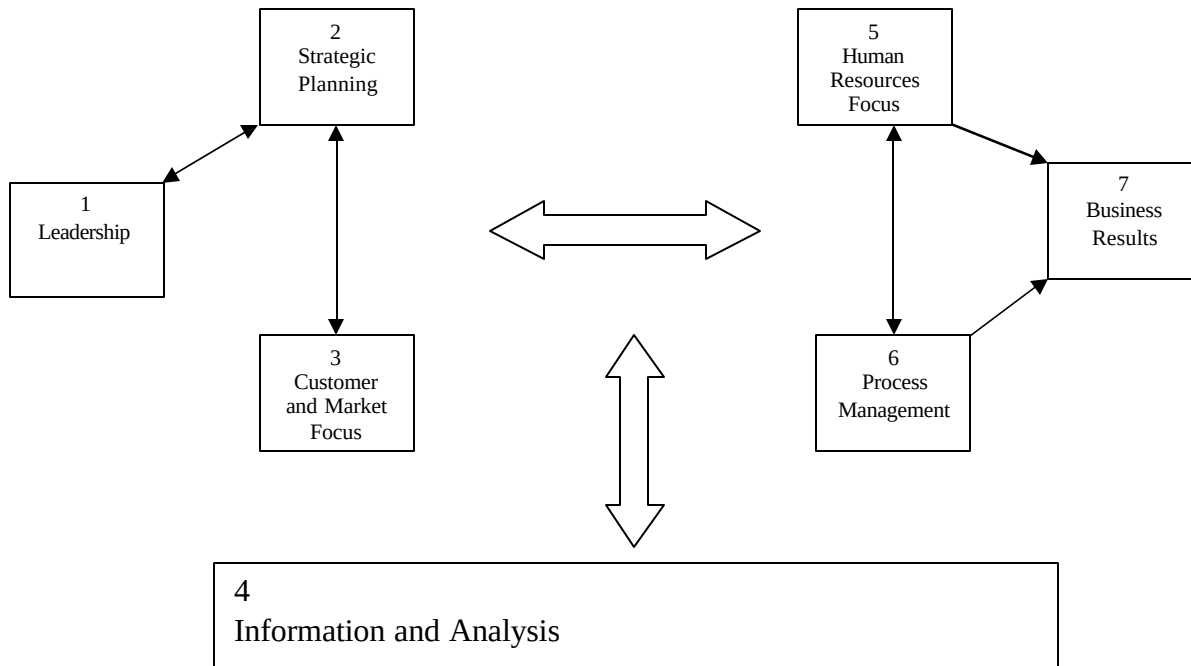
1. Leadership: This category examines how an organization's senior leaders address values, direction, and performance expectations, as well as their focus on customer s and other stakeholders, empowerment, innovation, and learning. Also examined is how an organization addresses its responsibilities to the public and supports its key communities.
2. Strategic Planning: this category examines how an organization develops strategic objectives and action plans. Also examined are how chosen strategic objectives and action plans are deployed and how progress is measured.
3. Customer and Market Focus: This category examines how an organization determines requirements, expectations, and preferences of customer s and markets. Also examined is how the organization builds relationships with customers and determines the key factors that lead to customer acquisition, satisfaction, and retention and to business expansion.
4. Information and Analysis: This category examines an organization's information management and performance measurement systems and how the organization analyzes performance data and ensures hardware and software quality.
5. Human Resource Focus: This category examines how an organization motivates and enables employees to develop and utilize their full potential in alignment with the organization's overall objectives and action plans. Also examined are the organization's efforts to build and maintain a work environment and an employee support climate conducive to performance excellence and to personal and organizational growth.
6. Process Management: This category examines the key aspects of an organization's process management, including customer-focused design, product and service delivery, key business, and support processes. This category encompasses all key processes and all work units.
7. Business Results: This category examines an organization's performance and improvement in key business areas – customer satisfaction, product and service performance, financial and marketplace performance, human resource results, and operational performance. Also examined are performance levels, relative to those of competitors.

The seven categories form an integrated management system. The umbrella over the seven categories reflects the focus that organizations must have on customers through their strategy and action plans for all key decisions. Leadership, Strategic Planning, and Customer and Market Focus represent the “leadership triad,” and suggest the importance of integrating these three functions. Human Resource focus and Process Management represent how the work in an organization is accomplished and leads to Business Results. These functions are linked to the leadership triad.

Malcolm Baldrige National Quality Award Criteria Framework

Baldrige Award Criteria Framework A systems Perspective

Organizational Profile: Environment, Relationships, and Challenges



Each category consists of several items (numbered 1.1, 1.2, 2.1, etc.) or major requirements on which businesses should focus. For example, the Leadership Category consists of the following items and areas to address:

- 1.1 Organizational Leadership
 - a. Senior Leadership Direction
 - b. Organizational Performance Review
- 1.2 Public Responsibility and Citizenship
 - a. Responsibilities to the Public
 - b. Support of Key Communities

The Senior Leadership Direction area asks organizations to answer the following questions:

- How do senior leaders set and deploy organizational values, short-and longer-term directions, and performance expectations, including a focus on creating and balancing value for customers and other stakeholders? Include how senior leaders communicate values, directions, and expectations through your leadership system and to all employees.
- How do senior leaders create an environment for empowerment, innovation, organizational agility, and organizational and employee learning?

One thing the criteria do not do is prescribe specific quality tolls, techniques, technologies, systems, or starting points. Companies are encouraged to develop and demonstrate creative, adaptive, and flexible

approaches to meeting basic requirements. Many innovative approaches have been developed by Baldrige winners and are now commonly used by many other companies.

Baldrige Award Item Point Values

2002 Categories/Items		Point Values
1.	Leadership	120
1.1	Organizational Leadership	80
1.2	Public Responsibility and Citizenship	40
2.	Strategic Planning	85
2.1	Strategy Development	40
2.2	Strategy Development.	45
3.	Customer and Market Focus	
3.1	Customer and Market Knowledge	40
3.2	Customer Satisfaction and Relationships.	45
4.	Information and Analysis	90
4.1	Measurement and Analysis of Organizational Performance	50
4.2	Information Management.	40
5.	Human Resources Focus	85
5.1	Work systems.	35
5.2	Employee Education, Training, and Development	25
5.3	Employee Well-Being and Satisfaction.	25
6.	Process Management	85
6.1	Product and Service Processes.	55
6.2	Business Process	15
6.3	Support Processes.	15
7.	Business Results	450
7.1	Customer-Focused Results	125
7.2	Financial and Market Results	125
7.3	Human Resource Results	80
7.4	Organizational Effectiveness Results	120
Total Points		1, 000

ISO 9000 (2000) International Quality Management System

As quality became a major focus of businesses throughout the world, various organizations developed standards and guidelines. Terms such as *quality management*, *quality control*, *quality system*, and *quality assurance* acquired different, and sometimes conflicting meanings from country to country, within a country, and even within an industry.¹⁰ As the European Community moved toward the European free trade agreement, which went into effect at the end of 1992, quality management became a key strategic objective. To standardize quality requirements for European countries within the common market and those wishing to do business with those countries, a specialized agency for standardization, the International Organization for Standardization, founded in 1946 and composed of representatives from the national standards bodies of 91 nations, adopted a series of written quality standards in 1987, which were revised in 1994, and again (significantly) in 2000. The most recent version is called the ISO 9000:2000 family of standards.

ISO 9000 is an international quality standard for goods and services. The term *quality standard* tends to be misleading. For example, ISO 9000 does not set any specifications for quality. Rather, it sets broad requirements for the assurance of quality and for management's involvement. The emphasis is on prevention rather than inspection and rework. In fact, this emphasis is placed not only on the production process but also on the product design process. The ISO 9000 approach is completely compatible with the total quality philosophy, though it is not as all encompassing. ISO 9000 is composed of three standards:

ISO 9000:2000 Quality Management Systems – Fundamentals and Vocabulary

ISO 9001:2000 Quality Management Systems – Requirements

ISO 9004:2000 Quality Management Systems – Guidelines for Performance Improvements

ISO 9000 CAN IMPROVE OPERATIONS IN A TRADITIONAL ENVIRONMENT

By “traditional environment,” we mean an organizational environment that has persisted in companies for decades, until the Total Quality Management movement began to change things. A traditional organizational environment is one which still operates according to the “old way of doing things” rather than according to the principles of Total Quality Management.

When ISO 9000 is implemented by a traditional organization, the company should be the better for it. We will not go so far as to say it will be the better for it, because much depends on the organization's reasons for adopting ISO 9000 and the degree of executive-level commitment to it. Put another way, if ISO 9000 is approached inappropriately and for the wrong reasons, it can become nothing more than a marketing ploy, and the organization's functional departments might develop even more problems than they had before ISO 9000.

ISO 9000 MAY BE REDUNDANT IN A MATURE TQM ENVIRONMENT

Just as ISO 9000 should help traditional organizations, it should also benefit TQM organizations. However, in an organization that has achieved a high level of maturity in its total quality journey, say in the 400-600 range on the Baldrige scale of 1,000 points, all ISO 9000 criteria may already be in place. In such a case, the only compelling reason for registration under ISO 9000 would be for marketing purposes. What would a company such as Toyota gain from ISO 9000 registration? Probably nothing. It already does everything required by ISO 9000. Its products and processes are recognized as world class. Consequently, It wouldn't gain even a marketing advantage. However, there are many fine TQM organizations that are not as well known as Toyota. Such organizations, even though they may already meet or exceed the requirements of ISO 9000, may find it necessary to register in order to let potential customers know that their products or services satisfy the international standard.

For Software and IT companies, the standard which is more acceptable and regarded worldwide is called CMMI levels 1-5.

ISO 9000 AND TQM ARE NOT IN COMPETITION

This is not case of one or the other. Organizations can adopt TQM or ISO 9000, or both. While there may be those who advocate one to the exclusion of the other, in the larger scheme of things, the two concepts fit well with each other. Both have worthwhile and similar aims. Our view is those not only are TQM and ISO 9000 compatible; they actually support each other and are complementary. There are good reasons for using both in a single management system.

MANAGEMENT MOTIVATION FOR REGISTRATION TO ISO 9000

Management motivation for adopting either ISO 9000 or TQM can vary widely. There are both appropriate and inappropriate motives. For example, if a company seeks ISO 9000 registration to obtain a marketing advantage, its motive is inappropriate. As a result, the organization will likely give mere lip service to adopting the standard. Appropriate motives for adopting ISO 9000 include the following:

- To improve operations by implementing a quality management system that satisfies the ISO 9000 requirements for management responsibility; resource management; product realization; and measurement, analysis, and improvement
- To create or improve a quality management system that will be recognized by customers worldwide.
- To improve product or service quality and the consistency of quality
- To improve customer satisfaction.
- To improve competitive posture
- To conform to the requirements of one or more major customers (although adoption would be better motivated by internal consideration, such as the preceding five)

What we are saying here is that, ideally, management will adopt ISO 9000 as a way to make real improvements in the company's operations, serve its customers in a more responsible way, and, as a result, be more successful. This approach is more likely to assure commitment and participation by top management. Approaching ISO 9000 from a strictly marketing perspective may result in a negative reaction to the amount of work required by the functional departments, and only enough management commitment to do the bare minimum for registration. In other words, if ISO 9000 is viewed as a necessary evil that one must adopt to compete in certain markets, every dollar and every hour spent on ISO 9000 will be seen as a burden to be endured rather than an investment in the organization's future. By definition, a burden is a load that is difficult to bear; the connotation is negative. When negative feelings abound among employees, commitment to ISO 9000 will suffer. It may be possible to fool the ISO 9000 registrar's auditor, but we guarantee that customers will not be fooled – at least not for long. Newfound markets will soon wither and disappear. If ISO 9000 is to have a real and permanent effect, it must be approached with a positive attitude and the unwavering commitment of top management.

Lesson # 21

DESIGNING ORGANIZATIONS FOR QUALITY

For design, development and implementation of a QMS, the ISO 9000 approach is completely compatible with the total quality philosophy, though it is not as all encompassing. ISO 9000 is composed of three standards:

ISO 9000:2000	Quality Management Systems – Fundamentals and Vocabulary
ISO 9001:2000	Quality Management Systems – Requirements
ISO 9004:2000	Quality Management Systems – Guidelines for Performance Improvements

ISO 9001 and ISO 9004 are known as Consistent Pair and are based and follow PDCA methodology. ISO system is about standardizing the approach organizations everywhere take in managing and improving the processes that ultimately result in producing better quality products and services. Specifically, ISO 9001(2000) establishes the requirements for quality management systems (QMS) that must be employed by all organizations registered to the standard. Registered organizations should enjoy:

- Wider customer acceptance of products and services
- Improved effectiveness and reliability of its processes
- Improved quality of products and services
- Improved organizational performance and competitiveness

By the time ISO 9000: 1987 was released, TQM was a mature management system, well understood by many in the West. It is clear that ISO's Technical Committee 176 (TC 176), which was charged with ISO 9000's development, borrowed some TQM elements, most notably its documentation requirements. ISO 9000: 1994 moved a bit closer to TQM, at least mentioning (though not requiring) continual improvement. ISO 9000:2000 made a giant leap in comparison, especially in the area of continual improvement, which has gone from receiving just cursory treatment to becoming a firm requirement. In addition, the standard now incorporates eight quality management principles that come directly from TQM. They are:

1. **Customer focus** – understanding customer's needs, striving to exceed their expectations.
2. **Leadership** – establishing direction, unity of purpose, and a supportive work environment.
3. **Involvement of people** – ensuring that all employees at all levels are able to fully use their abilities for the organization's benefit.
4. **Process approach** – recognizing that all work is done through processes, and managing them accordingly.
5. **System approach to management** – expands on the previous principle in that achieving any objective requires a system of interrelated processes.
6. **Continua improvement** – as a permanent organizational objective, recognizing and acting on the fact that no process is so good that further improvement is impossible.
7. **Factual approach to decision making** – acknowledging that sound decisions must be based on analysis of factual data and information.
8. **Mutually beneficial supplier relationships** – to take advantage of the synergy that can be found in such relationships.

By design, as a result of ISO 9000, any organization supplying products or service is able to develop and employ a quality management system that is recognized by customers worldwide. Customers around the

globe who deal with ISO 9000-registered organizations can expect that purchased goods or services will conform to a set of recognized standards.

ISO 9001's requirements for quality management systems are generic in nature, and are applicable to organizations in any industry or economic sector. Whether the organization manufactures a product or provides a service, whether it is a company or a governmental agency, whether it is large or small, ISO 9000 can apply, and be used to advantage.

To be registered the organization must go through a process that includes the following steps:

1. Develop (or upgrade) a quality manual that describes how the company will assure the quality of its products or services.
2. Document procedures (or upgrade existing documentation) that describe how the various processes for design, production, continual improvement, and so forth, will be operated. This must include procedures for management review/audits and the like.
3. The organization must provide evidence of top management's commitment to the QMS and its continual improvement.
4. The organization's top management must ensure that customer requirements are determined and met.
5. The organization must hire an accredited registrar company to examine its systems, processes, procedures, quality manual, and related items. If everything is in order, registration will be granted. Otherwise, the registrar will inform the company of which areas require work (but will not inform the company specifically what must be done), and a second visit will be scheduled.
6. Once registration is accomplished, the company will conduct its own internal audits to ensure that the systems, processes, and procedures are working as intended.
7. Also once registered, the outside registrar will make periodic audits for the same purpose. These audits must be passed to retain registration.

An important point to understand about ISO 9000 is that the organization has to respond to all ISO 9000 requirements and tell the registrar specifically what it is going to do and how. ISO does not tell the organization. Assuming the registrar agrees with the organization's plan, registration is awarded. To retain that registration, the *organization must do what it said it would do*.

Before the advent of the year 2000 release, ISO 9000 was concerned only with the standards which an organization could build its own version of a quality management system. ISO 9000:2000 has closed much of the gap that existed with TQM. The primary remaining difference between ISO 9000 and TQM is in the degree to which the total organization is involved, ISO 9000 does not require the QMS to include functions and levels that do not play a direct role in the management and execution of the product/service realization processes. Functions that are typically not involved under the QMS include human resources, finance (accounting), sales, and marketing.

Characteristics of Total Quality Management	ISO 9000:2000	TQM
Customer focus (internal and external)	✓	✓
Obsession with quality	✓	✓
Scientific approach to problem solving	✓	✓
Long-term commitment	Partial	✓
Teamwork	✓	✓
Continual process and product improvement	✓	✓
Education and training intensive	✓	✓

Freedom through control		
Unity of purpose		
Employee involvement and empowerment	Partial	

Total Quality Management Characteristics Compared with ISO 9000

In comparison, the ISO 9000 quality management system is designed to “provide the framework for continual improvement to increase the probability of enhancing customer satisfaction and the satisfaction of other interested parties. It provides confidence to the organization and its customers that it is able to provide products that consistently fulfill requirements.” ISO claims that beyond customer satisfaction, cost and risk-management benefits will also accrue to the organization. These benefits translate to improved competitiveness – the same as TQM’s objective. ISO claim these benefits result from emphasizing the eight quality management principles on which the standard is based. Following table provides a comparison of ISO’s eight quality management principles with Deming’s Fourteen Points and TQM.

ISO 9000’s Eight Quality Management Principles	Deming’s 14 Points	TQM
1. Customer focus		
2. Leadership	#1, #2, #7	
3. Involvement of people		
4. Process approach		
5. System approach to management		
6. Continual improvement	#5	
7. Factual approach to decision making		
8. Mutually beneficial supplier relationships	#4	

ISO 9000 IS COMPATIBLE WITH, AND CAN BE VIEWED AS A SUBSET OF, TQM

Clearly, TQM and ISO 9000 is not quite the same thing. However, there is nothing inherent in ISO 9000 that would prevent it from becoming part of a larger Total Quality Management environment. There are many examples today of companies that have

ISO 9000’s Quality Management Principles versus Deming’s Fourteen Points and TQM successfully included ISO 9000 as part of a larger total quality effort. Organizations that are already at some level of TQM maturity or CMMI level 3 maturities have typically found it easy to implement ISO 9000. This is because a TQM environment with its infrastructure of top management commitment, documented processes and procedures, continuous improvement, obsession with quality, and so on, easily supports the requirements of ISO 9001(2000).

DESIGNING AN ISO 9000 QMS CAN IMPROVE MARKET PERCEPTION IN GLOBAL POST WTO WORLD

A traditional organizational environment is one which still operates according to the “old way of doing things” rather than according to the principles of Total Quality Management and the technology based networked post WTO world. In Pakistan, you might say, an organization being run as an autocratic, non-participatory and “SAITH” like organization as many in Sialkot, Guranwala, Faisalabad, Lahore, and Karachi etc.

When ISO 9000 is implemented by a traditional organization, in its real spirit but much depends on the organization's reasons for adopting ISO 9000 and the degree of executive-level commitment to it. Let us take a look from a different view, if ISO 9000 is designed and developed for the wrong reasons, it can not become a good marketing tool, and the organization's functional departments especially operations and QA, might develop even more problems than they had before ISO 9000. Once again QMS principles are taken in its letter and spirit.

The eight quality management principles are defined and detailed in ISO 9004:2000, *Quality management systems Guidelines for performance improvements*.

- Principle 1 Customer focus
- Principle 2 Leadership
- Principle 3 Involvement of people
- Principle 4 Process approach
- Principle 5 System approach to management
- Principle 6 Continual improvement
- Principle 7 Factual approach to decision making
- Principle 8 Mutually beneficial supplier relationships

Principle 1 Customer focus :

Organizations depend on their customers and therefore should understand current and future customer needs, should meet customer requirements and strive to exceed customer expectations

Key benefits:

- **Increased revenue** and market share obtained through flexible and fast responses to market opportunities.
- Increased effectiveness in the use of the organization's resources to enhance customer satisfaction.
- Improved customer loyalty leading to repeat business.

Applying the principle of customer focus typically leads to:

- Researching and understanding customer needs and expectations.
- Ensuring that the objectives of the organization are linked to customer needs and expectations.
- Communicating customer needs and expectations throughout the organization.
- Measuring customer satisfaction and acting on the results.
- Systematically managing customer relationships.
- Ensuring a balanced approach between satisfying customers and other interested parties (such as owners, employees, suppliers, financiers, local communities and society as a whole).

Principle 2 Leadership:

Leaders establish unity of purpose and direction of the organization. They should create and maintain the internal environment in which people can become fully involved in achieving the organization's objectives.

Key benefits:

- People will understand and be motivated towards the organization's goals and objectives.
- Activities are evaluated, aligned and implemented in a unified way.
- Miscommunication between levels of an organization will be minimized.

Applying the principle of leadership typically leads to:

- Considering the needs of all interested parties including customers, owners, employees, suppliers, financiers, local communities and society as a whole.
- Establishing a clear vision of the organization's future.
- Setting challenging goals and targets.
- Creating and sustaining shared values, fairness and ethical role models at all levels of the organization.
- Establishing trust and eliminating fear.
- Providing people with the required resources, training and freedom to act with responsibility and accountability.
- Inspiring, encouraging and recognizing people's contributions.

Principle 3 Involvement of people:

People at all levels are the essence of an organization and their full involvement enables their abilities to be used for the organization's benefit.

Key benefits:

- Motivated, committed and involved people within the organization.
- Innovation and creativity in furthering the organization's objectives.
- People being accountable for their own performance.
- People eager to participate in and contribute to continual improvement.

Applying the principle of involvement of people typically leads to:

- People understanding the importance of their contribution and role in the organization.
- People identifying constraints to their performance.
- People accepting ownership of problems and their responsibility for solving them.

- People evaluating their performance against their personal goals and objectives.
- People actively seeking opportunities to enhance their competence, knowledge and experience.
- People freely sharing knowledge and experience.
- People openly discussing problems and issues.

Principle 4 Process approach:

A desired result is achieved more efficiently when activities and related resources are managed as a process.

Key benefits:

- Lower costs and shorter cycle times through effective use of resources.
- Improved, consistent and predictable results.
- Focused and prioritized improvement opportunities.

Applying the principle of process approach typically leads to:

- Systematically defining the activities necessary to obtain a desired result.
- Establishing clear responsibility and accountability for managing key activities.
- Analyzing and measuring of the capability of key activities.
- Identifying the interfaces of key activities within and between the functions of the organization.
- Focusing on the factors such as resources, methods, and materials that will improve key activities of the organization.
- Evaluating risks, consequences and impacts of activities on customers, suppliers and other interested parties.

Principle 5 System approach to management:

Identifying, understanding and managing interrelated processes as a system contributes to the organization's effectiveness and efficiency in achieving its objectives.

Key benefits:

- Integration and alignment of the processes that will best achieve the desired results.
- Ability to focus effort on the key processes.
- Providing confidence to interested parties as to the consistency, effectiveness and efficiency of the organization.

Applying the principle of system approach to management typically leads to:

- Structuring a system to achieve the organization's objectives in the most effective and efficient way.

- Understanding the interdependencies between the processes of the system.
- Structured approaches that harmonize and integrate processes.
- Providing a better understanding of the roles and responsibilities necessary for achieving common objectives and thereby reducing cross-functional barriers.
- Understanding organizational capabilities and establishing resource constraints prior to action.
- Targeting and defining how specific activities within a system should operate.
- Continually improving the system through measurement and evaluation.

Principle 6 Continual improvement:

Continual improvement of the organization's overall performance should be a permanent objective of the organization.

Key benefits:

- Performance advantage through improved organizational capabilities.
- Alignment of improvement activities at all levels to an organization's strategic intent.
- Flexibility to react quickly to opportunities.

Applying the principle of continual improvement typically leads to:

- Employing a consistent organization-wide approach to continual improvement of the organization's performance.
- Providing people with training in the methods and tools of continual improvement.
- Making continual improvement of products, processes and systems an objective for every individual in the organization.
- Establishing goals to guide, and measures to track, continual improvement.
- Recognizing and acknowledging improvements.

Principle 7 Factual approach to decision making:

Effective decisions are based on the analysis of data and information

Key benefits:

- Informed decisions.
- An increased ability to demonstrate the effectiveness of past decisions through reference to factual records.
- Increased ability to review, challenge and change opinions and decisions.

Applying the principle of factual approach to decision making typically leads to:

- Ensuring that data and information are sufficiently accurate and reliable.

- Making data accessible to those who need it.
- Analyzing data and information using valid methods.
- Making decisions and taking action based on factual analysis, balanced with experience and intuition.

Principle 8 Mutually beneficial supplier relationships:

An organization and its suppliers are interdependent and a mutually beneficial relationship enhances the ability of both to create value

Key benefits:

- Increased ability to create value for both parties.
- Flexibility and speed of joint responses to changing market or customer needs and expectations.
- Optimization of costs and resources.

Applying the principles of mutually beneficial supplier relationships typically leads to:

- Establishing relationships that balance short-term gains with long-term considerations.
- Pooling of expertise and resources with partners.
- Identifying and selecting key suppliers.
- Clear and open communication.
- Sharing information and future plans.
- Establishing joint development and improvement activities.
- Inspiring, encouraging and recognizing improvements and achievements by suppliers.

Lesson # 22**DEVELOPING ISO QMS FOR CERTIFICATION****DEVELOPMENT, IMPLEMENTATION AND REGISTRATION**

The ISO 9000 standards originally were intended to be advisory in nature and to be used for two-party contractual situations (between a customer and supplier) and for internal auditing. However, they quickly evolved into criteria for companies who wished to “certify” their quality management or achieve “registration” through a third-party auditor, usually a laboratory or some other accreditation agency (called a registrar). This process began in the United Kingdom. Rather than a supplier being audited for compliance to the standards by each customer, the registrar certifies the company, and this certification is accepted by all of the supplier’s customers.

The registration process includes document review by the registrar of the quality system documents or quality manual; pre-assessment, which identifies potential noncompliance in the quality system or in the documentation; assessment by a team of two or three auditors of the quality system and its documentation; and surveillance, or periodic re-audits to verify conformity with the practices and systems registered.

During the assessment, auditors might ask such question as (using management responsibility as an example): Does a documented policy on quality exist? Have job descriptions for people who manage or perform work affecting quality been documented? Are descriptions of functions that affect quality been documented? Are descriptions of functions that affect quality available? Has management designated a person or group with the authority to prevent nonconformities in products, identify and record quality problems, and recommend solutions? What means are used to verify the solutions?

Re-certification is required every three years. Individual sites – not entire companies – must achieve registration individually. All costs are borne by the applicant, so the process can be quite expensive.

PERSPECTIVES ON ISO 9000:2000

ISO 9000 provides a set of good basic practices for initiating a quality system, and is an excellent starting point for companies with no formal quality assurance program. In fact, it provides detailed guidance on process and product control. Thus, for companies in the early stages of developing a quality program, the standards enforce the discipline of control that is necessary before they can seriously pursue continuous improvement. The requirements of periodic audits reinforce the stated quality system until it becomes ingrained in the company. Thus, using ISO 9000 as a basis for a quality system can improve discipline, process, productivity, decrease costs, and increase customer satisfaction.

ISO 9000 AS A QUALITY JOURNEY AND NOT AS A DESTINATION

How to get started is always an issue for organizations just beginning their total quality journey.

The ISO 9000 development effort will benefit by having the following components: an executive-level steering committee, a vision with the guiding principles, a set of broad objectives, baselines on employee and customer satisfaction, an objective view of the organization’s strengths and weaknesses, and an indication of which employees at all levels can be counted on for support during the implementation. In addition, the organization will have a well thought out means of communicating with employees and all other stakeholders to keep them apprised of the changes taking place, why they are happening, and what they will mean to everyone.

The development and implementation of an organization's quality management system is influenced by its varying needs, its particular objectives, the products it provides, and the processes it employs. It is not the purpose of this ISO to imply uniformity of quality management systems.

The selection of the appropriate quality related processes described in this ISO Standard and the extent to which these processes are adopted and applied by an organization depends upon factors such as its size and structure, the market being served and the resources available.

The purpose of an organization is:

- a) To identify and meet the needs and expectations of its customers and other interested parties (i.e. employees, suppliers, owners, society), to achieve competitive advantage, and to do this in an effective and efficient manner;
- b) To achieve, maintain, and improve overall organizational performance and capabilities.

The application of quality management principles not only provides direct benefits but also makes an important contribution to managing costs and risks. Benefit, cost and risk considerations are important for the organization, its customers and other interested parties. These considerations on overall performance may impact on the following:

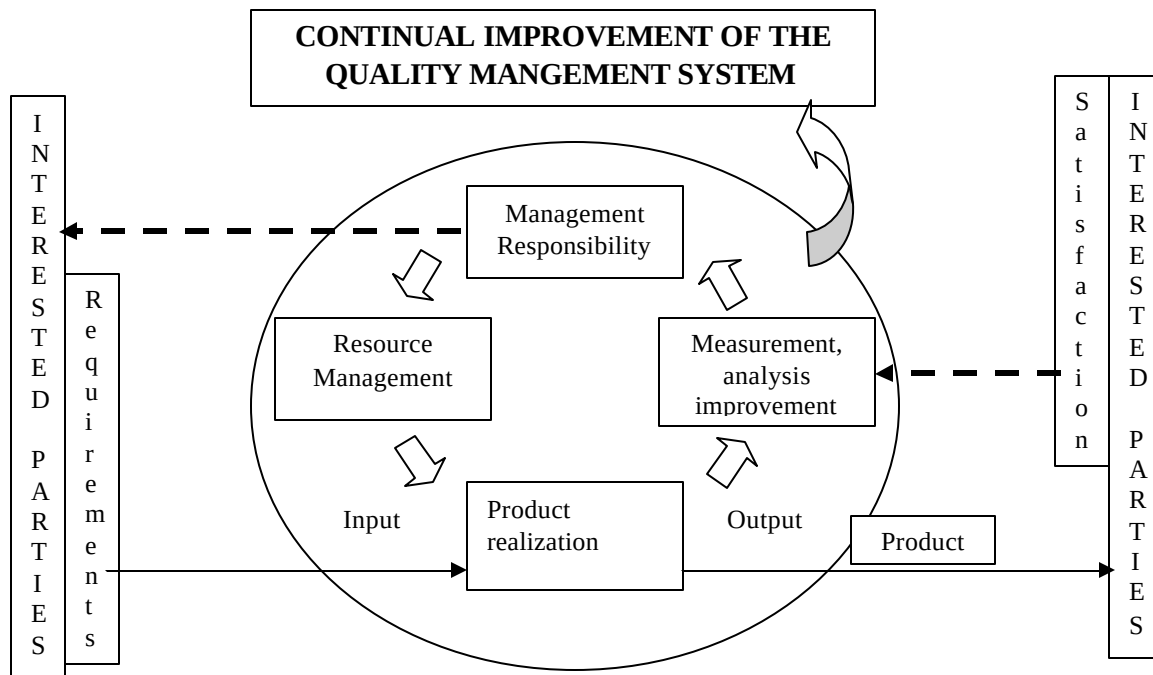
- a) Revenue (turnover), profits and market share; these may be increased by such aspects as leadership, increased efficiency, improved employee performance, and employee and customer satisfaction;
- b) Costs due to resources needed for business; inadequate resource funding is likely to cause losses and be a competitive disadvantage through the sale of deficient products.

Process approach

The ISO-9001(2000) Standard encourages the adoption of a process approach to quality management. Any activity which receives inputs and converts them to outputs can be considered as a process. For organizations to function effectively, they have to identify and manage numerous inter-linked processes. Often the output from one process will directly form the input into the next process. The systematic identification and management of the processes employed by an organization, and the interactions between such processes, may be referred to as the 'process approach'.

Following Figure is a conceptual illustration of one of the process approach. The model recognizes that customers and other interested parties play a significant role in defining inputs. Monitoring the satisfaction of customers and other interested parties is necessary to evaluate and validate whether the requirements of customers and other interested parties have been met. This model does not reflect processes at a detailed level, but covers all the contents of the ISO Standard.

The purpose of ISO 9001 is to define the minimum Quality Management System requirements needed to achieve customer satisfaction by meeting specified product requirements. Compliance with ISO 9001 may be used by an organization to demonstrate its capability to meet customer requirements.



Terms and definitions

For the purposes of this ISO 9001(2000) International Standard, the terms and definitions given in ISO 9000:2000, are applied in following way.

Supplier ----- >Organization ----- >Customer (Interested Parties)

The term “organization” replaces the previously used term “supplier”, to mean the unit to which this International Standard applies. The term “supplier” is now used instead of the previous term “subcontract”. The changes have been introduced to reflect the vocabulary used by organizations.

Product as a result of a Process

There are four agreed generic product categories as per ISO-9000:

- Hardware,
- Software,
- Services,
- Processed materials.

Most products are combinations of some of the four generic product categories.

Whether the combined product is then called hardware, processed material, software or service depends on the dominant element.

Following is one of the model methodology which can be used to develop the documentation for QMS and to get ready for certification by implementing clause by clause requirements (in the box) as given in ISO 9001(2000) as the minimum requirement for certification and one can also take help from the guidelines given in ISO-9004(2000) as presented below after the boxes.

ISO Quality Management System – Requirements /Guidelines

ISO 9001:2000 - Quality management systems -.Requirements

General requirements

The organization shall establish, document, implement, maintain a quality management and continually improve its effectiveness in accordance with the requirement of this International Standard.

The organization shall:

- a) Identify the processes needed for the quality management system and their application throughout the organization (see 1.2)
- b) Determine the sequence and interaction of these processes;
- c) Determine criteria and methods needed to ensure that both the operation and control of the processes are effective,
- d) Ensure the availability of resources and information necessary to support the operation and monitoring of these processes;
- e) Measure, monitor and analyze these processes, and
- f) Implement action necessary to achieve planned results and continual improvement these processes.

These processes shall be managed by the organization in accordance with the requirements of this International Standard.

Where an organization chooses to outsource any process that affects product conformity with requirements, the organization shall ensure control over such processes.

NOTE Processes needed for the quality management system referred to above should include processes for management activities, provision of resources, product realization and measurement.

Quality management system guidelines

Managing systems and processes

Leading and operating an organization successfully requires managing it in a systematic and visible manner. Success should result from implementing and maintaining a management system that is designed to continually improve performance by addressing the needs of all interested parties. Managing an organization encompasses quality management, among other management disciplines.

The quality management system of an organization is an important part of the overall management system. Organizations should define their systems and the processes contained within them to enable the systems and processes to be clearly understood, managed and improved. Management should ensure effective operation and control of processes and the measures and data used to determine satisfactory performance. The management of the organization should closely monitor the movement toward performance improvement. The activities and processes that can lead to performance improvement should be described and defined by the management.

In general, to fulfill the requirements of ISO standard;

First, Company should state (write/document) what do they want to do

Second, do the work as was stated and documented

Third, check them, whether the work is being carried out as was stated. See if there are any gaps .

Fourth, Show and prove to an external auditor that work is really being done as was stated in the first place.

So what in terms of documentation required is a manual altogether in one volume to be called as ISO 9000 QMS Manual or can be separated into following manuals:

1. Quality Policy Manual
2. Quality Procedures Manual
3. Quality Work Instructions
4. Quality Records Manual or Quality Data Collection Manual

Documentation and records may be in any form or in any media suitable for the needs of the organization.

Requirements for documentation and records may arise from

- contractual requirements from the customer or other interested parties,
- acceptance of international, national, regional and industry sector standards,
- relevant statutory and regulatory requirements, or
- decisions by the organization.

ISO 9001:2000 – Quality management systems -Requirements

Documentation Requirements

General

The quality management system documentation shall include:

- a) Documented statements of a quality policy and quality objectives
- b) A quality manual
- c) Documented procedures required by this International Standard;
- d) Documents needed by the organization to ensure the effective planning, operation and control

Use of quality management principles

ISO 9001:2000 – Quality management systems -Requirements

Documentation Requirements

Quality Manual

The organization shall establish and maintain a quality manual that includes

- a) The scope of the quality management system, including details of and justification for any exclusions (see 1.2),
- b) The documented procedures established for the quality management system, or reference to them, and
- c) a description of the interaction between the processes of the quality management system.

Control of Documents

Documents required by the quality management system shall be controlled. Records are a special type of document and shall be controlled according to the requirements given in 4.2.4.

A documented procedure shall be established to define the controls needed.

- a) to approve documents for adequacy prior to issue,
- b) to review and update as necessary and re-approve documents,
- c) to ensure that changes and the current revision status of documents are identified,
- d) to ensure that relevant versions of applicable documents are available at points of use,
- e) to ensure that documents remain legible and readily identifiable,
- f) to ensure that documents of external origin are identified and their distribution controlled, and

- g) to prevent the unintended use of obsolete documents, and to apply suitable identification to them if they are retained for any purpose.

Control of Records

Records shall be established and maintained to provide evidence of conformity to requirements and of the effective operation of the quality management system. Records shall remain legible, readily identifiable and retrievable. A documented procedure shall be established to define the controls needed for the identification, storage, protection, retrieval, retention time and disposition of records.

Documentation and Records

Management should define the documentation needed to support the quality management system. The nature and extent of the documentation should support the needs of the organization. The defined documentation should provide for implementation, maintenance and improvement of the system.

This documentation typically includes

- policy documents including the quality manual
- documentation for control of processes,
- work instructions for defined tasks,
- standard formats for collection and reporting of data and
- quality records.

The primary purpose of quality documentation is to express the quality policy and to describe the quality management system. This documentation serves as the basis for implementation and maintenance of the system. Suitable documentation should be available to achieve the effective operation of the quality management system.

Documentation control should be defined and implemented to ensure that correct documents are used. All obsolete documents should be promptly removed from all points of issue and use, or otherwise prevented from unintended use.

Documents to be retained, and records of quality performance, should be controlled, maintained and protected.

The organization should ensure that sufficient records be maintained to demonstrate conformance to requirements and verify effective operation of the quality management system. These records can also provide knowledge for maintenance and improvement of the quality management system.

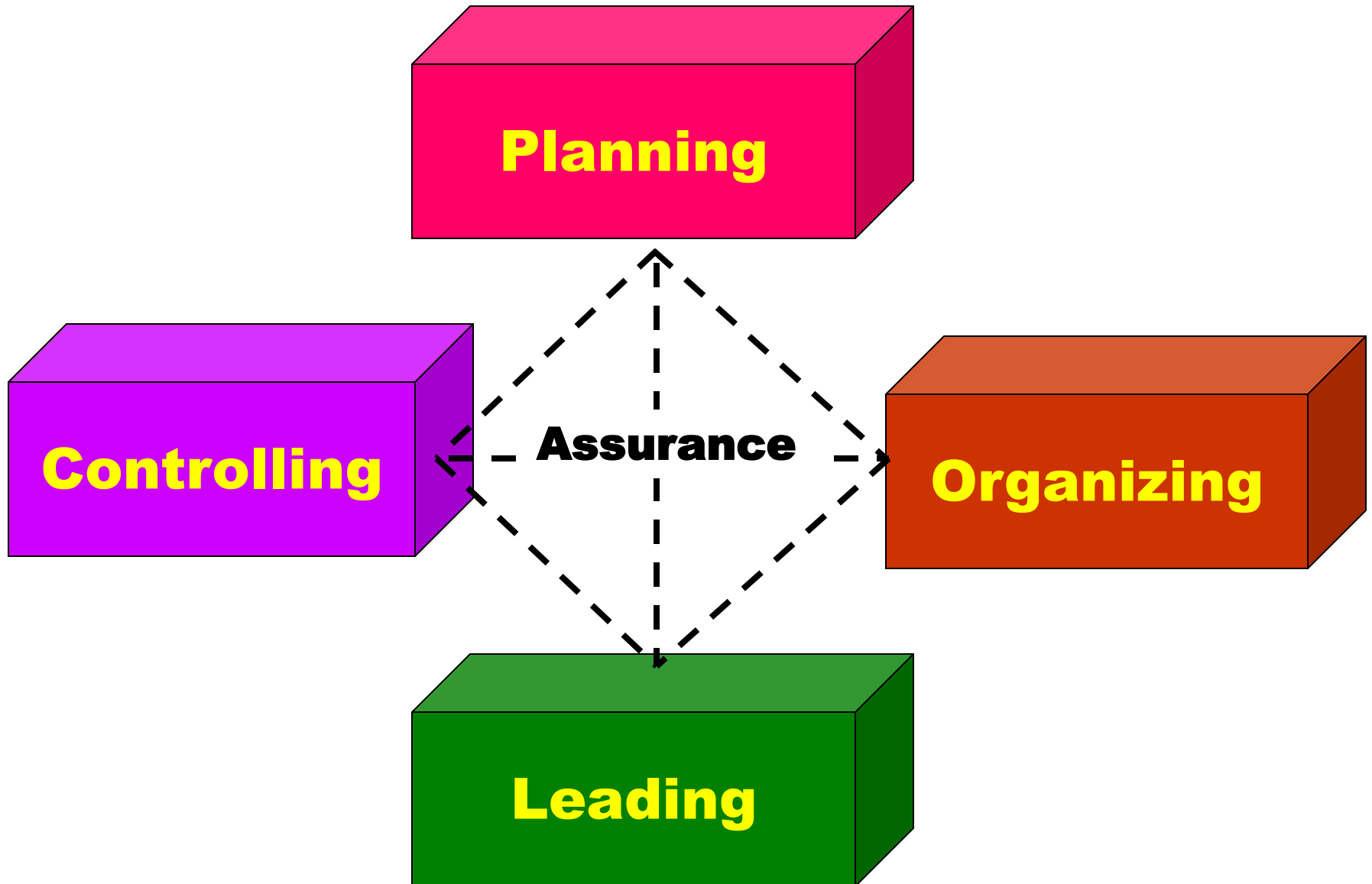
Quality records should be analyzed to provide inputs for corrective and preventive action, and process improvements. Analysis of records may also provide information for use in the improvement of the Quality management system.

TOTAL QUALITY MANAGEMENT (TQM)

Dr. A.Rashid Kausar
(UMT)

**TOTAL
QUALITY
Management,
Organization,
and
Strategy.**

Management Functions



Why Focus on Quality !

1. It's the moral thing to do
2. It's a glocal marketplace/
marketspace

Why Focus on Quality !

3. Need to meet govt. or market
requirements

Why Focus on Quality !

4. Essential for Competitive Success

Why Focus on Quality !

*5. Others think
Great
of my company*

Why Focus on Quality !

*6. For meaningful work and
integrity
in our life*

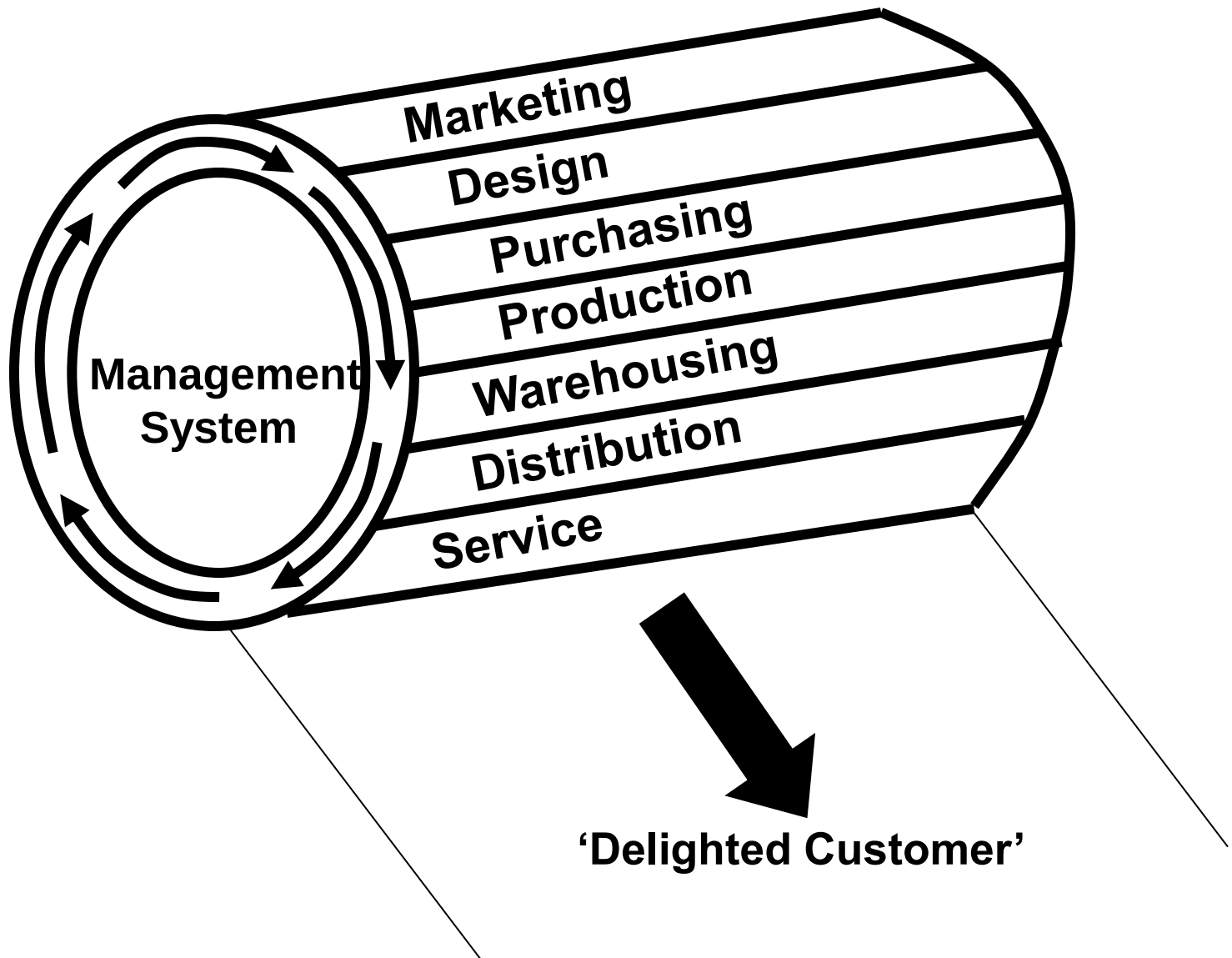
What is Quality?

1. Name some high quality goods and/or services
2. What makes them high quality?

TOTAL QUALITY MANAGEMENT (TQM)

Session # 2

**TOTAL
QUALITY
MANAGEMENT
(TQM)
(TOE)**



Systems Thinking

- A system is the functions or activities within an organization that work together for the aim of the organization.

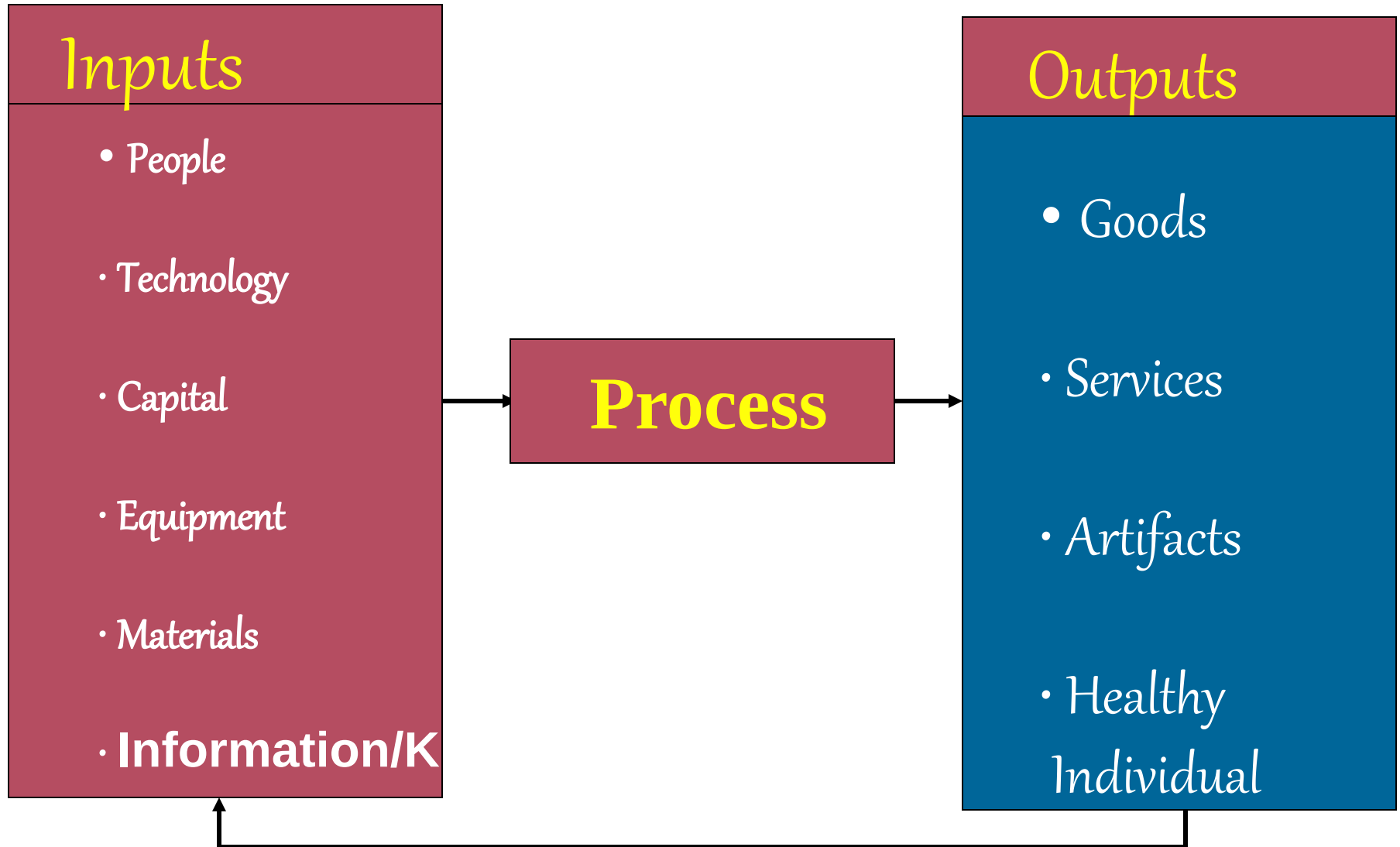
Systems Thinking

- Subsystems of an organization are linked together as internal customers and suppliers.

Systems Thinking

- A systems perspective acknowledges the importance of the interactions of subsystems, not the actions of them individually.

Systems Thinking



You can't improve
interdependent systems
and processes

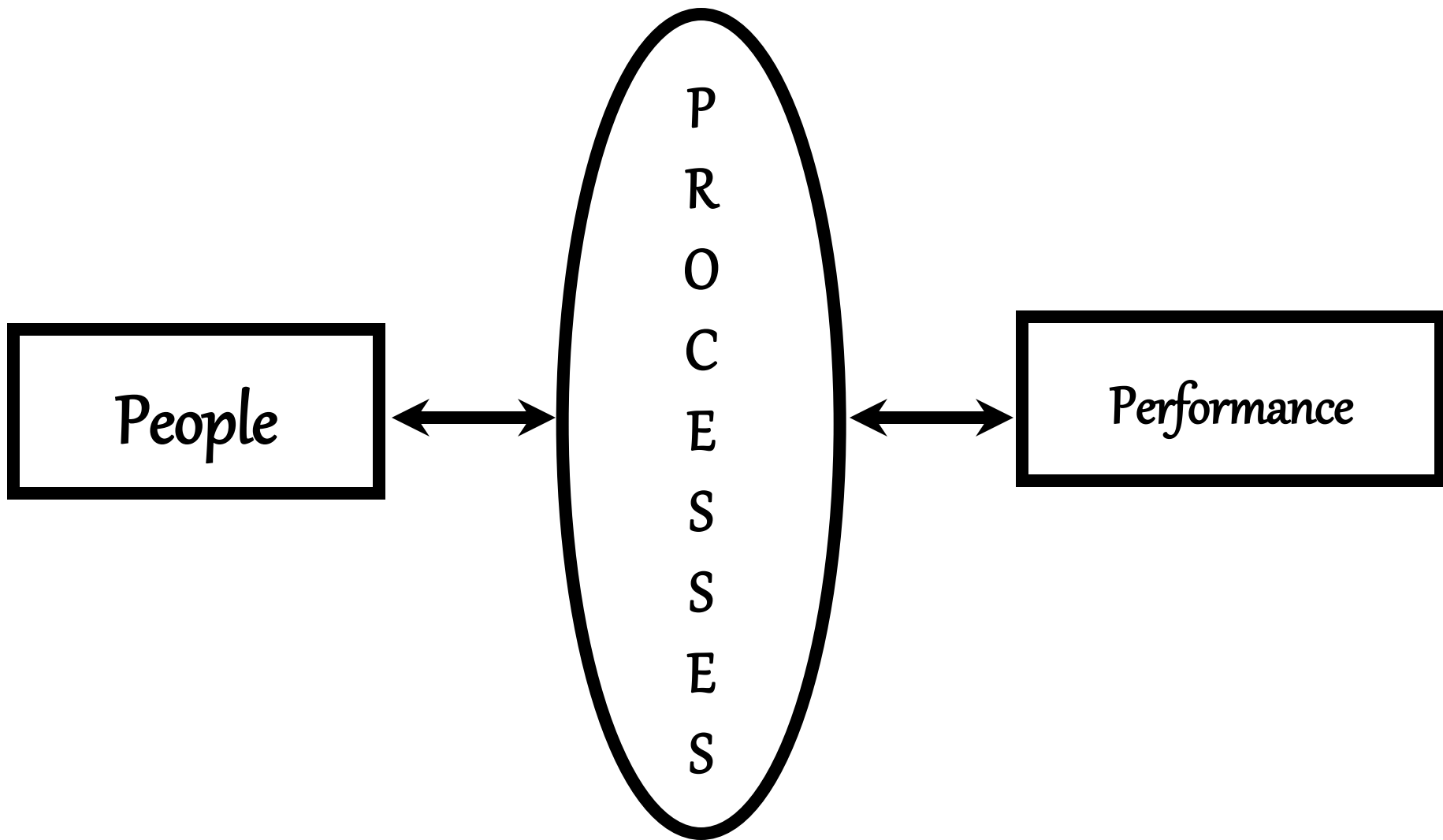
UNTIL

You progressively perfect the
interdependent interpersonal
relationships

(S.Covey)

TOTAL QUALITY MANAGEMENT (TQM)

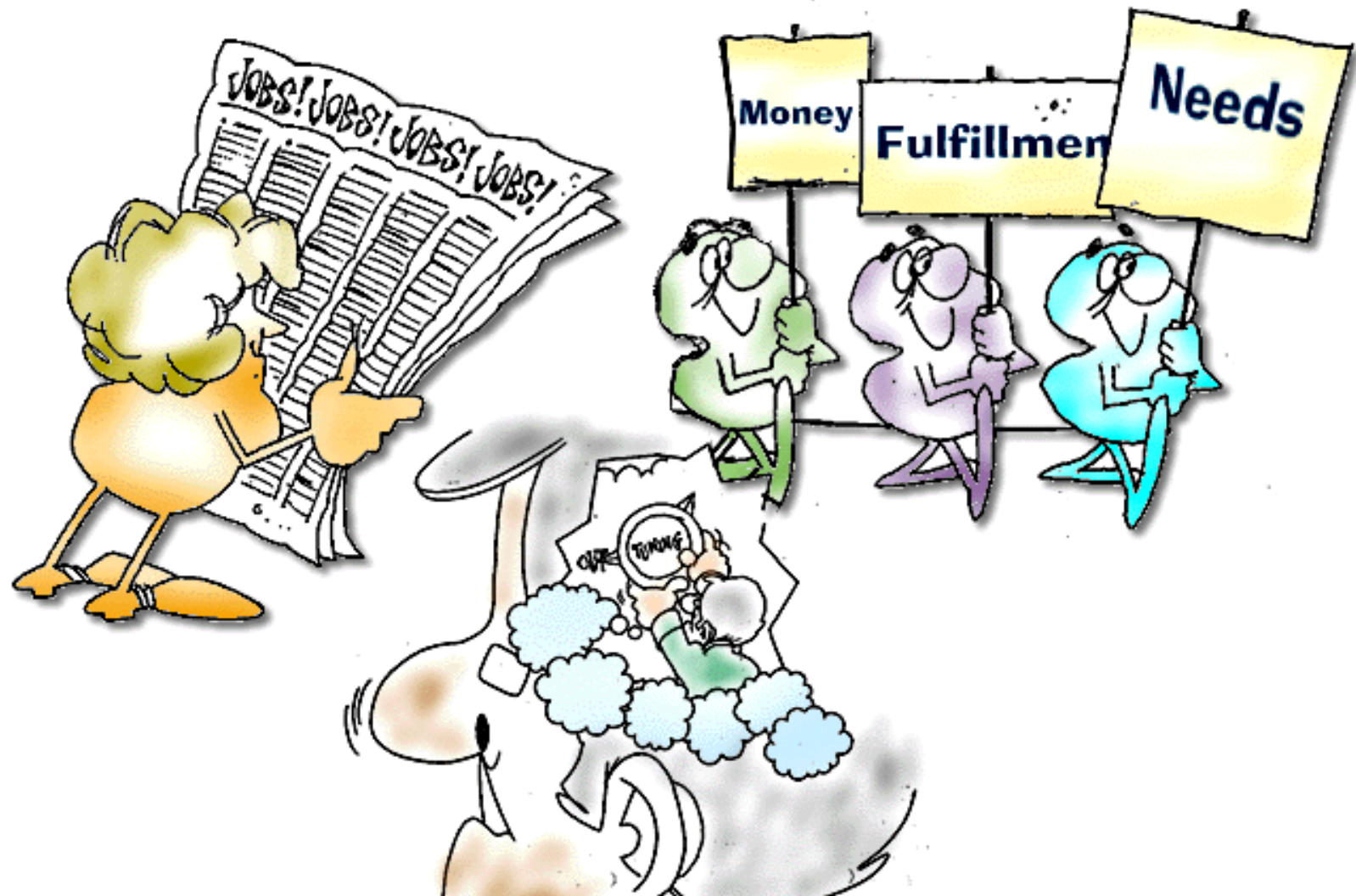
Session # 3



**In solving problems or
improving performance,
what do you usually pay
attention to?**



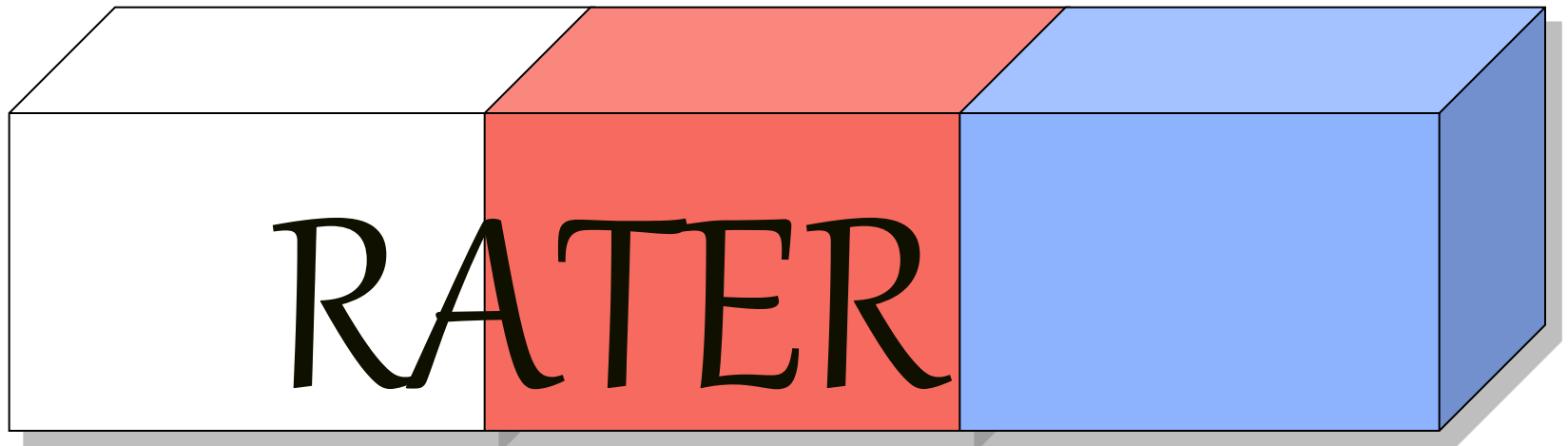
How do I motivate people in spite of the constraints?



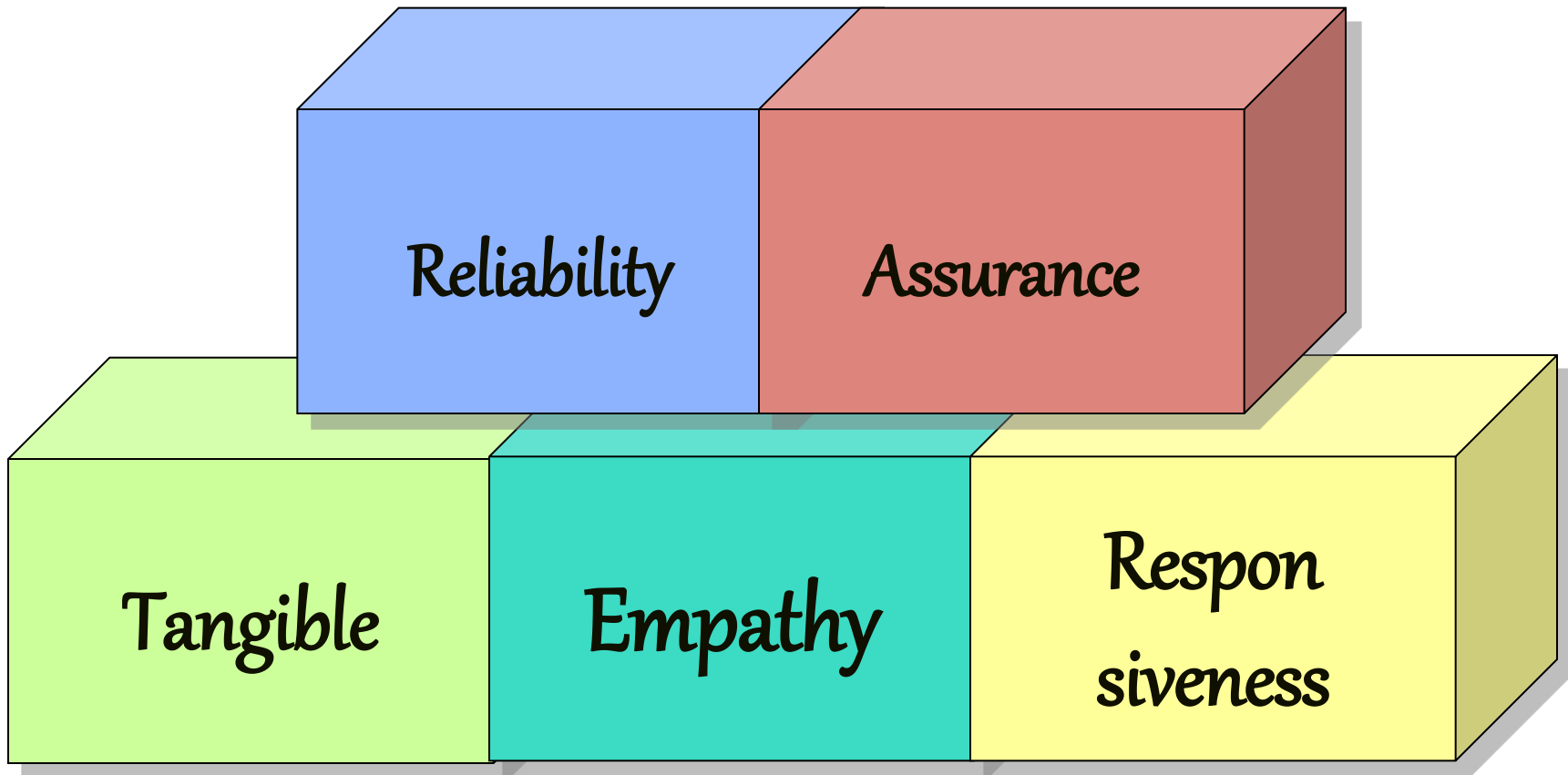
Quality =

*Performance /
Expectations*

Quality-Related Characteristics



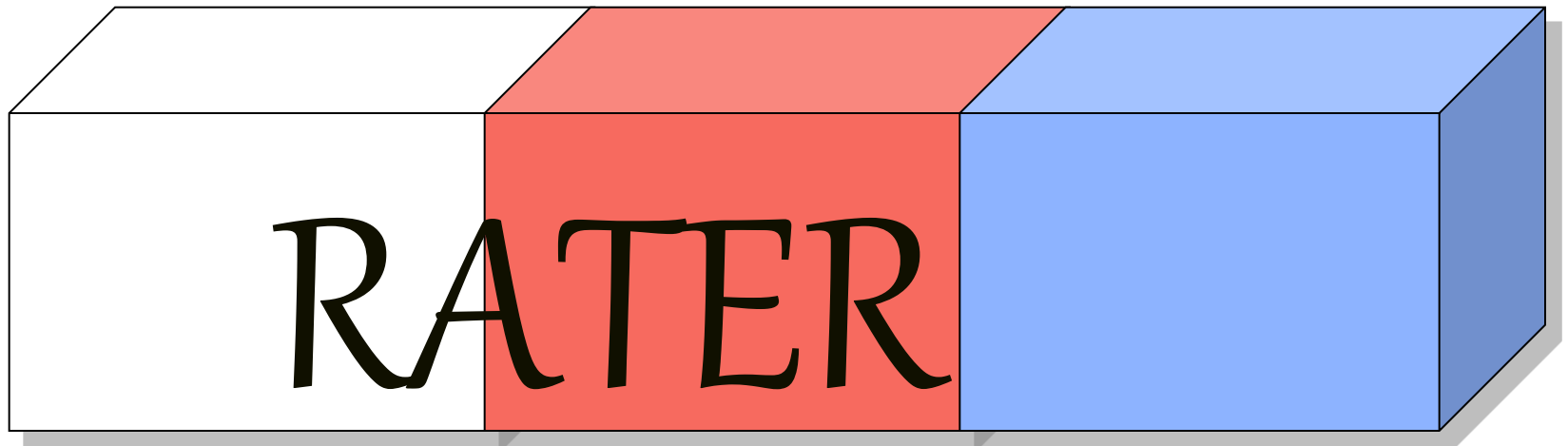
Quality-Related Characteristics



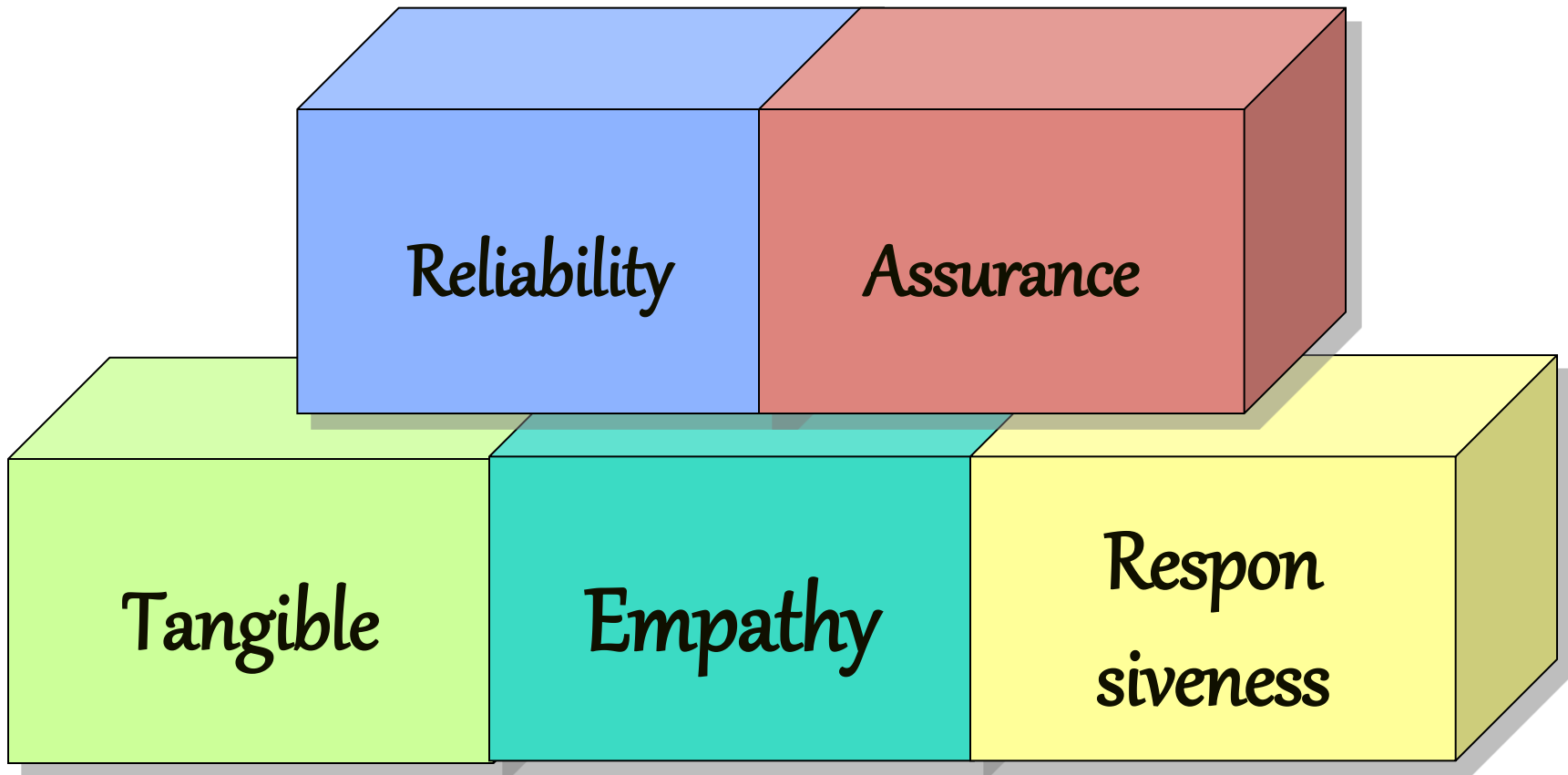
TOTAL QUALITY MANAGEMENT (TQM)

Session # 4

Quality-Related Characteristics



Quality-Related Characteristics



Dimensions of Quality in Manufacturing Sector

Dimensions of Quality in Service Sector

Dimensions of Quality in ?

*Quality
is a Race
without
a
Finish Line!*

Total Quality

1. Focus on increasing customer satisfaction and reducing costs

Total Quality

2. People-focused management System

Total Quality

3. A system's approach that integrates organizational functions and the entire supply chain

Total Quality

4. Based on Measurement of Facts

Total Quality

5. Stresses on Learning and adaptation to Change

Total Quality

6. *A system based on
Leadership and Strategic
Planning*

TOTAL QUALITY MANAGEMENT (TQM)

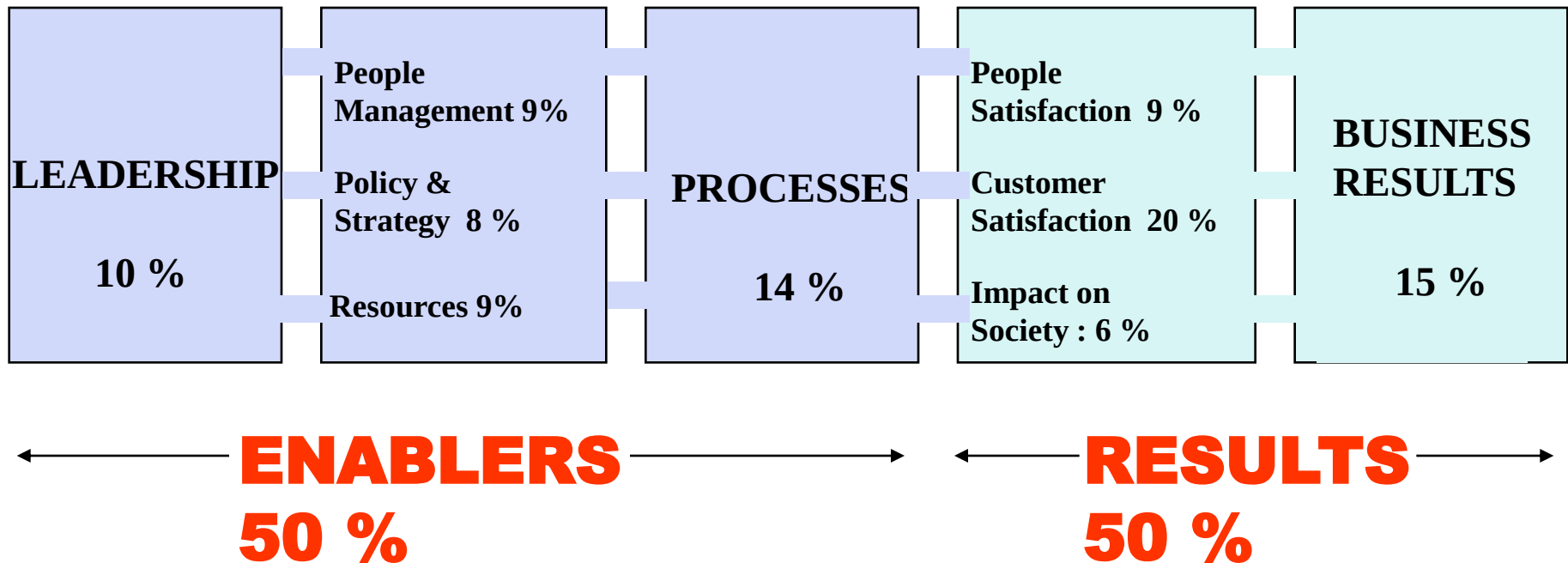
Session # 5

Quality Management Principles @ ISO



- 1. Customer focused Organization**
- 2. Leadership**
- 3. Involvement of People**
- 4. Process Orientation**
- 5. System Approach to Management**
- 6. Continual Improvement**
- 7. Factual approach to Decision Making**
- 8. Mutually beneficial Supplier Relationships.**

EFQM Model



The TQM System

Objective

***Continuous
Improvement***

Principles

**Customer
Focus**

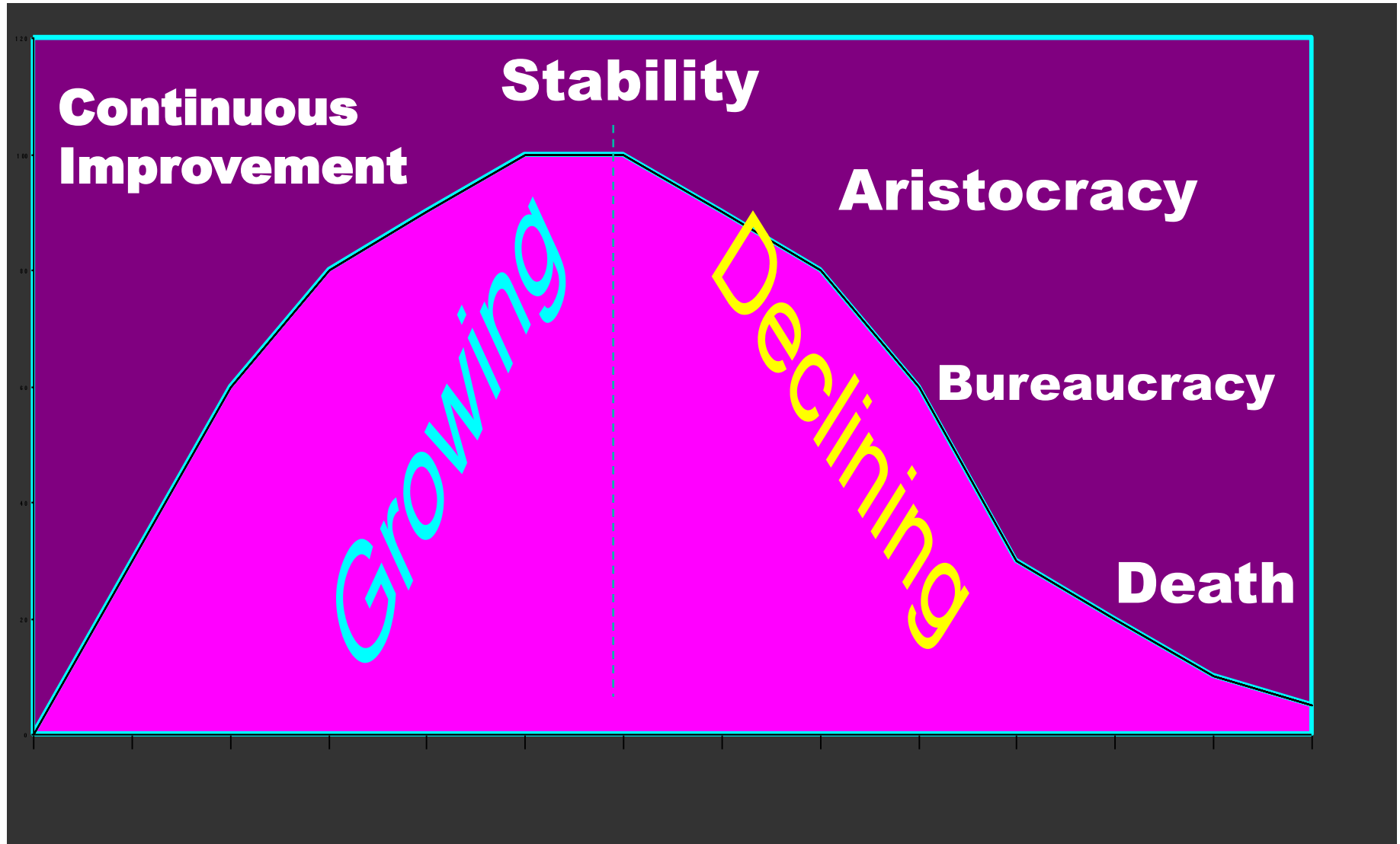
**Process
Improvement**

**Total
Involvement**

Elements

Leadership
Education and Training **Supportive Structure**
Communications **Reward and Recognition**
Measurement

Corporate Life-cycle





TQM

TQM's Customer Approach!

**“The customer
defines quality.”**

**“The customer is
always right.”**

TQM's Customer Approach!

**“The customer
always comes first.”**

**“The customer is a
king.”**

TQM's Customer Approach

**“Quality begins
and ends with
the customer”**

The Customer

External Customer

Internal Customer

Investor Customer

Social Customer

Need for a New Strategy

**1.Foreign markets
have grown**

**2.Import barriers and
protection are not
the answer.**

Need for a New Strategy

**3. Consumers are more sophisticated
They demand new and better products.**

Need for a New Strategy

**4. Consumers are
offered more choices
They have become
more discriminating.**

Competitive Advantage

**--Is driven by
customer wants
and needs**

Competitive Advantage

**--Match
organization's
unique Resources
with Opportunities**

Competitive Advantage

- Provides basis for
further improvement**
- Provides direction
and motivation**

TQ Infrastructure

**1.Customer Relationship
Management**

**2.Leadership and Strategic
Planning**

3.Human Capital Management

4.Process/System Management

5.Data and Information and

Knowledge Management

Quality Meetings

- **Nine Discussion Skills**
 - **Open the discussion**
 - **Listen**
 - **Ask for clarification**
 - **Manage participation**
 - **Summarize**
 - **Manage time**
 - **Contain digressions**
 - **Test for agreement**
 - **Close the discussion**

TOTAL QUALITY MANAGEMENT (TQM)

Session # 6

The background is a vibrant orange color, featuring several large, overlapping circles in various shades of orange and yellow. These circles create a layered, abstract effect. In the center of the image, the word "Recap" is written in a bold, white, sans-serif font. The text has a subtle black outline, making it stand out against the busy background.

Recap

Quality Meetings

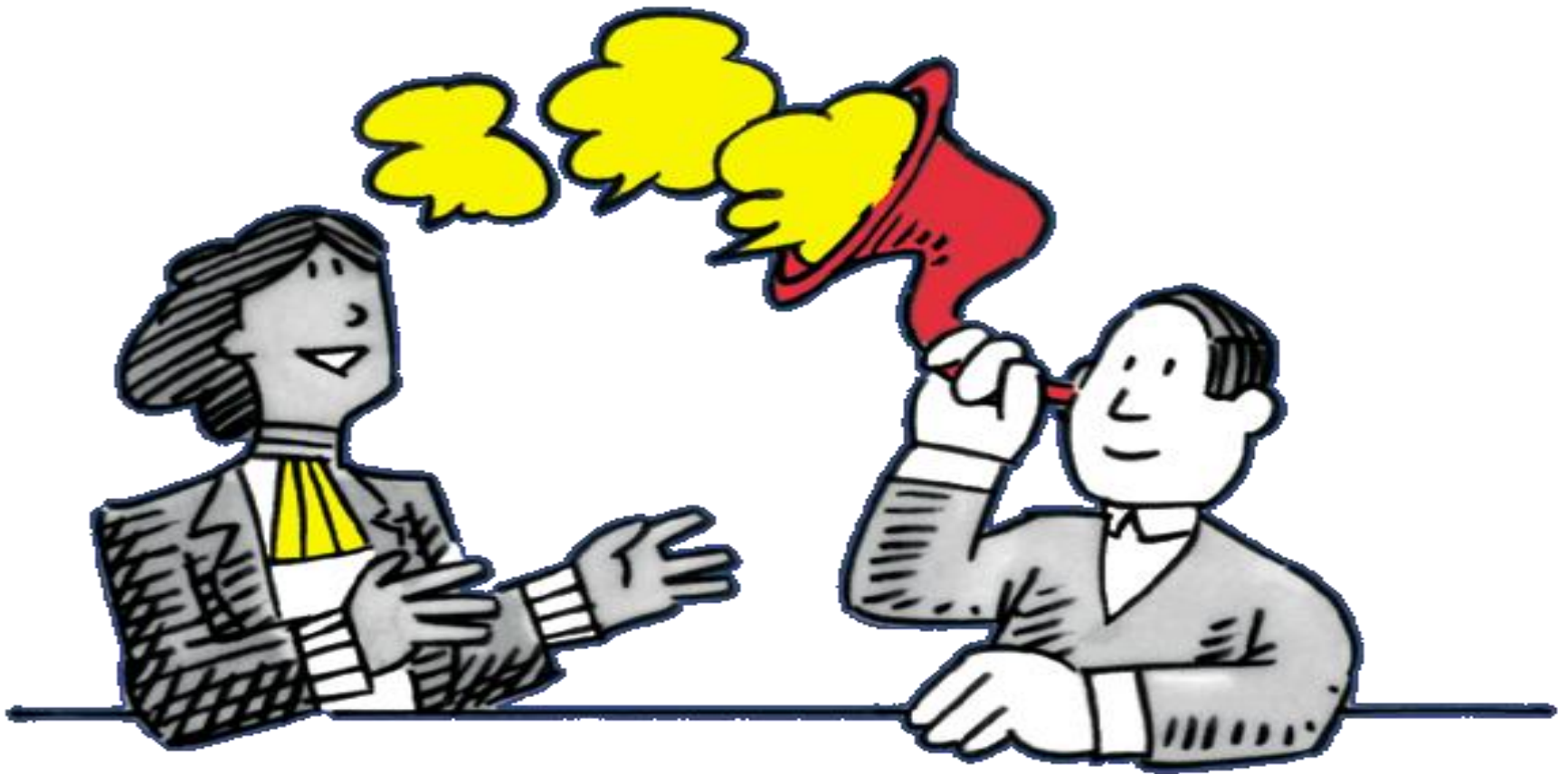
- **Nine Discussion Skills**

- **Open the discussion**
- **Listen**
- **Ask for clarification**
- **Manage participation**
- **Summarize**
- **Manage time**
- **Contain digressions**
- **Test for agreement**
- **Close the discussion**

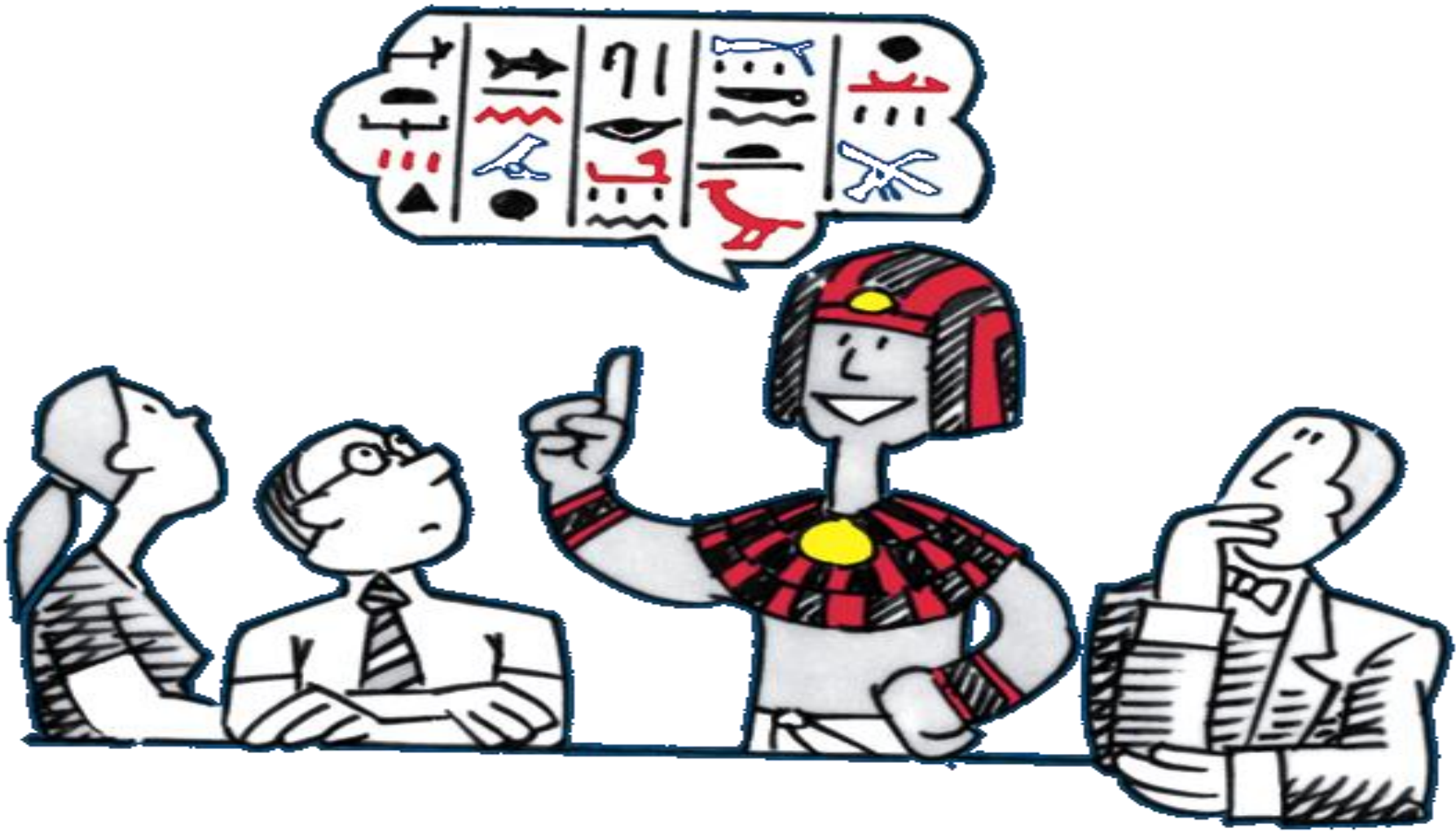
1. Open the Discussion



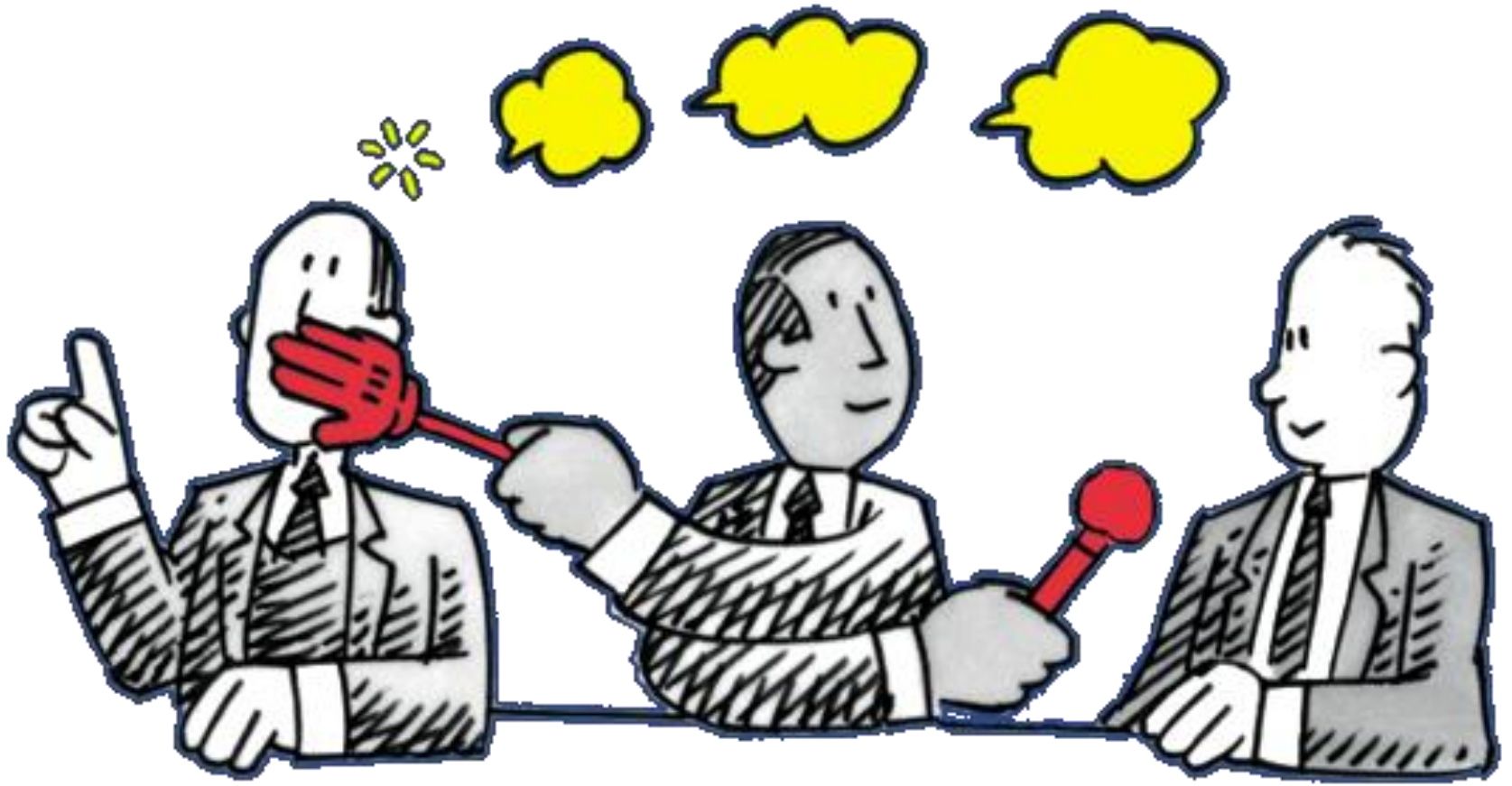
2.Listen



3. Ask for Clarification



4. Manage Participation



5. Summarize



6. Manage Time



7. Contain Digressions



8. Test for Agreement



9. Close the Discussion



Examples of Quality Dimensions

Dimension	(Product) Automobile	(Service) Auto Repair
1. Performance	Everything works, Fit & Finish, Ride, Handling,	All Work done at agreed Price, Friendliness, Courtesy, Competency, Quickness.
2. Aesthetics	Grade of materials used, Interior design, Soft touch	Clean work/Waiting Area
3. Special Features	Gauge/Control placement Cellular Phone /Cup Holder , CD/VCD/DVD Player	Location, Call when Ready Computer Diagnostics

Examples of Quality Dimensions

<u>Dimension</u>	<u>(Product)</u> <u>Automobile</u>	<u>(Service)</u> <u>Auto Repair</u>
5. Reliability	Infrequency of Breakdowns	Work done correctly, Ready when promised
6. Durability	Useful Life in KM, Resistance to Rust & Corrosion	Work Holds up over Time
7. Perceived Quality	Top-Rated Car	Award-winning Service Department
8. Serviceability	Handling of Complaints and/or Requests for Information	Handling of Complaints

Costs Per Hour of Downtime Caused by Software Defects

Business	Cost (Rs .X60)
ATM (Medium-sized Bank)	\$14,500
Package Shipping Service	\$28,250
Telephone Ticket Sales	\$69,000
Catalog Sales Center	\$90,000
Airline Reservation Center (Small Airline)	\$89,500

Assignment

Determining Quality

- For any 3 product or service , list the top 4 criteria for determining Quality.**
 - Be prepared to share your conclusions with the class.**
- 1. Desktop Computer**
 - 2. Wedding Clothing**
 - 3. A Painting**
 - 4. Workshop on Leadership**
 - 5. A Workout Gym**
 - 6. BBQ Restaurant**

TOTAL QUALITY MANAGEMENT (TQM)

Session # 7

The background is a vibrant orange color with several overlapping, semi-transparent circles of varying shades of orange, creating a layered, abstract effect. The text is centered in the middle of the image.

**Just What is
Quality?**

What is Quality?

- **Excellent Product**
 - **Low price**
- **Long life of Items**
 - **Fast Service**
- **Conformance to Requirements**
- **A Product or Service Free of Deficiencies**

What is Quality?

“The Quality of a product or service is a Customer’s Perception of the degree to which the product or service meets his or her Expectations.”

Customer Service Experience

**Case of an Airline
Reservation
Agency.**

Customer Service Experience

Discussion Questions:

- 1. Summarize the service failures associated with this customer service experience.**
- 2. What might airline agency have done to provide better service experience for Mr. Ali.**
- 3. How do your suggestions relate to the TQ principles?**

The Concept

of Quality

Quality by Definition

Quality by Philosophy

Quality by Attitude

Quality by System/Tools

Quality by Latent Needs

**Please now
YOU
define
Quality?**

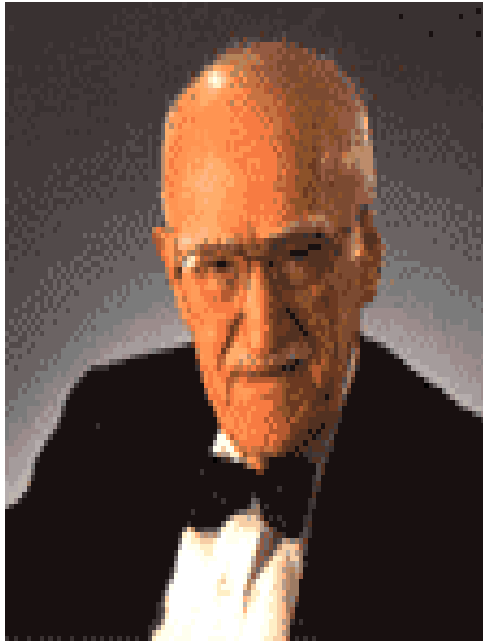
TQM

- **Is Management Led**
- **Is Customer / Client Focused**
- **Is about Prevention not Detection**
- **Right First Time and Every Time**
- **Everyone is Responsible for Quality**
- **Organisation Wide**
- **Continuous Improvement Culture**

Leaders in Quality Revolution

- **W. Edwards DEMING**
- **Joseph M. JURAN**
- **Philip B. CROSBY**
- **K ISHIKAWA**
- **G TAGUCHI**

Who's Who?





Dr. K. Ishikawa

Defining Quality

- **Deming - “Non-Faulty Systems”**
- **Juran - “Fitness for Use”**
- **Crosby - “Conformance to Requirements”**

TOTAL QUALITY MANAGEMENT (TQM)

Session # 8

**Please now
YOU
define
Quality?**

My definition of Quality?

$$Q = \frac{\text{Total Get}}{\text{Total Give}}$$

**Q = Level of Sustainable
and
Emotional Benefits**

Definitions of Quality

- **Product-based : Product Attributes**
- **User-based: Fitness for intended use**
 - **Value-based : Quality vs. Price**
- **Transcendent : Level of Excellence**

Manufacturing-based: Conformance to Specifications

Quality and Personal Values

- **Personal initiative has a positive impact on business success**
 - **Quality begins with personal attitudes**
- **Quality-focused individuals often exceed customer expectations**
- **Attitudes can be changed through awareness, action and awards!**

Leaders in Quality Revolution

- **W. Edwards DEMING**
- **Joseph M. JURAN**
- **Philip B. CROSBY**
- **K ISHIKAWA**
- **G TAGUCHI**

Defining Quality

- **Deming - “Non-Faulty Systems”**
- **Juran - “Fitness for Use”**
- **Crosby - “Conformance to Requirements”**

Kaoru Ishikawa

- Student of Deming**
- Respect for humanity as a management philosophy - full participation.**
- Advanced the use of quality circles (worker quality teams).**
- Cross-functional management.**
- Advocate of the use of the 7 tools (e.g., cause-and-effect diagram or fishbone diagram)**

Defining Quality

Ishikawa

**Degree of match between
True and Substitute
Quality Characteristics
defines
Customer Delight!**

Genichi Taguchi

Taguchi methods emphasize consistency of performance and reduced variation

- Quality Loss Function (deviation from target is a loss to society).**
- Parameter design (Robust Engineering) which is an application of Design of Experiments (DOE).**

Defining Quality

Taguchi

**Stresses on engineering
approach to Quality!**

**Produce with minimum
variation to target goals!**

Focus on Design for Q

Quality is Loss Function!

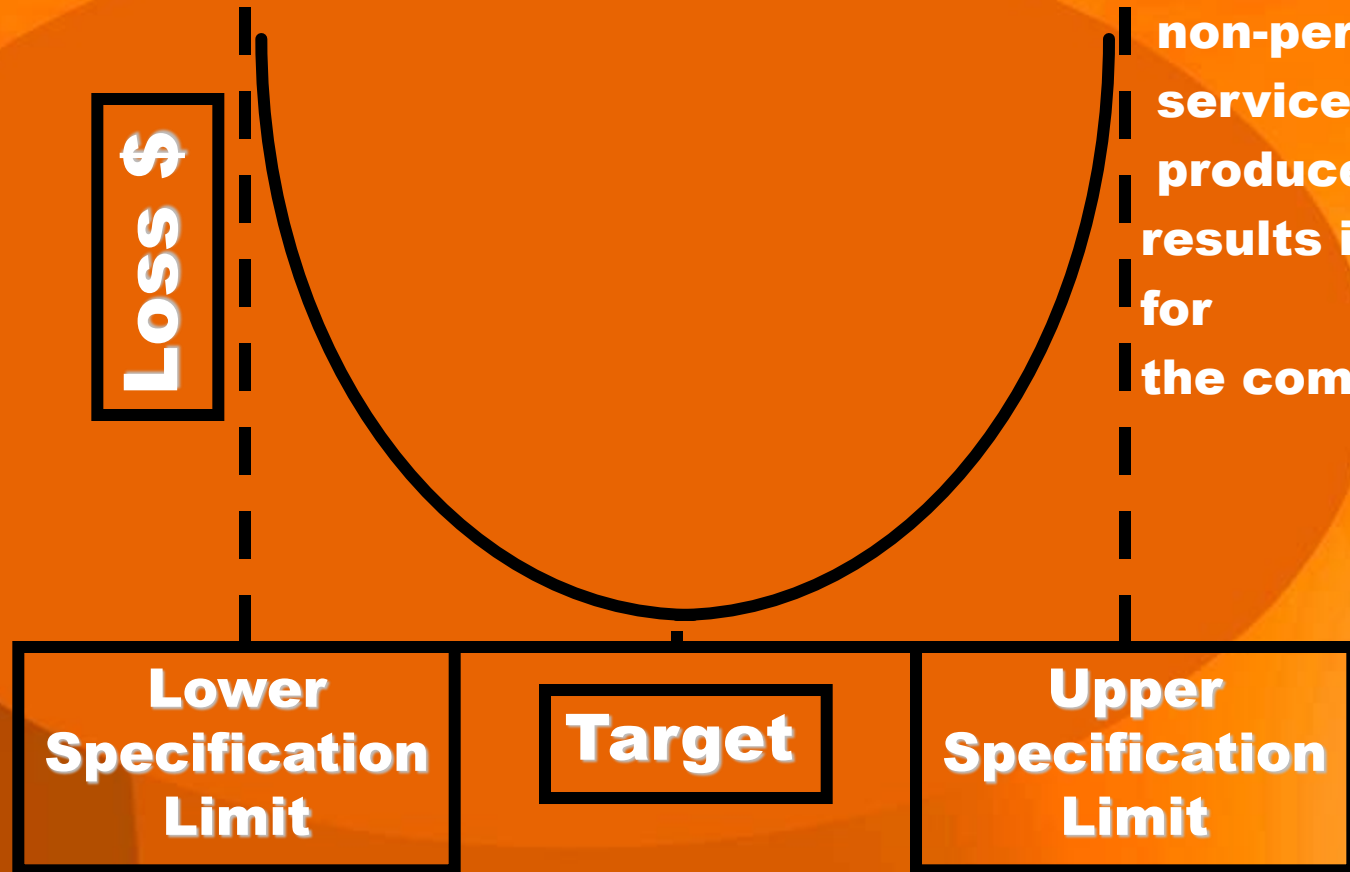
Genichi Taguchi

- **Quality is based on the economic value of being on target and reducing variation.**



Taguchi Loss Function

It is a way to show how each non-perfect service or unit produced, results in a loss for the company.



It's a way to show how each non-perfect service or unit produced, results in a Loss for the company or the customer.

TOTAL QUALITY MANAGEMENT (TQM)

Session # 9

Quality @

ISO 9000(2K)

**Ability of a set of inherent
CHARACTERISTICS of a
PRODUCT,**

**SYSTEM or PROCESS to
fulfill**

**REQUIREMENTS of
CUSTOMERS and other
INTERESTED PARTIES**

A Tale of Two Restaurants

Tale of Two Restaurants

- 1. Contrast these two restaurants from the perspective of TQM. How do they follow or not follow the fundamental principles of TQM?**
- 2. What advice would you recommend to the owners?**

WTO (in 20XX)

The Shifting Focus of Corporate Culture

From	To
Thinking Global	Being Global
Focus on Local Market	Focus on Glocal Market
MNC in manufacturing	MNC in Services
Antagonistic towards Rivals	Collaboration with Rivals
Work with Hands	Work with 3H & H³
Upgrade Technology	Upgrade People
Focus on Process Quality	Focus on Design Quality
Mass Mindset	Lean and Me Mindset
Manage by Control	Manage by Commitment
Individual Values	Shared Values

Organizational Models for Management

Dimension	Mechanistic Model	Organisimic Model	TQ Model	Cultural Model
Goal	Organizational efficiency and performance	Organizational survival	Long-term survival	Meet individual needs; human development
Definition of quality	Conformance to standards	Customer satisfaction	Satisfying or delighting the customer	Constituent satisfaction
Role/nature of environment	Objective; outside boundary	Objective; inside boundary	Blurred organization and environmental boundaries	Enacted/boundaries defined through relationships
Role of management	Coordinate and provide visible control	Coordinate and provide invisible control by creating vision	Focus on improvement and creating a system that can produce quality outcomes	Coordinate and mediate negotiations
Structural rationality	Chain of command (vertical) Technical rationality	Process flow (horizontal and vertical) Organizational rationality	Horizontal processes beginning with suppliers and ending with customers and supported by teams	Mutual adjustment in any direction Political rationality
Philosophy toward change	Stability is valued; learning arises from specialization	Change and learning assist adaptation	Change, continuous improvement, and learning are encouraged	Change and learning are valued in themselves

TOTAL QUALITY MANAGEMENT (TQM)

Session # 10

History of Quality

- **When do you think the Focus on Quality started?**

a. 3000 B.C.

b. Craftsmen Guilds(1400s)

c. Late 1800s

d. Mid 1980

e. From 2000 onward

History of Quality Paradigms

1. Customer-Craft Quality Paradigm:

1. Design and Build each product for a particular Customer.

2. Producer knows the Customer directly.

History of Quality Paradigms

2. Mass Production and Inspection Quality Paradigm:

- 1. Focus on designing and building products for Mass Consumption.**
- 2. Larger Volumes will Reduce Costs and Increases Profits.**

History of Quality Paradigms

2. Mass Production and Inspection Quality Paradigm:

3. Push products on the customer

4. Quality is maintained by Inspecting and Detecting Bad Products.

History of Quality Paradigms

3. TQM or

“Customer Driven Quality” Paradigm:

1. Potential Customers determine what to Design and Build.

2. Higher Quality will be obtained by Preventing Problems

Shift to Quality



TOTAL QUALITY MANAGEMENT (TQM)

Session # 11

Modern History of Quality

- **Industrial Revolution 1800's.**
 - **Scientific Management - -
Frederick Taylor 1900's.**
- **Statistical Quality Control 1920's
at Bell Labs.**
- **International Competition...
1970's-1980's @ Japan.**
 - **Global Quality Revolution
1980-1990 @ Deming / ISO.**

Modern History of Quality

- **EU market entry and ISO-9000 (1994)**
- **Quality became a requirement for success in global market 1997.**
- **ISO-9000 (2000) CMM.... Six Sigma ... KM and Learning Organization**
- **ISO-9000 (2007) emphasizing KM**

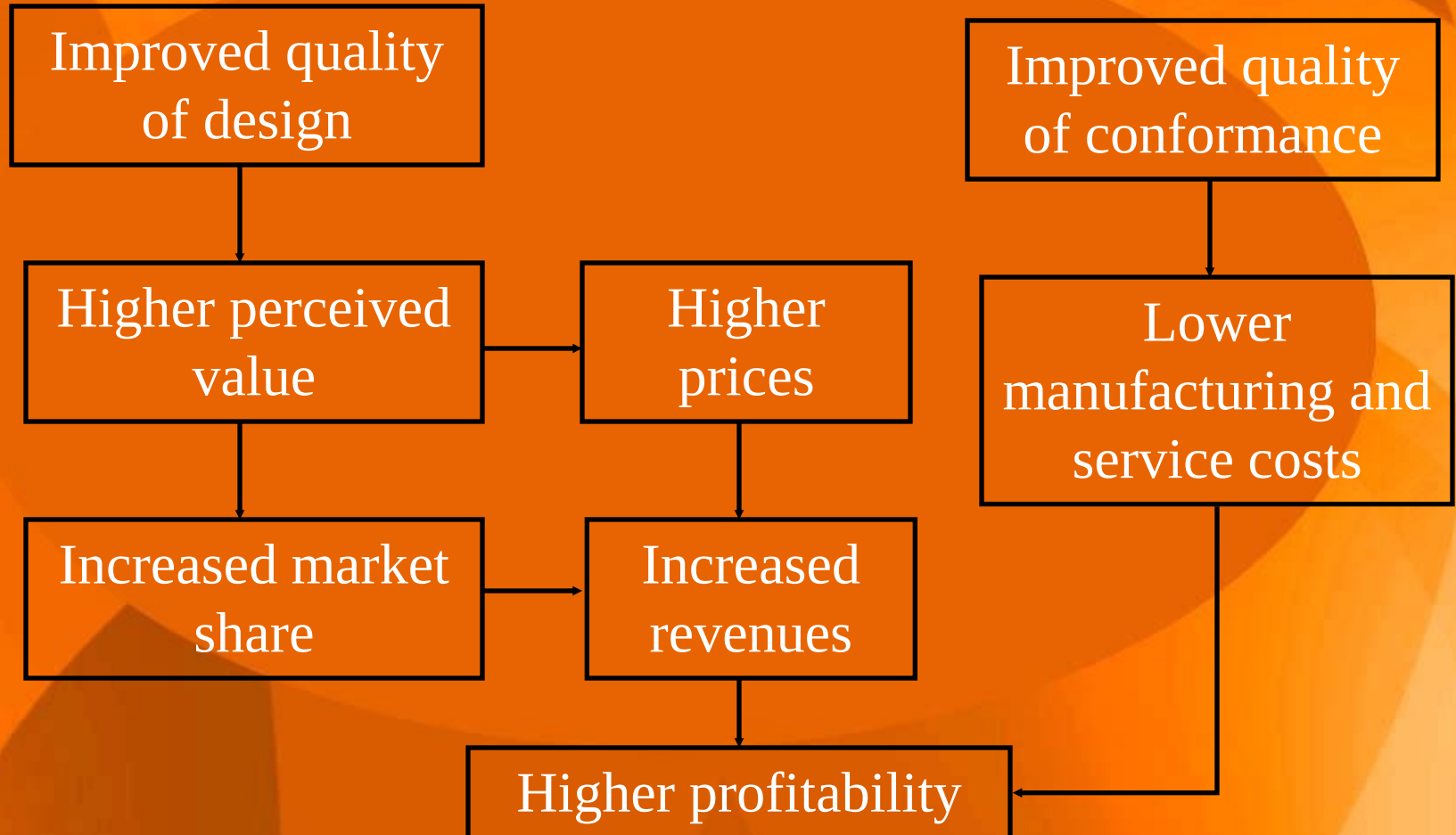
Three Levels of Quality

- 1. Organizational level: meeting external customer requirements**
- 2. Process level: linking external and internal customer requirements**
- 3. Performer level: meeting internal customer requirements**

Deming Chain Reaction

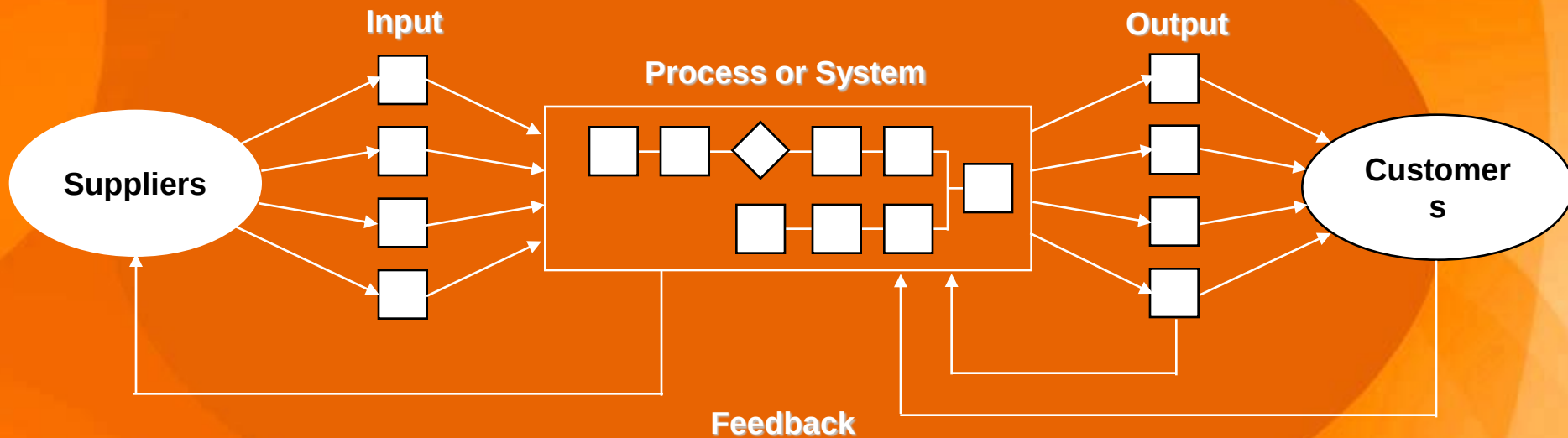


Quality and Profitability



SIPOC Diagram

Suppliers, Inputs, Processes, Outputs and Customers.



Q

QC

SPC

QM / QA

QMS

TQM

Six Sigma & TOE / BSC

KM / Learning Org.

TOTAL QUALITY MANAGEMENT (TQM)

Session # 12

Q

QC

SPC

QM / QA

QMS

TQM

Six Sigma & TOE / BSC

KM / Learning Org.

QM

- **Quality Control (QC) - “the operational techniques and activities that are used to fulfil requirements for quality”.**

QM

- **Quality Management (QM)**
 - “that aspect of the overall management function which determines and implements the quality policy’.

QM

QUALITY POLICY - should be developed and endorsed by senior executives and management, understood and implemented at all levels in the organisation, and be relevant to the organisational goals and the needs and expectations of its customers.

QM

- **Quality Assurance (QA) -**
“all planned and systemic activities necessary to provide adequate confidence that a product or service will satisfy given requirements for quality”.

QMS

- **Quality Management Systems (QMS) -“organisational structure, responsibilities, procedures, processes and resources for implementing quality management”. e.g. ISO 9000, EFQM, MBNQA etc .**

TQM

- **A management philosophy focusing on customers' expectations, preventing problems, building commitment to best practice in the entire workforce, and promoting open decision making with a will to improve.**

Evolution of Quality Management

Inspection

Sorting, grading, corrective actions, identify sources of non-conformance

Quality Control

Develop quality manual, process performance data, self-inspection, product testing, basic quality planning, use of basic statistics, paperwork control.

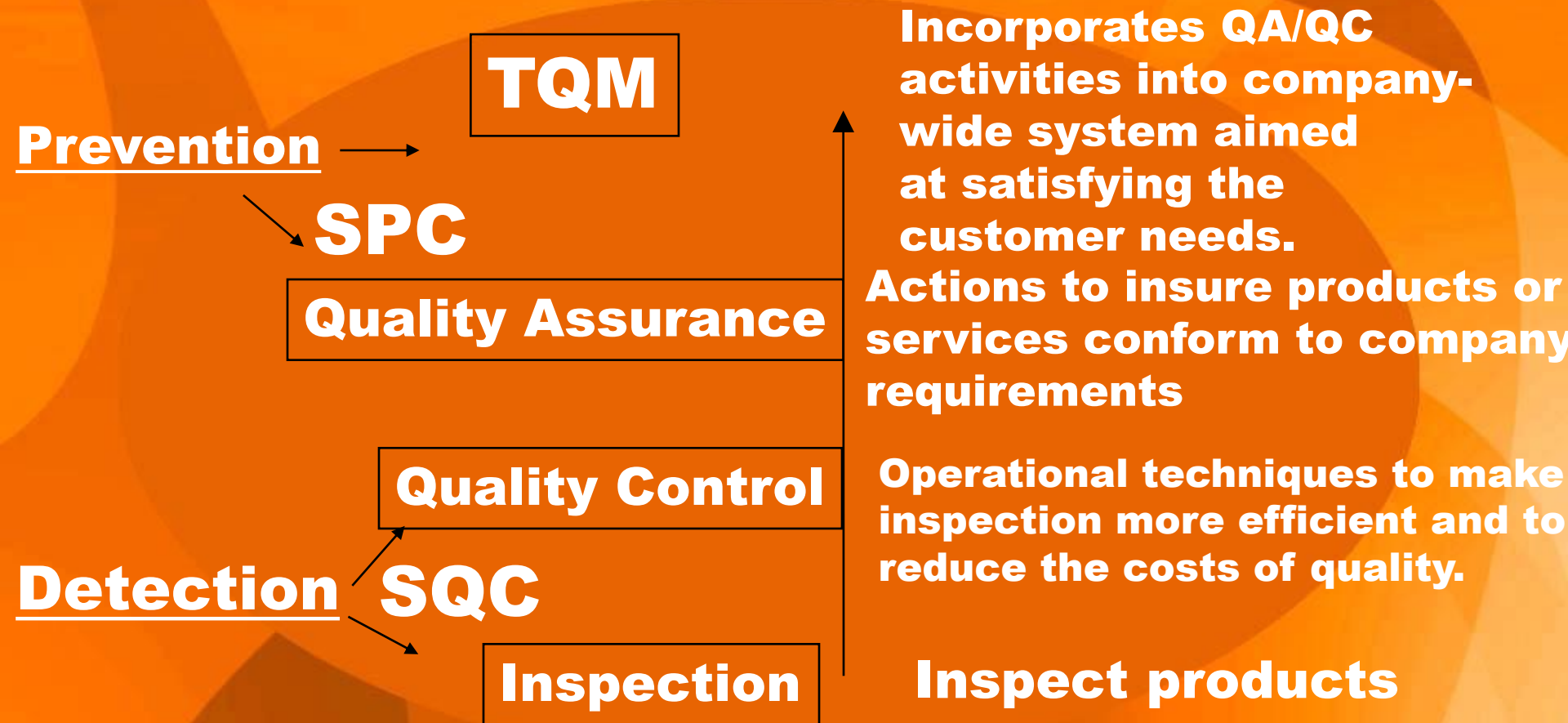
Quality Assurance

Quality systems development, advanced quality planning, comprehensive quality manuals, use of quality costs, involvement of non-production operations, failure mode and effects analysis, SPC.

TQM

Policy deployment, involve supplier & customers, involve all operations, process management, performance measurement, teamwork, employee involvement.

The Quality Hierarchy



The Concept

of Quality

Quality by Definition

Quality by Philosophy

Quality by Attitude

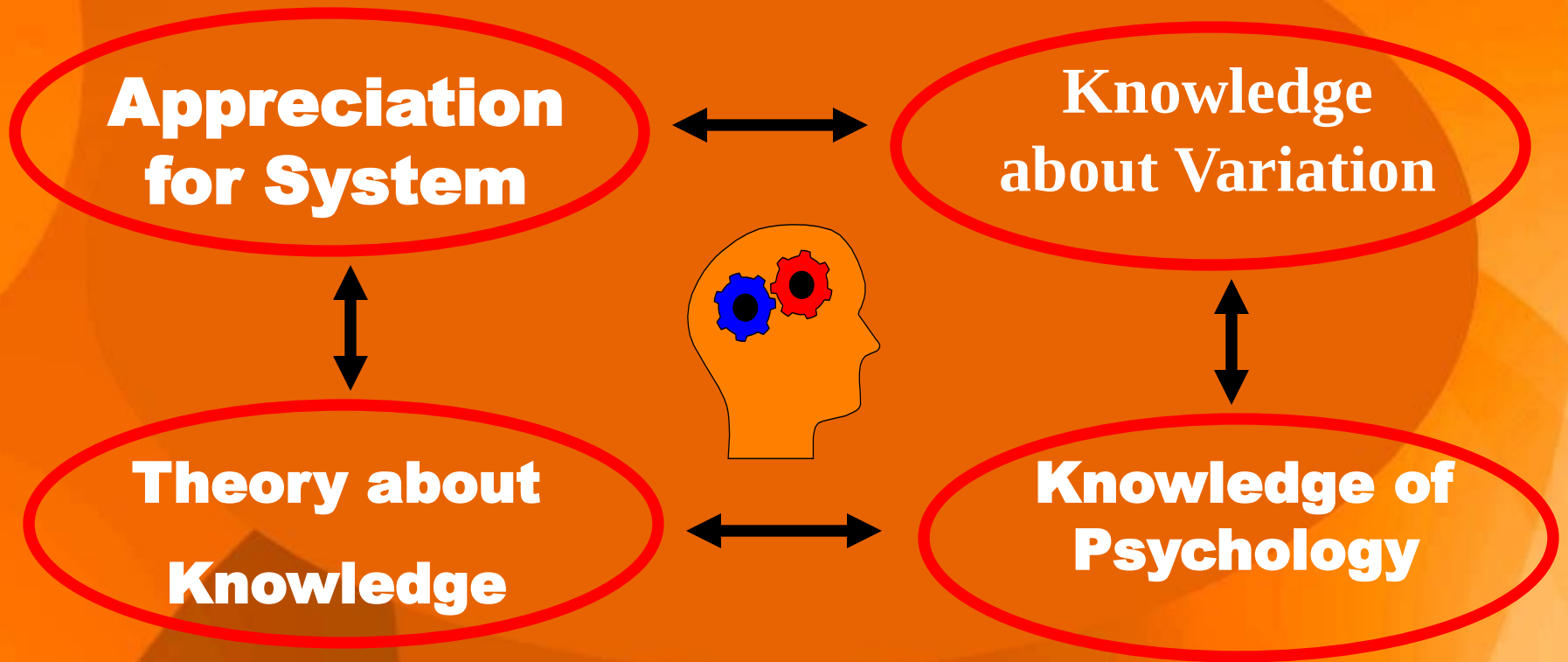
Quality by System/Tools

Quality by Latent Needs

Deming's System of Profound Knowledge

- **Appreciation for a system**
- **Understanding variation**
- **Theory of knowledge**
- **Psychology**

Deming's System of Profound Knowledge



TOTAL QUALITY MANAGEMENT (TQM)

Session # 13

Systems

- **Most organizational processes are cross-functional**
- **Parts of a system must work together**
- **Every system must have a purpose**
- **Management must optimize the system as a whole**

Variation

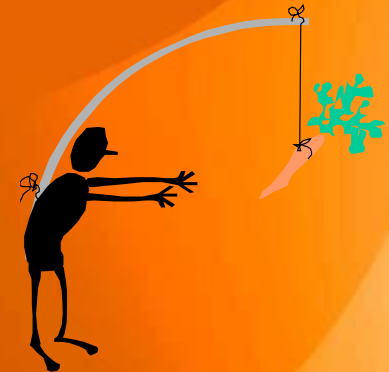
- **Many sources of uncontrollable variation exist in any process**
- **Excessive variation results in product failures, unhappy customers, and unnecessary costs**
- **Statistical methods can be used to identify and quantify variation to help understand it and lead to improvements**

Theory of Knowledge

- **Knowledge is not possible without theory**
- **Experience alone does not establish a theory, it only *describes***
- **Theory shows cause-and-effect relationships that can be used for prediction**

Psychology

- **People are motivated intrinsically and extrinsically**
- **Fear is demotivating**
- **Managers should develop pride and joy in work**



Deming's 14 Points

1. Create constancy of purpose towards improvement.

2. Adopt the new philosophy

3. Cease dependence on inspection.

TOTAL QUALITY MANAGEMENT (TQM)

Session # 14

**Deming's System of
Profound Knowledge
When People
don't
understand
Systems !!**

**Deming's System of
Profound Knowledge
When People
don't
understand
Variations !!**

**Deming's System of
Profound Knowledge
When People
don't
understand
Psychology!!**

**Deming's System of
Profound Knowledge
When People
don't
understand
Theory of K !!**

Deming's 14 Points

1. Create constancy of purpose towards improvement.

2. Adopt the new philosophy

3. Cease dependence on inspection.

Deming's 14 Points

4. Move towards a single supplier for any one item

5. Improve constantly and forever.

Deming's 14 points

6. Institute training on the job

7. Institute leadership

8. Drive out fear

**9. Break down barriers
between departments.**

10. Eliminate slogans

Deming's 14 points

**11. Eliminate
Management by
Objectives (MBO)**

**12. Remove Barriers
to Pride of
Workmanship**

Deming's 14 points

**13. Institute
Education and Self-
Improvement.**

**14.The
Transformation is
Everyone's Job.**

TOTAL QUALITY MANAGEMENT (TQM)

Session # 15

Deming's Seven Deadly Diseases

**1. Lack of Constancy of
Purpose**

**2. Emphasis on Short-term
Profits**

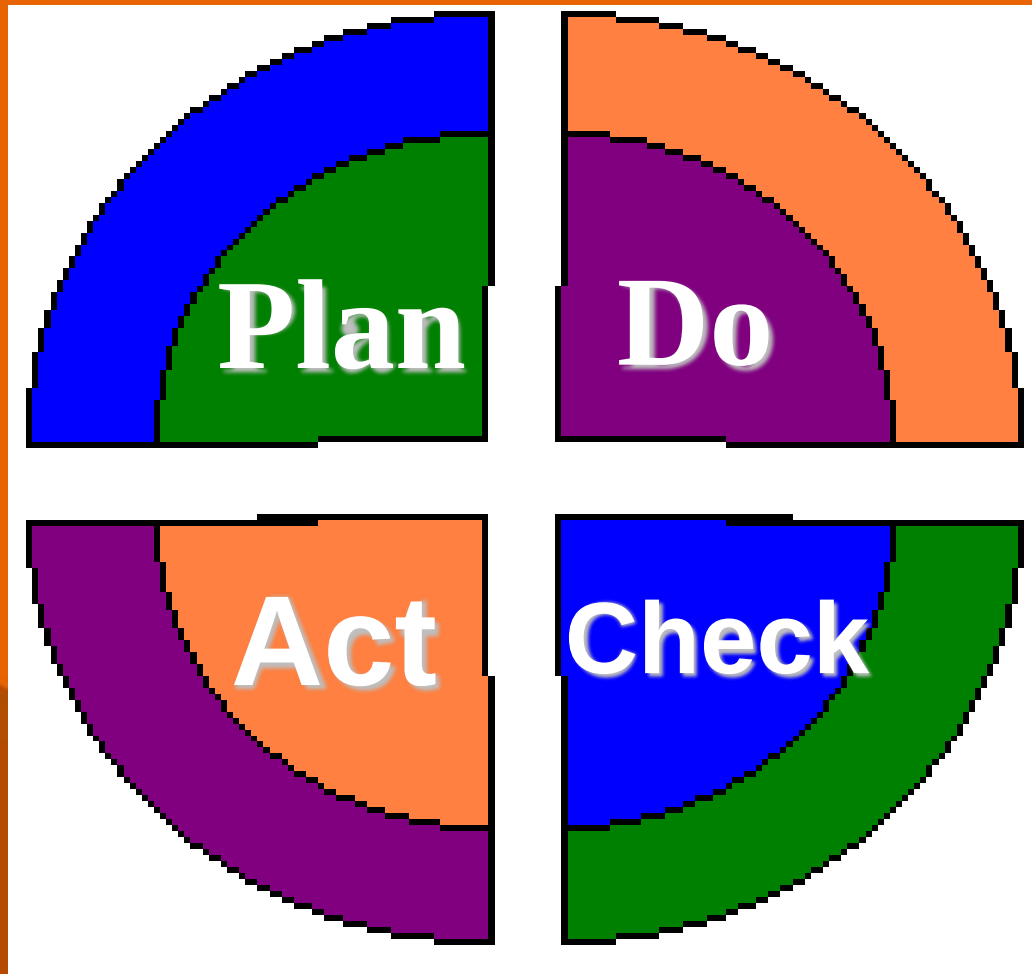
**3. Evaluation by Performance
rating, Merit rating, or Annual
performance review**

4. Mobility of Management

Deming's Seven Deadly Diseases

- 5. Running a company on
visible figures alone**
- 6. Excessive medical costs**
- 7. Excessive costs of
warranty**

Deming Cycle



The Deming Cycle



The Deming Cycle

- 1. Conduct consumer research and use it in planning the product (PLAN).**
- 2. Produce the product (DO).**
- 3. Check the product to make sure it was produced in accordance with the plan (CHECK).**
- 4. Market the product (ACT).**
- 5. Analyze how the product is received in the marketplace in terms of quality, cost, and other criteria (EVALUATE).**

The Deming Cycle



Juran's TEN Points

1. Build awareness of the need and opportunity for improvement.

2. Set goals for improvement.

3. Organise to reach the goals by deploying quality councils.

Juran's TEN Points

4. Provide training.

**5. Carry out projects
to solve problems**

6. Report progress.

7. Give recognition.

Juran's TEN Points

8. Communicate results.

9. Keep score.

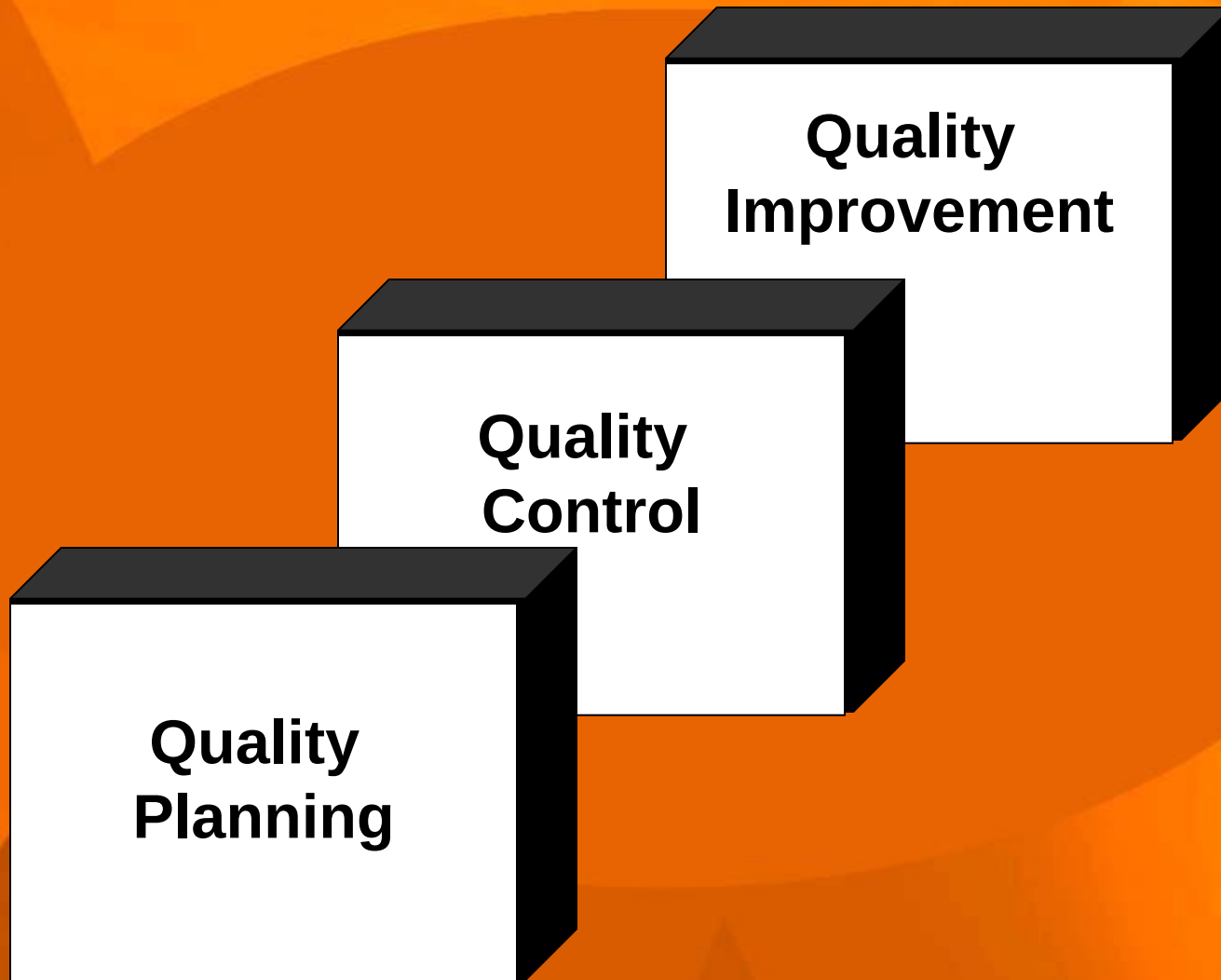
**10. Maintain momentum
by making annual
improvement part of the
regular systems .**

Juran's Quality Trilogy

- **Quality planning**
- **Quality control**
- **Quality improvement**



The Juran Trilogy



TOTAL QUALITY MANAGEMENT (TQM)

Session # 16

The Juran Trilogy



Quality Planning

1. Determine who the customers are.

2. Identify customers' needs.

3. Develop products with features that respond to customer needs.

Quality Planning

4. Develop systems and processes that allow the organization to produce these features.

5. Deploy the plans to operational levels.

Quality Control

1. Assess actual quality performance

2. Compare Performance with goals

3. Act on differences between performance and goals.

Quality Improvement

- 1. Develop the infrastructure necessary to make annual quality improvements.**
- 2. Identify specific areas in need of improvement, and implement improvement projects.**

Quality Improvement

3. Establish a project team with responsibility for completing each improvement project.

4. Provide teams with what they need to be able to diagnose problems to determine root causes, develop solutions, and establish controls that will maintain the gains made.

Phillip Crosby

Quality is Free . . . :

“Quality is free. It’s not a gift, but it is free.

What costs money are the unquality things -- all the actions that involve not doing jobs right the first time.”

Philip Crosby

4 Absolutes of Quality

**–#1- Quality is defined by
Conformance to
Requirements,
“Not Goodness”.**

**–#2 - System for causing
quality is Prevention not
Appraisal.**

TOTAL QUALITY MANAGEMENT (TQM)

Session # 17

Philip Crosby

Four Absolutes of Quality

**-#3 - Performance Standard is
Zero Defects.**

**-#4 - Measurement of quality
is the Price of
Nonconformance.**

The Concept

of Quality

Quality by Definition

Quality by Philosophy

Quality by Attitude

Quality by System/Tools

Quality by Latent Needs

Costs of Quality in Life

- Failure Attitude**
- Appraisal Attitude**
- Preventive Attitude**
- Prerogative Attitude**

Costs of Quality

- **Prevention**
- **Appraisal**
- **Internal Failures**
- **External Failures**

Quality Cost

A quality cost is considered to be any cost that the company would not have incurred if the quality of the product or service were perfect.

TOTAL QUALITY MANAGEMENT (TQM)

Session # 18

Quality Cost

A quality cost is considered to be any cost that the company would not have incurred if the quality of the product or service were perfect.

Quality Dimensions

- **Design Quality**
 - **Characteristics of the product's original design**
- **Conformance Quality**
 - **Building products (or delivering services) to the specifications of the product designers**

Costs of Quality

- **External Failures**
- **Internal Failures**
 - **Appraisal**
 - **Prevention**

Costs of Quality

- **External Failures**

- 1. Cost of Processing
Customer Complaints**

- 2. Customer Returns**

- 3. Field Service**

- 4. Warranty Claims**

- 5. Products Recalls**

Costs of Quality

- **Internal Failures**

- 1. Costs of Rework & Repair**

- 2. Retesting and Re-inspecting Costs**

- 3. Downtime**

- 4. Internal miscommunications that results in delays**

Costs of Quality

- **Appraisal**

(Costs associated with Measuring, Evaluating, or Auditing)

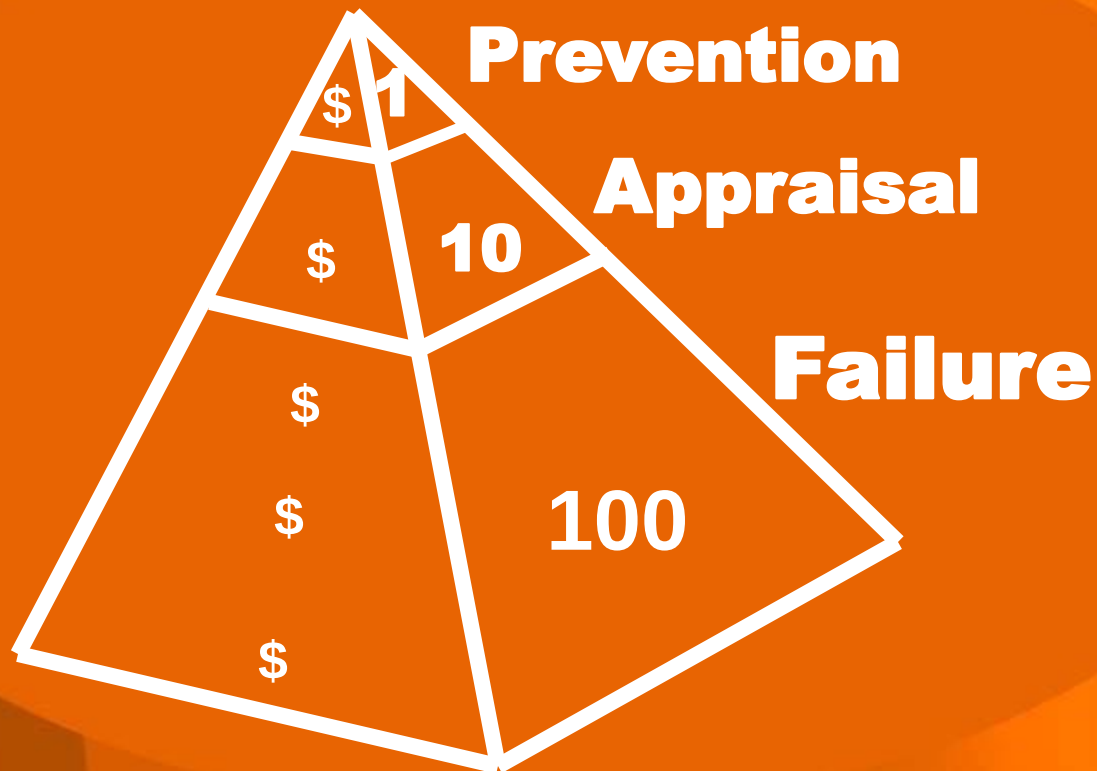
- **Incoming inspection**
- **Test of purchased materials**
 - **Verification and checking activities**
- **In-process and final inspection**
 - **Calibration Costs**

Costs of Quality

- **Prevention**

- 1. New products design reviews**
- 2. Quality planning**
- 3. Supplier Quality Survey**
- 4. Quality Improvement Meetings**
- 5. Quality Education and Training**

The 1-10-100 Rule



Return on Quality (ROQ)

- **ROQ – measure of revenue gains against costs associated with quality efforts**
- **Principles**
 - **Quality is an investment**
 - **Quality efforts must be made financially accountable**

Juran and the Cost Of Quality

TWO Types of Costs:

Unavoidable Costs: preventing defects (inspection, sampling, sorting, QC)

Avoidable Costs: defects and product failures (scrapped materials, labor for re-work, complaint processing, losses from unhappy customers

“Gold in the Mine”



Quality is a Journey,
not a Destination



Quality

“Quality is never an accident; it is always the result of high intention, sincere effort, intelligent direction and skillful execution. It represents the wise choice of many alternatives.”

TOTAL QUALITY MANAGEMENT (TQM)

Session # 19

The Concept

of Quality

Quality by Definition

Quality by Philosophy

Quality by Attitude

Quality by System/Tools

Quality by Latent Needs

Which Q Approach?

None of Q Theories and Philosophies provide a Framework

- 1. HOW to implement TQM in a Systemic way**
- 2. How to assess one's Q performance against a peer or a world class company!**

Which Q Approach?

- **Quality @ Little q**
 - **Organizations focusing on QC and inspection only!**
- **Quality @ Big Q**
 - **Must transform their thinking to Quality across the whole organization!**

Which Q Approach?

- **M Baldrige Quality Award**
- **EFQM Excellence Model (EQA)**
- **National Awards:**
 - Pak Q Award ?**
 - Sh. Hamdan Arab Award,**
 - Australian / Singapore Q Award ..**
- **ISO-9000(2000) International QMS Award**

Which Q Approach?

- **Other ISO-9000 Related Awards**
 - 1. QS-9000 for Auto Industry**
 - 2. ISO-9003(2004) Guide for SE**
 - 3. ISO-10006(2003) Guide for QM in Projects**
 - 4. ISO-14000(2004) EMS**
 - 5. ISO-22000 Food Safety MS**

Which Q Approach?

- **Other Related Standards:**
 - 1. *CE* Mark (a French word for *Conformite Europeenne*)**
 - 2. SA-8000 Social Accountability**
 - 3. GMP**
 - 4. *Tick IT (9001 in SW Dev. Sector)***
 - 5. SEI-CMM (Levels 1-5)**

Which Q Approach?

- **Continuous i.e. (Non-stop) OR
Continual i.e. (Recurring often)
Improvement Frameworks !!!**
 - 1. KAIZEN**
 - 2. SIX SIGMA**
 - 3. BENCHMARKING**

**ISO-9000
QMS
International
Award**

Who is ISO?

- **QMS standards were first published in 1987 by a technical committee of the International Organization for Standardization (ISO).**
- **Based in Geneva, Switzerland, ISO is a consortium of the national standard bodies of more than 100 countries.**

TOTAL QUALITY MANAGEMENT (TQM)

Session # 20

The background is a vibrant orange with several overlapping, semi-transparent circles of varying shades, creating a dynamic, layered effect. The text is centered and reads:

**TOTAL
QUALITY
SYSTEMS
FOR
BUSINESS
EXCELLENCE**

BUSINESS EXCELLENCE



BQA Framework

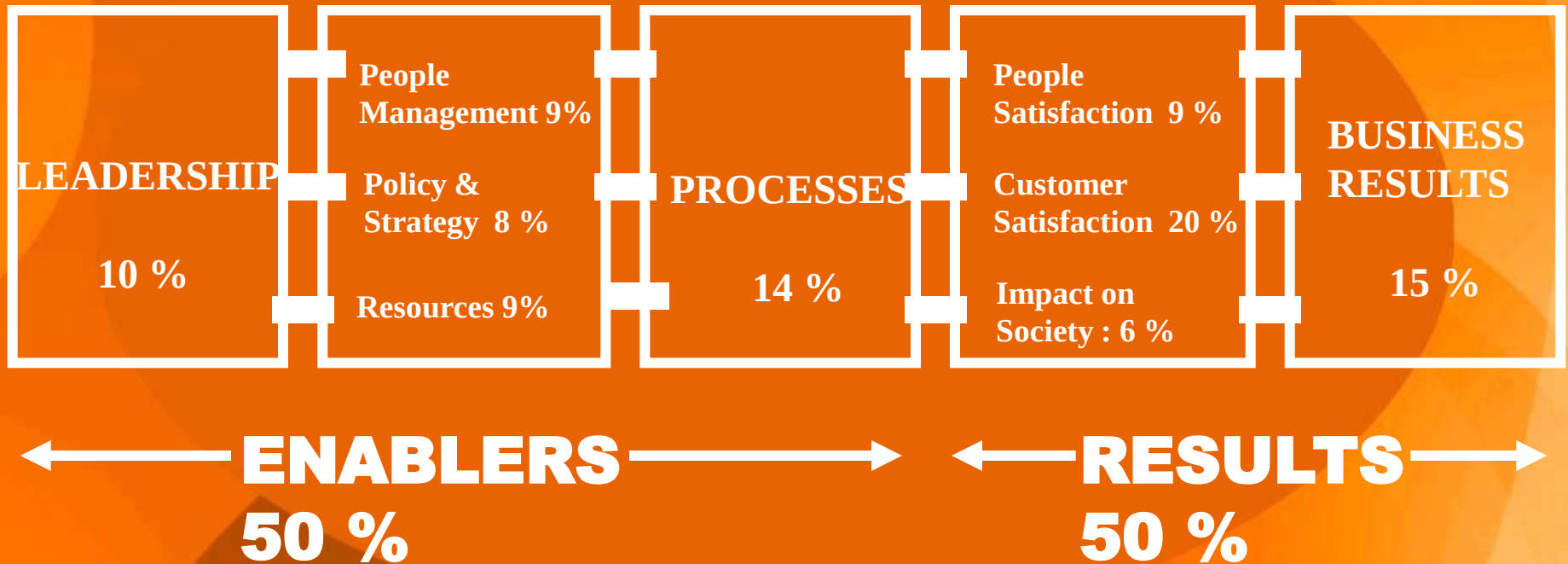


BQA 200X- Categories

- 1. Leadership (12%)**
- 2. Strategic Planning (8.5%)**
- 3. Customer and Market Focus 8.5%**
- 4. Information and Analysis (9%)**
- 5. Human Resource Focus (8.5%)**
- 6. Process Management (8.5%)**
- 7. Business Results (45%)**

EFQM (EQA) Model

Innovation & Learning Focus



ISO-9000(2000) Quality Management System

What is ISO 9000?

- ISO 9000 is a series of international standards establishing global requirements for QMS.**
- It is not a product standard, but a Quality System Standard .**
 - It applies not to products or services, but to the Processes which creates them.**

Objectives of ISO 9000!

- 1. Provide confidence that QS requirements are being met**
- 2. Provide confidence to customers and other stakeholders that quality requirements are being achieved in the delivered products/services/artifacts**

Objectives of ISO 9000!

- 3. Provide confidence to internal customers that q / Q improvements are taking place**
- 4. Improve the quality of operations to continually meet customers and other stakeholders stated and implied needs**

Objectives of ISO 9000!

5. Achieve, maintain, and seek to continuously improve PRODUCT quality in relationship to Needs / Wants / Demands;

- **Stated**
- **Expected / Implied**
 - **Latent**

Benefits of ISO 9000!

- 1.Reduction in Waste,
Rework and Redundancy**
- 2.Reduction in potential for
External Failure**
- 3.Cost Reductions**
- 4.Increased Productivity**
- 5.Improved Performance**

Benefits of ISO 9000!

- 6. Improved Customer Satisfaction / Retention**
- 7. Reduction in On-site Customer (Second Party) Audits**
- 8. Marketing Edge**
- 9. Increased Workers Pride**

Benefits

- **20-30% lower manufacturing costs**
- **20-75% shorter lead times**
- **30-60% lower WIP inventory**
- **50% lower quality costs**
- **30-60% less floor space**
- **5-10% lower purchasing costs**
- **10-20% lower design costs**
- **10-20% fewer design changes**
- **Stock out-performance (4.4 to 1)**
- **20-50% less labor costs**

Basic Requirements

- **A documented quality system**
- **Consistent adherence to the documented quality system**
- **Management by facts**
- **Check to see that the plan was carried out**
- **Customers and processes be happy**

Quality @ ISO 9000(2K)

**Ability of a set of realized and
inherent CHARACTERISTICS
of a PRODUCT,
SYSTEM or PROCESS to fulfill
REQUIREMENTS of
CUSTOMERS and other
INTERESTED PARTIES**

TOTAL QUALITY MANAGEMENT (TQM)

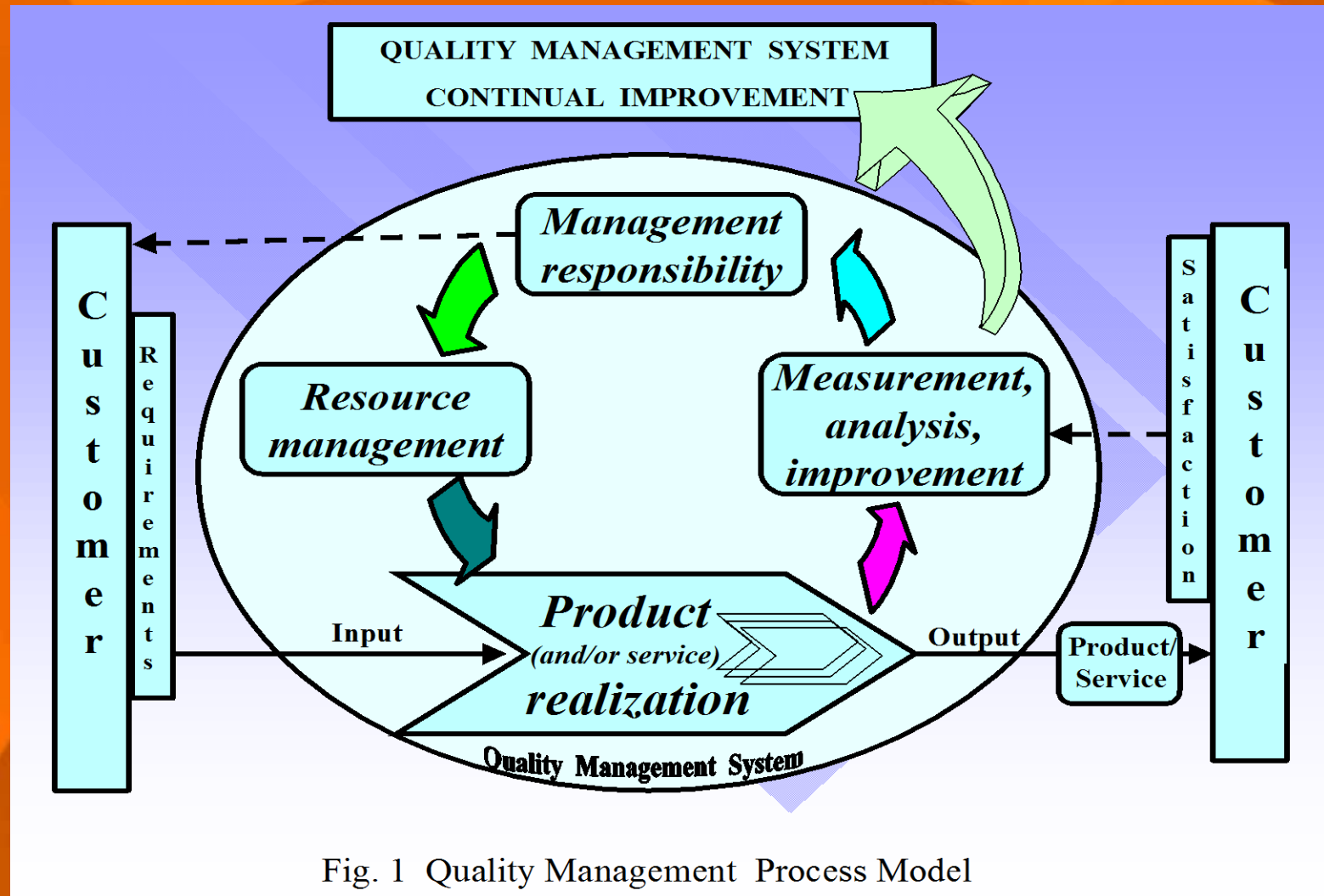
Session # 21

ISO 9000(2000)

Family of Standards

- **ISO 9000 – Quality Management Systems-Fundamentals & Vocabulary**
- **ISO 9001 – Quality Management Systems-Requirements**
- **ISO 9004 – Quality Management Systems-Guidelines for Performance**
- **ISO 19011 Quality Management Systems- Guidelines for Auditing**

ISO 9001(2000) QMS



ISO 9001:2000 Requirements

- 1. Processes to describe and monitor what each customer wants
(customer requirements drive inputs)**
- 2. Processes to measure and analyze customer satisfaction
(outputs drive customer satisfaction)**
- 3. Demonstration of continual improvement of internal process and customer satisfaction**

Management Structure of ISO 9001(2000)



ISO 9001 & 9004

“consistent pair”



- **Numbering structure of ISO 9001 & 9004 aligned.**
- **Both standards closely follows Plan-Do-Check-Act cycle.**
- **Make standard more understandable**

The “Consistent Pair”

ISO 9001 - remains a requirements standard focusing only on those processes necessary to give confidence that customer product's Quality requirements will be met

ISO 9004 - takes the Organization beyond limited scope and focuses on continual improvement to quality management Processes that lead towards models of Business Excellence

ISO 900X:2000 Quality Management Principles

- 1. Customer Focus**
- 2. Leadership**
- 3. Involvement of People**
- 4. Process Approach**
- 5. System Approach to Management**
- 6. Continual Improvement**
- 7. Factual Approach to Decision Making**
- 8. Mutually Beneficial Supplier Relationships**

QUALITY MANAGEMENT PRINCIPLES

PRINCIPLE 1 - CUSTOMER- FOCUSED ORGANISATION

Organisations depend on their customers, therefore should understand current and future customer needs, meet customer requirements and strive to exceed customer expectations.

QUALITY MANAGEMENT PRINCIPLES

PRINCIPLE 2 - LEADERSHIP

Leaders establish unity of purpose and direction of the organisation. They should create and maintain the internal environment in which people can become fully involved in achieving the organisation's objectives.

QUALITY MANAGEMENT PRINCIPLES

PRINCIPLE 3 - INVOLVEMENT OF PEOPLE

People at all levels are the essence of an organisation and their full involvement enables their abilities to be used for the organisation's benefit.

QUALITY MANAGEMENT PRINCIPLES

PRINCIPLE 4 - PROCESS APPROACH

**A desired result is achieved
more efficiently when
related resources and
activities are managed as a
process**

QUALITY MANAGEMENT PRINCIPLES

PRINCIPLE 5 - SYSTEM APPROACH TO MANAGEMENT

Identifying, understanding and managing a system of interrelated processes for a given objective improves the organization's effectiveness and efficiency

QUALITY MANAGEMENT PRINCIPLES

PRINCIPLE 6 - CONTINUAL IMPROVEMENT

**Continual improvement
should be a permanent
objective of the organisation**

QUALITY MANAGEMENT PRINCIPLES

PRINCIPLE 7 - FACTUAL APPROACH TO DECISION MAKING

**Effective decisions are based
on the analysis of data and
information**

QUALITY MANAGEMENT PRINCIPLES

PRINCIPLE 8 - MUTUAL BENEFICIAL SUPPLIER RELATIONSHIPS

**An organization and its
suppliers are interdependent,
and a mutually beneficial
relationship enhances the
ability of both to create value**

ISO 9001(2000) QMS

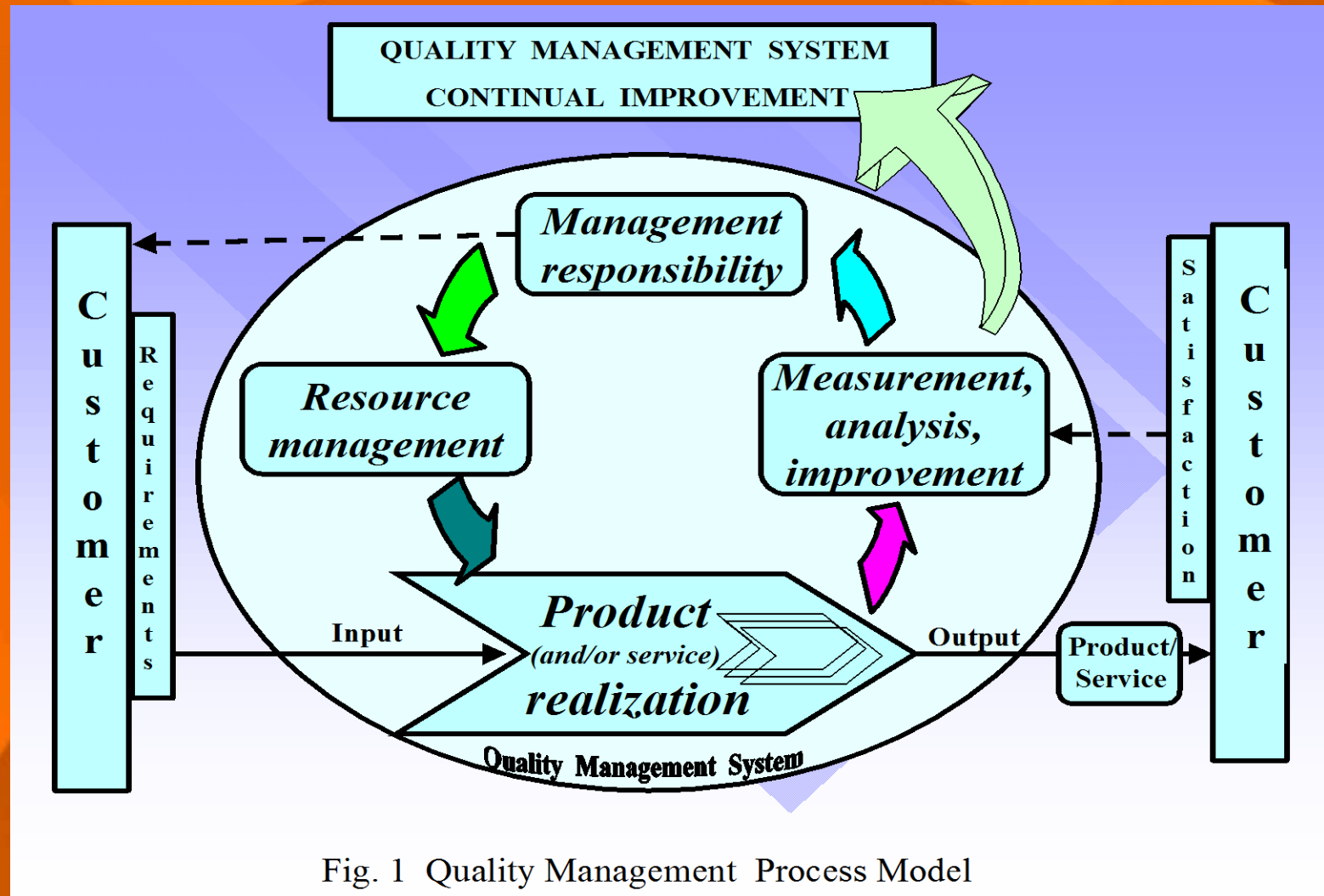


Fig. 1 Quality Management Process Model

Management Structure of ISO 9001(2000)



ISO 9001(2000) QMS

ISO QM System must be documented ;

Therefore

Think , what you want to say

Say , what you want to Do

Do, what you have Said

Record, what you have Done

Prove, that you Did what you Said

Reflect and Refine your Thinking

So you continually improve in

Thinking, Saying and Doing

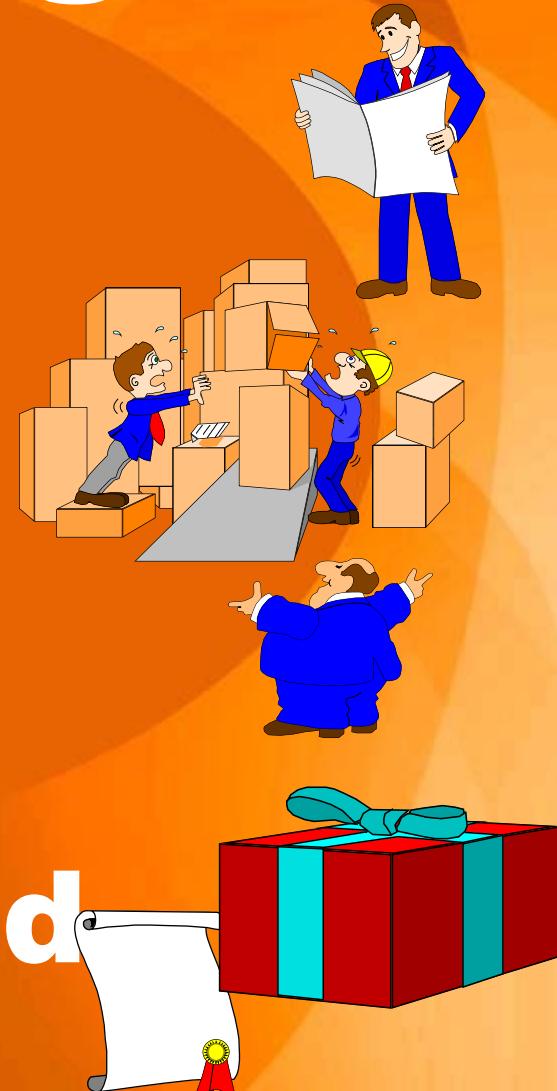
Documenting Procedures

- Say what you do

- Do what you say

and where necessary

- Show what you did



TOTAL QUALITY MANAGEMENT (TQM)

Session # 22

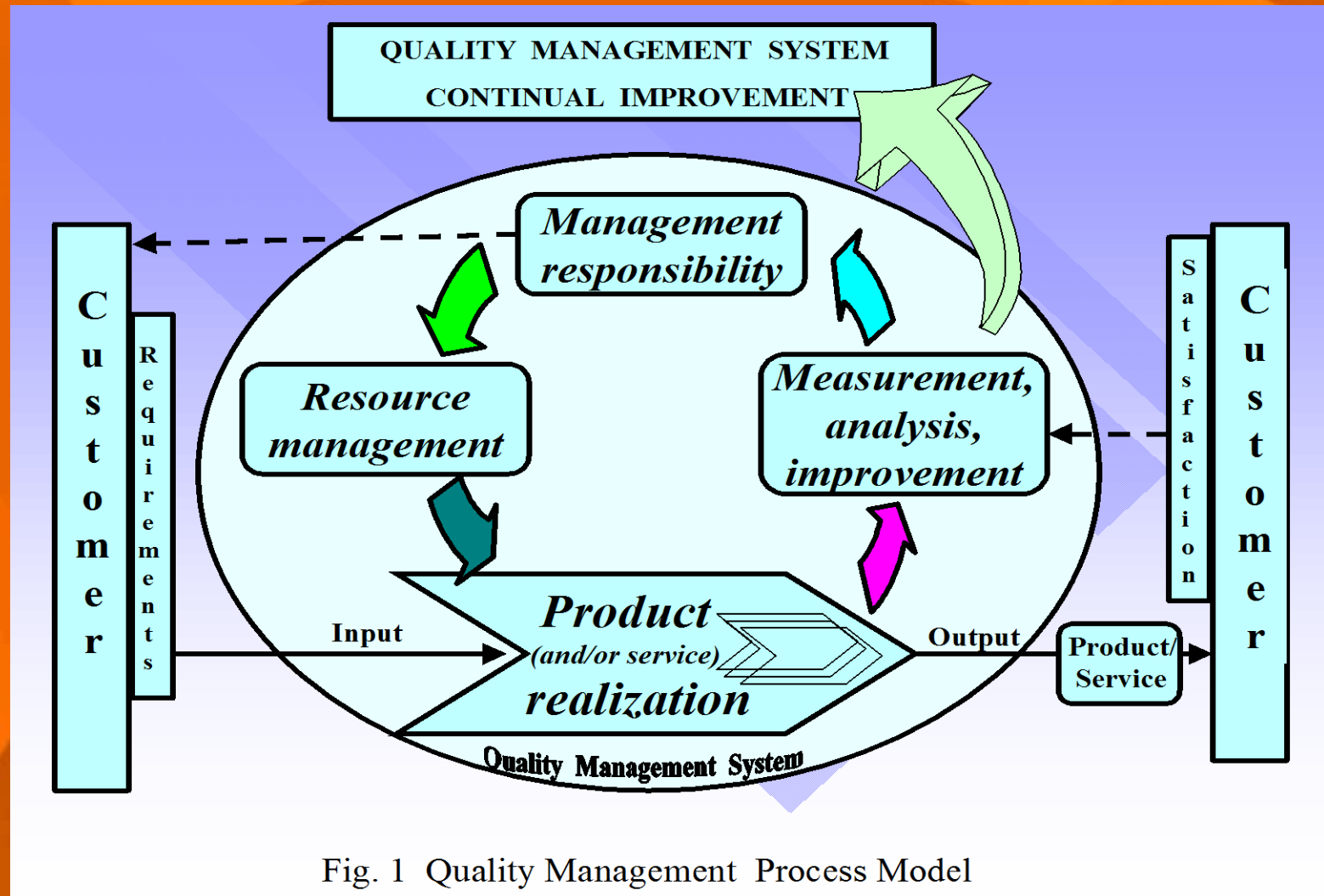
ISO-9001, 9004(2K) and SEI-CMM

**Performance Maturity Level
through
Self Assessment Approach
of
ISO-9004(2000)**

ISO-9001, 9004(2K) and SEI-CMM

Level # 1	No Systemic/Formal Approach
Level # 2	Reactive Approach
Level # 3	Stable/Formal Processes Approach
Level # 4	Continual Improvement ..Must
Level # 5	Best in Class

ISO 9001(2000) QMS



Management Structure of ISO 9001(2000)



4■ Quality Management System

4.1 General Requirements

- Ensure availability of resources and information.**
- Monitor ,measure and analyze the processes**
- Implement actions necessary to achieve planned results and continual improvement of these processes.**

4.2 Document Requirements

4.2.1 General

- **Documented Q Policy and Q Objectives Manual**
- **Documented Q Processes, Procedures and Instruction Manuals**
- **Documents to ensure effective planning, operation and control of its processes**
- **Records**

4.2.3 Control of Documents

1. Documents shall be controlled

2. A documented procedure shall be established to define the controls needed ...

- to approve documents for adequacy prior to issue.**
- to review and update, and re-approve documents.**

Control of Documents

(Contd..)

3. Changes and revisions are identified.

4. **Availability of documents at all points of use.**

5. Documents of external origin are identified and their distribution controlled.

6. Suitable identification of obsolete documents and unintended usage.

4.2.4 Control of Records

**Records shall be established
and maintained**

- **Documented procedures
for identification, storage,
retrieval, protection,
retention time and
disposition of quality
records.**

ISO 9000 QMS Documentation

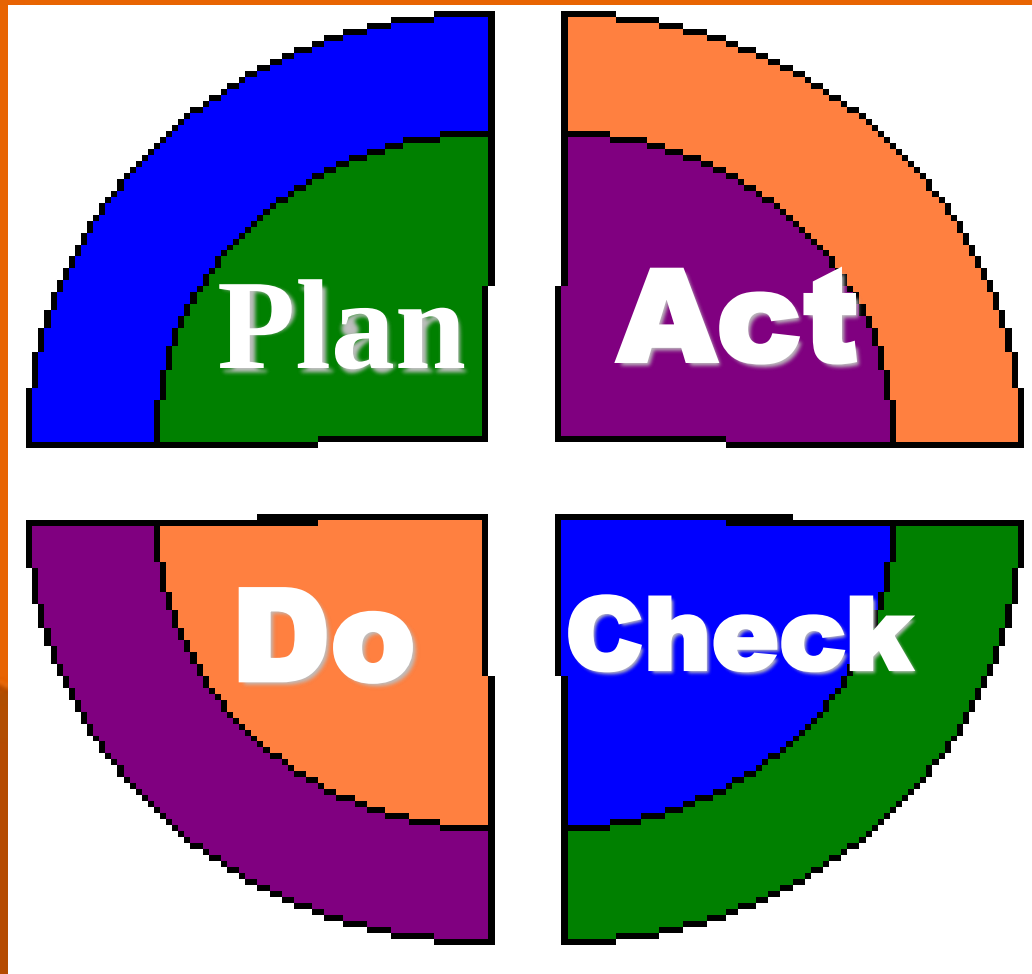
**Quality
Policy
Manual**

**Quality
Processes
Manual**

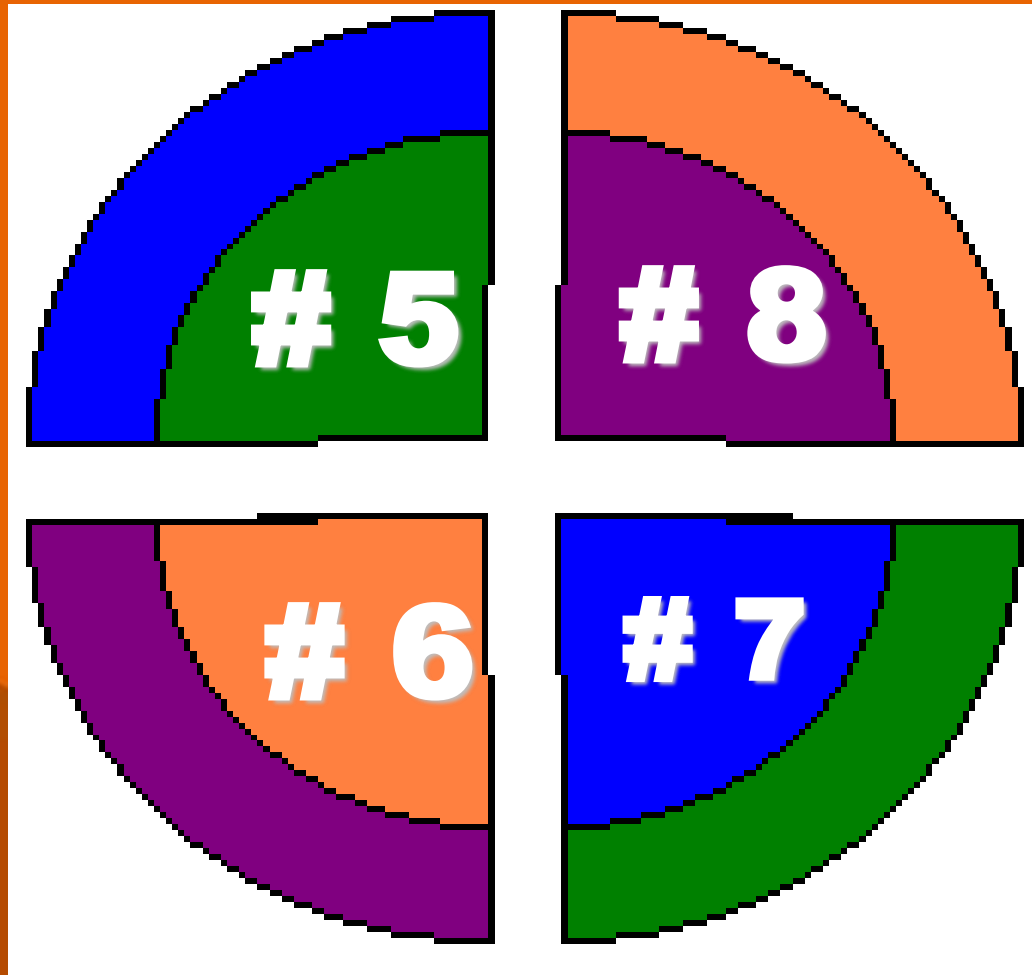
**Work
Instru-
ctions**

**Forms
for
Records**

Deming Cycle and ISO-9001(2000) QMS



Deming Cycle and ISO-9001(2000) QMS



ISO 9001(2000) QMS

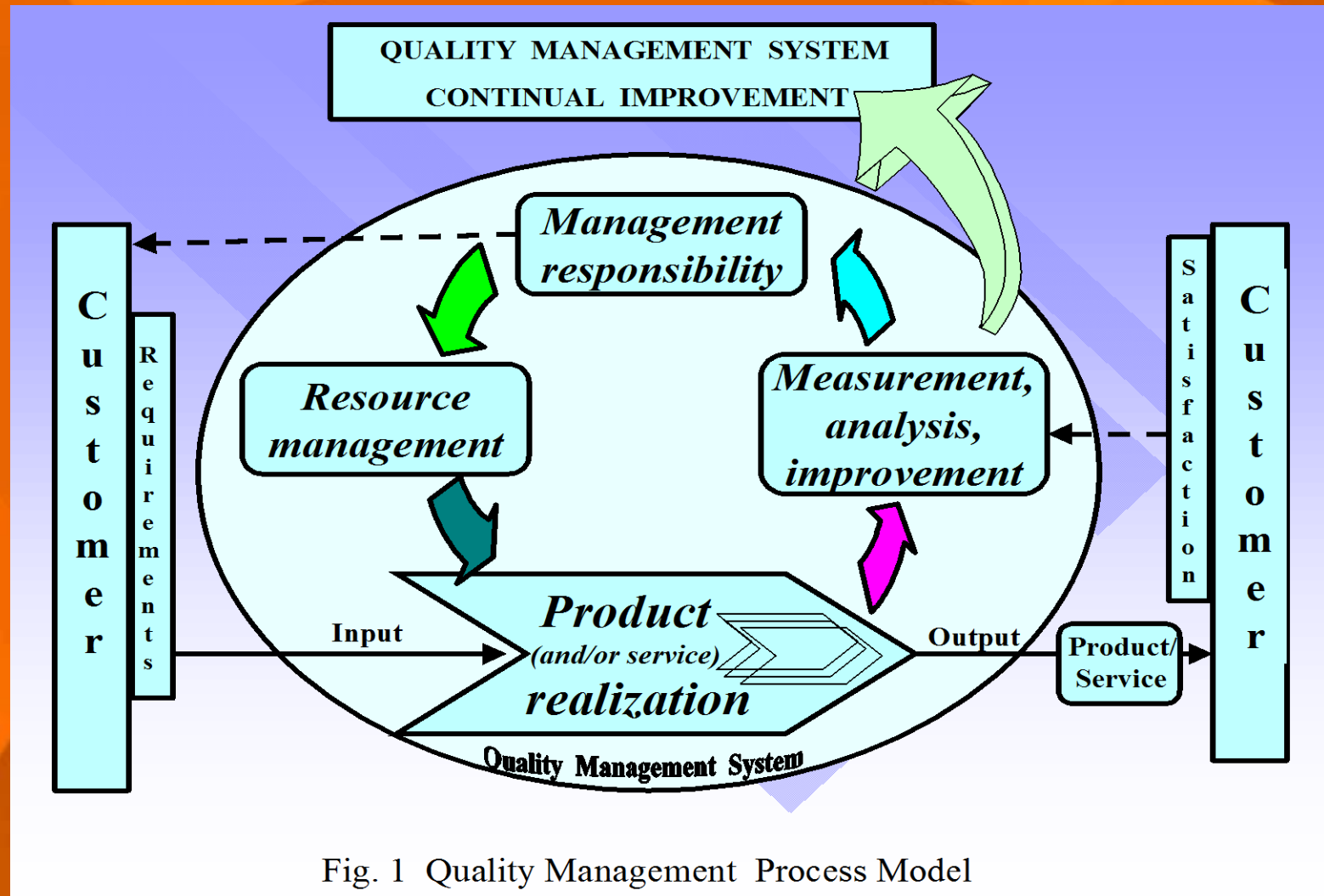


Fig. 1 Quality Management Process Model

Management Structure of ISO 9001(2000)



Total Quality Management – MGT510

<http://www.vustudents.net>

Note: Choices with marked * are verified answers

~~Quiz no.1~~

Question # 1 of 15 (Start time: 11:02:05 PM) Total Marks: 1

In total quality perspective, quality planning, and strategic business planning are treated as which of the following?

Different

***Consistent**

Indistinguishable

Dissimilar

Question # 2 of 15 (Start time: 11:03:13 PM) Total Marks: 1

The marketing team of a company carried out a survey concerning expectations of the customers from the product. What should be the product development department of the company design?

***The product meeting requirements of the customers**

The product meeting requirements of the distributors

The product meeting requirements of the management

The product meeting requirements of the shareholders

Question # 3 of 15 (Start time: 11:04:29 PM) Total Marks: 1

Which of the following results in low costs?

***High productivity and high capacity utilization**

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

Question # 4 of 15 (Start time: 11:05:35 PM) Total Marks: 1

Which of the following are generally easy to detect using statistical methods, and it is usually economical to remove them?

***Common causes**

Special causes

Implicit causes

Assignable causes

Question # 5 of 15 (Start time: 11:07:10 PM) Total Marks: 1

Which one of the following depicts feature, which is a dimension of quality?

Product looks and sounds

Brand name and price

Bells and whistles of the product

***Ability to repair a product**

Question # 6 of 15 (Start time: 11:08:37 PM) Total Marks: 1

During which century, the manufacturers began to include quality processes in quality practices?

17th century

18th century

19th century

***20th century**

Question # 7 of 15 (Start time: 11:09:47 PM) Total Marks: 1

Which of the following models has the viewpoint that change and learning are valued in themselves?

***Organism model**

Mechanistic model

Cultural model

Total Quality model

Question # 8 of 15 (Start time: 11:11:19 PM) Total Marks: 1

After the war, Deming was invited to which of the following countries?

Germany

***Japan**

France

China

Question # 9 of 15 (Start time: 11:12:46 PM) Total Marks: 1

Which of the following statements is TRUE for the organizations focusing problem-solving and continuous improvement?

Fail to emphasize error prevention and design quality

***Fail to focus on what is truly important to the customer**

Fail to identify the best times about the situation in the past

Fail to integrate all organizational functions

Question # 10 of 15 (Start time: 11:14:06 PM) Total Marks: 1

Quality applies to which aspect(s) of the organization?

Products only

Products and services

People and product

***Products, services, and people**

Question # 11 of 15 (Start time: 11:14:38 PM) Total Marks: 1

Which of the following statements holds true for quality assurance?

Top management's intention regarding quality

Functions determining implementation of the quality policy

***Actions to provide confidence of satisfying quality requirements**

Responsibilities and processes, which implement quality management

Question # 12 of 15 (Start time: 11:16:08 PM)

Total Marks: 1

Which of the following models reflect horizontal processes beginning with suppliers and ending with customers?

Organism model

Mechanistic model

Cultural model

***Total Quality model**

Question # 13 of 15 (Start time: 11:17:14 PM)

Total Marks: 1

Which of the following is concerned about quality for achievement of TQM?

The Managing Director

The Operative

The Quality Manager

***Everyone in the organization**

Question # 14 of 15 (Start time: 11:18:39 PM)

Total Marks: 1

Which one of the following depicts durability – a dimension of quality?

Exterior finish

Quality of work

***Useful life**

Ease of repair

Question # 15 of 15 (Start time: 11:19:36 PM) Total Marks: 1

"Not giving respect to other people's contributions", depicts which of the following self-oriented behavior?

Diverting

Excluding

***Belittling**

Blocking

Quiz No .2

which century, the manufacturers began to include quality processes in quality practices?

17th century

18th century

19th century

***20th century**

In the new paradigm, managers view error as which one of the following?

A threat

A substitute

***An opportunity**

An end

Whose concepts are referred to as statistical quality control (SQC)?

***Shewhart's**

Deming's

Juran's

Crosby's

In which of the following operations great deal of variations can occur?

***Manufacturing**

Distribution

Purchasing

Selling

Which of the following are generally easy to detect using statistical methods, and it is usually economical to remove them?

***Common causes**

Special causes

Implicit causes

Assignable causes

Who argued that beauty could be understood only after exposure to a series of objects that display its characteristics?

Drucker

Crosby

Deming

***Plato**

During which war, Deming taught quality control courses as part of the national defense efforts?

World War I

***World War II**

Mexican-American War

The Civil War

Which one of the following depicts feature, which is a dimension of quality?

Product looks and sounds

Brand name and price

Bells and whistles of the product

***Ability to repair a product**

The majority of advertisers appeal the public on the basis of which of the following?

***Quality of product**

Quality of staff

Inferiority of product

Inferiority of service

The standards for output based on quality and customer-service requirements that originate at the organizational and process levels include requirements for all of the following EXCEPT:

Accuracy

Innovation

Timeliness

Turnover

"Not giving respect to other people's contributions", depicts which of the following self-oriented behavior?

Diverting

Excluding

***Belittling**

Blocking

After the war, Deming was invited to which of the following countries?

Select correct option:

Germany

***Japan**

France

China

Which of the following results in low costs?

***High productivity and high capacity utilization**

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

The customer value strategy may be addressed in which of the following topics?

Projects

Products

*** Services**

Measurement

Who developed Quality Function Deployment (QFD)?

Frederick Taylor

Joseph M. Juran

***Y. Akao**

Armand Feigenbaum

TQM Quiz no 3

<http://www.vustudents.net>

Question # 1 of 15 (Start time: 12:44:29 PM) Total Marks: 1

How the Baldrige Award recognizes the importance of total quality principles?

Through independence with overall business planning

Through a no relation with overall business planning

***Through integration with overall business planning**

Through dependence with overall business planning

Question # 2 of 15 (Start time: 12:45:33 PM) Total Marks: 1

Which of the following models considers the goal of meeting individual needs?

Organism model

Mechanistic model

***Cultural model**

Total Quality model

Question # 3 of 15 (Start time: 12:46:26 PM) Total Marks: 1

All of the following are the initial considerations in the Hoshin Kanri approach to business system change EXCEPT:

Measuring the business system as a whole

Setting core objectives of the business

Understanding the environmental situation in which the business operates

***Ceasing resources for performing activities to achieve business objectives**

Question # 4 of 15 (Start time: 12:47:48 PM) Total Marks: 1

Who argued that beauty could be understood only after exposure to a series of objects that display its characteristics?

Drucker

Crosby

Deming

***Plato**

Question # 5 of 15 (Start time: 12:48:17 PM) Total Marks: 1

Which of the following statements is TRUE for ISO 9000?

***Describes the principles of a quality management system and defines the terminology**

Describes the requirements relative to a quality management system either for internal use or for contractual or certification purposes

Intended for internal use and not for contractual purposes, focuses particularly on continually improving performance

Contains the guidelines for auditing quality management and/or environmental management systems

Question # 6 of 15 (Start time: 12:49:39 PM) Total Marks: 1

Which of the following statements holds true for quality assurance?

Top management's intention regarding quality

Functions determining implementation of the quality policy

***Actions to provide confidence of satisfying quality requirements**

Responsibilities and processes, which implement quality management

Question # 7 of 15 (Start time: 12:51:05 PM) Total Marks: 1

Which of the following is an answer to the Second Absolute of Quality Management as summarized by Crosby; what system is needed to cause quality?

Conformance to requirements

***Prevention**

Zero defects

Price of nonconformance

Question # 8 of 15 (Start time: 12:52:13 PM) Total Marks: 1

All of the following are the reasons why Total Quality Management programs are not prolific for many companies, EXCEPT:

TQM focuses on minimum standards

TQM develops its own cumbersome bureaucracy

TQM has no place for love and passion

***TQM focuses on customers and stakeholders**

Question # 9 of 15 (Start time: 12:53:15 PM) Total Marks: 1

What is the other name used for Deming Cycle?

***Shewhart cycle**

Product life cycle

Data life cycle management

Information life cycle management

Question # 10 of 15 (Start time: 12:54:15 PM) Total Marks: 1

What is necessary for a system to be effective?

***The components of any system must work together for the system**

The components of any system must not work together for the system

The components of any system should not be organized for the system

The components of any system should provide disruption for the system

Question # 11 of 15 (Start time: 12:55:08 PM) Total Marks: 1

Which of the following is NOT the criterion under Canadian Awards for Business Excellence?

Leadership

***Planning for contraction**

Process optimization

Supplier focus

Question # 12 of 15 (Start time: 12:56:32 PM) Total Marks: 1

Which of the following is true for subsystems?

These are not linked together as internal customers and suppliers

These are linked together as external customers and suppliers

These are linked together as external and internal customers and suppliers

***These are linked together as internal customers and suppliers**

Question # 13 of 15 (Start time: 12:58:00 PM) Total Marks: 1

What is the relationship between Ishikawa and Deming?

Son and father

Nephew and uncle

***Student and teacher**

Grandson and grandfather

Question # 14 of 15 (Start time: 12:58:51 PM) Total Marks: 1

Which of the following statements is TRUE for little q?

***Striving for quality in a limited or specific area**

Searching for specific lines of text containing a particular pattern

Matching rules encoded in a string according to certain syntax rules

Giving combination of values for the free variables

Question # 15 of 15 (Start time: 01:00:14 PM) Total Marks: 1

The Capability Maturity Model can help in all of the following EXCEPT:

Setting process improvement goals and priorities

Providing guidance for quality processes

Providing a yardstick for assessing current practices

Documenting procedures as to how the various processes for design operate

Quiz no ---3

Question # 1 of 15 (Start time: 11:56:02 PM) Total Marks: 1

The customer value strategy may be addressed in which of the following topics?

Select correct option:

Projects

Products

Services

***Measurement**

Question # 2 of 15 (Start time: 11:57:11 PM) Total Marks: 1

Who is known as a father of modern-day statistical quality control?

Select correct option:

***Shewhart**

Deming

Juran

Crosby

Question # 3 of 15 (Start time: 11:57:40 PM) Total Marks: 1

Which of the following forms the basis of TQM as an organization wide process?

Select correct option:

Not using the resources of the total organization properly

Organizational flexibility and response to change

Not bound in business relationships

Not measuring continuous improvement

Question # 4 of 15 (Start time: 11:58:54 PM) Total Marks: 1

How should the benchmarking of individual processes as well as customer satisfaction be employed?

Select correct option:

As a standard tool

As a defective tool

***As an improvement tool**

As an instrumental tool

Question # 5 of 15 (Start time: 11:59:39 PM) Total Marks: 1

Identify the example of prevention costs.

Select correct option:

***Quality planning**

Re-inspection

Product recalls

Customer returns

Question # 6 of 15 (Start time: 12:00:51 AM) Total Marks: 1

Which of the following is NOT included in the ISO registration process?

Select correct option:

Document review by the registrar of the quality system documents or quality manual

Pre-assessment, which identifies potential noncompliance in the quality system or in the documentation

***Review the program descriptions and select the program(s) that will enable you to contribute more to your organization**

Assessment by a team of two or three auditors of the quality system and its documentation

Question # 7 of 15 (Start time: 12:02:13 AM) Total Marks: 1

Who made some salient observations about the failure to understand the components of Profound Knowledge?

Select correct option:

***Scholtes**

Deming

Juran

Crosby

Question # 8 of 15 (Start time: 12:02:56 AM) Total Marks: 1

Which of the following results in low costs?

Select correct option:

***High productivity and high capacity utilization**

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

Question # 9 of 15 (Start time: 12:03:47 AM) Total Marks: 1

How can the quality be computed?

Select correct option:

Quality = Expectation/Performance

Quality = Performance/Expectation

Quality = Performance+Expectation

Quality = Performance-Expectation

Question # 10 of 15 (Start time: 12:05:14 AM) Total Marks: 1

Documentation and data on existing products, are an example which of the following?

Select correct option:

External inputs

***Internal inputs**

Either internal or external inputs

Neither internal nor external inputs

Question # 11 of 15 (Start time: 12:06:19 AM) Total Marks: 1

Which of the following focuses on results, not process, and encourages short-term behavior?

Select correct option:

***Typical American MBO system**

Typical Japanese MBO system

Typical Chinese MBO system

Uncommon Chinese MBO system

Question # 12 of 15 (Start time: 12:07:08 AM) Total Marks: 1

Who developed Quality Function Deployment (QFD)?

Select correct option:

Frederick Taylor

Joseph M. Juran

***Y. Akao**

Armand Feigenbaum

Question # 13 of 15 (Start time: 12:08:06 AM) Total Marks: 1

All of the following are the examples of appraisal costs EXCEPT:

Select correct option:

<http://www.vustudents.net>

Quality measurement criteria

Quality improvement projects

Laboratory acceptance testing

Inspection and testing

Question # 14 of 15 (Start time: 12:09:33 AM) Total Marks: 1

In order to achieve the objectives and stimulate innovation, what the organization should encourage?

Select correct option:

The involvement of its human resource

The involvement of its process

The involvement of its system

The involvement of its capital

Question # 15 of 15 (Start time: 12:10:49 AM) Total Marks: 1

How the requirements for documentation and records may not arise?

Select correct option:

Contractual requirements from the customer or other interested parties

Acceptance of international, national, regional and industry sector standards

***Irrelevant statutory and regulatory requirements**

Decisions by the organization

Quiz no 3

Question # 1 of 15 (Start time: 12:18:45 AM) Total Marks: 1

At which level, does the quality concern centers on meeting internal customer requirements?

Select correct option:

Organizational level

Performer level

Process level

Production level

Question # 2 of 15 (Start time: 12:20:12 AM) Total Marks: 1

Who is known as a father of modern-day statistical quality control?

Select correct option:

***Shewhart**

Deming

Juran

Crosby

Question # 3 of 15 (Start time: 12:20:43 AM) Total Marks: 1

Which of the following is TRUE for higher costs of quality?

Select correct option:

Doing the wrong thing in the wrong way

Doing the wrong thing in the right way

Doing the right thing in the wrong way

Doing the right thing in the right way

Question # 4 of 15 (Start time: 12:22:09 AM) Total Marks: 1

Which of the following concepts and tools can be used to assess internal customer needs and to communicate internal supplier capabilities?

Select correct option:

Financial

Management

***Marketing**

Production

Question # 5 of 15 (Start time: 12:23:02 AM) Total Marks: 1

The management representatives appointed and given authority by the senior management are for all of the following EXCEPT:

Select correct option:

Managing

Mishandling

Evaluating

Coordinating

Question # 6 of 15 (Start time: 12:24:20 AM) Total Marks: 1

The management preserves the conformity of products during internal processing and final delivery. Which of the following is included in such preservation?

Select correct option:

Handling

Abolition

Risk

Shrinking

Question # 7 of 15 (Start time: 12:25:36 AM) Total Marks: 1

All of the following are the basic requirements of ISO 9000 EXCEPT:

Select correct option:

A documented quality system

Consistent adherence to the documented quality system

Check to see that the plan was carried out

Customers and processes be displeased

Question # 8 of 15 (Start time: 12:26:50 AM) Total Marks: 1

During 1990s, the following institution(s) began to pay increase attention to quality.

Select correct option:

Education and health care only

***Education, health care, and government**

Health care and government only

Government and financial institutions only

Question # 9 of 15 (Start time: 12:27:47 AM) Total Marks: 1

Which one of the following depicts reliability – a dimension of quality?

Select correct option:

Bells and whistles of the product

Meeting specifications

Average time for the unit to fail

Quality of work

Question # 10 of 15 (Start time: 12:29:07 AM) Total Marks: 1

Which of the following models considers the goal of organizational efficiency and performance?

Select correct option:

Organism model

***Mechanistic model**

Cultural model

Total Quality model

Question # 11 of 15 (Start time: 12:29:56 AM) Total Marks: 1

All of the following are the examples of external failure costs EXCEPT:

Select correct option:

Processing customer complaints

Customer returns

Warranty claims

***Quality planning**

Question # 12 of 15 (Start time: 12:30:34 AM) Total Marks: 1

Identify the example of external failure costs.

Select correct option:

Quality planning

Re-inspection

Material review

***Customer returns**

Question # 13 of 15 (Start time: 12:31:35 AM) Total Marks: 1

All of the following are the grounds to give birth to total quality approach EXCEPT:

Select correct option:

Obsession with quantity

Confusion and poor leadership

One employee in charge

Political disorder

Question # 14 of 15 (Start time: 12:32:59 AM) Total Marks: 1

Which of the following is viewed as a Consistent Pair?

Select correct option:

ISO 9000 and ISO 9004

ISO 1987 and ISO 1992

ISO 1992 and ISO 1997

***ISO 9001 and ISO 9004**

Question # 15 of 15 (Start time: 12:33:40 AM) Total Marks: 1

In which century, the US broke further from the European tradition and adopted a new management approach developed by Frederick W. Taylor?

Select correct option:

17th century

18th century

***19th century**

20th century

Next Quiz

1. All of the following are the grounds to give birth to total quality approach EXCEPT:
 - A. Obsession with quantity
 - B. Confusion and poor leadership
 - C. One employee in charge
 - D. Political disorder**

2. Which of the following was developed by Motorola to improve its processes by minimizing defects?
 - A. ISO 9000
 - B. *Six sigma**
 - C. QS 9000
 - D. TQM

3. Quality applies to which aspect(s) of the organization?
 - A. Products only
 - B. Products and services
 - C. People and product
 - D. *Products, services, and people**

4. Which of the following statements is TRUE about Deming's philosophy?
 - A. *It is based on improving products and services by reducing uncertainty and variability in the design and manufacturing processes**
 - B. Quality is either or not present in the whole organization; that quality is the responsibility of everyone in the organization
 - C. Increasing conformance to specifications through elimination of defects, supported extensively by statistical tools for analysis
 - D. Increasing loss, for the producer, the customer, and society, associated with increasing variability from a target value

5. Which of the following statements, in relation to system, is TRUE for people failing to understand the components of Profound Knowledge?
 - A. *They see the symptoms but not the deep causes of problems**
 - B. They do not know when expectations are realistic
 - C. They do not understand motivation/why people do what they do
 - D. They do not know how to plan and accomplish learning and improvement

6. "Problems will remain unsolved, despite their best efforts." When this will happen?
 - A. When people don't understand systems
 - B. When people don't understand psychology
 - C. When people don't understand variation

D. *When people don't understand the theory of knowledge

7. Which stage in the Shewhart Cycle implements the change on a small scale?

A. *Do stage

B. Act stage

C. Plan stage

D. Check stage

8. Which of the following is determining what to control, establishing units of measurement so that data may be objectively evaluated?

A. Quality improvement

B. Quality planning

C. *Quality control

D. Quality process

9. All of the following steps are required for quality planning EXCEPT:

A. Determine who the customers are

B. Deploy the plans to operational levels

C. Identify customers' needs

D. *Assess actual quality performance

10. Which of the following is an answer to the First Absolute of Quality Management

as summarized by Crosby; what is quality?

A. *Conformance to requirements

B. Prevention

C. Zero defects

D. Price of nonconformance

Next Quiz:

Question # 1 of 15 (Start time: 09:54:15 PM) Total Marks: 1

Quality applies to which aspect(s) of the organization?

Select correct option:

Products only

Products and services

People and product

Products, services, and people

Question # 2 of 15 (Start time: 09:55:01 PM) Total Marks: 1

Juran defines quality as “fitness for use”. This can be broken down into how many categories?

Select correct option:

Five

***Four**

Three

Two

Question # 3 of 15 (Start time: 09:56:24 PM) Total Marks: 1

How many maturity levels are there in the Capability Maturity Model?

Select correct option:

Four

***Five**

Six

Seven

Question # 4 of 15 (Start time: 09:57:11 PM) Total Marks: 1

Which of the following is a unit for measuring process performance according to traditional view of quality?

Select correct option:

Effective parts per thousand produced

Defective parts per thousand produced

Effective parts per hundred produced

***Defective parts per hundred produced**

Question # 5 of 15 (Start time: 09:58:27 PM) Total Marks: 1

Which of the following statements, in relation to system, is TRUE for people failing to understand the components of Profound Knowledge?

Select correct option:

***They see the symptoms but not the deep causes of problems**

They do not know when expectations are realistic

They do not understand motivation/why people do what they do

They do not know how to plan and accomplish learning and improvement

Question # 6 of 15 (Start time: 09:59:10 PM) Total Marks: 1

Which of the following statements, in relation to variation, is TRUE for people failing to understand the components of Profound Knowledge?

Select correct option:

<http://www.vustudents.net>

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

***They do not know the difference between prediction, forecasting, and guesswork**

They create cynicism, demoralization, de-motivation, guilt, resentment, burnout, craziness, and turnover

They do not understand the difference between improvement and change

Question # 7 of 15 (Start time: 10:00:18 PM) Total Marks: 1

Which of the following models has a vertical chain of command?

Select correct option:

Organism model

***Mechanistic model**

Cultural model

Total Quality model

Question # 8 of 15 (Start time: 10:01:19 PM) Total Marks: 1

What was the role of quality professionals during 1800s?

Select correct option:

Troubleshooting and the application of statistical methods

***Inspection, sorting, counting, and grading**

Quality measurement, quality planning, and program design

Goal setting, education and training, and program design

Question # 9 of 15 (Start time: 10:02:25 PM) Total Marks: 1

All of the following are the major categories and items within each category of Canadian Awards for Business Excellence EXCEPT:

Select correct option:

Customer focus

Distributor focus

People focus

Supplier focus

Question # 10 of 15 (Start time: 10:03:44 PM) Total Marks: 1

How the Baldrige Award recognizes the importance of total quality principles?

Select correct option:

Through independence with overall business planning

Through a no relation with overall business planning

***Through integration with overall business planning**

Through dependence with overall business planning

Question # 11 of 15 (Start time: 10:04:16 PM) Total Marks: 1

Which of the following involves determining what to control, establishing units of measurement so that data may be objectively evaluated?

Select correct option:

Quality improvement

Quality planning

***Quality control**

Quality process

Question # 12 of 15 (Start time: 10:04:55 PM) Total Marks: 1

After the war, Deming was invited to which of the following countries?

Select correct option:

Germany

***Japan**

France

China

Question # 13 of 15 (Start time: 10:05:25 PM) Total Marks: 1

Which of the following organizations has a little chance of survival?

Select correct option:

With no corporate philosophy about how to create long-term customer loyalty

With no corporate objectives about how to create long-term customer loyalty

With no mission about how to create long-term customer loyalty

***With no vision about how to create long-term customer loyalty**

Question # 14 of 15 (Start time: 10:06:47 PM) Total Marks: 1

Who suggested that there are three categories of performance?

Select correct option:

Taguchi

Taylor

Ishikawa

***Deming**

Question # 15 of 15 (Start time: 10:07:19 PM) Total Marks: 1

Which of the following statements holds true for quality assurance?

Select correct option:

Top management's intention regarding quality

Functions determining implementation of the quality policy

***Actions to provide confidence of satisfying quality requirements**

Responsibilities and processes, which implement quality management

Quiz No. 3 (Talha)

Question # 1 of 15 (Start time: 11:43:33 AM) Total Marks: 1

Which of the following statements is TRUE about Deming's philosophy?

Select correct option:

***It is based on improving products and services by reducing uncertainty and variability in the design and manufacturing processes**

Quality is either or not present in the whole organization; that quality is the responsibility of everyone in the organization

Increasing conformance to specifications through elimination of defects, supported extensively by statistical tools for analysis

Increasing loss, for the producer, the customer, and society, associated with increasing variability from a target value

Question # 2 of 15 (Start time: 11:44:22 AM) Total Marks: 1

All of the following are the major decision areas in supply chain management EXCEPT:

Select correct option:

Location

***Production**

Inventory

Pricing

Question # 3 of 15 (Start time: 12:11:54 PM) Total Marks: 1

By the mid 1970s, which markets were penetrated by the Japanese companies?

Select correct option:

***Western markets**

Eastern markets

Northern markets

Southern markets

Question # 4 of 15 (Start time: 12:12:33 PM) Total Marks: 1

The input for management review to examine the quality management system shall include information about all of the following EXCEPT:

Select correct option:

Status of preventive and corrective actions

Follow-up actions from previous management reviews

Changes that could affect the quality management system

***Improvement of product related to customer requirements**

Question # 5 of 15 (Start time: 12:13:47 PM) Total Marks: 1

Juran advocates a never-ending spiral of activities that includes all of the following EXCEPT:

Select correct option:

Product development

***Workers**

Inspection and testing

Purchasing

Question # 6 of 15 (Start time: 12:14:42 PM) Total Marks: 1

Which of the following statements is TRUE for little q?

Select correct option:

***Striving for quality in a limited or specific area**

Searching for specific lines of text containing a particular pattern

Matching rules encoded in a string according to certain syntax rules

Giving combination of values for the free variables

Question # 7 of 15 (Start time: 12:15:19 PM) Total Marks: 1

Which of the following is the Total Quality Tool that is simply a graphical representation of an outline that presents a chain of causes and effects?

Select correct option:

Check sheet

Fishbone diagram

Scatter diagram

Pareto chart

Question # 8 of 15 (Start time: 12:16:32 PM) Total Marks: 1

Who found that teams are seldom fully used, and their individual members are often complacent?

Select correct option:

Oakland and Schein

***Newall and Dale**

Lehr and Cox

Juran and Schein

Question # 9 of 15 (Start time: 12:17:14 PM) Total Marks: 1

The customer value strategy may be addressed in which of the following topics?

Select correct option:

Projects

Products

Services

***Measurement**

Question # 10 of 15 (Start time: 12:18:10 PM) Total Marks: 1

Which of the following should be reviewed by the organization to identify requirements for operations that realize products or deliver services in order to ensure compliance with specifications?

Select correct option:

Inability to comply with contractual requirements

***Training and competence of people**

Problem conception

Irrelevant statutory and regulatory requirements

Question # 11 of 15 (Start time: 12:19:19 PM) Total Marks: 1

Who developed Quality Function Deployment (QFD)?

Select correct option:

Frederick Taylor

Joseph M. Juran

***Y. Akao**

Armand Feigenbaum

Question # 12 of 15 (Start time: 12:20:15 PM) Total Marks: 1

What are the ISO (International Organization for Standardization) series standards for industries such as automobiles?

Select correct option:

ISO 9000

ISO 14000

***QS 9000**

AS 9000

Question # 13 of 15 (Start time: 12:21:43 PM) Total Marks: 1

Which of the following should be used as input to management review in order to ensure that continual improvement is the driver for organizational development?

Select correct option:

***Performance information**

Performance management

Ethical performance

Performance development

Question # 14 of 15 (Start time: 12:23:02 PM) Total Marks: 1

Which of the following are generally easy to detect using statistical methods, and it is usually economical to remove them?

***Common causes**

Special causes

Implicit causes

Assignable causes

Question # 15 of 15 (Start time: 12:24:27 PM) Total Marks: 1

What is the objective of a TQM system?

Select correct option:

***Continuous improvement**

Continual improvement

Business improvement

Process improvement

Next Quiz

Question # 1 of 15 (Start time: 09:21:59 PM) Total Marks: 1

Based in Geneva, Switzerland, International Organization for Standardization is a consortium of the national standard bodies, which has how many countries?

More than 100

Less than 100

Less than 90

Less than 80

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Question # 2 of 15 (Start time: 09:23:27 PM) Total Marks: 1

Which of the following is the contributor to the work method cause in the house paint peeling cause-and-effect diagram?

Select correct option:

Cheap solvents

Temperature

Acid rain

Paint over dirt

Question # 3 of 15 (Start time: 09:24:28 PM) Total Marks: 1

Which of the following concepts and tools can be used to assess internal customer needs and to communicate internal supplier capabilities?

Select correct option:

Financial

Management

***Marketing**

Production

Question # 4 of 15 (Start time: 09:25:27 PM) Total Marks: 1

The input for management review to examine the quality management system shall include information about all of the following EXCEPT:

Select correct option:

Status of preventive and corrective actions

Follow-up actions from previous management reviews

Changes that could affect the quality management system

***Improvement of product related to customer requirements**

Question # 5 of 15 (Start time: 09:26:28 PM) Total Marks: 1

What the organization must accomplish to achieve the mission, by examination and categorization of the impacts?

Select correct option:

Critical success factors

Key process indicators

Illegal and policy factors

The enemy indicator

Question # 6 of 15 (Start time: 09:27:43 PM) Total Marks: 1

Who suggested, "A quality product is one that provides performance or conformance at an acceptable price or cost"?

Select correct option:

Crosby

Ishikawa

***Garvin**

Shewhart

Question # 7 of 15 (Start time: 09:28:29 PM) Total Marks: 1

Marketing research carried out to know whether the particular area should be covered in this course. What the course instructor should develop?

Select correct option:

***The course meeting requirements of the university**

The course meeting requirements of the instructor

The course meeting requirements of the students

The course meeting requirements of the bookseller

Question # 8 of 15 (Start time: 09:30:00 PM) Total Marks: 1

Under which perspective of the balance scorecard, would you classify the operating cost management measurement?

Select correct option:

The customer perspective

The business process perspective

[The financial perspective](#)

The learning & growth perspective

Question # 9 of 15 (Start time: 09:31:24 PM) Total Marks: 1

Which of the following is the purpose of ISO 9001?

Select correct option:

***To define the minimum Quality Management System requirements needed to achieve customer satisfaction by meeting specified product requirements**

To determine metals and metalloids in airborne particulate matter by inductively coupled plasma atomic emission spectrometry

To ensure that the IT service management processes are aligned both with the needs of the business and with international best practice

To evaluate ultimate aerobic biodegradability of organic compounds in aqueous medium by determination of oxygen demand in a closed respirometer

Question # 10 of 15 (Start time: 09:32:16 PM) Total Marks: 1

Which of the following is a four-step quality model amongst the most widely used tools for continuous improvement?

Select correct option:

***Deming Cycle**

Juran Trilogy

Taguchi's Loss Function

Quality Circles

Question # 11 of 15 (Start time: 09:33:10 PM) Total Marks: 1

What was the transcendent view of quality?

Select correct option:

Satisfying customers

Meeting needs and wants

***Innate excellence**

Conformance

Question # 12 of 15 (Start time: 09:34:18 PM) Total Marks: 1

Which of the following focuses on results, not process, and encourages short-term behavior?

Select correct option:

***Typical American MBO system**

Typical Japanese MBO system

Typical Chinese MBO system

Uncommon Chinese MBO system

Question # 13 of 15 (Start time: 09:35:25 PM) Total Marks: 1

Which of the following is NOT the category in Baldrige Award?

Select correct option:

Leadership

Business results

***Supply chain**

Process management

Question # 14 of 15 (Start time: 09:36:30 PM) Total Marks: 1

What is the number of principles in Dr. W. Edwards Deming's quality principles?

8

10

12

***14**

Question # 15 of 15 (Start time: 09:37:03 PM) Total Marks: 1

Which of the following is NOT an advantage of functional structure?

Select correct option:

It provides organizations with a clear chain of command

***It enables employees to learn about customer expectations**

It allows evaluating people based on clear set of responsibilities

It allows people to specialize in the aspect of the work

Quiz No. 4

Question # 1 of 15 (Start time: 06:46:03 PM) Total Marks: 1

Which of the following is a requirement of ISO 9000:2000?

Select correct option:

***Processes to measure and analyze customer satisfaction**

An Environmental Policy to be in existence within the organization

Procedures for registration of font-related identifiers

Irregular improvement of internal process and customer satisfaction

Question # 2 of 15 (Start time: 06:47:17 PM) Total Marks: 1

The management responsible for the area being audited is responsible for ensuring that actions are taken without undue delay to eliminate detected nonconformities and their causes. Which of the following is a follow-up activity?

Select correct option:

***The verification of the actions taken**

The reporting of contradiction results

The contradiction of the actions taken

The verification of contradiction results

Question # 3 of 15 (Start time: 06:57:48 PM) Total Marks: 1

Which of the following total quality tools is employed in the step “plan and implement a solution” in problem solving?

Select correct option:

4W and 1H matrix

Pareto chart

***Control chart**

Scatter plot

Question # 4 of 15 (Start time: 06:59:05 PM) Total Marks: 1

Under which perspective of the balance scorecard, would you classify the customer retention measurement?

Select correct option:

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Question # 5 of 15 (Start time: 06:59:42 PM) Total Marks: 1

What was the role of quality professionals during 1950s?

Select correct option:

***Troubleshooting and the application of statistical methods**

Inspection, sorting, counting, and grading

Quality measurement, quality planning, and program design

Goal setting, education and training, and program design

Question # 6 of 15 (Start time: 07:00:46 PM) Total Marks: 1

What is quality?

Select correct option:

***Meeting and exceeding the customer's expectations**

Adding factors to make the customer happy

The degree to which the project meets requirements

Conformance to management's objectives

Question # 7 of 15 (Start time: 07:01:11 PM) Total Marks: 1

The quality initiatives focused initially on reducing defects and errors in products and services with the help of which of the following?

Select correct option:

***Measurement, statistics, and other problem solving tools**

Measurement, statistics and cause and effect analysis

Statistics and cause and effect analysis only

Failure modes and cause and effect analysis only

Question # 8 of 15 (Start time: 07:01:59 PM) Total Marks: 1

Which of the following performance levels have good results and have emphasized sustained improvement trends?

Select correct option:

Reactive approach

Stable formal system approach

Best-in-class performance

***Continual improvement emphasized**

Question # 9 of 15 (Start time: 07:02:42 PM) Total Marks: 1

Do not permit overlong examples or irrelevant discussion. This refers to which of the following discussion skills in meetings?

Select correct option:

Summarizing

***Managing time**

Closing the discussion

Containing digression

Question # 10 of 15 (Start time: 07:03:32 PM) Total Marks: 1

Which of the following statements, in relation to psychology, is TRUE for people failing to understand the components of Profound Knowledge?

Select correct option:

***They see the symptoms but not the deep causes of problems**

They do not know when expectations are realistic

They do not understand motivation/why people do what they do

They do not know how to plan and accomplish learning and improvement

Question # 11 of 15 (Start time: 07:04:21 PM) Total Marks: 1

Which of the following results in low costs?

Select correct option:

***High productivity and high capacity utilization**

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

Question # 12 of 15 (Start time: 07:05:03 PM) Total Marks: 1

Under which perspective of the balance scorecard, would you classify motivated workforce measurement?

Select correct option:

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The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Question # 13 of 15 (Start time: 07:05:40 PM) Total Marks: 1

Which of the following models value stability?

Select correct option:

Organism model

***Mechanistic model**

Cultural model

Total Quality model

Question # 14 of 15 (Start time: 07:06:33 PM) Total Marks: 1

Who stressed that systems must be focused toward a purpose?

Select correct option:

Crosby

Ishikawa

Taguchi

***Deming**

Question # 15 of 15 (Start time: 07:07:58 PM) Total Marks: 1

Which of the following leads to questioning and can be tested and validated?

Select correct option:

***Theory of knowledge**

Psychology

Understanding of variation

Appreciation for a system

Question # 1 of 15 (Start time: 07:10:06 PM) Total Marks: 1

Which of the following statements, in relation to variation, is TRUE for people failing to understand the components of Profound Knowledge?

Select correct option:

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

***They do not know the difference between prediction, forecasting, and guesswork**

They create cynicism, demoralization, de-motivation, guilt, resentment, burnout, craziness, and turnover

They do not understand the difference between improvement and change

Question # 2 of 15 (Start time: 07:11:25 PM) Total Marks: 1

Which of the following represent short-term perspectives and do not encourage long-term improvement as stated by Deming?

Select correct option:

***Numerical quotas**

Constitutional quotas

Political party quotas

Mechanical quotas

Question # 3 of 15 (Start time: 07:12:41 PM) Total Marks: 1

All of the followings are true for zero defects concept EXCEPT:

Select correct option:

A performance standard for management

A standard of the craftsperson regardless of his or her assignment

A tool of management to communicate to do things right the first time

A motivational, widely understood and accepted program

Question # 4 of 15 (Start time: 07:14:02 PM) Total Marks: 1

Who suggested, "A quality product is one that provides performance or conformance at an acceptable price or cost"?

Select correct option:

Crosby

Ishikawa

***Garvin**

Shewhart

Question # 5 of 15 (Start time: 07:14:35 PM) Total Marks: 1

Which of the following tools plots the relationship of one numeric variable to another on a horizontal and vertical axis and determines the degree of dependency or interdependency of those variables?

Select correct option:

***Scatter diagram**

Check sheet

Fishbone diagram

Pareto chart

Question # 6 of 15 (Start time: 07:15:52 PM) Total Marks: 1

Who warned against the use of extrinsic controls that compel behavioral changes in people when rewards and incentive programs are seldom shifted to favor quality improvement?

Select correct option:

Oakland

Schein

***Cox**

Dale

Question # 7 of 15 (Start time: 07:16:38 PM) Total Marks: 1

How many maturity levels are there in the Capability Maturity Model?

Select correct option:

Four

***Five**

Six

Seven

Question # 8 of 15 (Start time: 07:17:11 PM) Total Marks: 1

Which one of the following depicts aesthetics, which is a dimension of quality?

Select correct option:

***Exterior finish**

Quality of work

Ranking first

Ease of repair

Question # 9 of 15 (Start time: 07:18:28 PM) Total Marks: 1

Which of the following firms solely owns the entire material flow and each channel member operates independently?

Select correct option:

Horizontal integrated firms

Backward vertical integrated firms

Vertical integrated firms

Conglomerate integrated firms

Quiz Start Time: 07:10 PM Time Left 85

sec(s)

Question # 10 of 15 (Start time: 07:19:52 PM) Total Marks: 1

Which of the following is NOT the deadly disease according to Deming?

Select correct option:

Excessive medical costs

***Excessive transportation costs**

Lack of constancy of purpose

Mobility of management

Question # 11 of 15 (Start time: 07:21:21 PM) Total Marks: 1

Which stage in the Shewhart Cycle uses data and facts to analyze the results of the change?

Select correct option:

Do stage

Act stage

Plan stage

***Check stage**

Question # 12 of 15 (Start time: 07:21:54 PM) Total Marks: 1

Why product-based definitions of quality suit engineers?

Select correct option:

They are concerned with designing of the products

They can judge quality

They can reduce specific standards

***They are concerned with translating product requirements into specific components and physical dimensions thereby produced**

Question # 13 of 15 (Start time: 07:23:08 PM) Total Marks: 1

All of the following are the elements of a TQM system EXCEPT:

Select correct option:

Leadership

Communications

Measurement

***Detentions**

Question # 14 of 15 (Start time: 07:24:03 PM) Total Marks: 1

Which of the following is the contributor to the equipment cause in the house paint peeling cause-and-effect diagram?

Select correct option:

Cheap solvents

Temperature

Bad brush bristle

Paint over dirt

Question # 15 of 15 (Start time: 07:24:24 PM) Total Marks: 1

Which of the following is viewed as a Consistent Pair?

Select correct option:

ISO 9000 and ISO 9004

ISO 1987 and ISO 1992

ISO 1992 and ISO 1997

***ISO 9001 and ISO 9004**

Question # 1 of 15 (Start time: 07:17:17 PM) Total Marks: 1

All of the following are the objectives achieved simultaneously for the sustained growth to be recognized EXCEPT:

Select correct option:

Customer satisfaction

Effective leadership

Effective human resources

Integration with the supplier base

Question # 2 of 15 (Start time: 07:18:10 PM) Total Marks: 1

Some organizations emphasizes the use of quality tools but failed to do which of the following?

Select correct option:

Focus on what is truly important to the distributors

***Incorporate continuous improvement efforts**

Make fundamental changes in their processes and culture

Focus on what is truly important to the customer

Question # 3 of 15 (Start time: 07:18:53 PM) Total Marks: 1

Which of the following tools has a weakness that it does not tell whether the variation is the result of special causes or common causes?

Select correct option:

Histogram

Check sheet

Pareto chart

Run chart

All of the following are Juran's basic steps to progress EXCEPT:

Select correct option:

Achieve structured improvements on a continual basis combined with dedication and a sense of urgency

Establish an extensive training program

***Establish an extensive marketing program**

Establish commitment and leadership on the part of higher management

Question # 5 of 15 (Start time: 07:20:14 PM) Total Marks: 1

Which of the following statements, in relation to theory of knowledge, is TRUE for people failing to understand the components of Profound Knowledge?

Select correct option:

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

***They do not know the difference between prediction, forecasting, and guesswork**

They create cynicism, demoralization, de-motivation, guilt, resentment, burnout, craziness, and turnover

They do not understand the difference between improvement and change

Question # 6 of 15 (Start time: 07:21:09 PM) Total Marks: 1

Which of the following performance levels is a systematic process-based approach, early stage of systematic improvements?

Select correct option:

No formal approach

Reactive approach

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***Stable formal system approach**

Best-in-class performance

Question # 7 of 15 (Start time: 07:21:47 PM) Total Marks: 1

In creative problem solving, which of the following refers to brainstorming?

Select correct option:

Mess Finding

Fact Finding

Problem Finding

Idea Finding

Question # 9 of 15 (Start time: 07:22:33 PM) Total Marks: 1

Which of the following describes a complete data set?

Select correct option:

Measures of dispersion

Measures of central tendency

Deductive statistics

Inductive statistics

Question # 10 of 15 (Start time: 07:22:59 PM) Total Marks: 1

Which of the following tools represents the entire process at a high or detailed (depending on your use) level of observation, allowing analysis and optimization of workflow?

Select correct option:

Check sheet

Pareto diagram

Flow chart

Histogram

Question # 11 of 15 (Start time: 07:23:33 PM) Total Marks: 1

Which of the following is NOT the characteristic of the organizations operating according to the principles of quality management?

Select correct option:

A total commitment to continually increasing value for customers, investors, and employees

A firm understanding that market driven means that quality is defined by customers, not the company

A commitment to leading people with a bias for broken improvement and miscommunication

***A commitment to fundamental improvement through knowledge, skills, problem solving, and teamwork**

Question # 12 of 15 (Start time: 07:24:12 PM) Total Marks: 1

Ahmad has organized a manual data collection in such a way as to be easily used, understood and analyzed automatically. Which of the following Ahmad has prepared?

Select correct option:

Check sheet

Pareto diagram

Flow chart

Histogram

Question # 13 of 15 (Start time: 07:24:53 PM) Total Marks: 1

Which of the following is TRUE for design and development outputs under ISO 9001:2000 - Quality Management Systems ?

Select correct option:

It Identifies of and corrects against potential conformance

It is used where excess in product will only be apparent in use

It is Insignificance and concealment of significant features of the processes

***It meets the input requirements for design and development**

Question # 14 of 15 (Start time: 07:25:26 PM) Total Marks: 1

What is the purpose of the Control Chart?

Select correct option:

***To indicate variations, trends, and the amount of change from one time period to another**

To show where to apply resources by revealing the significant few from the trivial many

To provide a scientific approach to gathering information to improve the validity of decisions and interventions

To monitor effects of the variables on the difference between target and actual performance

Question # 15 of 15 (Start time: 07:25:52 PM) Total Marks: 1

Which of the following is the contributor to the work method cause in the house paint peeling cause-and-effect diagram?

Select correct option:

Cheap solvents

Temperature

Acid rain

Paint over dirt

..Mega Quiz File..

(MGT510)

Shahzad Sadiq

lushahz@gmail.com

<http://www.facebook.com/LushahzGoldsmithsCompany>

*(*Please correct me, if you find any mistake)*

.....Highlighted in Red are not sure or unsolved.....

What does the transcendent view imply for high quality?

It is something everlasting

It is something timeless and enduring

It is something priceless

It is something perfect

Taguchi suggested that loss in a process is increased with increase in which of the following?

Specifications

Standards

Competition

Variability

The products manufactured during 1800s were unique. How quality was ensured in this era?

Through renovation

Through inspection

Through calculation

Through repair

Which of the following is typically viewed to lie between fully vertically integrated firms?

Supply chain management

Production chain management

Value chain management

Demand chain management

The majority of advertisers appeal the public on the basis of which of the following?

Quality of product

Quality of staff

Inferiority of product

Inferiority of service

Which of the following models value stability?

Organism model

Mechanistic model

Cultural model

Total Quality model

What is the relationship between Ishikawa and Deming?

Son and father

Nephew and uncle

Student and teacher

Grandson and grandfather

Which of the following was developed by Motorola to improve its processes by minimizing defects?

ISO 9000

Six sigma

QS 9000

TQM

What was the primary concern of managers during 1980s?

Detection

Strategic impact

Control

Coordination

You can ask direct questions to quiet people and try to control talkative people", for which of the following purpose?

Mobilizing the group

Keeping the group focused

Restating purpose

Starting punctually

Some organizations emphasizes the use of quality tools but failed to do which of the following?

Focus on what is truly important to the distributors

Incorporate continuous improvement efforts

Make fundamental changes in their processes and culture

Focus on what is truly important to the customer

Learn how to tell when nothing can be gained from further discussion. This refers to which of the following discussion skills in meetings?

Acting as gatekeepers

Closing the discussion

Asking for clarification

Testing for consensus

Which of the following models has the viewpoint that change and learning are valued in themselves?

Organism model

Mechanistic model

Cultural model

Total Quality model

Marketing research carried out to know whether the particular area should be covered in this course. What the course instructor should develop?

The course meeting requirements of the university

The course meeting requirements of the instructor

The course meeting requirements of the students

The course meeting requirements of the bookseller

Which of the following statements is TRUE about Deming's philosophy?

It is based on improving products and services by reducing uncertainty and variability in the design and manufacturing processes

Quality is either or not present in the whole organization; that quality is the responsibility of everyone in the organization

Increasing conformance to specifications through elimination of defects, supported extensively by statistical tools for analysis

Increasing loss, for the producer, the customer, and society, associated with increasing

variability from a target value

All of the following are the elements of a TQM system EXCEPT:

Leadership
Communications
Measurement
Detentions

The standards for output based on quality and customer-service requirements that originate at the organizational and process levels include requirements for all of the following EXCEPT:

Accuracy
Innovation
Timeliness
Turnover

Which of the following refers to SIPOC (Supplier-Input-Process-Output-Customer) diagram?

A high-level process map
A low-level process map
A high-level performer map
A low-level performer map

Which of the following is concerned about quality for achievement of TQM?

The Managing Director
The Operative
The Quality Manager
Everyone in the organization

Which of the following is required for meeting and exceeding customer's need?

Planning
Control
Execution

Innovation

Which one of the following depicts feature, which is a dimension of quality?

Product looks and sounds

Brand name and price

Bells and whistles of the product

Ability to repair a product

Which of the following can help to eliminate fear and encourage teamwork?

Management

Workmanship

Leadership

Risk-taking

Which of the following focuses on results, not process, and encourages short-term behavior?

Typical American MBO system

Typical Japanese MBO system

Typical Chinese MBO system

Uncommon Chinese MBO system

Which of the following statements, in relation to theory of knowledge, is TRUE for people failing to understand the components of Profound Knowledge?

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

They do not know the difference between prediction, forecasting, and guesswork

They create cynicism, demoralization, de-motivation, guilt, resentment, burnout, craziness, and turnover

They do not understand the difference between improvement and change

Who made some salient observations about the failure to understand the components of Profound Knowledge?

Scholtes

Deming
Juran
Crosby

Do not permit overlong examples or irrelevant discussion. This refers to which of the following discussion skills in meetings?

Summarizing
Managing time
Closing the discussion
Containing digression

Which of the following decisions assume the existence of the facilities, but determine the exact path(s) through which a product flows to and from these facilities?

Location decisions
Production decisions
Inventory decisions
Transportation decisions

If the change was successful, implement it on a wider scale and continuously assess your results. Which stage represents this statement in the Shewhart Cycle?

Do stage
Act stage
Plan stage
Check stage

Which of the following is TRUE about entry-level managers as noted by Deming in Japan?

They spend 5 to 20 years on the factory floor
They spend 4 to 12 years on the factory floor
They spend 4 to 17 years on the factory floor
They spend 7 to 22 years on the factory floor

Which of the following statements is TRUE for total quality?

Stresses on learning and adaptation to change

Ability to repair a product quickly and easily
Difficulty in establishing and measuring service levels
A customer focused management system

What is necessary for a system to be effective?

The components of any system must work together for the system

The components of any system must not work together for the system
The components of any system should not be organized for the system
The components of any system should provide disruption for the system

The concept of common and special causes of variations was developed by:

Deming
Walter Shewhart
Juran
Taguchi

MBO stands for:

Management by Objective

Management by Organization
Maintenance by Objective
Maintenance by Organization

Which of the following statements is true about routine inspection:

It adds value to the product
It acknowledges defects as well as adds value to the product
It acknowledges defects but does not add value to the product
All of the given options

The success of Japanese manufacturers is due to _____, small and incremental improvements in design and production.

Continuous
Abrupt

Continual
None of the given options

All of the following statements are true about institute training except:

It improves quality
It improves productivity
It enhances workers' morale
It certainly results in salary increment

One of the advantages of team work is:

It breaks down barriers between internal customers and suppliers
It results in promotion
It results in salary increment
None of the given options

Reworking _____ the cost of quality.

Decreases
Increases
Neutralizes

_____ refers to general processes of improvement and encompasses discontinuous improvements

Continuous improvement
Continual improvement
Constant improvement
Consecutive improvement

Total quality costs include:

Prevention costs
Appraisal costs
Failure costs
All of the given options

The no. of principles in Dr. W. Edwards Deming's quality principles is

_____.

8

10

12

14

During 1980s, the job characteristic of quality professionals was:

Educating others

Achieving personal targets

Consultative work with other departments

None of the above

During 1800s, the quality was viewed as:

A problem that can never be solved

A problem to be solved

A tool for measuring system

A remedy to control decline

Why the factory managers created inspection departments?

To keep defective products aside, ensuring they do not reach the customers

To ensure quantity of goods/services

To count, grade, and rework

All of the above

During 1950s, controlling quality meant:

To understand and address the causes of poor quality

To simplify and speed up process

To link outcomes to uses

Acting on defects

The 'father' of statistical quality control is:

F. W. Taylor

Joseph M. Juran

Philip Crosby

Walter Shewhart

The Six Sigma is a new file separate from the total quality philosophy.

True

False

Today's consumers define quality as being flawless and having all of the attributes they want in just the way they want them.

True

False

The Deming Chain Reaction theory states that one of the main purposes of having quality products and processes is more rework.

True

False

The theory of variation helps us understand when there has not been a genuine improvement in performance

True

False

A SIPOC diagram is a tool used by a team to identify all relevant elements of a process improvement project before work begins.

True

False

Nonconformance is an expense of:

Profit of quality
Defects of quality
Quality of product
Cost of quality

Crosby's approach to management is:

A problem that can never be solved
Absolutes of Quality Management
Interim Management
ISO

Best price or zero cost is:

To keep defective products aside, ensuring they do not reach the customers
Prerogative cost
To count, grade, and rework
Cost leadership

Example of prevention costs is:

Quality improvement projects
Downgrading
To link outcomes to uses
Warranty claims

Big Q is:

Quality of services
Quality of people
Quality of processes
All of the above

Zero Defects is a widely accepted and understandable concept.

True

False

Managers assumed that they should not improve quality because defects are less costly.

True

False

Quality is a destination.

True

False

“Narrowly interpreted, quality means quality of a product,” was stated by Ishikawa.

True

False

ISO was first released in 1882.

True

False

Which of the following document(s) is (are) included in the quality system?

A quality policy

Customer focus

Commitment

All of the given options

Appropriate motives for adopting ISO 9000 include which of the following?

To create a quality management system

To improve operations

To conform to the requirements of customers

All of the given options

Quality checklists are used to:

Ensure that Quality Assurance steps were followed

Keep quality inspectors busy

Inform upper management where failures occur

Prevent project audits

The ISO 9001:2000 standards are

More widely adopted in the U.S. than in Europe

Quality management procedures that address leadership, documentation, and record-keeping

Are known as the Baldrige standards in the U.S

A set of product specifications for all industries

Documentation and data on existing products are the examples of _____

External inputs

Internal inputs

Other inputs

None of the above options

Which of the following statements outline the relationship factor?

ISO 9000 and total quality are not in competition

ISO 9000 and total quality are not interchangeable

ISO 9000 is compatible with total quality

All of the above

Which of the following is not an example of physical factors affecting work environment?

Noise

Cleanliness

Air flow

None of the given options

“Performance comes effortlessly” refers to _____ stage of learning according to psychologists.

Unconscious incompetence

Conscious incompetence

Conscious competence

Unconscious competence

There are _____ levels of the CMM.

Four

Five

Six

Three

The underlying cause(s) of TQM deficiencies is (are):

The failure of managers to understand the concept of quality

Costly implementation of quality management system

Lack of employee involvement

All of the given options

Leaders:

Push rather than pull

Must be trustworthy but not necessarily courageous

Know where they want to go

Need to dictate

The Steering Committee must develop:

Mission statement

Continual improvement

Vision statement

All of the above

One thing that the implementation plan must provide is:

Short-term focus

Leadership delegation

Vision statement

Training

_____ provides information on how productivity and quality can be continuously improved through problem identification.

Statistical quality control

Statistical process control

Tree Diagram

None of the given option

_____ is a graphic tool for defining the relationship between customer desires and the firm/product capabilities.

House of Quality

Affinity diagram

Arrow diagram

None of the given option

The affinity diagram is used to bring logic to the process of identifying relationships among the various ideas recorded on the tree diagram.

False

True

Total quality implementation must not be delegated.

True

False

Competitive and benchmark data are essential for planning quality leadership because they make possible clear and objective quality comparisons.

True

False

Deployment refers to the extent to which the approaches are applied to all relevant areas and activities addressed and implied in the examination items.

True

False

It is best to apply SPC to all processes at once.

False

True

What is quality?

Meeting and exceeding the customer's expectations

Adding extras to make the customer happy

The degree to which the project meets requirements

Conformance to management's objectives

All of the following are the grounds to give birth to total quality approach EXCEPT:

Obsession with quantity

Confusion and poor leadership

One employee in charge

Political disorder

All of the following are the major decision areas in supply chain management EXCEPT:

Location

Production

Inventory

Pricing

Which of the following is true for subsystems?

These are not linked together as internal customers and suppliers

These are linked together as external customers and suppliers

These are linked together as external and internal customers and suppliers

These are linked together as internal customers and suppliers

In which of the following operations great deal of variations can occur?

Manufacturing

Financing

Purchasing

Selling

With total quality perspective, employees are empowered to think and make recommendations for which of the following?

Continual improvement

One time check

After completion check

Incremental improvement

Which one of the following depicts durability – a dimension of quality?

Exterior finish

Quality of work

Useful life

Ease of repair

Which of the following statements is TRUE for total quality?

Stresses on learning and adaptation to change

Ability to repair a product quickly and easily

Establishing and measuring service levels may be difficult

A primary operating characteristic of product

What is the objective of a TQM system?

Continuous improvement

Continual improvement
Business improvement
Process improvement

Which of the following results in low costs?

High productivity and high capacity utilization

Low productivity and low capacity utilization
Low productivity and high capacity utilization
High productivity and low capacity utilization

In the process of acquiring the certificate of ISO 9000, which step follows the step of setting the registration objective?

Develop and implement the QMS
Perform self-analysis audit
Submit quality manual for approval
Select the appropriate standards

In the process of acquiring the certificate of ISO 9000, which step follows the step of pre-assessment by registrar?

Take corrective action

Obtain ISO registration
Perform self-analysis audit
Select a third party registrar

In the process of acquiring the certificate of ISO 9000, which step follows the step of developing and implementing the quality and management system?

Set the registration objectives
Select a third party registrar
Pre-assessment by registrar
Take corrective action

In the process of acquiring the certificate of ISO 9000, which step follows the step of

selecting the appropriate standards?

Perform self-analysis audit

Submit quality manual for approval

Develop and implement the QMS

Take corrective action

In the process of acquiring the certificate of ISO 9000, which step follows the step of selecting a third party registrar?

Take corrective action

Obtain ISO registration

Perform self-analysis audit

Final assessment by registrar

In the process of acquiring the certificate of ISO 9000, which step follows the step of performing self-analysis audit?

Final assessment by registrar

Set the registration objectives

Develop and implement the QMS

Submit quality manual for approval

In the process of acquiring the certificate of ISO 9000, which step follows the step of submitting quality manual for approval?

Pre-assessment by registrar

Take corrective action

Perform self-analysis audit

Develop and implement the QMS

Which of the following is TRUE for higher costs of quality?

Doing the wrong thing in the wrong way

Doing the wrong thing in the right way

Doing the right thing in the wrong way

Doing the right thing in the right way

For what duration is the ISO certification valid for any organization?

5 years

4 years

3 years

2 years

Identify the core leadership skills among the following.

a. Vision

b. Empowerment

c. Value congruence

d. Intuition

e. Skilled labor

f. Gross motor

a, b, c and f

b, c, d, and e

c, d, e, and f

a, b, c, and d

The primary requirement for organizational excellence is strategic focus.

True

False

The primary requirement for organizational excellence is tactical focus.

True

False

Japanese companies mostly are not ISO certified because they think they are beyond the requirements of ISO.

True

False

The means for staff to perform their tasks right the first time is a benefit of

implementing the quality management system.

True

False

The means for documenting the company's experience in a structured manner is not a benefit of implementing the quality management system.

True

False

The implementation of quality management system does not bring clarity and transparency to duties and responsibilities.

True

False

The implementation of quality management system does not make export marketing easier.

True

False

With the implementation of quality management system the company cannot secure greater customer loyalty.

True

False

With the implementation of quality management system the company's customer will not receive products or services of a known level of quality that is independently audited.

True

False

What is quality for any organization?

A solution

A problem

A specification

A project

What kind of decisions did the managers make in the old paradigm?

Strategic positioning based on warfare models

Product positioning based on warfare models

Market positioning based on warfare models

Strategic and product positioning based on warfare models

What is the traditional view of quality?

Quality is measured by establishing an acceptable level of conformance

Quality is measured by establishing an acceptable level of employee satisfaction

Quality is measured by establishing an acceptable level of nonconformance

Quality is measured by establishing an acceptable level of both conformance and nonconformance

When was the definition of total quality endorsed?

In 1950

In 1970

In 1985

In 1992

Customers want an excellent product or service from an organization that also provides which of the following?

A heartiest welcome

Explicit costs

Reliable delivery and after purchase support

Implicit costs

Service standards are usually hard to measure because of the following reason?

Service standards are intangible
Service standards are unaccountable
Service standards are judgmental
Service standards are invariable

Which of the following is required to ensure quality?

Absence of wastage
Absence of defects
Absence of variations
Absence of specifications

Asking for examples, pictures, diagrams, and data in meetings, refers to which of the following?

Acting as gatekeepers
Closing the discussion
Asking for clarification
Testing for consensus

The user-based view of quality is popular among the people of which of the following area?

Accounting
Banking
Physics
Marketing

Motorola was committed to which of the following campaigns in 1980s?

Zero defect
Six sigma
Benchmarking
Cause-and-effect analysis

What are the ISO (International Organization for Standardization) series standards for industries such as aerospace?

QS 9000

AS 9000

TL 9000

ISO/TS 16949

Juran advocates a never-ending spiral of activities that includes all of the following EXCEPT:

Product development

Workers

Inspection and testing

Purchasing

To increase the awareness and involvement of people, employee training should cover all of the following EXCEPT:

The vision for the future of the organization

The organization's policies and objectives

Organizations's intangible resources such as intellectual property

The initiation and implementation of improvement activities

Which of the following is a follow up activity relating to internal audit?

Collection of quality cost data

Questionnaires and surveys

Verification of implementation

Utilization of technologies

All of the following are the consideration of the organizations when analyzing how all the processes interact to make sure that all processes operate as an efficient network EXCEPT:

Ensuring inputs, activities and outputs are clearly defined and controlled

Managing risks and opportunities

Monitoring inputs and outputs to confirm that, group processes are unrelated

Establishing data analysis that facilitates continual improvement across all processes

Which of the following is a valid statement about design and development outputs?

It identifies and rectifies against potential conformance
It is used where excess in product will only be apparent in use
It signifies and conceals the significant features of the processes
It meets the input requirements for design and development

Against which of the following scales, the Capability Maturity Model can be used to assess an organization?

Three process maturity levels
Four process maturity levels
Five process maturity levels
Six process maturity levels

In the levels of organizational maturity, the dysfunctional system precedes which of the following systems?

Awakening System
Developing System
Maturing System
World Class System

Which of the following is TRUE for lower costs of quality?

Doing the wrong thing in the wrong way
Doing the wrong thing in the right way
Doing the right thing in the wrong way
Doing the right thing in the right way

Which of the following defines the boundaries of the business in which the organization operates?

Vision
Mission
Plan
Objectives

What are the core competencies of leadership?

Activities that get the opportunity to meet many people with similar interests
Activities and processes needed to meet customers' minimum requirements
Activities that gives you an opportunity to learn and experience new things
Activities that underpin competitive advantage and are difficult for competitors to copy

Which of the following is NOT an organizational weakness?

Obsolete facilities
Obsolete processes
Weak management team
Higher quality competitors

Which of the following base all actions, activities, and decisions on what is most likely within an ethical framework to ensure successful performance in the marketplace?

Operation Management
Strategic Management
Financial Management
Network Management

Total quality relates to which of the following which enhances an organization's ability to gain a sustainable competitive advantage in the marketplace?

Operation Management
Strategic Management
Financial Management
Network Management

Which of the following is a communication tool used to tell a story of how value is created for the organization?

Injective map
Strategy map
Surjective map
Bijective map

Under which perspective of the balance scorecard, would you classify inventory management measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Under which perspective of the balance scorecard, would you classify corporate goals measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Under which perspective of the balance scorecard, would you classify operating cost management measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Under which perspective of the balance scorecard, would you classify average time to process orders measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Under which perspective of the balance scorecard, would you classify number of strategic skills learned measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

All of the following are the examples of product concerns from customers and end-users EXCEPT:

Conformance

Transportation

Availability

Delivery

Brainstorming refers to which of the following?

Mess Finding

Fact Finding

Problem Finding

Idea Finding

The Pareto Principle is a technique used by quality managers to determine the quality control problems concerning a particular service or manufacturing process to be corrected. Which of the following statements best represents the problems employed by this principle?

In order to minimize financial losses, all problems should be corrected

The majority of defects are caused by a small percentage of the identifiable problems.

In order to achieve zero defects, all quality control problems should be corrected

Generally, 80% of the quality control problems are justifiable for correction via cost-benefit analysis.

Which of the following helps you distinguish between normal and unusual variation in a process?

Control Chart

Cause and effect diagram

Pareto Chart

Scatter Diagram

The observation in the ‘middle’ of sorted data is called:

Mode

Median

Range

Variance

Which of the following is the contributor to materials as a cause in free-throwing cause-and-effect diagram?

Basketball

Games

Video camera

Touched rim

Which of the following is the contributor to equipments as a cause in free-throwing cause-and-effect diagram?

Basketball

Games

Video camera

Touched rim

In which of the following teams, the team members communicate by computers, take turns as leaders, and jump in and out as necessary?

Virtual teams

Quality circles

Management teams

Natural work teams

Which of the following is performed in the process of benchmarking?

To collect related and structured activities that produce a specific product

To plan and control the performance of any type of activity

To compare external data collected to the internal data collected

To create goods and services that meet needs and wants of customers

Which of the following is NOT the responsibility of organization regarding Competence, Awareness and Training of employees?

To implement the country's legal requirements for efficient transportation of people

To make training available to employees or take other actions to satisfy competency needs

To determine the necessary competence for employees performing work affecting product quality

To ensure that employees are aware of the relevance and importance of their activities

You know how to paint, but your reactions are not automatic---- you have to work at it. This statement refers to which of the following stages of learning?

Unconscious incompetence

Conscious incompetence

Conscious competence

Unconscious competence

Which of the following is the odd one out?

Peter Drucker

Philip Crosby

W. Edwards Deming

Kaoro Ishikawa

One of the most effective means of implementing quality initiatives for executive management is to:

Establish quality goals tied to organizational performance

Conduct meetings on quality and demonstrate support for initiatives

Make public announcements explaining the company's quality goals

Hire a quality consultant to develop a total quality plan and lead its implementation

ISO 9000 certification:

Aims to promote top management's commitment to TQM

Certifies products and their manufacturing processes

Does not apply to service industries

Is seen as a customer requirement

According to Crosby, most companies spend _____ percent of their sales dollars on quality costs.

15 to 20

20 to 25

25 to 30

None of the given options

The control of quality involves the following processes except:

Assess actual quality performance

Compare performance with goals

Act on differences between performance and goals

None of the given options

Methods of continuous improvement include _____.

Six Sigma

Lean Production

Kaizen

All of the given options

Shareholder is an example of _____.

Internal customer

External customer

Investor customer

Social customer

_____ concepts are referred to as statistical quality control (SQC).

Shewhart's

Deming's

Juran's

Crosby's

_____ is the judge of quality.

Customer

Supplier

Producer

None of the given options

_____ is not an example of organization.

Factory

Mosque

Hospital

None of the given options

Deming cycle was developed to link the production of a product with consumer needs.

True

False

Avoidable costs are related to preventing defects.

True

False

Total quality requires a set of guiding principles.

True

False

Crosby is best known for his advocacy of zero defects management.

True

False

The Six Sigma is a new field separate from the total quality philosophy.

True

False

Which of the following statements best describe the acronym SWOT?

Specific, Witness, Opportunities, and Time

Single, Weaknesses, Occasion, and Tactics

Strengths, Weaknesses, Opportunities, and Threats

Strategic, Weaknesses, Objectives, and Tactics

Which of the following is NOT a law of organizational change?

Involve everyone affected by change in making it

Be prepared to listen and observe

Trust people with dignity and respect

Understand the history behind the current culture

The most effective leaders can be found in what part of an organization?

Top-level management

Mid-level management

Lowest paid wage earner

All of the given options

Which of the following character traits promote successful teamwork?

Honesty

Controlling

Accountability

Bossy

Which of the following statements outlines the relationship factor?

ISO 9000 and total quality are not in competition

ISO 9000 and total quality are not interchangeable

ISO 9000 is compatible with total quality

All of the given options

What is the only tool of the seven tools that is not based on statistics?

Pareto Chart

Fishbone Diagram

Scatter Diagram

Histogram

What portion of the QFD house explanation is where customer requirements are converted into manufacturing terms?

The wall

The roof

The middle

The bottom

The most common inhibitor of SPC is:

Inadequate training

Lack of resources

Failure to have processes under control

Low production rates

Little Q regards the customer as:

The person who buys the product

All people involved, internal, and external

The decision-maker

Having continually changing needs

Which of the following behavior best describes the concept of benchmarking?

Unethical

Illegal

Industrial espionage

None of the given options

Which of the following statements is considered true concerning the affects of global integration?

It has helped fuel economic growth throughout the industrialized world

It can be the Achilles heel of economic growth

Downturns in one country can now have a ripple effect that quickly spreads to other countries

All of the given options

Which of the following statements is not true about Total Quality Management (TQM)?

TQM is focused on improving product and customer service quality

TQM is expensive to initiate

TQM requires top management support

TQM requires ISO 9000 certification

Which of the following may be a dimension of quality?

Performance

Conformance

Reliability

Durability

According to Juran trilogy, quality control involves which of the following process(es).

Assessment of actual quality performance

Comparison of performance with goals

Action on differences between performance and goals

All of the given options

is best known for his zero defects management.

Crosby

Deming

Juran

None of the given options

The need to improve an organization's financial condition correlated directly with the process of making and measuring quality improvements.

True

False

SWOT Analysis is the only step needed in the strategic planning process.

True

False

The concept of quality has often been defined, from a transcendent view, as innate excellence.

True

False

Quality materials produced by Juran have been translated into 12 different languages.

True

False

Philosophy helps us to understand people, interactions between people and circumstances.

True

False

In total quality perspective, quality planning, and strategic business planning are treated as which of the following?

Different

Consistent

Indistinguishable

Dissimilar

The marketing team of a company carried out a survey concerning expectations of the customers from the product. What should be the product development department of the company design?

The product meeting requirements of the customers

The product meeting requirements of the distributors
The product meeting requirements of the management
The product meeting requirements of the shareholders

Which of the following results in low costs?

High productivity and high capacity utilization

Low productivity and low capacity utilization
Low productivity and high capacity utilization
High productivity and low capacity utilization

Which of the following are generally easy to detect using statistical methods, and it is usually economical to remove them?

Common causes

Special causes
Implicit causes
Assignable causes

During which century, the manufacturers began to include quality processes in quality practices?

17th century
18th century
19th century
20th century

Which of the following models has the viewpoint that change and learning are valued in themselves?

Organism model

Mechanistic model
Cultural model
Total Quality model

After the war, Deming was invited to which of the following countries?

Germany

Japan

France

China

Which of the following statements is TRUE for the organizations focusing problem-solving and continuous improvement?

Fail to emphasize error prevention and design quality

Fail to focus on what is truly important to the customer

Fail to identify the best times about the situation in the past

Fail to integrate all organizational functions

Quality applies to which aspect(s) of the organization?

Products only

Products and services

People and product

Products, services, and people

Which of the following statements holds true for quality assurance?

Top management's intention regarding quality

Functions determining implementation of the quality policy

Actions to provide confidence of satisfying quality requirements

Responsibilities and processes, which implement quality management

Which of the following models reflect horizontal processes beginning with suppliers and ending with customers?

Organism model

Mechanistic model

Cultural model

Total Quality model

Which of the following is concerned about quality for achievement of TQM?

The Managing Director

The Operative

The Quality Manager

Everyone in the organization

Which one of the following depicts durability – a dimension of quality?

- Exterior finish
- Quality of work
- Useful life**
- Ease of repair

"Not giving respect to other people's contributions", depicts which of the following self-oriented behavior?

- Diverting
- Excluding
- Belittling**
- Blocking

which century, the manufacturers began to include quality processes in quality practices?

- 17th century
- 18th century
- 19th century
- 20th century**

In the new paradigm, managers view error as which one of the following?

- A threat
- A substitute
- An opportunity**
- An end

Whose concepts are referred to as statistical quality control (SQC)?

- Shewhart's**
- Deming's
- Juran's
- Crosby's

In which of the following operations great deal of variations can occur?

Manufacturing

Distribution

Purchasing

Selling

Which of the following are generally easy to detect using statistical methods, and it is usually economical to remove them?

Common causes

Special causes

Implicit causes

Assignable causes

Who argued that beauty could be understood only after exposure to a series of objects that display its characteristics?

Drucker

Crosby

Deming

Plato

During which war, Deming taught quality control courses as part of the national defense efforts?

World War I

World War II

Mexican-American War

The Civil War

The majority of advertisers appeal the public on the basis of which of the following?

Quality of product

Quality of staff

Inferiority of product

Inferiority of service

Which of the following results in low costs?

High productivity and high capacity utilization

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

The customer value strategy may be addressed in which of the following topics?

Projects

Products

Services

Measurement

Who developed Quality Function Deployment (QFD)?

Frederick Taylor

Joseph M. Juran

Y. Akao

Armand Feigenbaum

How the Baldrige Award recognizes the importance of total quality principles?

Through independence with overall business planning

Through a no relation with overall business planning

Through integration with overall business planning

Through dependence with overall business planning

Which of the following models considers the goal of meeting individual needs?

Organism model

Mechanistic model

Cultural model

Total Quality model

All of the following are the initial considerations in the Hoshin Kanri approach to business system change EXCEPT:

Measuring the business system as a whole

Setting core objectives of the business

Understanding the environmental situation in which the business operates

Ceasing resources for performing activities to achieve business objectives

Which of the following statements is TRUE for ISO 9000?

Describes the principles of a quality management system and defines the terminology

Describes the requirements relative to a quality management system either for internal use or for contractual or certification purposes

Intended for internal use and not for contractual purposes, focuses particularly on continually improving performance

Contains the guidelines for auditing quality management and/or environmental management systems

Which of the following statements holds true for quality assurance?

Top management's intention regarding quality

Functions determining implementation of the quality policy

Actions to provide confidence of satisfying quality requirements

Responsibilities and processes, which implement quality management

Which of the following is an answer to the Second Absolute of Quality Management as summarized by Crosby; what system is needed to cause quality?

Conformance to requirements

Prevention

Zero defects

Price of nonconformance

All of the following are the reasons why Total Quality Management programs are not prolific for many companies, EXCEPT:

TQM focuses on minimum standards

TQM develops its own cumbersome bureaucracy

TQM has no place for love and passion

TQM focuses on customers and stakeholders

What is the other name used for Deming Cycle?

Shewhart cycle

Product life cycle

Data life cycle management

Information life cycle management

What is necessary for a system to be effective?

The components of any system must work together for the system

The components of any system must not work together for the system

The components of any system should not be organized for the system

The components of any system should provide disruption for the system

Which of the following is NOT the criterion under Canadian Awards for Business Excellence?

Leadership

Planning for contraction

Process optimization

Supplier focus

Which of the following is true for subsystems?

These are not linked together as internal customers and suppliers

These are linked together as external customers and suppliers

These are linked together as external and internal customers and suppliers

These are linked together as internal customers and suppliers

What is the relationship between Ishikawa and Deming?

Son and father

Nephew and uncle

Student and teacher

Grandson and grandfather

Which of the following statements is TRUE for little q?

Striving for quality in a limited or specific area

Searching for specific lines of text containing a particular pattern

Matching rules encoded in a string according to certain syntax rules

Giving combination of values for the free variables

The Capability Maturity Model can help in all of the following EXCEPT:

Setting process improvement goals and priorities

Providing guidance for quality processes

Providing a yardstick for assessing current practices

Documenting procedures as to how the various processes for design operate

The customer value strategy may be addressed in which of the following topics?

Projects

Products

Services

Measurement

Who is known as a father of modern-day statistical quality control?

Shewhart

Deming

Juran

Crosby

Which of the following forms the basis of TQM as an organization wide process?

Not using the resources of the total organization properly

Organizational flexibility and response to change

Not bound in business relationships

Not measuring continuous improvement

How should the benchmarking of individual processes as well as customer satisfaction be employed?

As a standard tool

As a defective tool

As an improvement tool

As an instrumental tool

Identify the example of prevention costs.

Quality planning

Re-inspection

Product recalls

Customer returns

Which of the following is NOT included in the ISO registration process?

Document review by the registrar of the quality system documents or quality manual

Pre-assessment, which identifies potential noncompliance in the quality system or in the documentation

Review the program descriptions and select the program(s) that will enable you to contribute more to your organization

Assessment by a team of two or three auditors of the quality system and its documentation

Who made some salient observations about the failure to understand the components of Profound Knowledge?

Scholtes

Deming

Juran

Crosby

Which of the following results in low costs?

High productivity and high capacity utilization

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

How can the quality be computed?

Quality = Expectation/Performance

Quality = Performance/Expectation

Quality = Performance+Expectation

Quality = Performance-Expectation

Documentation and data on existing products, are an example which of the following?

External inputs

Internal inputs

Either internal or external inputs

Neither internal nor external inputs

Which of the following focuses on results, not process, and encourages short-term behavior?

Typical American MBO system

Typical Japanese MBO system

Typical Chinese MBO system

Uncommon Chinese MBO system

Who developed Quality Function Deployment (QFD)?

Frederick Taylor

Joseph M. Juran

Y. Akao

Armand Feigenbaum

All of the following are the examples of appraisal costs EXCEPT:

Quality measurement criteria

Quality improvement projects

Laboratory acceptance testing

Inspection and testing

In order to achieve the objectives and stimulate innovation, what the organization should encourage?

The involvement of its human resource

The involvement of its process

The involvement of its system

The involvement of its capital

How the requirements for documentation and records may not arise?

Contractual requirements from the customer or other interested parties

Acceptance of international, national, regional and industry sector standards

Irrelevant statutory and regulatory requirements

Decisions by the organization

At which level, does the quality concern centers on meeting internal customer requirements?

Organizational level

Performer level

Process level

Production level

Who is known as a father of modern-day statistical quality control?

Shewhart

Deming

Juran

Crosby

Which of the following is TRUE for higher costs of quality?

Doing the wrong thing in the wrong way

Doing the wrong thing in the right way

Doing the right thing in the wrong way

Doing the right thing in the right way

Which of the following concepts and tools can be used to assess internal customer needs and to communicate internal supplier capabilities?

Financial

Management

Marketing

Production

The management representatives appointed and given authority by the senior management are for all of the following EXCEPT:

Managing

Mishandling

Evaluating

Coordinating

The management preserves the conformity of products during internal processing and final delivery. Which of the following is included in such preservation?

Handling

Abolition

Risk

Shrinking

All of the following are the basic requirements of ISO 9000 EXCEPT:

A documented quality system
Consistent adherence to the documented quality system
Check to see that the plan was carried out

Customers and processes be displeased

During 1990s, the following institution(s) began to pay increase attention to quality.

Education and health care only
Education, health care, and government
Health care and government only
Government and financial institutions only

Which one of the following depicts reliability – a dimension of quality?

Bells and whistles of the product
Meeting specifications
Average time for the unit to fail
Quality of work

Which of the following models considers the goal of organizational efficiency and performance?

Organism model
Mechanistic model
Cultural model
Total Quality model

All of the following are the examples of external failure costs EXCEPT:

Processing customer complaints
Customer returns
Warranty claims
Quality planning

Identify the example of external failure costs.

Quality planning
Re-inspection
Material review
Customer returns

Which of the following is viewed as a Consistent Pair?

- ISO 9000 and ISO 9004
- ISO 1987 and ISO 1992
- ISO 1992 and ISO 1997
- ISO 9001 and ISO 9004**

In which century, the US broke further from the European tradition and adopted a new management approach developed by Frederick W. Taylor?

- 17th century
- 18th century
- 19th century**
- 20th century

Quality applies to which aspect(s) of the organization?

- Products only
- Products and services
- People and product
- Products, services, and people**

Which of the following statements, in relation to system, is TRUE for people failing to understand the components of Profound Knowledge?

- They see the symptoms but not the deep causes of problems**
- They do not know when expectations are realistic
- They do not understand motivation/why people do what they do
- They do not know how to plan and accomplish learning and improvement

"Problems will remain unsolved, despite their best efforts." When this will happen?

- When people don't understand systems
- When people don't understand psychology
- When people don't understand variation
- When people don't understand the theory of knowledge**

Which stage in the Shewhart Cycle implements the change on a small scale?

Do stage

Act stage

Plan stage

Check stage

Which of the following s determining what to control, establishing units of measurement so that data may be objectively evaluated?

Quality improvement

Quality planning

Quality control

Quality process

All of the following steps are required for quality planning EXCEPT:

Determine who the customers are

Deploy the plans to operational levels

Identify customers' needs

Assess actual quality performance

Which of the following is an answer to the First Absolute of Quality Management as summarized by Crosby; what is quality?

Conformance to requirements

Prevention

Zero defects

Price of nonconformance

Juran defines quality as “fitness for use”. This can be broken down into how many categories?

Five

Four

Three

Two

How many maturity levels are there in the Capability Maturity Model?

Four

Five

Six

Seven

Which of the following is a unit for measuring process performance according to traditional view of quality?

- Effective parts per thousand produced
- Defective parts per thousand produced
- Effective parts per hundred produced
- Defective parts per hundred produced**

Which of the following statements, in relation to system, is TRUE for people failing to understand the components of Profound Knowledge?

- They see the symptoms but not the deep causes of problems**
- They do not know when expectations are realistic
- They do not understand motivation/why people do what they do
- They do not know how to plan and accomplish learning and improvement

Which of the following statements, in relation to variation, is TRUE for people failing to understand the components of Profound Knowledge?

- They see events as individuals' incidents rather than the net result of many interactions and interdependent forces
- They do not know the difference between prediction, forecasting, and guesswork**
- They create cynicism, demoralization, de-motivation, guilt, resentment, burnout, craziness, and turnover
- They do not understand the difference between improvement and change

Which of the following models has a vertical chain of command?

- Organism model
- Mechanistic model**
- Cultural model
- Total Quality model

What was the role of quality professionals during 1800s?

- Troubleshooting and the application of statistical methods
- Inspection, sorting, counting, and grading**
- Quality measurement, quality planning, and program design
- Goal setting, education and training, and program design

All of the following are the major categories and items within each category of Canadian Awards for Business Excellence EXCEPT:

Customer focus

Distributor focus

People focus

Supplier focus

How the Baldrige Award recognizes the importance of total quality principles?

Through independence with overall business planning

Through a no relation with overall business planning

Through integration with overall business planning

Through dependence with overall business planning

Which of the following involves determining what to control, establishing units of measurement so that data may be objectively evaluated?

Quality improvement

Quality planning

Quality control

Quality process

After the war, Deming was invited to which of the following countries?

Germany

Japan

France

China

Which of the following organizations has a little chance of survival?

With no corporate philosophy about how to create long-term customer loyalty

With no corporate objectives about how to create long-term customer loyalty

With no mission about how to create long-term customer loyalty

With no vision about how to create long-term customer loyalty

Who suggested that there are three categories of performance?

Taguchi
Taylor
Ishikawa
Deming

Which of the following statements holds true for quality assurance?

Top management's intention regarding quality
Functions determining implementation of the quality policy
Actions to provide confidence of satisfying quality requirements
Responsibilities and processes, which implement quality management

Which of the following statements is TRUE about Deming's philosophy?

It is based on improving products and services by reducing uncertainty and variability in the design and manufacturing processes
Quality is either or not present in the whole organization; that quality is the responsibility of everyone in the organization
Increasing conformance to specifications through elimination of defects, supported extensively by statistical tools for analysis
Increasing loss, for the producer, the customer, and society, associated with increasing variability from a target value

By the mid 1970s, which markets were penetrated by the Japanese companies?

Western markets
Eastern markets
Northern markets
Southern markets

The input for management review to examine the quality management system shall include information about all of the following EXCEPT:

Status of preventive and corrective actions
Follow-up actions from previous management reviews
Changes that could affect the quality management system
Improvement of product related to customer requirements

Juran advocates a never-ending spiral of activities that includes all of the following EXCEPT:

Product development

Workers

Inspection and testing

Purchasing

Which of the following statements is TRUE for little q?

Striving for quality in a limited or specific area

Searching for specific lines of text containing a particular pattern

Matching rules encoded in a string according to certain syntax rules

Giving combination of values for the free variables

Which of the following is the Total Quality Tool that is simply a graphical representation of an outline that presents a chain of causes and effects?

Check sheet

Fishbone diagram

Scatter diagram

Pareto chart

Who found that teams are seldom fully used, and their individual members are often complacent?

Oakland and Schein

Newall and Dale

Lehr and Cox

Juran and Schein

The customer value strategy may be addressed in which of the following topics?

Projects

Products

Services

Measurement

Which of the following should be reviewed by the organization to identify requirements for operations that realize products or deliver services in order to ensure compliance with specifications?

Inability to comply with contractual requirements

Training and competence of people

Problem conception
Irrelevant statutory and regulatory requirements

What are the ISO (International Organization for Standardization) series standards for industries such as automobiles?

ISO 9000
ISO 14000
QS 9000
AS 9000

Which of the following should be used as input to management review in order to ensure that continual improvement is the driver for organizational development?

Performance information

Performance management
Ethical performance
Performance development

Based in Geneva, Switzerland, International Organization for Standardization is a consortium of the national standard bodies, which has how many countries?

More than 100

Less than 100
Less than 90
Less than 80

Which of the following is the contributor to the work method cause in the house paint peeling cause-and-effect diagram?

Cheap solvents
Temperature
Acid rain
Paint over dirt

Which of the following concepts and tools can be used to assess internal customer needs and to communicate internal supplier capabilities?

Financial
Management

Marketing

Production

The input for management review to examine the quality management system shall include information about all of the following EXCEPT:

Status of preventive and corrective actions

Follow-up actions from previous management reviews

Changes that could affect the quality management system

Improvement of product related to customer requirements

What the organization must accomplish to achieve the mission, by examination and categorization of the impacts?

Critical success factors

Key process indicators

Illegal and policy factors

The enemy indicator

Who suggested, "A quality product is one that provides performance or conformance at an acceptable price or cost"?

Crosby

Ishikawa

Garvin

Shewhart

Under which perspective of the balance scorecard, would you classify the operating cost management measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Which of the following is the purpose of ISO 9001?

To define the minimum Quality Management System requirements needed to achieve customer satisfaction by meeting specified product requirements

To determine metals and metalloids in airborne particulate matter by inductively coupled plasma atomic emission spectrometry

To ensure that the IT service management processes are aligned both with the needs of the business and with international best practice
To evaluate ultimate aerobic biodegradability of organic compounds in aqueous medium by determination of oxygen demand in a closed respirometer

Which of the following is a four-step quality model amongst the most widely used tools for continuous improvement?

Deming Cycle

Juran Trilogy

Taguchi's Loss Function

Quality Circles

What was the transcendent view of quality?

Satisfying customers

Meeting needs and wants

Innate excellence

Conformance

Which of the following focuses on results, not process, and encourages short-term behavior?

Typical American MBO system

Typical Japanese MBO system

Typical Chinese MBO system

Uncommon Chinese MBO system

Which of the following is NOT the category in Baldrige Award?

Leadership

Business results

Supply chain

Process management

What is the number of principles in Dr. W. Edwards Deming's quality principles?

8

10

12

Which of the following is NOT an advantage of functional structure?

It provides organizations with a clear chain of command

It enables employees to learn about customer expectations

It allows evaluating people based on clear set of responsibilities

It allows people to specialize in the aspect of the work

Which of the following is a requirement of ISO 9000:2000?

Processes to measure and analyze customer satisfaction

An Environmental Policy to be in existence within the organization

Procedures for registration of font-related identifiers

Irregular improvement of internal process and customer satisfaction

The management responsible for the area being audited is responsible for ensuring that actions are taken without undue delay to eliminate detected nonconformities and their causes. Which of the following is a follow-up activity?

The verification of the actions taken

The reporting of contradiction results

The contradiction of the actions taken

The verification of contradiction results

Which of the following total quality tools is employed in the step “plan and implement a solution” in problem solving?

4W and 1H matrix

Pareto chart

Control chart

Scatter plot

Under which perspective of the balance scorecard, would you classify the customer retention measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

What was the role of quality professionals during 1950s?

Troubleshooting and the application of statistical methods

Inspection, sorting, counting, and grading

Quality measurement, quality planning, and program design

Goal setting, education and training, and program design

What is quality?

Meeting and exceeding the customer's expectations

Adding factors to make the customer happy

The degree to which the project meets requirements

Conformance to management's objectives

The quality initiatives focused initially on reducing defects and errors in products and services with the help of which of the following?

Measurement, statistics, and other problem solving tools

Measurement, statistics and cause and effect analysis

Statistics and cause and effect analysis only

Failure modes and cause and effect analysis only

Which of the following performance levels have good results and have emphasized sustained improvement trends?

Reactive approach

Stable formal system approach

Best-in-class performance

Continual improvement emphasized

Do not permit overlong examples or irrelevant discussion. This refers to which of the following discussion skills in meetings?

Summarizing

Managing time

Closing the discussion

Containing digression

Which of the following statements, in relation to psychology, is TRUE for people failing to understand the components of Profound Knowledge?

They see the symptoms but not the deep causes of problems

They do not know when expectations are realistic

They do not understand motivation/why people do what they do

They do not know how to plan and accomplish learning and improvement

Under which perspective of the balance scorecard, would you classify motivated workforce measurement?

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The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Which of the following models value stability?

Organism model

Mechanistic model

Cultural model

Total Quality model

Who stressed that systems must be focused toward a purpose?

Crosby

Ishikawa

Taguchi

Deming

Which of the following leads to questioning and can be tested and validated?

Theory of knowledge

Psychology

Understanding of variation

Appreciation for a system

Which of the following statements, in relation to variation, is TRUE for people failing to understand the components of Profound Knowledge?

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

They do not know the difference between prediction, forecasting, and guesswork

They create cynicism, demoralization, de-motivation, guilt, resentment, burnout, craziness, and turnover

They do not understand the difference between improvement and change

Which of the following represent short-term perspectives and do not encourage long-term improvement as stated by Deming?

Numerical quotas

Constitutional quotas

Political party quotas

Mechanical quotas

All of the followings are true for zero defects concept EXCEPT:

A performance standard for management

A standard of the craftsperson regardless of his or her assignment

A tool of management to communicate to do things right the first time

A motivational, widely understood and accepted program

Which of the following tools plots the relationship of one numeric variable to another on a horizontal and vertical axis and determines the degree of dependency or interdependency of those variables?

Scatter diagram

Check sheet

Fishbone diagram

Pareto chart

Who warned against the use of extrinsic controls that compel behavioral changes in people when rewards and incentive programs are seldom shifted to favor quality improvement?

Oakland

Schein

Cox

Dale

How many maturity levels are there in the Capability Maturity Model?

Four

Five

Six
Seven

Which one of the following depicts aesthetics, which is a dimension of quality?

Exterior finish

Quality of work
Ranking first
Ease of repair

Which of the following firms solely owns the entire material flow and each channel member operates independently?

Horizontal integrated firms
Backward vertical integrated firms
Vertical integrated firms
Conglomerate integrated firms

Which of the following is NOT the deadly disease according to Deming?

Excessive medical costs
Excessive transportation costs
Lack of constancy of purpose
Mobility of management

Which stage in the Shewhart Cycle uses data and facts to analyze the results of the change?

Do stage
Act stage
Plan stage
Check stage

Why product-based definitions of quality suit engineers?

They are concerned with designing of the products
They can judge quality
They can reduce specific standards
They are concerned with translating product requirements into specific components and physical dimensions thereby produced

All of the following are the objectives achieved simultaneously for the sustained growth to be recognized EXCEPT:

Customer satisfaction

Effective leadership

Effective human resources

Integration with the supplier base

Some organizations emphasizes the use of quality tools but failed to do which of the following?

Focus on what is truly important to the distributors

Incorporate continuous improvement efforts

Make fundamental changes in their processes and culture

Focus on what is truly important to the customer

Which of the following tools has a weakness that it does not tell whether the variation is the result of special causes or common causes?

Histogram

Check sheet

Pareto chart

Run chart

All of the following are Juran's basic steps to progress EXCEPT:

Achieve structured improvements on a continual basis combined with dedication and a sense of urgency

Establish an extensive training program

Establish an extensive marketing program

Establish commitment and leadership on the part of higher management

Which of the following statements, in relation to theory of knowledge, is TRUE for people failing to understand the components of Profound Knowledge?

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

They do not know the difference between prediction, forecasting, and guesswork

They create cynicism, demoralization, de-motivation, guilt, resentment, burnout, craziness, and turnover

They do not understand the difference between improvement and change

Which of the following performance levels is a systematic process-based approach, early stage of systematic improvements?

No formal approach

Reactive approach

Stable formal system approach

Best-in-class performance

Which of the following describes a complete data set?

Measures of dispersion

Measures of central tendency

Deductive statistics

Inductive statistics

Which of the following tools represents the entire process at a high or detailed (depending on your use) level of observation, allowing analysis and optimization of workflow?

Check sheet

Pareto diagram

Flow chart

Histogram

Which of the following is NOT the characteristic of the organizations operating according to the principles of quality management?

A total commitment to continually increasing value for customers, investors, and employees

A firm understanding that market driven means that quality is defined by customers, not the company

A commitment to leading people with a bias for broken improvement and miscommunication

A commitment to fundamental improvement through knowledge, skills, problem solving, and teamwork

Ahmad has organized a manual data collection in such a way as to be easily used, understood and analyzed automatically. Which of the following Ahmad has prepared?

Check sheet

Pareto diagram

Flow chart

Histogram

Which of the following is TRUE for design and development outputs under ISO 9001:2000 - Quality Management Systems ?

It Identifies of and corrects against potential conformance

It is used where excess in product will only be apparent in use

It is Insignificance and concealment of significant features of the processes

It meets the input requirements for design and development

What is the purpose of the Control Chart?

To indicate variations, trends, and the amount of change from one time period to another

To show where to apply resources by revealing the significant few from the trivial many

To provide a scientific approach to gathering information to improve the validity of decisions and interventions

To monitor effects of the variables on the difference between target and actual performance

You have to identify which of the following quality leader said that quality is “non faulty system”.

W. Edwards Deming

Joseph M. Juran

Philip B. Crosby

Ishikawa

Goods, healthy individuals and services come under which of the following sections of the organization.

Input

Process

Output

None of the given

What is the objective of a TQM system?

Continuous improvement

Continual improvement

Business improvement

Process improvement

From system perspective, Deming said that which one of the following should inter connected?

Function

Process

Department

Worker

How can the quality be computed?

Quality = Expectation/Performance

Quality = Performance/Expectation

Quality = Performance + Expectation

Quality = Performance-Expectation

In quality meeting, give everyone opportunity to take part in the discussion is called

Manage Time

Manage Meeting

Manage Participants

None of the above

Which of the following statements, in relation to variation, is TRUE for people failing to understand the components of Profound Knowledge?

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

They do not know the difference between prediction, forecasting, and guesswork

They create cynicism, demoralization, de-motivation, guilt, resentment, burnout, craziness. and turnover

They do not understand the difference between improvement and change

Internal customers of the organization are

Customers
Stakeholders
Employees
Suppliers

Whose efforts at quality improvement were supported by massive training programs and top management leadership?

Lebanese
Chinese
Japanese
Vietnamese

Long term growth is the goal of which of the following model

Mechanistic Model
Organic Model
Structural Model
Unstructured Model

Which of the following is a set of two or more interrelated elements of any kind, not being an ultimate indivisible element but a whole that can be divided into parts?

Process thinking
Systems thinking
Analytical thinking
Divergent thinking

When a firm focuses on the whole world as market but does adapt to the local requirements, the firm is called

Global
Glocal
Local
Regional

What was the primary concern of managers during 1980s?

Detection

Strategic impact

Control
Coordination

Which of the following is not included in external failure?

Customer Service

Downtime

Field Service
Warranty Claims

Which of the following forms the basis of TQM as an organization wide process?

Not using the resources of the total organization property
Organizational flexibility and response to change
Not bound in business relationships

Not measuring continuous improvement

In which of the following operations great deal of variations can occur?

Manufacturing

Distribution
Purchasing
Selling

Which of the following is an answer to the Second Absolute of Quality Management as summarized by Crosby; what system is needed to cause quality?

Conformance to requirements

Prevention

Zero defects
Price of nonconformance

Which of the following is an answer to the Third Absolute of Quality Management as summarized by Crosby; what performance standard should be used?

Conformance to requirements
Prevention

Zero defects

Price of nonconformance

Who argued that "the quality of a manufactured product might be described in terms of a set of characteristics"?

Shewhart

Drucker

Crosby

Ishikawa

Environmental management system is discussed in which of the following ISO family of standard.

ISO-14000 (2004)

ISO-3CCC

ISO-10006

ISO-22000

With the total quality perspective, there is an emphasis on continuous improvement of which one of the following?

Products and people only

Processes and people only

Products, processes, and purpose

Products, processes, and people

To beat the competition, organizations often must exceed which of the following?

Customer specifications

Customer expectations

Customer knowledge

Customer beliefs

According to Shewhart, how the fraction of nonconforming items produced by a manufacturing process is studied?

Mathematically

Scientifically

Statistically

Physically

Motorola was committed to which of the following campaigns in 1980s?

Zero defect

Six sigma

Benchmarking

Cause-and-effect analysis

Who has formally linked inspection to quality control?

G.S. Radford

Fredrick W. Taylor

C.T. Friel

F.S. Brackett

Which of the following statements is TRUE about most of the quality problems?

These originate in the quality department where the ultimate responsibility for quality rests

These originate on the shop floor because of waste and product rework

These are the results of management's lack of attention to potential quality improvement ideas

These could be eliminated if shop supervisors monitor their workers more closely

Which of the following are the factors that are present as a natural part of a process?

Common causes of variation

Special causes of variation

Red tape causes

Root causes

When was Systems theory refined?

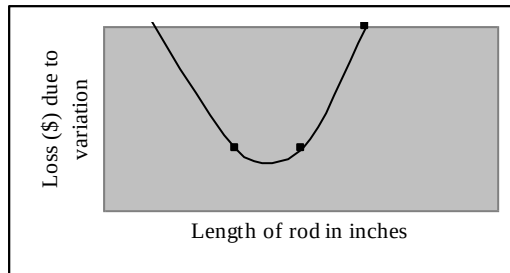
1910s to 1930s

1930s to 1950s

1950s to 1970s

1970s to 1990s

Who represented the following graph?



Feigenbaum

Taguchi

Deming

Shewhart

Which of the following is the arrangement established for validation of processes for production and service provision?

Use of suitable equipment

Requirements for records

Undefined criteria for review

Packaging materials

In the levels of organizational maturity, awakening system precedes which of the following systems?

Dysfunctional System

Developing System

Maturing System

World Class System

In the levels of organizational maturity, developing system precedes which of the following systems?

Dysfunctional System

Awakening System

Maturing System

World Class System

Which of the following statements is TRUE about mission statement?

"The how-to" statement

It precedes the vision statement

It is the picture of company

“What do we want to become” statement

Which of the following is NOT an organizational external opportunity?

Availability of new customers

Removal of barriers that inhibit growth

Entry of lower cost competitors

Market expansion for existing products

Under which perspective of the balance scorecard, would you classify percent sales from new products measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Under which perspective of the balance scorecard, would you classify reduction in unit costs measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Brainstorming refers to which of the following?

Mess Finding

Fact Finding

Problem Finding

Idea Finding

The Pareto Principle is a technique used by quality managers to determine the quality control problems concerning a particular service or manufacturing process to be corrected. Which of the following statements best represents the problems employed by this principle?

In order to minimize financial losses, all problems should be corrected

The majority of defects are caused by a small percentage of the identifiable problems.

In order to achieve zero defects, all quality control problems should be corrected

Generally, 80% of the quality control problems are justifiable for correction via cost-benefit analysis.

What should be the responsibility of managers regarding Quality?

To ensure that purchased product conforms to specified purchase requirements

To act as government officials in the cases of investigation

To assist in securing their performance by other lawyers

To enroll in the courses for which they are asking

Customer satisfaction is the measurements of the performance of the quality management system. What should be the responsibility of organization regarding Customer Satisfaction?

To enforce federal, state laws along with city ordinances

To implement the country's legal requirements for efficient transportation of people

To improve the effectiveness of the quality management system

To apply new professional engineering approaches and techniques

Which of the following stages of learning refers that "You does not know that you do not know"?

Unconscious incompetence

Conscious incompetence

Conscious competence

Unconscious competence

Which of the following stages of learning refers that "You learn to do, but with a conscious effort"?

Unconscious incompetence

Conscious incompetence

Conscious competence

Unconscious competence

The Sitara Corporation has an established "game plan" for its business operations. This game plan reflects the company's awareness of how and where it should compete and against whom the competition should take place. It can be stated that this organization has established which of the following?

Goal
Vision
Policy
Strategy

In which of the following perspectives of a balanced scorecard, the metrics allow the managers to know how well their business is running, and whether the products and services conform to customer's requirements?

The learning & growth perspective
The business process perspective
The customer perspective
The financial perspective

ABC organization has selected the objectives for ISO registration, now what should be the next step.

Submit quality manual for approval
Develop and implement the QMS
Set the registration objectives
Selection of appropriate standard

Being a manager, you are handling customers' complaints, for this purpose you are using different quality tools. Which of the following sequence is most suitable in this regard?

Scatter diagram, Flow chart, Histogram
Fishbone diagram, Check sheet, Pareto diagram
Pareto diagram, Fishbone diagram, Flow chart
None of the given options

There is a high probability for getting a value closer to the mean if:

s is equal to m
s is small
s is large
None of the given options

Which of the following levels of management is mostly a part of steering committee?.

Middle Level Management

Top Level Management

Operational Level Management

None of the given options

Population of Pakistan is increasing at a faster rate. Which of the following theories relates to the mindset to utilize a large number of human resources for increasing the productivity?

Theory of Scarcity

Theory of Abundance

Theory of Constraints

None of given options

1 In the process of acquiring the certificate of ISO 9000, which step follows the step of setting the registration objective?

- a. Develop and implement the QMS
- b. Perform self-analysis audit
- c. Submit quality manual for approval

d. Select the appropriate standards

2 In the process of acquiring the certificate of ISO 9000, which step follows the step of pre-assessment by registrar?

a. Take corrective action

- b. Obtain ISO registration
- c. Perform self-analysis audit
- d. Select a third party registrar

3 In the process of acquiring the certificate of ISO 9000, which step follows the step of developing and implementing the quality and management system?

a. Set the registration objectives

b. Select a third party registrar

- c. Pre-assessment by registrar
- d. Take corrective action

4 In the process of acquiring the certificate of ISO 9000, which step follows the step of selecting the appropriate standards?

- a. Perform self-analysis audit
- b. Submit quality manual for approval

c. Develop and implement the QMS

d. Take corrective action

5 In the process of acquiring the certificate of ISO 9000, which step follows the step of selecting a third party registrar?

a. Take corrective action

b. Obtain ISO registration

c. Perform self-analysis audit

d. Final assessment by registrar

6 In the process of acquiring the certificate of ISO 9000, which step follows the step of performing self-analysis audit?

- a. Final assessment by registrar
- b. Set the registration objectives
- c. Develop and implement the QMS
- d. Submit quality manual for approval**

7 In the process of acquiring the certificate of ISO 9000, which step follows the step of submitting quality manual for approval?

a. Pre-assessment by registrar

b. Take corrective action

c. Perform self-analysis audit

d. Develop and implement the QMS

8 Which of the following is TRUE for higher costs of quality?

a. Doing the wrong thing in the wrong way

b. Doing the wrong thing in the right way

c. Doing the right thing in the wrong way

d. Doing the right thing in the right way

9 For what duration is the ISO certification valid for any organization?

a. 5 years

b. 4 years

c. 3 years

d. 2 years

10 During 1980s, the job characteristic of quality professionals was:

A Educating others

B Achieving personal targets

C Consultative work with other departments

D None of the above

11 During 1800s, the quality was viewed as:

A A problem that can never be solved

B A problem to be solved

C A tool for measuring system

D A remedy to control decline

12 During 1950s, controlling quality meant:

A To understand and address the causes of poor quality

B To simplify and speed up process

C To link outcomes to uses

D Acting on defects

13. Which of the following document(s) is (are) included in the quality system?

a.A quality policy

b.Customer focus

c.Commitment

d.All of the given options

14. Appropriate motives for adopting ISO 9000 include which of the following?

a.To create a quality management system

b.To improve operations

c.To conform to the requirements of customers

d.All of the given options

15. Quality checklists are used to:

a. **Ensure that Quality Assurance steps were followed**

b. Keep quality inspectors busy

c. Inform upper management where failures occur

d. Prevent project audits

Taguchi suggested that loss in a process is increased with increase in which of the following?

Specifications

Standards

Competition

Variability

The products manufactured during 1800s were unique. How quality was ensured in this era?

Through renovation

Through inspection

Through calculation

Through repair

Which of the following is typically viewed to lie between fully vertically integrated firms?

Supply chain management

Production chain management

Value chain management

Demand chain management

Which of the following is not a benefit of long-term supplier relationships?

supplier involvement in design

information system linkages to customer facilities

capacity and facility plans favorable to their customers

all of the above are benefits

The majority of advertisers appeal the public on the basis of which of the following?

Quality of product

Quality of staff

Inferiority of product

Inferiority of service

Which of the following models value stability?

Organism model

Mechanistic model

Cultural model

Total Quality model

What is the relationship between Ishikawa and Deming?

Son and father

Nephew and uncle

Student and teacher

Grandson and grandfather

Which of the following was developed by Motorola to improve its processes by minimizing defects?

ISO 9000

Six sigma

QS 9000

TQM

What was the primary concern of managers during 1980s?

Detection

Strategic impact

Control

Coordination

You can ask direct questions to quiet people and try to control talkative people", for which of the following purpose?

Mobilizing the group

Keeping the group focused

Restating purpose

Starting punctually

Some organizations emphasizes the use of quality tools but failed to do which of the following?

Focus on what is truly important to the distributors

Incorporate continuous improvement efforts

Make fundamental changes in their processes and culture

Focus on what is truly important to the customer

All of the following are the elements of a TQM system EXCEPT:

Leadership

Communications

Measurement

Detentions

The standards for output based on quality and customer- service requirements that originate at the organizational and process levels include requirements for all of the following EXCEPT:

Accuracy

Innovation

Timeliness

Turnover

One of the advantages of team work is:

It breaks down barriers between internal customers and suppliers

It results in promotion

It results in salary increment
None of the given options

Reworking _____ the cost of quality.

Decreases
Increases
Neutralizes

_____ refers to general processes of improvement and encompasses discontinuous improvements

Continuous improvement
Continual improvement
Constant improvement
Consecutive improvement

Total quality costs include:

Prevention costs
Appraisal costs
Failure costs
All of the given options

the job characteristic of quality professionals is:

Educating others
Achieving personal targets
Consultative work with other departments
None of the above

Why the factory managers created inspection departments?

To keep defective products aside, ensuring they do not reach the customers
To ensure quantity of goods/services
To count, grade, and rework

All of the above

The "Father" of statistical quality control is:

F. W. Taylor

Joseph M. Juran

Philip Crosby

Walter Shewhart

1. While setting Quality objective_____ to be considered.

a. Customer need

b. Organizational need

c. Supplier need

d. Worker need

(a)

2. Which of the following is for Environment management?

a. ISO-9000

b. ISO-14000

c. ISO-26000

d. ISO-31000

(b)

3. According to Deming, Quality problems are

a. Due to management

b. Due to method

c. Due to machine

d. Due to material

(a)

4. While setting Quality objective, _____ to be considered.

a. Material quality

b. Customer need

c. Market demand

d. All of the above

(b)

5. The objective of ISO-9000 family of Quality management is

a. Customer satisfaction

b. Employee satisfaction

c. Skill enhancement

d. Environmental issues

(a)

6. Total Quality Management (TQM) focuses on

a. Employee

b. Customer

c. Both (a) and (b)

d. None of the above

(c)

7. The role of management is to

a. provide Resources

b. define EMS

c. monitor the effectiveness of the system

d. All of the above

(d)

8. Tally chart is

a. Process monitoring tool

b. Data collection tool

c. Process planning tool

d. None of the above

(b)

9. Control chart is a

a. Process monitoring tool

b. Process control tool

c. Both (a) and (b)

d. None of the above

(c)

10. Which ISO standard is used in international automobile companies to set automotive quality system standards ?

- a. ISO 14000
- b. TS 16949
- c. ISO 9000
- d. none of the above

(B)

11. ISO 14000 quality standard is related with

- a. Environmental management systems
- b. Automotive quality standards
- c. Eliminating poor quality
- d. Customer satisfaction

(a)

1-Malcolm Baldrige national quality award is for (MBNQA)

- a. Total Quality Management
- b. International Standard Organization
- c. Total Productive Maintenance
- d. Total Quality Control

(Ans:a)

2-The process mapping is a _____ diagram.

- a. Data flow
- b. Work flow

- c. Circular
- d. Audit

(Ans:b)

-

3-Control chart is a

- a. Process monitoring tool
- b. Process control tool
- c. Both (a) and (b)
- d. None of the above

(Ans:c)

4-The objective of ISO-9000 family of Quality management is

- a. Customer satisfaction
- b. Employee satisfaction
- c. Skill enhancement
- d. Environmental issues

(Ans:a)

5-Total Quality Management (TQM) focuses on

- a. Employee
- b. Customer
- c. Both (a) and (b)
- d. None of the above

(Ans:c)

6-Which of the following is responsible for quality objective?

- a. Top level management
- b. Middle level management
- c. Frontline management
- d. All of the above

(Ans:a)

7-The following is (are) the machine down time.

- a. Waste
- b. No material
- c. Breakdown
- d. All of the above

(Ans:d)

8-TQM & ISO both focuses on

- a. Customer
- b. Employee
- c. Supplier
- d. All of the above

(Ans:a)

9-According to Deming, Quality problems are

- a. Due to management
- b. Due to method
- c. Due to machine
- d. Due to material

(Ans:a)

10-While setting Quality objective, _____ to be considered.

- a. Material quality
- b. Customer need
- c. Market demand
- d. All of the above

(Ans:b)

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MGT510- Total Quality Management

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Mega File of Final term Papers & Solved Quizzes

For more Help Rep At

Faheem_saqib2003@yahoo.com

Faheem.saqib2003@gmail.com

0334-6034849



FINAL TERM EXAMINATION

Fall 2009

MGT510- Total Quality Management

Question No: 1 (Marks: 1) - Please choose one

With the total quality perspective, there is an emphasis on continuous improvement of which one of the following?

- ▶ Products and people only
- ▶ Processes and people only
- ▶ Products, processes, and purpose
- ▶ Products, processes, and people **Correct**

Question No: 2 (Marks: 1) - Please choose one

To beat the competition, organizations often must exceed which of the following?

- ▶ Customer specifications
- ▶ Customer expectations **Correct**
- ▶ Customer knowledge
- ▶ Customer beliefs

Question No: 3 (Marks: 1) - Please choose one

According to Shewhart, how the fraction of nonconforming items produced by a manufacturing process is studied?

- ▶ Mathematically
- ▶ Scientifically
- ▶ Statistically **Correct**

- ▶ Physically

Question No: 4 (Marks: 1) - Please choose one

Motorola was committed to which of the following campaigns in 1980s?

- ▶ Zero defect
- ▶ Six sigma **Correct**
- ▶ Benchmarking
- ▶ Cause-and-effect analysis

Question No: 5 (Marks: 1) - Please choose one

Who has formally linked inspection to quality control?

- ▶ G.S. Radford **Correct**
- ▶ Fredrick W. Taylor
- ▶ C.T. Friel
- ▶ F.S. Brackett

Question No: 6 (Marks: 1) - Please choose one

Which of the following statements is TRUE about most of the quality problems?

- ▶ These originate in the quality department where the ultimate responsibility for quality rests
- ▶ These originate on the shop floor because of waste and product rework
- ▶ These are the results of management's lack of attention to potential quality improvement ideas **Correct**
- ▶ These could be eliminated if shop supervisors monitor their workers more closely

Question No: 7 (Marks: 1) - Please choose one

Which of the following are the factors that are present as a natural part of a process?

- ▶ Common causes of variation **Correct**
- ▶ Special causes of variation
- ▶ Red tape causes
- ▶ Root causes

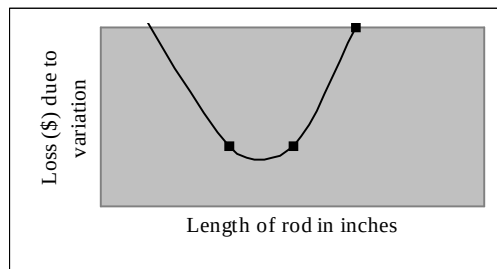
Question No: 8 (Marks: 1) - Please choose one

When was Systems theory refined?

- ▶ 1910s to 1930s
- ▶ 1930s to 1950s
- ▶ 1950s to 1970s **Correct**
- ▶ 1970s to 1990s

Question No: 9 (Marks: 1) - Please choose one

Who represented the following graph?



- ▶ Feigenbaum
- ▶ Taguchi **Correct**
- ▶ Deming
- ▶ Shewhart

Question No: 10 (Marks: 1) - Please choose one

Which of the following is the arrangement established for validation of processes for production and service provision?

- ▶ Use of suitable equipment
- ▶ Requirements for records
- ▶ Undefined criteria for review **Correct**
- ▶ Packaging materials

Question No: 11 (Marks: 1) - Please choose one

All of the following are the consideration of the organizations when analyzing how all the processes interact to make sure that all processes operate as an efficient network EXCEPT:

- ▶ Ensuring inputs, activities and outputs are clearly defined and controlled
- ▶ Managing risks and opportunities
- ▶ Monitoring inputs and outputs to confirm that, group processes are unrelated
- ▶ Establishing data analysis that facilitates continual improvement across all processes **Correct**

Question No: 12 (Marks: 1) - Please choose one

In the levels of organizational maturity, awakening system precedes which of the following systems?

- ▶ Dysfunctional System **Correct**
- ▶ Developing System
- ▶ Maturing System
- ▶ World Class System

Question No: 13 (Marks: 1) - Please choose one

In the levels of organizational maturity, developing system precedes which of the following systems?

- ▶ Dysfunctional System
- ▶ Awakening System
- ▶ Maturing System **Correct**
- ▶ World Class System

Question No: 14 (Marks: 1) - Please choose one

Which of the following statements is TRUE about mission statement?

- ▶ "The how-to" statement
- ▶ It precedes the vision statement **Correct**
- ▶ It is the picture of company
- ▶ "What do we want to become" statement

Question No: 15 (Marks: 1) - Please choose one

Which of the following is NOT an organizational external opportunity?

- ▶ Availability of new customers
- ▶ Removal of barriers that inhibit growth
- ▶ Entry of lower cost competitors **Correct**
- ▶ Market expansion for existing products

Question No: 16 (Marks: 1) - Please choose one

Under which perspective of the balance scorecard, would you classify inventory management measurement?

- ▶ The customer perspective
- ▶ The business process perspective
- ▶ The financial perspective **Correct**
- ▶ The learning & growth perspective

Question No: 17 (Marks: 1) - Please choose one

Under which perspective of the balance scorecard, would you classify percent sales from new products measurement?

- ▶ The customer perspective
- ▶ The business process perspective
- ▶ The financial perspective
- ▶ The learning & growth perspective **Correct**

Question No: 18 (Marks: 1) - Please choose one

Under which perspective of the balance scorecard, would you classify reduction in unit costs measurement?

- ▶ The customer perspective **Correct**
- ▶ The business process perspective
- ▶ The financial perspective
- ▶ The learning & growth perspective

Question No: 19 (Marks: 1) - Please choose one

Brainstorming refers to which of the following?

- ▶ Mess Finding
- ▶ Fact Finding
- ▶ Problem Finding **Correct**
- ▶ Idea Finding

Question No: 20 (Marks: 1) - Please choose one

The Pareto Principle is a technique used by quality managers to determine the quality control problems concerning a particular service or manufacturing process to be corrected. Which of the following statements best represents the problems employed by this principle?

- ▶ In order to minimize financial losses, all problems should be corrected
- ▶ The majority of defects are caused by a small percentage of the identifiable problems.
- ▶ In order to achieve zero defects, all quality control problems should be corrected
- ▶ Generally, 80% of the quality control problems are justifiable for correction via cost-benefit analysis. **Correct**

Question No: 21 (Marks: 1) - Please choose one

Which of the following is the contributor to materials as a cause in free-throwing cause-and-effect diagram?

- ▶ Basketball
- ▶ Games
- ▶ Video camera **Correct**
- ▶ Touched rim

Question No: 22 (Marks: 1) - Please choose one

What should be the responsibility of managers regarding Quality?

- ▶ To ensure that purchased product conforms to specified purchase requirements **Correct**
- ▶ To act as government officials in the cases of investigation
- ▶ To assist in securing their performance by other lawyers
- ▶ To enroll in the courses for which they are asking

Question No: 23 (Marks: 1) - Please choose one

Customer satisfaction is the measurements of the performance of the quality management system. What should be the responsibility of organization regarding Customer Satisfaction?

- ▶ To enforce federal, state laws along with city ordinances
- ▶ To implement the country's legal requirements for efficient transportation of people **Correct**
- ▶ To improve the effectiveness of the quality management system
- ▶ To apply new professional engineering approaches and techniques

Question No: 24 (Marks: 1) - Please choose one

Which of the following stages of learning refers that "You does not know that you do not know"?

- ▶ Unconscious incompetence **Correct**
- ▶ Conscious incompetence
- ▶ Conscious competence
- ▶ Unconscious competence

Question No: 25 (Marks: 1) - Please choose one

Which of the following stages of learning refers that "You learn to do, but with a conscious effort"?

- ▶ Unconscious incompetence
- ▶ Conscious incompetence
- ▶ Conscious competence **Correct**
- ▶ Unconscious competence

Question No: 26 (Marks: 1) - Please choose one

The Sitara Corporation has an established "game plan" for its business operations. This game plan reflects the company's awareness of how and where it should compete and against whom the competition should take place. It can be stated that this organization has established which of the following?

- ▶ Goal
- ▶ Vision
- ▶ Policy
- ▶ Strategy **Correct**

Question No: 27 (Marks: 1) - Please choose one

In which of the following perspectives of a balanced scorecard, the metrics allow the managers to know how well their business is running, and whether the products and services conform to customer's requirements?

- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ The learning & growth perspective
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ The business process perspective
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ The customer perspective **Correct**
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ The financial perspective

Question No: 28 (Marks: 1) - Please choose one

ABC organization has selected the objectives for ISO registration, now what should be the next step.

- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Submit quality manual for approval
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Develop and implement the QMS
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Set the registration objectives **Correct**
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Selection of appropriate standard

Question No: 29 (Marks: 1) - Please choose one

Being a manager, you are handling customers' complaints, for this purpose you are using different quality tools. Which of the following sequence is most suitable in this regard?

- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Scatter diagram, Flow chart, Histogram
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Fishbone diagram, Check sheet, Pareto diagram
- ☐ ☐ ☐ ☐ ☐ ☐ ▶ ☐ Pareto diagram, Fishbone diagram, Flow chart

☐ ☐ ☐ ☐ ☐ ☐ ☒ None of the given options **Correct**

Question No: 30 (Marks: 1) - Please choose one

There is a high probability for getting a value closer to the mean if:

- ▶ s is equal to m
- ▶ s is small
- ▶ s is large
- ▶ None of the given options **Correct**

Question No: 31 (Marks: 1) - Please choose one

Which of the following levels of management is mostly a part of steering committee?.

- [illegible]

Question No: 32 (Marks: 1) - Please choose one

Population of Pakistan is increasing at a faster rate. Which of the following theories relates to the mindset to utilize a large number of human resources for increasing the productivity?

- ☐ ☐ ☐ ☐ ☐ ☐ ☒ Theory of Scarcity
☐ ☐ ☐ ☐ ☐ ☐ ☒ Theory of Abundance
☐ ☐ ☐ ☐ ☐ ☐ ☒ Theory of Constraints
☐ ☐ ☐ ☐ ☐ ☐ ☒ None of given options **Correct**

Question No: 33 (Marks: 3)

The President of Samsung Electronics has appointed the Manager Quality as 'Management Representative'. Discuss the authority and responsibility of the Management Representative.

Authority and Responsibility:

- a) To ensure that needed process for the quality management system are established, maintained and implemented.
- b) To report to top management regarding the performance of the quality management system and any improvement. if required.
- c) To ensure the awareness of customer requirements throughout the organization.

Question No: 34 (Marks: 3)

Flow chart is an effective tool for data collection and analysis. Comment on this statement.

Ans:

This is right flow chart is effective tool for data collection and analysis but it not gives a detailed picture. It normally helps people to visualize simple but important changes that could be happen in process. It also uses to identify the quality problems and areas for improvement.

Question No: 35 (Marks: 3)

What are the qualities of an effective team and the circumstances in which teams realize that the plan is not improved? (Marks=2+1)

Ans:

Qualities

- 1- Effective team must achieve its goals
- 2- It improves the quality performance quickly.
- 3- Team must maintain and increase it strength as a unit.

Plan is not improved in below circumstances if it happens.

- 1- Goals are not clear
- 2- Improvement plan is not available
- 3- Roles are not defined properly
- 4- Loose communication

Question No: 36 (Marks: 5)

The policy of ABC Machine Works Ltd. is to provide products and services that meet the requirements of our customers in accordance with our quality policy. How the management and employees of this company are committed to achieving this policy?

Ans:

When a company is committed to provide the products and services that meet the requirements of our customers in accordance with their quality policy then it means company has some quality policy and certifications from third party due to which are providing they are assured of product and services.

Because in company quality policy it is stated that company will follow the international and health related standards to provide the best product and services to others.

Like in ISO 9001:2000, quality management system is described and company have to follow it to remain ISO certified.

Question No: 37 (Marks: 5)

Performing is one of the stages of Team Growth. What can be the feelings of leader during this stage? Also discuss the role of a leader in “Performing”? (Marks=3+2)

Ans:

Performing stage is very critical state for a leader because this is the time of to show and deliver something to others. To prove himself as leader.

In this stage leader has to proven his role as a leader. As a leader he has to guide his team, motivate his team, take the responsibility and finish the work with team work.

Communicate everything clearly to team to perform at this stage. A leader should take his team as a unit with increasing strength with the passage of time.

Question No: 38 (Marks: 5)

Explain how the solution is planned and implemented in the seven-step method for problem solving.

Question No: 39 (Marks: 10)

Like many other companies, Toppan Printing Company wants to adopt policy deployment. What will be the procedure to adopt such process?

Ans

This policy is deployed by Japanese. Name is Hoshin Planning which also named as Policy Deployment.

Policy deployment is a systems approach to managing change in critical business processes. It emphasizes organization-wide planning and setting of priorities, providing resources to

meet objectives, and measuring performance as a basis for improving performance.

Policy deployment is essentially a TQ-based approach to executing a strategy.

In policy deployment top management is responsible for developing and communicating a vision,

then building organization-wide commitment to its achievement.

Policy deployment provides frequent evaluation and modification based on feedback from regularly scheduled audits of the process. Plans and actions are developed based on analysis of the root causes of a problem, rather than only on the symptoms. Planning has a high degree of detail, including the anticipation of possible problems during implementation. The emphasis is on the improvement of the process, as opposed to a results-only orientation.

For example, to illustrate the need for policy deployment, let us consider the following case:

A president of an airline company believes in safety and makes it as corporate goal and announced that make sure that safety is maintained throughout the company. Now every

one is believe in safety as a job. But in this policy deployment safety is not practicing it is only lipsing. But if president states that safety is a company policy and one safety manger will look after this then this policy should be practicing.

If we see the procedure than it starts with the senior managers of the company. The senior managers establish the vision and core objectives of the company.

“Middle management negotiates with senior management regarding the goals that will achieve the objectives. Goals specify numerically the degree of change that is expected. Strategies specify the means to achieve the goals. They include more specific actions to be taken.

Middle managers are responsible for managing the resources to accomplish the goals.

Middle

Management then negotiates with the implementation teams regarding the performance measures that are used to indicate progress toward accomplishing the strategies.

Question No: 40 (Marks: 10)

Individuals must continue to learn. Complete understanding of how things work involves learning disciplines. What are those disciplines? Explain them briefly.

Ans

As per the current period of learning, an individual should continue to learn, gain knowledge. Prof of MIT portrays, Peter Senge, said organizational learning is not only linked with knowledge, it is beyond that and organizational learning needs a deeper and a complete understanding of how things work involving learning discipline. It involves five learning disciplines.

1- Team Learning:

In this learning discipline, use of dialogue is very much important. A person should learn that how to deliver the dialogue and better conversation to understand the things to others.

2- Shared Vision:

In this discipline, all members are working jointly for a common goal or view for which all are seeking. All members shared their visions and then commonly working on common goal.

3- Mental Models:

In this model, we have to understand that how our cognitive schema affects our view of the world. Because after training of this learning discipline our view of world is changed. In continues learning model could be more accurate.

4- Personal Mastery:

In this discipline, both type of continuous improvement is involved i.e. a personal and organizational. A person is continuing to drive himself for improvement and also for organization development.

5- System Thinking

In this, we have to learn the multiple cause effect relationships, how and why they occurred. We also have to learn that how they are interconnected in organization and society. How organization and society affect each other.

Mgt510 Total Quality Management Spring Fall Final Term 2011 19-02-2011

1. What should be the responsibility of managers regarding Identification and Traceability of product?

- To enforce federal, state laws along with city ordinances
- **To identify the product by suitable means throughout product realization**
- To implement the country's legal requirements for efficient transportation of people
- To apply new professional engineering approaches and techniques

2. Why is there a need for identification and traceability for control of products?

- Due to packaging materials
- Due to use of suitable equipment
- Due to requirements for records
- **Due to risk mitigation**

3. What is the traditional view of quality?

- Quality is measured by establishing an acceptable level of conformance
- Quality is measured by establishing an acceptable level of employee satisfaction
- **Quality is measured by establishing an acceptable level of nonconformance**
- Quality is measured by establishing an acceptable level of both conformance and employee satisfaction

4. Customer needs must be linked to which of the following?

- Organization's strategic planning, process improvement and finances
- Product design, product development and process improvement and tax returns
not sure
- Organization's strategic planning, product design and process improvement
- Process improvement, managerial planning, accounting tactics, and finances

5. When was the definition of total quality endorsed?

- In 1950
- In 1970
- In 1985
- **In 1992**

6. When were the behavioral theories developed?

- 1930s
- 1940s
- 1950s
- **1960s**

7. Edward Deming is best known for which of the following?

- Acquisition Integration Approaches
- **Seven Deadly Diseases**
- Brand Personality
- ERG Theory

8. By whom was the Deming Application Prize instituted in 1951?

- **Union of Japanese Scientists and Engineers**
-

9. Which of the following is NOT a requirement of ISO 9001:2000?

- Processes to describe and monitor, what each customer wants
- Processes to measure and analyze customer satisfaction
- **Demonstration of continual improvement of internal process and customer satisfaction**
- Environmental Policy to be in existence within the organization

10. All of the following are the responsibilities of top management EXCEPT:

- Setting policies and objectives and leading the organization
- Provision of organization structure and resources
- Maintenance of current performance
- **Implementation of short-term plans and dealing with the routine work of their team**

11. Which of the following is a follow up activity relating to internal audit?

- Collection of quality cost data
- Questionnaires and surveys
- **Verification of implementation**
- Utilization of technologies

12. All of the following are the consideration of the organizations when analyzing how all the processes interact to make sure that all processes operate as an efficient network EXCEPT:

- Ensuring inputs, activities and outputs are clearly defined and controlled
- Managing risks and opportunities
- **Monitoring inputs and outputs to confirm that, group processes are unrelated**
- Establishing data analysis that facilitates continual improvement across all processes

13. What should be the responsibility of managers regarding Identification and Traceability of product?

- To enforce federal, state laws along with city ordinances
- **To identify the product by suitable means throughout product realization**
- To implement the country's legal requirements for efficient transportation of people
- To apply new professional engineering approaches and techniques

14. SPC focuses on the process to produce the output. The use of SPC results in the form of

- **Both the consistency in product and satisfied customer**
- Consistency in product only
- Satisfied customer only
- Fluctuation in customer's demands only

15. Attribute process control charts are used for:

- Defect Prevention
- Process improvement
- Defect Deductions
- None of the given options

16. Which of the following is used to detect changes in variation within sub-groups?

- X-Bar Chart
- R- Chart
- P-Chart
- C-Chart

17. Which of the following will fall in the category of non conformity if the range for the length of a yard stick is ± 1000 mm?

- Yard stick which is 999 mm long
- Yard stick which is 1001 mm long
- Yard stick which is 1000.99 mm long
- Yard stick which is 1001.01 mm long

18. An effective team has all the characteristics EXCEPT:

- Goal centered
- Supportive
- Openness
- Self centered

19. Which of the following teams should be formulated if marketing department of ABC Company reports some problem regarding product design?

- Steering committee
- Self managed team
- Cross functional team
- Project team

20. Which of the following skills is most important for Self Managed Teams?

- Conceptual skills
- Technical skills
- Social Networking skills
- Writing skills

21. Implementation of Enterprise Resource Planning (ERP) system to gain a competitive advantage is an example of

- Vision
- Mission
- **Strategies**
- Action plans

22. Balance score card is a quality planning tool which is implemented to the companies intending to reach at

- Level 1
- Level 2
- Level 4
- **Level 7**

23. You are working in an organization which is manufacturing leather bags. The purchase of leather from supplier will be the

- Key performance indicator
- Critical success factor
- Core process
- **None**

24. How can the quality be computed?

- $\text{Quality} = \text{Expectation/Performance}$
- **$\text{Quality} = \text{Performance/ Expectation}$**
- $\text{Quality} = \text{Performance} + \text{Expectation}$
- $\text{Quality} = \text{Performance} - \text{Expectation}$

25. What is the traditional view of quality?

- Quality is measured by establishing an acceptable level of conformance
- Quality is measured by establishing an acceptable level of employee satisfaction
- Quality is measured by establishing an acceptable level of nonconformance
- **Quality is measured by establishing an acceptable level of both conformance and nonconformance**

26. Soft system thinking is used in which of the following situations?

- Friendly, certainty or ease in relationships
- Certain and distinct relationships
- **Disciplined, organized and lively culture**
- Conflict, uncertainty or unease in relationships

27. Total quality organizations expect to make at least improvements

➤ **10 or more improvements per employee per year**

- 40 or more improvements per employee per year
- 60 or more improvements per employee per year
- 80 or more improvements per employee per year

28. Which of the following is not a tenet of Crosby?

- Conformance to requirements.
- Zero defects
- Fitness for use
- Quality is free

29. The quality dimension that relates to the expected operational life of a product is:

- Durability.
- Serviceability.
- Perceived Quality.
- **Reliability.**

Short Questions

1. **How can we differentiate b/w leadership and management in an organization? 3**
2. **What are the various responsibilities of a steering committee and how can the leader of a team make the team members aware of the group process? 3**
3. **What are the basic types of statistical process control charts and how can you compute the upper control limit and the lower control limit? 3**
4. **Explain appraisal costs with the help of examples. 3**
5. **Using the quality control tool of run chart, discuss that how it can help in managing absenteeism in a production based organization. 5**

6. What are the requirements of total quality for human resources and how can the managers ensure empowerment in the organization? **5**
7. Discuss the motivational factors that stimulate companies to obtain ISO certification. **5**
8. What are the various approaches that the organizations use to achieve cost leadership for high volume products? How can the benefits resulting from cost leadership be cancelled out? **5**

MGT510 Total Quality Management (5-12-2010)

Paper2

Q.No.1: Why did Deming believe that managers should stop decision purely based on cost? **3marks**

Q.No.2: The European Quality Award is divided into two parts to whom these are awarded? **3marks**

Q.No.3: What were the structured rationality and philosophy towards change in the mechanistic model? **(2.5+2.5) 5marks**

Q.No.4: The (SPIOC) Supplier-Input-Process-Output-Customer diagram induced a high levelmap the process that “map out its basic steps” Explain the element of this process? **5marks**

Q.No.5: Briefly explain as to how a management can be improved? **5marks**

MGT510 Total Quality Management (5-12-2010)

Paper3

Q.No.1: How does Juran's Philosophy fit wall into existing management? **3Marks**

Q.No.2: How quantity is defined by the five leaders in quantity revolution? **3marks**

Q.No.3: Mutually beneficial supplier's relationship and process approach benefits? **5marks**

Q.No.4: Briefly explain as to how a management can be improved? **5marks**

Q.No.5: Why did Deming believe the business should not exist simply for profit? **5marks**

Total Quality Management – MGT510

Note: Choices with marked * are verified answers

Quiz no.1

Question # 1 o f 15 (Start time: 11:02:05 PM) Total Marks: 1

In total quality perspective, quality planning, and strategic business planning are treated as which of the following ?

Different

***Consistent**

Indistinguishable

Dissimilar

Question # 2 of 15 (Start time: 11:03:13 PM)

Total Marks: 1

The marketing team of a company carried out a survey concerning expectations of the customers from the product. What should be the product development department of the company design?

***The product meeting requirements of the customers**

The product meeting requirements of the distributors

The product meeting requirements of the management

The product meeting requirements of the shareholders

Question # 3 of 15 (Start time: 11:04:29 PM)

Total Marks: 1

Which of the following results in low costs?

***High productivity and high capacity utilization**

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

Question # 4 of 15 (Start time: 11:05:35 PM)
Total Marks: 1

Which of the following are generally easy to detect using statistical methods, and it is usually economical to remove them?

***Common causes**

Special causes

Implicit causes

Assignable causes

Question # 5 of 15 (Start time: 11:07:10 PM)
Total Marks: 1

Which one of the following depicts feature, which is a dimension of quality?

Product looks and sounds

Brand name and price

Bells and whistles of the product

***Ability to repair a product**

Question # 6 of 15 (Start time: 11:08:37 PM)

Total Marks: 1

During which century, the manufacturers began to include quality processes in quality practices?

17th century

18th century

19th century

***20th century**

Question # 7 of 15 (Start time: 11:09:47 PM)
Total Marks: 1

Which of the following models has the viewpoint that change and learning are valued in themselves?

***Organism model**

Mechanistic model

Cultural model

Total Quality model

Question # 8 of 15 (Start time: 11:11:19 PM)
Total Marks: 1

After the war, Deming was invited to which of the following countries?

Germany

***Japan**

France

China

Question # 9 of 15 (Start time: 11:12:46 PM)
Total Marks: 1

Which of the following statements is TRUE for the organizations focusing problem-solving and continuous improvement?

Fail to emphasize error prevention and design quality

***Fail to focus on what is truly important to the customer**

Fail to identify the best times about the situation in the past

Fail to integrate all organizational functions

Question # 10 of 15 (Start time: 11:14:06 PM)
Total Marks: 1

Quality applies to which aspect(s) of the organization?

Products only

Products and services

People and product

***Products, services, and people**

Question # 11 of 15 (Start time: 11:14:38 PM)
Total Marks: 1

Which of the following statements holds true for quality assurance?

To top management's intention regarding quality

Functions determining implementation of the quality policy

***Actions to provide confidence of satisfying quality requirements**

Responsibilities and processes, which implement quality management

Question # 12 of 15 (Start time: 11:16:08 PM)
Total Marks: 1

Which of the following models reflect horizontal processes beginning with suppliers and ending with customers?

Organism model

Mechanistic model

Cultural model

***Total Quality model**

Question # 13 of 15 (Start time: 11:17:14 PM)
Total Marks: 1

Which of the following is concerned about quality for achievement of TQM?

The Managing Director

The Operative

The Quality Manager

***Everyone in the organization**

Question # 14 of 15 (Start time: 11:18:39 PM)
Total Marks: 1

Which one of the following depicts durability – a dimension of quality?

Exterior finish

Quality of work

***Useful life**

Ease of repair

Question # 15 of 15 (Start time: 11:19:36 PM)
Total Marks: 1

"Not giving respect to other people's contributions", depicts which of the following self-oriented behavior?

Diverting
Excluding

***Belittling**

Blocking

Quiz No .2

which century, the manufacturers began to include quality processes in quality practices?

17th century

18th century

19th century

***20th century**

In the new paradigm, managers view error as which one of the following?

A threat

A substitute

***An opportunity**

An end

Whose concepts are referred to as statistical quality control (SQC)?

***Shewhart's**

Deming's

Juran's

Crosby's

In which of the following operations great deal of variations can occur?

***Manufacturing**

Distribution

Purchasing

Selling

Which of the following are generally easy to detect using statistical methods, and it is usually economical to remove them?

***Common causes**

Special causes

Implicit causes

Assignable causes

Who argued that beauty could be understood only after exposure to a series of objects that display its characteristics?

Drucker

Crosby

Deming

***Plato**

During which war, Deming taught quality control courses as part of the national defense efforts?

World War I

***World War II**

Mexican-American War

The Civil War

Which one of the following depicts feature, which is a dimension of quality?

Product looks and sounds

Brand name and price
Bells and whistles of the product

***Ability to repair a product**

The majority of advertisers appeal the public on the basis of which of the following?

***Quality of product**

Quality of staff

Inferiority of product

Inferiority of service

The standards for output based on quality and customer-service requirements that originate at the organizational and process levels include requirements for all of the following EXCEPT:

Accuracy

Innovation

Timeliness

Turnover

"Not giving respect to other people's contributions", depicts which of the following self-

oriented behavior?

Diverting
Excluding

***Belittling**

Blocking

After the war, Deming was invited to which of the following countries?

Select correct option:

Germany

***Japan**

France

China

Which of the following results in low costs?

***High productivity and high capacity utilization**

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

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The customer value strategy may be addressed in which of the following topics?

Projects
Products

*** Services**

Measurement

Who developed Quality Function Deployment (QFD)?

Frederick Taylor

Joseph M. Juran

***Y. Akao**

Armand Feigenbaum

TQM Quiz no 3

Question # 1 of 15 (Start time: 12:44:29 PM)
Total Marks: 1

How the Baldrige Award recognizes the importance of total quality principles?

Through independence with overall business planning

Through a no relation with overall business planning

***Through integration with overall business planning**

Through dependence with overall business planning

Question # 2 of 15 (Start time: 12:45:33 PM)
Total Marks: 1

Which of the following models considers the goal of meeting individual needs?

Organism model

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Mechanistic model

***Cultural model**

Total Quality model

Question # 3 of 15 (Start time: 12:46:26 PM)
Total Marks: 1

All of the following are the initial considerations in the Hoshin Kanri approach to business system change EXCEPT:

Measuring the business system as a whole

Setting core objectives of the business

Understanding the environmental situation in which the business operates

***Ceasing resources for performing activities to achieve business objectives**

Question # 4 of 15 (Start time: 12:47:48 PM)
Total Marks: 1

Who argued that beauty could be understood only after exposure to a series of objects that display its characteristics?

Drucker

Crosby
Deming

***Plato**

Question # 5 of 15 (Start time: 12:48:17 PM)
Total Marks: 1

Which of the following statements is TRUE for ISO 9000?

***Describes the principles of a quality management system and defines the terminology**

Describes the requirements relative to a quality management system either for internal use or for contractual or certification purposes

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Intended for internal use and not for contractual purposes, focuses particularly on continually improving performance

Contains the guidelines for auditing quality management and/or environmental management systems

Question # 6 of 15 (Start time: 12:49:39 PM)
Total Marks: 1

Which of the following statements holds true for quality assurance?

To p management's intention regarding quality

Functions determining implementation of the quality policy

***Actions to provide confidence of satisfying quality requirements**

Responsibilities and processes, which implement quality management

Question # 7 of 15 (Start time: 12:51:05 PM)
Total Marks: 1

Which of the following is an answer to the Second Absolute of Quality Management

as summarized by Crosby; what system is needed to cause quality?

Conformance to requirements

***Prevention**

Zero defects

Price of nonconformance

Question # 8 of 15 (Start time: 12:52:13 PM)
Total Marks: 1

All of the following are the reasons why Total Quality Management programs are not prolific for many companies, EXCEPT:

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TQM focuses on minimum standards

TQM develops its own cumbersome bureaucracy
TQM has no place for love and passion

***TQM focuses on customers and stakeholders**

Question # 9 of 15 (Start time: 12:53:15 PM)
Total Marks: 1

What is the other name used for Deming Cycle?

***Shewhart cycle**

Product life cycle

Data life cycle management

Information life cycle management

Question # 10 of 15 (Start time: 12:54:15 PM)
Total Marks: 1

What is necessary for a system to be effective?

***The components of any system must work together for the system**

The components of any system must not work together for the system

The components of any system should not be organized for the system

The components of any system should provide disruption for the system

Question # 11 of 15 (Start time: 12:55:08 PM)
Total Marks: 1

Which of the following is NOT the criterion

under Canadian Awards for Business
Excellence?

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Leadership

***Planning for contraction**

Process optimization

Supplier focus

**Question # 12 of 15 (Start time: 12:56:32 PM)
Total Marks: 1**

Which of the following is true for subsystems?

These are not linked together as internal customers and suppliers

These are linked together as external customers and suppliers

These are linked together as external and internal customers and suppliers

***These are linked together as internal customers and suppliers**

Question # 13 of 15 (Start time: 12:58:00 PM)

Total Marks: 1

What is the relationship between Ishikawa and Deming?

Son and father

Nephew and uncle

***Student and teacher**

Grandson and grandfather

Question # 14 of 15 (Start time: 12:58:51 PM)
Total Marks: 1

Which of the following statements is TRUE for little q?

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***Striving for quality in a limited or specific area**

Searching for specific lines of text containing a particular pattern

Matching rules encoded in a string according to certain syntax rules

Giving combination of values for the free variables

Question # 15 of 15 (Start time: 01:00:14 PM)
Total Marks: 1

The Capability Maturity Model can help in all of the following EXCEPT:

Setting process improvement goals and priorities

Providing guidance for quality processes

Providing a yardstick for assessing current practices

Documenting procedures as to how the various processes for design operate

Quiz no - --3

Question # 1 of 15 (Start time: 11:56:02 PM)
Total Marks: 1

The customer value strategy may be addressed in which of the following topics?

Select correct option:

Projects

Products

Services

***Measurement**

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Question # 2 of 15 (Start time: 11:57:11 PM)

Total Marks: 1

Who is known as a father of modern-day statistical quality control?

Select correct option:

***Shewhart**

Deming

Juran

Crosby

Question # 3 of 15 (Start time: 11:57:40 PM)

Total Marks: 1

Which of the following forms the basis of TQM as an organization wide process?

Select correct option:

Not using the resources of the total organization

properly

Organizational flexibility and response to change

Not bound in business relationships

Not measuring continuous improvement

Question # 4 of 15 (Start time: 11:58:54 PM)
Total Marks: 1

How should the benchmarking of individual processes as well as customer satisfaction be employed?

Select correct option:

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As a standard tool

As a defective tool

***As an improvement tool**

As an instrumental tool

Question # 5 of 15 (Start time: 11:59:39 PM)
Total Marks: 1

Identify the example of prevention costs.

Select correct option:

***Quality planning**

Re-inspection

Product recalls

Customer returns

Question # 6 of 15 (Start time: 12:00:51 AM)
Total Marks: 1

Which of the following is NOT included in the ISO registration process?

Select correct option:

Document review by the registrar of the quality system documents or quality manual

Pre-assessment, which identifies potential noncompliance in the quality system or in the documentation

***Review the program descriptions and select the program(s) that will enable you to contribute more to your organization**

Assessment by a team of two or three auditors of the quality system and its documentation

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Question # 7 of 15 (Start time: 12:02:13 AM)
Total Marks: 1

Who made some salient observations about the failure to understand the components of Profound Knowledge?

Select correct option:

***Scholtes**

Deming

Juran

Crosby

Question # 8 of 15 (Start time: 12:02:56 AM)
Total Marks: 1

Which of the following results in low costs?
Select correct option:

***High productivity and high capacity utilization**

Low productivity and low capacity utilization

Low productivity and high capacity utilization
High productivity and low capacity utilization

Question # 9 of 15 (Start time: 12:03:47 AM)
Total Marks: 1

How can the quality be computed?

Select correct option:

Quality = Expectation/Performance

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Quality = Performance/Expectation

Quality = Performance+Expectation

Quality = Performance-Expectation

Question # 10 of 15 (Start time: 12:05:14 AM)
Total Marks: 1

Documentation and data on existing products, are an example which of the following ?

Select correct option:

External inputs

***Internal inputs**

Either internal or external inputs

Neither internal nor external inputs

Question # 11 of 15 (Start time: 12:06:19 AM)
Total Marks: 1

Which of the following focuses on results, not process, and encourages short-term behavior?

Select correct option:

***Typical American MBO system**

Typical Japanese MBO system

Typical Chinese MBO system

Uncommon Chinese MBO system

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Question # 12 of 15 (Start time: 12:07:08 AM)
Total Marks: 1

Who developed Quality Function Deployment (QFD)?

Select correct option:

Frederick Taylor

Joseph M. Juran

***Y. Akao**

Armand Feigenbaum

Question # 13 of 15 (Start time: 12:08:06 AM)

Total Marks: 1

All of the following are the examples of appraisal costs EXCEPT:

Select correct option:

Quality measurement criteria

Quality improvement projects

Laboratory acceptance testing

Inspection and testing

Question # 14 of 15 (Start time: 12:09:33 AM)

Total Marks: 1

In order to achieve the objectives and stimulate innovation, what the organization should

encouragement?

Select correct option:

The involvement of its human resource

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The involvement of its process

The involvement of its system

The involvement of its capital

Question # 15 of 15 (Start time: 12:10:49 AM)
Total Marks: 1

How the requirements for documentation and records may not arise?

Select correct option:

Contractual requirements from the customer or other interested parties

Acceptance of international, national, regional and industry sector standards

***Irrelevant statutory and regulatory requirements**

Decisions by the organization

Quiz no 3

Question # 1 of 15 (Start time: 12:18:45 AM)

Total Marks: 1

At which level, does the quality concern centers on meeting internal customer requirements?

Select correct option:

Organizational level

Performer level

Process level

Production level

Question # 2 of 15 (Start time: 12:20:12 AM)

Total Marks: 1

Who is known as a father of modern-day statistical quality control?

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Select correct option:

***Shewhart**

Deming

Juran

Crosby

Question # 3 of 15 (Start time: 12:20:43 AM)

Total Marks: 1

Which of the following is TRUE for higher costs of quality?

Select correct option:

Doing the wrong thing in the wrong way

Doing the wrong thing in the right way

Doing the right thing in the wrong way

Doing the right thing in the right way

Question # 4 of 15 (Start time: 12:22:09 AM)

Total Marks: 1

Which of the following concepts and tools can be used to assess internal customer needs and to communicate internal supplier capabilities?

Select correct option:

Financial

Management

***Marketing**

Production

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Question # 5 of 15 (Start time: 12:23:02 AM)
Total Marks: 1

The management representatives appointed and given authority by the senior management are for all of the following EXCEPT:

Select correct option:

Managing

Mishandling

Evaluating

Coordinating

Question # 6 of 15 (Start time: 12:24:20 AM)
Total Marks: 1

The management preserves the conformity of products during internal processing and final delivery. Which of the following is included in such preservation?

Select correct option:

Handling

Abolition

Risk

Shrinking

Question # 7 of 15 (Start time: 12:25:36 AM)
Total Marks: 1

All of the following are the basic requirements of ISO 9000 EXCEPT:

Select correct option:

A documented quality system

Consistent adherence to the documented quality system

Check to see that the plan was carried out

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Customers and processes be displeased

Question # 8 of 15 (Start time: 12:26:50 AM)
Total Marks: 1

During 1990s, the following institution(s) began to pay increase attention to quality.

Select correct option:

Education and health care only

***Education, health care, and government**

Health care and government only

Government and financial institutions only

Question # 9 of 15 (Start time: 12:27:47 AM)

Total Marks: 1

Which one of the following depicts reliability – a dimension of quality?

Select correct option:

Bells and whistles of the product

Meeting specifications

Average time for the unit to fail

Quality of work

Question # 10 of 15 (Start time: 12:29:07 AM)

Total Marks: 1

Which of the following models considers the goal of organizational efficiency and performance?

Select correct option:

Organism model

***Mechanistic model**

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Cultural model

Total Quality model

Question # 11 of 15 (Start time: 12:29:56 AM)
Total Marks: 1

All of the following are the examples of external failure costs EXCEPT:

Select correct option:

Processing customer complaints

Customer returns

Warranty claims

***Quality planning**

Question # 12 of 15 (Start time: 12:30:34 AM)
Total Marks: 1

Identify the example of external failure costs.

Select correct option:

Quality planning

Re- inspection
Material review

***Customer returns**

Question # 13 of 15 (Start time: 12:31:35 AM)
Total Marks: 1

All of the following are the grounds to give birth to total quality approach EXCEPT:
Select correct option:

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Obsession with quantity

Confusion and poor leadership

One employee in charge
Political disorder

Question # 14 of 15 (Start time: 12:32:59 AM)
Total Marks: 1

Which of the following is viewed as a Consistent Pair?
Select correct option:

ISO 9000 and ISO 9004

ISO 1987 and ISO 1992

ISO 1992 and ISO 1997

***ISO 9001 and ISO 9004**

Question # 15 of 15 (Start time: 12:33:40 AM)

Total Marks: 1

In which century, the US broke further from the European tradition and adopted a new management approach developed by Frederick W. Taylor ?

Select correct option:

17th century

18th century

***19th century**

20th century

Next Quiz

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1. All of the following are the grounds to give

birth to total quality approach
EXCEPT:

- A. Obsession with quantity
- B. Confusion and poor leadership
- C. One employee in charge
- D. Political disorder**

2. Which of the following was developed by Motorola to improve its processes by minimizing defects?

- A. ISO 9000
- B. *Six sigma**
- C. QS 9000
- D. TQM

3. Quality applies to which aspect(s) of the organization?

- A. Products only
- B. Products and services
- C. People and product
- D. *Products, services, and people**

4. Which of the following statements is TRUE about Deming's philosophy?

- A. *It is based on improving products and services by reducing uncertainty and variability in the design and manufacturing processes**
- B. Quality is either or not present in the whole organization; that quality is the responsibility of everyone in the organization
- C. Increasing conformance to specifications through elimination of defects, supported extensively by statistical tools for analysis
- D. Increasing loss, for the producer, the

customer, and society, associated with increasing variability from a target value

5. Which of the following statements, in relation to system, is TRUE for people failing to understand the components of Profound Knowledge?

A. *They see the symptoms but not the deep causes of problems

B. They do not know when expectations are realistic

C. They do not understand motivation/why people do what they do

D. They do not know how to plan and accomplish learning and improvement

6. "Problems will remain unsolved, despite their best efforts." When will this happen?

A. When people don't understand systems

B. When people don't understand psychology

C. When people don't understand variation

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D. *When people don't understand the theory of knowledge

7. Which stage in the Shewhart Cycle implements the change on a small scale?

A. *Do stage

- B. Act stage
- C. Plan stage
- D. Check stage

8. Which of the following is determining what to control, establishing units of measurement so that data may be objectively evaluated?

- A. Quality improvement
- B. Quality planning
- C. *Quality control**
- D. Quality process

9. All of the following steps are required for quality planning EXCEPT:

- A. Determine who the customers are
- B. Deploy the plans to operational levels
- C. Identify customers' needs
- D. *Assess actual quality performance**

10. Which of the following is an answer to the First Absolute of Quality

Management

as summarized by Crosby; what is quality?

- A. *Conformance to requirements**
- B. Prevention
- C. Zero defects
- D. Price of nonconformance

Next Quiz:

Question # 1 of 15 (Start time: 09:54:15 PM)
Total Marks: 1

Quality applies to which aspect(s) of the organization?

Select correct option:

Products only

Products and services

People and product

Products, services, and people

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Question # 2 of 15 (Start time: 09:55:01 PM)
Total Marks: 1

Juran defines quality as “fitness for use”. This can be broken down into how many categories?

Select correct option:

Five

***Four**

Three

Two

Question # 3 of 15 (Start time: 09:56:24 PM)

Total Marks: 1

How many maturity levels are there in the Capability Maturity Model?

Select correct option:

Four

***Five**

Six

Seven

Question # 4 of 15 (Start time: 09:57:11 PM)

Total Marks: 1

Which of the following is a unit for measuring process performance according to traditional view of quality?

Select correct option:

Effective parts per thousand produced

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Defective parts per thousand produced

Effective parts per hundred produced

***Defective parts per hundred produced**

Question # 5 of 15 (Start time: 09:58:27 PM)
Total Marks: 1

Which of the following statements, in relation to system, is TRUE for people failing to understand the components of Profound Knowledge?
Select correct option:

***They see the symptoms but not the deep causes of problems**

They do not know when expectations are realistic

They do not understand motivation/why people do what they do

They do not know how to plan and accomplish learning and improvement

Question # 6 of 15 (Start time: 09:59:10 PM)
Total Marks: 1

Which of the following statements, in relation to variation, is TRUE for people failing to understand the components of Profound Knowledge?
Select correct option:

They see events as individuals' incidents rather than the net result of many interactions

and interdependent forces

***They do not know the difference between prediction, forecasting, and guesswork**

They create cynicism, demoralization, demotivation, guilt, resentment, burnout, craziness, and turnover

They do not understand the difference between improvement and change

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Question # 7 of 15 (Start time: 10:00:18 PM)

Total Marks: 1

Which of the following models has a vertical chain of command?

Select correct option:

Organism model

***Mechanistic model**

Cultural model

Total Quality model

Question # 8 of 15 (Start time: 10:01:19 PM)

Total Marks: 1

What was the role of quality professionals during 1800s?

Select correct option:

Troubleshotting and the application of statistical methods

***Inspection, sorting, counting, and grading**

Quality measurement, quality planning, and program design

Goal setting, education and training, and program design

Question # 9 of 15 (Start time: 10:02:25 PM)
Total Marks: 1

All of the following are the major categories and items within each category of Canadian Awards for Business Excellence EXCEPT:

Select correct option:

Customer focus

Distributor focus

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People focus

Supplier focus

Question # 10 of 15 (Start time: 10:03:44 PM)

Total Marks: 1

How the Baldrige Award recognizes the importance of total quality principles?

Select correct option:

Through independence with overall business planning

Through a no relation with overall business planning

***Through integration with overall business planning**

Through dependence with overall business planning

Question # 11 of 15 (Start time: 10:04:16 PM)

Total Marks: 1

Which of the following involves determining what to control, establishing units of measurement so that data may be objectively evaluated?

Select correct option:

Quality improvement

Quality planning

***Quality control**

Quality process

Question # 12 of 15 (Start time: 10:04:55 PM)

Total Marks: 1

After the war, Deming was invited to which of the following countries?

Select correct option:

Germany

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***Japan**

France

China

Question # 13 of 15 (Start time: 10:05:25 PM)

Total Marks: 1

Which of the following organizations has a little chance of survival?

Select correct option:

With no corporate philosophy about how to create long-term customer loyalty

With no corporate objectives about how to

create long-term customer loyalty

With no mission about how to create long-term customer loyalty

***With no vision about how to create long-term customer loyalty**

Question # 14 of 15 (Start time: 10:06:47 PM)
Total Marks: 1

Who suggested that there are three categories of performance?

Select correct option:

Taguchi
Taylor

Ishikawa

***Deming**

Question # 15 of 15 (Start time: 10:07:19 PM)
Total Marks: 1

Which of the following statements holds true for quality assurance?

Select correct option:

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To p management's intention regarding quality
Functio ns determining implementatio n o f the
quality po licy

***Actions to provide confidence of satisfying
quality requirements**

Responsibilities and processes, which implement
quality management

Qui z No. 3 (Talha)

Question # 1 o f 15 (Start time: 11:43:33 AM)
Total Marks: 1

Which of the fo llowing statements is TRUE abo ut
Deming's philo sophy?

Select correct option:

***It is based on improving product s and services
by reducing uncertainty and
variability in the design and manufacturing
processes**

Quality is either or not present in the whole o
rganization; that quality is the
responsibility of everyo ne in the organization

Increasing confor mance to specifications through
eliminatio n of defects, supported
extensively by statistical tools for analysis

Increasing loss, for the producer, the customer, and society, associated with increasing variability from a target value

Question # 2 of 15 (Start time: 11:44:22 AM)
Total Marks: 1

All of the following are the major decision areas in supply chain management EXCEPT:

Select correct option:

Location

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***Production**

Inventory
Pricing

Question # 3 of 15 (Start time: 12:11:54 PM)
Total Marks: 1

By the mid 1970s, which markets were penetrated by the Japanese companies?

Select correct option:

***Western markets**

Eastern markets

Northern markets

Southern markets

Question # 4 of 15 (Start time: 12:12:33 PM)
Total Marks: 1

The input for management review to examine the quality management system shall include information about all of the following EXCEPT:

Select correct option:

Status of preventive and corrective actions

Follow-up actions from previous management reviews

Changes that could affect the quality management system

***Improvement of product related to customer requirements**

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Question # 5 of 15 (Start time: 12:13:47 PM)
Total Marks: 1

Juran advocates a never-ending spiral of activities that includes all of the following EXCEPT:

Select correct option:

Product development

***Workers**

Inspection and testing

Purchasing

Question # 6 of 15 (Start time: 12:14:42 PM)
Total Marks: 1

Which of the following statements is TRUE for little q?

Select correct option:

***Striving for quality in a limited or specific area**

Searching for specific lines of text containing a particular pattern

Matching rules encoded in a string according to certain syntax rules

Giving combination of values for the free variables

Question # 7 of 15 (Start time: 12:15:19 PM)
Total Marks: 1

Which of the following is the Total Quality Tool that is simply a graphical representation of an outline that presents a chain of causes and effects?

Select correct option:

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Check sheet

Fishbone diagram
Scatter diagram

Pareto chart

Question # 8 of 15 (Start time: 12:16:32 PM)
Total Marks: 1

Who found that teams are seldom fully used, and their individual members are often co mp lacent?

Select correct option:

Oakland and Schein

***Newall and Dale**

Lehr and Cox
Juran and Schein

Question # 9 of 15 (Start time: 12:17:14 PM)
Total Marks: 1

The customer value strategy may be addressed in which of the following topics?

Select correct option:

Projects

Products

Services

***Measurement**

Question # 10 of 15 (Start time: 12:18:10 PM)
Total Marks: 1

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Which of the following should be reviewed by the organization to identify requirements for operations that realize products or deliver

services in order to ensure compliance with specifications?

Select correct option:

Inability to comply with contractual requirements

***Training and competence of people**

Problem conception

Irrelevant statutory and regulatory requirements

Question # 11 of 15 (Start time: 12:19:19 PM)

Total Marks: 1

Who developed Quality Function Deployment (QFD)?

Select correct option:

Frederick Taylor

Joseph M. Juran

***Y. Akao**

Armand Feigenbaum

Question # 12 of 15 (Start time: 12:20:15 PM)

Total Marks: 1

What are the ISO (International Organization for Standardization) series standards for

industries such as automobiles?
Select correct option:

ISO 9000

ISO 14000

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***QS 9000**

AS 9000

Question # 13 of 15 (Start time: 12:21:43 PM)
Total Marks: 1

Which of the following should be used as input to management review in order to ensure that continual improvement is the driver for organizational development?

Select correct option:

***Performance information**

Performance management

Ethical performance

Performance development

Question # 14 of 15 (Start time: 12:23:02 PM)
Total Marks: 1

Which of the following are generally easy to detect using statistical methods, and it is usually economical to remove them?

***Common causes**

Special causes

Implicit causes

Assignable causes

Question # 15 of 15 (Start time: 12:24:27 PM)
Total Marks: 1

What is the objective of a TQM system?

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Select correct option:

***Continuous improvement**

Continual improvement

Business improvement

Process improvement

Next Quiz

Question # 1 of 15 (Start time: 09:21:59 PM)

Total Marks: 1

Based in Geneva, Switzerland, International Organization for Standardization is a consortium of the national standard bodies, which has how many countries?

More than 100

Less than 100

Less than 90

Less than 80

Question # 2 of 15 (Start time: 09:23:27 PM)

Total Marks: 1

Which of the following is the contributor to the work method cause in the house paint peeling cause-and-effect diagram?

Select correct option:

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Cheap so lvents

Temperature

Acid rain

Paint over dirt

Question # 3 o f 15 (Start time: 09:24:28 PM)

Total Marks: 1

Which of the fo llowing co ncepts and tools can be used to assess internal custo mer needs and to commu nicate internal supplier capabilities?

Select correct option:

Financial

Management

***Marketing**

Productio n

Question # 4 o f 15 (Start time: 09:25:27 PM)

Total Marks: 1

The input for management review to examine the

quality management system shall include information about all of the following EXCEPT:

Select correct option:

Status of preventive and corrective actions

Follow-up actions from previous management reviews

Changes that could affect the quality management system

***Improvement of product related to customer requirements**

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Question # 5 of 15 (Start time: 09:26:28 PM)
Total Marks: 1

What the organization must accomplish to achieve the mission, by examination and categorization of the impacts?

Select correct option:

Critical success factors

Key process indicators

Illegal and policy factors

The enemy indicator

Question # 6 of 15 (Start time: 09:27:43 PM)

Total Marks: 1

Who suggested, "A quality product is one that provides performance or conformance at an acceptable price or cost"?

Select correct option:

Crosby

Ishikawa

***Garvin**

Shewhart

Question # 7 of 15 (Start time: 09:28:29 PM)

Total Marks: 1

Marketing research carried out to know whether the particular area should be covered in this course. What the course instructor should develop?

Select correct option:

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***The course meeting requirements of the university**

The course meeting requirements of the instructor
The course meeting requirements of the students

The course meeting requirements of the bookseller

Question # 8 of 15 (Start time: 09:30:00 PM)
Total Marks: 1

Under which perspective of the balance scorecard, would you classify the operating cost management measurement?

Select correct option:

The customer perspective

The business process perspective

[The financial perspective](#)

The learning & growth perspective

Question # 9 of 15 (Start time: 09:31:24 PM)

Total Marks: 1

Which of the following is the purpose of ISO 9001?

Select correct option:

***To define the minimum Quality Management System requirements needed to achieve customer satisfaction by meeting specified product requirements**

To determine metals and metalloids in airborne particulate matter by inductively coupled plasma atomic emission spectrometry

To ensure that the IT service management processes are aligned both with the needs of the business and with international best practice

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To evaluate ultimate aerobic biodegradability of organic compounds in aqueous medium by determination of oxygen demand in a closed respirometer

Question # 10 of 15 (Start time: 09:32:16 PM)
Total Marks: 1

Which of the following is a four-step quality

model among the most widely used tools for continuous improvement?

Select correct option:

***Deming Cycle**

Juran Trilogy

Taguchi's Loss Function
Quality Circles

Question # 11 of 15 (Start time: 09:33:10 PM)
Total Marks: 1

What was the transcendent view of quality?

Select correct option:

Satisfying customers

Meeting needs and wants

***Innate excellence**

Conformance

Question # 12 of 15 (Start time: 09:34:18 PM)
Total Marks: 1

Which of the following focuses on results, not process, and encourages short-term behavior?

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Select correct option:

***Typical American MBO system**

Typical Japanese MBO system

Typical Chinese MBO system

Uncommon Chinese MBO system

Question # 13 of 15 (Start time: 09:35:25 PM)

Total Marks: 1

Which of the following is NOT the category in Baldrige Award?

Select correct option:

Leadership

Business results

***Supply chain**

Process management

Question # 14 of 15 (Start time: 09:36:30 PM)

Total Marks: 1

What is the number of principles in Dr. W. Edwards Deming's quality principles?

8

10

12

***14**

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Question # 15 of 15 (Start time: 09:37:03 PM)

Total Marks: 1

Which of the following is NOT an advantage of functional structure?

Select correct option:

It provides organizations with a clear chain of command

***It enables employees to learn about customer expectations**

It allows evaluating people based on clear set of responsibilities

It allows people to specialize in the aspect of the

work

Quiz No. 4

Question # 1 of 15 (Start time: 06:46:03 PM)
Total Marks: 1

Which of the following is a requirement of ISO 9000:2000?

Select correct option:

***Processes to measure and analyze customer satisfaction**

An Environmental Policy to be in existence within the organization

Procedures for registration of font-related identifiers

Irregular improvement of internal process and customer satisfaction

Question # 2 of 15 (Start time: 06:47:17 PM)
Total Marks: 1

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The management responsible for the area being audited is responsible for ensuring that actions are taken without undue delay to eliminate detected nonconformities and their causes. Which of the following is a follow-up activity?

Select correct option:

***The verification of the actions taken**

The reporting of contradiction results
The contradiction of the actions taken

The verification of contradiction results

Question # 3 of 15 (Start time: 06:57:48 PM)
Total Marks: 1

Which of the following total quality tools is employed in the step “plan and implement a solution” in problem solving?

Select correct option:

4W and 1H matrix

Pareto chart

***Control chart**

Scatter plot

Question # 4 of 15 (Start time: 06:59:05 PM)
Total Marks: 1

Under which perspective of the balance scorecard, would you classify the customer retention measurement?

Select correct option:

The customer perspective

The business process perspective

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The financial perspective

The learning & growth perspective

Question # 5 of 15 (Start time: 06:59:42 PM)
Total Marks: 1

What was the role of quality professionals during 1950s?

Select correct option:

***Troubleshooting and the application of statistical methods**

Inspection, sorting, counting, and grading

Quality measurement, quality planning, and program design

Goal setting, education and training, and program design

Question # 6 of 15 (Start time: 07:00:46 PM)

Total Marks: 1

What is quality?

Select correct option:

***Meeting and exceeding the customer's expectations**

Adding factors to make the customer happy

The degree to which the project meets requirements

Conformance to management's objectives

Question # 7 of 15 (Start time: 07:01:11 PM)

Total Marks: 1

The quality initiatives focused initially on reducing defects and errors in products and

services with the help of which of the following?

Select correct option:

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***Measurement, statistics, and other problem solving tools**

Measurement, statistics and cause and effect analysis

Statistics and cause and effect analysis only

Failure modes and cause and effect analysis only

Question # 8 of 15 (Start time: 07:01:59 PM)
Total Marks: 1

Which of the following performance levels have good results and have emphasized sustained improvement trends?

Select correct option:

Reactive approach

Stable formal system approach

Best-in-class performance

***Continuous improvement emphasized**

Question # 9 of 15 (Start time: 07:02:42 PM)
Total Marks: 1

Do not permit overlong examples or irrelevant discussion. This refers to which of the following discussion skills in meetings?

Select correct option:

Summarizing

***Managing time**

Closing the discussion

Containing digression

Question # 10 of 15 (Start time: 07:03:32 PM)

Total Marks: 1

Which of the following statements, in relation to psychology, is TRUE for people failing to understand the components of Profound Knowledge?

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Select correct option:

***They see the symptoms but not the deep causes of problems**

They do not know when expectations are realistic

They do not understand motivation/why people do what they do

They do not know how to plan and accomplish learning and improvement

Question # 11 of 15 (Start time: 07:04:21 PM)
Total Marks: 1

Which of the following results in low costs?

Select correct option:

***High productivity and high capacity utilization**

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

Question # 12 of 15 (Start time: 07:05:03 PM)
Total Marks: 1

Under which perspective of the balance scorecard, would you classify motivated workforce measurement?

Select correct option:

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

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Question # 13 of 15 (Start time: 07:05:40 PM)
Total Marks: 1

Which of the following models value stability?
Select correct option:

Organism model

***Mechanistic model**

Cultural model

Total Quality model

Question # 14 of 15 (Start time: 07:06:33 PM)
Total Marks: 1

Who stressed that systems must be focused toward a purpose?

Select correct option:

Crosby

Ishikawa

Taguchi

***Deming**

Question # 15 of 15 (Start time: 07:07:58 PM)
Total Marks: 1

Which of the following leads to questioning and can be tested and validated?

Select correct option:

***Theory of knowledge**

Psychology

Understanding of variation

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Appreciation for a system

Question # 1 of 15 (Start time: 07:10:06 PM)
Total Marks: 1

Which of the following statements, in relation to variation, is TRUE for people failing to understand the components of Profound Knowledge?

Select correct option:

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

***They do not know the difference between prediction, forecasting, and guesswork**

They create cynicism, demoralization, demotivation, guilt, resentment, burnout, craziness, and turnover

They do not understand the difference between improvement and change

Question # 2 of 15 (Start time: 07:11:25 PM)
Total Marks: 1

Which of the following represent short-term perspectives and do not encourage long-term improvement as stated by Deming ?

Select correct option:

***Numerical quotas**

Constitutional quotas

Political party quotas

Mechanical quotas

Question # 3 of 15 (Start time: 07:12:41 PM)
Total Marks: 1

All of the following are true for zero defects concept EXCEPT:

Select correct option:

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A performance standard for management
A standard of the craftsman regardless of his or her assignment

A tool of management to communicate to do things right the first time

A motivational, widely understood and accepted program

Question # 4 of 15 (Start time: 07:14:02 PM)
Total Marks: 1

Who suggested, "A quality product is one that provides performance or conformance at an acceptable price or cost"?

Select correct option:

Crosby
Ishikawa

***Garvin**

Shewhart

Question # 5 of 15 (Start time: 07:14:35 PM)
Total Marks: 1

Which of the following tools plots the relationship of one numeric variable to another on a horizontal and vertical axis and determines the degree of dependency or

interdependency of those variables?

Select correct option:

***Scatter diagram**

Check sheet

Fishbone diagram

Pareto chart

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Question # 6 of 15 (Start time: 07:15:52 PM)

Total Marks: 1

Who warned against the use of extrinsic controls that compel behavioral changes in people when rewards and incentive programs are seldom shifted to favor quality improvement?

Select correct option:

Oakland

Schein

***Cox**

Dale

Question # 7 of 15 (Start time: 07:16:38 PM)

Total Marks: 1

How many maturity levels are there in the Capability Maturity Model?

Select correct option:

Four

***Five**

Six

Seven

Question # 8 of 15 (Start time: 07:17:11 PM)

Total Marks: 1

Which one of the following depicts aesthetics, which is a dimension of quality?

Select correct option:

***Exterior finish**

Quality of work

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Ranking first

Ease of repair

Question # 9 of 15 (Start time: 07:18:28 PM)
Total Marks: 1

Which of the following firms solely owns the entire material flow and each channel member operates independently?

Select correct option:

Horizontal integrated firms

Backward vertical integrated firms

Vertical integrated firms

Conglomerate integrated firms

Quiz Start Time: 07:10 PM Time Left 85
sec(s)

Question # 10 of 15 (Start time: 07:19:52 PM)
Total Marks: 1

Which of the following is NOT the deadly disease according to Deming?

Select correct option:

Excessive medical costs

***Excessive transportation costs**

Lack of constancy of purpose

Mobility of management

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Question # 11 of 15 (Start time: 07:21:21 PM)
Total Marks: 1

Which stage in the Shewhart Cycle uses data and facts to analyze the results of the change?

Select correct option:

Do stage

Act stage

Plan stage

***Check stage**

Question # 12 of 15 (Start time: 07:21:54 PM)
Total Marks: 1

Why product-based definitions of quality suit engineers?

Select correct option:

They are concerned with designing of the products

They can judge quality

They can reduce specific standards

***They are concerned with translating product requirements into specific components and physical dimensions thereby produced**

Question # 13 of 15 (Start time: 07:23:08 PM)

Total Marks: 1

All of the following are the elements of a TQM system EXCEPT:

Select correct option:

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Leadership

Communications

Measurement

***Detentions**

Question # 14 of 15 (Start time: 07:24:03 PM)
Total Marks: 1

Which of the following is the contributor to the equipment cause in the house paint peeling cause-and-effect diagram?

Select correct option:

Cheap solvents

Temperature

Bad brush bristle

Paint over dirt

Question # 15 of 15 (Start time: 07:24:24 PM)
Total Marks: 1

Which of the following is viewed as a Consistent Pair?

Select correct option:

ISO 9000 and ISO 9004

ISO 1987 and ISO 1992

ISO 1992 and ISO 1997

***ISO 9001 and ISO 9004**

Question # 1 of 15 (Start time: 07:17:17 PM)

Total Marks: 1

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All of the following are the objectives achieved simultaneously for the sustained growth to be recognized EXCEPT:

Select correct option:

Customer satisfaction

Effective leadership

Effective human resources

Integration with the supplier base

Question # 2 of 15 (Start time: 07:18:10 PM)
Total Marks: 1

Some organizations emphasize the use of quality tools but failed to do which of the following?

Select correct option:

Focus on what is truly important to the distributors

***Incorporate continuous improvement efforts**

Make fundamental changes in their processes and culture

Focus on what is truly important to the customer

Question # 3 of 15 (Start time: 07:18:53 PM)
Total Marks: 1

Which of the following tools has a weakness that it does not tell whether the variation is the result of special causes or common causes?

Select correct option:

Histogram

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Check sheet

Pareto chart

Run chart

All of the following are Juran's basic steps to progress EXCEPT:

Select correct option:

Achieve structured improvements on a continual basis combined with dedication and a sense of urgency

Establish an extensive training program

***Establish an extensive marketing program**

Establish commitment and leadership on the part of higher management

Question # 5 of 15 (Start time: 07:20:14 PM)
Total Marks: 1

Which of the following statements, in relation to theory of knowledge, is TRUE for people failing to understand the components of Profound Knowledge?

Select correct option:

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

***They do not know the difference between prediction, forecasting, and guesswork**

They create cynicism, demoralization, demotivation, guilt, resentment, but not, craziness, and turnover

They do not understand the difference between improvement and change

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Question # 6 of 15 (Start time: 07:21:09 PM)
Total Marks: 1

Which of the following performance levels is a systematic process-based approach, early stage of systematic improvements?

Select correct option:

No formal approach

Reactive approach

***Stable formal system approach**

Best-in-class performance

Question # 7 of 15 (Start time: 07:21:47 PM)
Total Marks: 1

In creative problem solving, which of the following refers to brainstorming?

Select correct option:

Mess Finding

Fact Finding

Problem Finding

Idea Finding

Question # 9 of 15 (Start time: 07:22:33 PM)
Total Marks: 1

Which of the following describes a complete data set?

Select correct option:

Measures of dispersion

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Measures of central tendency

Deductive statistics

Inductive statistics

Question # 10 of 15 (Start time: 07:22:59 PM)
Total Marks: 1

Which of the following tools represents the entire

process at a high or detailed
(depending on your use) level of observation,
allowing analysis and optimization of
workflow?

Select correct option:

Check sheet

Pareto diagram

Flow chart

Histogram

Question # 11 of 15 (Start time: 07:23:33 PM)
Total Marks: 1

Which of the following is NOT the characteristic
of the organizations operating
according to the principles of quality
management?

Select correct option:

A total commitment to continually increasing
value for customers, investors, and
employees

A firm understanding that market driven means
that quality is defined by customers, not
the company

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A commitment to leading people with a bias for broken improvement and miscommunication

***A commitment to fundamental improvement through knowledge, skills, problem solving, and teamwork**

Question # 12 of 15 (Start time: 07:24:12 PM)
Total Marks: 1

Ahmad has organized a manual data collection in such a way as to be easily used, understood and analyzed automatically. Which of the following Ahmad has prepared?

Select correct option:

Check sheet

Pareto diagram
Flow chart

Histogram

Question # 13 of 15 (Start time: 07:24:53 PM)
Total Marks: 1

Which of the following is TRUE for design and

development outputs under ISO
9001:2000 - Quality Management Systems ?

Select correct option:

It Identifies of and corrects against potential conformance

It is used where excess in product will only be apparent in use

It is Insignificance and concealment of significant features of the processes

***It meets the input requirements for design and development**

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Question # 14 of 15 (Start time: 07:25:26 PM)

Total Marks: 1

What is the purpose of the Control Chart?

Select correct option:

***To indicate variations, trends, and the amount of change from one time period to**

another

To show where to apply resources by revealing the significant few from the trivial many

To provide a scientific approach to gathering information to improve the validity of decisions and interventions

To monitor effects of the variables on the difference between target and actual performance

Question # 15 of 15 (Start time: 07:25:52 PM)
Total Marks: 1

Which of the following is the contributor to the work method cause in the house paint peeling cause-and-effect diagram?

Select correct option:

Cheap solvents

Temperature

Acid rain

Paint over dirt

Question # 1 of 15 (Start time: 06:05:27 PM)

Total Marks: 1

Which of the following tools is a graphical representation of a process, depicting inputs, outputs and units of activity?

Select correct option:

- ☐ Flow chart
- ☐ Check sheet
- ☐ Pareto diagram
- ☒ Histogram

Question # 3 of 15 (Start time: 06:07:39 PM)

Total Marks: 1

Which of the following concepts and tools can be used to assess internal customer needs and to communicate internal supplier capabilities?

Select correct option:

- ☐ Financial
- ☐ Management
- ☒ Marketing
- ☐ Production

Question # 4 of 15 (Start time: 06:08:32 PM)**Total Marks: 1**

Which of the following best describes the acronym SWOT?

Select correct option:

- ☐ Specific, Witness, Opportunities, and Time
- ☐ Single, Weaknesses, Occasion, and Tactics
- ☒ Strengths, Weaknesses, Opportunities, and Threats
- ☐ Strategic, Weaknesses, Objectives, and Tactics

Question # 5 of 15 (Start time: 06:09:35 PM)**Total Marks: 1**

All of the following are the trends explained by Armand Feigenbaum in an article for Quality Digest that will shape the future of quality management EXCEPT:

Select correct option:

- ☐ Demanding global customers
- ☐ Shifting customer expectations
- ☐ Opposing economic pressures
- ☒ Old approaches to management

MGMT 510 TQM All Quiz

by Admin on Sat Jan 23, 2010 12:37 pm

TQM Quiz 5 L
1-12 November 2009

What does the transcendent view imply for high quality?

It is something everlasting

It is
something timeless and enduring

It is something priceless

It is something perfect

During which century, the manufacturers began to include quality processes in quality practices?

17th century

18th century

19th century

20th
century

Taguchi suggested that loss in a process is increased with increase in which of the following?

Specifications

Standards

Competition

Variability

The products manufactured during 1800s were unique. How quality was ensured in this era?

Through renovation

Through inspection

Through calculation

Through repair

Which of the following is typically viewed to lie between fully vertically integrated firms?

Supply chain management ??

Production chain management

Value chain management

Demand chain management

The majority of advertisers appeal the public on the basis of which of the following?

Quality of product

Quality of staff

Inferiority of product

Inferiority of service

Which of the following models value stability?

Organism model

Mechanistic model

Cultural model

Total Quality model

What is the relationship between Ishikawa and Deming?

Son and father

Nephew and uncle

**Student
and teacher**

Grandson and grandfather

Which of the following was developed by Motorola to improve its processes by minimizing defects?

ISO 9000

**Six
sigma**

QS 9000

TQM

What was the primary concern of managers during 1980s?

Detection

**Strategic
impact**

Control

Coordination

You can ask direct questions to quiet people and try to control talkative people", for which of the following purpose?

**Mobilizing
the group**

Keeping the group focused

Restating purpose

Starting punctually

Some organizations emphasizes the use of quality tools but failed to do which of the following?

Focus on what is truly important to the distributors

**Incorporate
continuous improvement efforts**

Make fundamental changes in their processes and culture

Focus on what is truly important to the customer

Learn how to tell when nothing can be gained from further discussion. This refers to which of the following discussion skills in meetings?

Acting as gatekeepers

Closing the discussion

Asking for clarification

Testing for consensus

Which of the following models has the viewpoint that change and learning are valued in themselves?

Organism model

Mechanistic model

**Cultural
model**

Total Quality model

Marketing research carried out to know whether the particular area should be covered in this course. What the course instructor should develop?

The
course meeting requirements of the university

The course meeting requirements of the instructor

The course meeting requirements of the students

The course meeting requirements of the bookseller

Which of the following statements is TRUE about Deming's philosophy?

It is based on improving products and services by reducing uncertainty and variability in the design and manufacturing processes

Quality
is either or not present in the whole organization; that quality is the responsibility of everyone in the organization ??

Increasing conformance to specifications through elimination of defects, supported extensively by statistical tools for analysis

Increasing loss, for the producer, the customer, and society, associated with increasing variability from a target value

All of the following are the elements of a TQM system EXCEPT:

Leadership

Communications

Measurement

Detentions

The standards for output based on quality and customer-service requirements that originate at the organizational and process levels include requirements for all of the following EXCEPT:

Accuracy

Innovation

Timeliness

Turnover
??

Quiz No 2

Nov 20, 2009 11:59 PM

Which of the following refers to SIPOC
(Supplier-Input-Process-Output-Customer) diagram?

A high-level process map

A low-level
process map

A
high-level performer map

A
low-level performer map

Which of the following is concerned about quality for
achievement of TQM?

The
Managing Director

The
Operative

The

Quality Manager

Everyone in the organization

Which of the following is required for meeting and exceeding customer's need?

Planning

Control

Execution

Innovation

Which one of the following depicts feature, which is a dimension of quality?

Product
looks and sounds

Brand
name and price

Bells
and whistles of the product

Ability
to repair a product

Which of the following can help to eliminate fear and encourage teamwork?

Management

Workmanship

Leadership

Risk-taking

Which of the following focuses on results, not process, and encourages short-term behavior?

Typical American MBO system

Typical
Japanese MBO system

Typical
Chinese MBO system

Uncommon
Chinese MBO system

Which of the following statements, in relation to theory of knowledge, is TRUE for people failing to understand the components of Profound Knowledge?

They see events as individuals' incidents rather than the net result of many interactions and interdependent forces

They do not know the difference between prediction, forecasting, and guesswork

They create cynicism, demoralization, de-motivation, guilt, resentment, burnout, craziness, and turnover

They do not understand the difference between improvement and change

Who made some salient observations about the failure to understand the components of Profound Knowledge?

Scholtes

Deming

Juran

Crosby

Do not permit overlong examples or irrelevant discussion. This refers to which of the following discussion skills in meetings?

Summarizing

Managing time

Closing
the discussion

Containing
digression

Which of the following decisions assume the existence of the facilities, but determine the exact path(s) through which a product flows to and from these facilities?

Location
decisions

Production
decisions

Inventory
decisions

Transportation decisions

If the change was successful, implement it on a wider scale and continuously assess your results. Which stage represents this statement in the Shewhart Cycle?

Select correct option:

Do
stage

Act stage

Plan
stage

Check
stage

Which of the following is TRUE about entry-level managers as noted by Deming in Japan?

They
spend 5 to 20 years on the factory floor

**They spend 4 to 12 years on the
factory floor**

They

spend 4 to 17 years on the factory floor

They
spend 7 to 22 years on the factory floor

Which of the following statements is TRUE about
Deming's philosophy?

It is based on improving
products and services by reducing uncertainty and variability in the design and
manufacturing processes

Quality is either or not
present in the whole organization; that quality is the responsibility of everyone
in the organization

Increasing conformance to
specifications through elimination of defects, supported extensively by
statistical tools for analysis

Increasing loss, for the
producer, the customer, and society, associated with increasing variability from
a target value

Which of the following statements is TRUE for total
quality?

**Stresses on learning and adaptation
to change**

Ability
to repair a product quickly and easily

Difficulty
in establishing and measuring service levels

A
customer focused management system

What is necessary for a system to be effective?

**The components of any system must
work together for the system**

The
components of any system must not work together for the system

The
components of any system should not be organized for the system

The
components of any system should provide disruption for the system

Admin
Administrator

Number of posts: 8058
Age: 36
Location: Sharqi Abadi Mustafabad Kasur
Job/hobbies: Do more for other
Mode (i.e. cool, angry etc): Serious

Warning:



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Registration date: 2007-10-23



Re: MGMT 510 TQM All Quiz

by **Admin** on Sat Jan 23, 2010 12:56 pm

1)

The concept of common and special causes of variations was developed by:

.

Deming

.

**Walter
Shewhart**

.

Juran

.

Taguchi

2)

MBO stands for:

.

Management by Objective

.

Management by
Organization

.

Maintenance
by Objective

.

Maintenance
by Organization

3)

Which of the following statements is true about routine inspection:

.

It adds value to the product

.

It acknowledges defects as well as adds value to the product

.

**It
acknowledges defects but does not add value to the product**

.

All of the given options

4)

The success of Japanese manufacturers is due to _____, small and incremental

improvements in design and production.

.

Continuous

.

Abrupt

.

Continual

.

None of the given options

5)

All of the following statements are true about institute training except:

.

It improves quality

.

It improves productivity

.

It enhances workers' morale

.

**It
certainly results in salary increment**

6)
One of the advantages of team work is:

.

**It
breaks down barriers between internal customers and suppliers**

.

It results in promotion

.

It results in salary increment

.

None of the given options

7)
Reworking _____ the cost of quality.

.

Decreases

.

Increases

.

Neutralizes



_____ refers to general processes of improvement and encompasses discontinuous improvements

.

Continuous improvement

.

**Continual
improvement**

.

Constant improvement

.

Consecutive improvement

9)

Total quality costs include:

.

Prevention costs

.

Appraisal costs

.

Failure costs

.

All
of the given options

10) The
no. of principles in Dr. W. Edwards Deming's quality principles is _____.

.

8

.

10

.

12

.

14

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Re: MGMT 510 TQM All Quiz

by **Admin** on Sat Jan 23, 2010 12:57 pm

Total Quality Management (MGMT 510) Quiz: 01

Marks: 10 Semester: Spring 2008

This Quiz covers lessons: 10-12.

1 During 1980s,
the job characteristic of quality professionals was:

A Educating
others

B Achieving
personal targets

C Consultative
work with other departments

D None of the above

2 During 1800s,
the quality was viewed as:

A A problem
that can never be solved

B A problem to be solved

C A tool for
measuring system

D A remedy to
control decline

3 Why the
factory managers created inspection departments?

**A To keep defective products aside, ensuring
they do not reach the customers**

B To ensure
quantity of goods/services

C To count,
grade, and rework

D All of the
above

4 During 1950s,
controlling quality meant:

A To understand
and address the causes of poor quality

B To simplify
and speed up process

C To link
outcomes to uses

D Acting on defects

5 The 'father'
of statistical quality control is:

A F. W. Taylor

B Joseph M.
Juran

C Philip Crosby

D Walter Shewhart

6 The Six Sigma
is a new file separate from the total quality philosophy.

True

False

7 Today's consumers define quality as being flawless and having all of the attributes they want in just the way they want them.

True

False

8 The Deming Chain Reaction theory states that one of the main purposes of having quality products and processes is more rework.

True

False

9 The theory of variation helps us understand when there has not been a genuine improvement in performance

True

False

10 A SIPOC diagram is a tool used by a team to identify all relevant elements of a process improvement project before work begins.

True

False

Total Quality Management (MGMT 510) Quiz: 02 Marks: 10
Semester: Spring 2008

This Quiz
covers lessons: 17-19.

1
Nonconformance is an expense of:

A Profit of
quality

B Defects of quality

C Quality of product

D Cost of quality

2 Crosby's approach to management is:

A A problem that can never be solved

B Absolutes of Quality Management

C Interim Management

D ISO

3 Best price or zero cost is:

A To keep defective products aside, ensuring they do not reach the customers

B Prerogative cost

C To count,
grade, and rework

D Cost
leadership

4 Example of
prevention costs is:

A Quality improvement projects

B Downgrading

C To link
outcomes to uses

D Warranty
claims

5 Big Q is:

A Quality of
services

B Quality of
people

C Quality of
processes

D All of the above

6 Zero Defects
is a widely accepted and understandable concept.

True

False

7 Managers
assumed that they should not improve quality because defects are
less costly.

True

False

8 Quality is a
destination.

True

False

9 “Narrowly interpreted, quality means quality of a product,” was stated by

Ishikawa.

True

False

10 ISO was
first released in 1882.

True

False

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Re: MGMT 510 TQM All Quiz

by **Admin** on Sat Jan 23, 2010 12:57 pm

L#23

1. Which of the following document(s) is (are) included in the quality system?

a. A quality policy

b. Customer focus

c. Commitment

d. All of the given options

2. Appropriate motives for adopting ISO 9000 include which of the following?

a.

To create a quality management system

b.

To improve operations

c.

To conform to the requirements of customers

d.

All

of the given options

3.

Quality checklists are used to:

a. Ensure that Quality Assurance steps were followed

b. Keep quality inspectors busy

c. Inform upper management where failures occur

d. Prevent project audits

4.

The ISO 9001:2000 standards are

a. More widely adopted in the U.S. than in Europe

b.

Quality

management procedures that address leadership, documentation, and record-keeping

c. Are known as the Baldrige standards in the U.S

d. A set of product specifications for all industries

5. Documentation and data on existing products are the examples of _____

External inputs

Internal inputs

Other inputs

None of the
above options

6. Which of the following
statements outline the relationship factor?

a.

ISO 9000 and total quality are not in competition

b.

ISO 9000 and total quality are not interchangeable

c.

ISO 9000 is compatible with total quality

d.

All of the above

7. Which of the following is not an example of physical factors affecting work environment?

Noise

Cleanliness

Air

flow

None

of the given options

8. “Performance comes effortlessly” refers to _____ stage of learning according to psychologists.

Unconscious

incompetence

Conscious

incompetence

Conscious

competence

Unconscious

competence

9. There are _____ levels of the CMM.

Four

Five

Six
Three

10.
The underlying cause(s) of TQM deficiencies is (are):

The
failure of managers to understand the concept of quality
Costly
implementation of quality management system
Lack
of employee involvement
All
of the given options

1.
Leaders:

a. Push
rather than pull

b. Must
be trustworthy but not necessarily courageous

c. Know
where they want to go

d. Need
to dictate

2.

The Steering Committee must develop:

a.

Mission statement

b.

Continual improvement

c.

Vision statement

d.

All of the above

3.

One thing that the implementation plan must provide is:

a.

Short-term focus

b.

Leadership delegation

c.

Vision statement

d.

Training

provides information on how productivity and quality can be continuously improved through problem identification.

Statistical
quality control
**Statistical
process control**
Tree
Diagram
None
of the given option

_____ is a graphic tool for
defining the relationship between customer desires and the firm/product
capabilities.

House of Quality
Affinity diagram
Arrow diagram
None of the given option

**TRUE
OR FALSE**

6.
The affinity diagram is used to bring logic to the
process of identifying relationships among the various ideas recorded on the
tree diagram. **False**

7. Total
quality implementation must not be delegated. **True**

8. Competitive and benchmark data are essential for
planning quality leadership because they make possible clear and objective
quality comparisons. **True**

9. Deployment refers to the extent to which the approaches
are applied to all relevant areas and activities addressed and implied in the
examination items. **True**

10. It is
best to apply SPC to all processes at once. **False**

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Re: MGMT 510 TQM All Quiz

by **Admin** on Sat Jan 23, 2010 12:58 pm
Solution of Quiz

1 What is
quality?

**Meeting
and exceeding the customer's expectations**

Adding extras to make the customer happy

The degree to which the project meets requirements

Conformance to management's objectives

2 All of the
following are the grounds to give birth to total quality approach

EXCEPT:

Obsession with quantity

Confusion and poor leadership

One employee in charge

**Political
disorder**

3 All of the following are the major decision areas in supply chain management

EXCEPT:

Location

Production

Inventory

Pricing

4 Which of the following is true for subsystems?

These are not linked together as internal customers and suppliers

These are linked together as external customers and suppliers

These are linked together as external and internal customers and suppliers

These
are linked together as internal customers and suppliers

5 In which of the following operations great deal of variations can occur?

Manufacturing

Financing

Purchasing

Selling

6 With total quality perspective, employees are empowered to think and make

recommendations for which of the following?

Continual
improvement

One time check

After completion check

Incremental improvement

7 Which one of the following depicts durability – a dimension of quality?

Exterior finish

Quality of work

**Useful
life**

Ease of repair

8 Which of the following statements is TRUE for total quality?

**Stresses
on learning and adaptation to change**

Ability to repair a product quickly and easily

Establishing and measuring service levels may be difficult

A primary operating characteristic of product

9 What is the objective of a TQM system?

Continuous improvement

Continual improvement

Business improvement

Process improvement

10 Which of the following results in low costs?

High productivity and high capacity utilization

Low productivity and low capacity utilization

Low productivity and high capacity utilization

High productivity and low capacity utilization

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Re: MGMT 510 TQM All Quiz

by **Admin** on Sat Jan 23, 2010 12:58 pm

Total Quality Management (MGMT 510) Quiz: 03

Marks: 20 Semester: Spring 2008

This Quiz covers lessons: 29-32.

1 In the process of acquiring the certificate of ISO 9000, which step follows the

step of setting the registration objective?

a. Develop and implement the QMS

b. Perform
self-analysis audit

c. Submit
quality manual for approval

d. Select the appropriate standards

2 In the
process of acquiring the certificate of ISO 9000, which step follows the

step of pre-assessment by registrar?

a. Take corrective action

b. Obtain ISO
registration

c. Perform
self-analysis audit

d. Select a
third party registrar

3 In the
process of acquiring the certificate of ISO 9000, which step follows the

step of developing and implementing the quality and
management system?

a. Set the
registration objectives

b. Select a third party registrar

c.
Pre-assessment by registrar

d. Take
corrective action

4 In the
process of acquiring the certificate of ISO 9000, which step follows the
step of selecting the appropriate standards?

a. Perform
self-analysis audit

b. Submit
quality manual for approval

c. Develop and implement the QMS

d. Take
corrective action

5 In the

process of acquiring the certificate of ISO 9000, which step follows the

step of selecting a third party registrar?

a. Take
corrective action

b. Obtain ISO
registration

c. Perform self-analysis audit

d. Final
assessment by registrar

6 In the
process of acquiring the certificate of ISO 9000, which step follows the

step of performing self-analysis audit?

a. Final
assessment by registrar

b. Set the
registration objectives

c. Develop and
implement the QMS

d. Submit quality manual for approval

7 In the process of acquiring the certificate of ISO 9000, which step follows the step of submitting quality manual for approval?

a. Pre-assessment by registrar

b. Take corrective action

c. Perform self-analysis audit

d. Develop and implement the QMS

8 Which of the following is TRUE for higher costs of quality?

a. Doing the wrong thing in the wrong way

b. Doing the wrong thing in the right way

c. Doing the right thing in the wrong way

d. Doing the right thing in the right way

9 For what duration is the ISO certification valid for any organization?

a. 5 years

b. 4 years

c. 3 years

d. 2 years

10 Identify the core leadership skills among the following.

a. Vision

b. Empowerment

c. Value
congruence

d. Intuition

e. Skilled
labor

f. Gross motor

a. a, b, c and
f

b. b, c, d, and
e

c. c, d, e, and
f

d. a, b, c, and d

11 The primary
requirement for organizational excellence is strategic focus.

True

False

12 The primary
requirement for organizational excellence is tactical focus.

True

False

13 Japanese companies mostly are not ISO certified because they think they are

beyond the requirements of ISO.

True

False

14 The means for staff to perform their tasks right the first time is a benefit of

implementing the quality management system.

True

False

15 The means for documenting the company's experience in a structured manner is

not a benefit of implementing the quality management system.

True

False

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Re: MGMT 510 TQM All Quiz

by **Admin** on Sat Jan 23, 2010 12:59 pm

16 The implementation of the quality management system requires objective evidence.

True

False

17 The implementation of quality management system does not bring clarity and transparency to duties and responsibilities.

True

False

18 The implementation of quality management system does not make export marketing easier.

True

False

19 With the implementation of quality management system the company cannot secure greater customer loyalty.

True

False

20 With the implementation of quality management system the company's

customer will not receive products or services of a known level of quality that is

independently audited.

True

False

MGMT510 - TOTAL QUALITY MANAGEMENT - IMPORTANT QUESTIONS

What is quality for any organization?

A
solution

A problem

A specification

A project

What kind of decisions did the managers make in the old paradigm?

Strategic
positioning based on warfare models

Product positioning based on warfare models

Market positioning based on warfare models

Strategic and product positioning based on warfare models

What is the traditional view of quality?

Quality is measured by establishing an acceptable level of conformance

Quality is measured by establishing an acceptable level of employee satisfaction

Quality
is measured by establishing an acceptable level of nonconformance

Quality is measured by establishing an acceptable level of both conformance and nonconformance

When was the definition of total quality endorsed?

In 1950

In 1970

In 1985

In
1992

Customers want an excellent product or service from an organization that also provides which of the following?

A heartiest welcome

Explicit costs

Reliable
delivery and after purchase support

Implicit costs

Service standards are usually hard to measure because of the following reason?

Service standards are intangible

Service standards are unaccountable

Service
standards are judgmental

Service standards are invariable

Which of the following is required to ensure quality?

Absence of wastage

**Absence
of defects**

Absence of variations

Absence of specifications

Asking for examples, pictures, diagrams, and data in meetings, refers to which of the following?

Acting as gatekeepers

Closing the discussion

**Asking
for clarification**

Testing for consensus

The user-based view of quality is popular among the people of which of the following area?

Accounting

Banking

Physics

Marketing

Motorola was committed to which of the following campaigns in 1980s?

Zero defect

Six
sigma

Benchmarking

Cause-and-effect analysis

What are the ISO (International Organization for Standardization) series standards for industries such as aerospace?

QS 9000

AS
9000

TL 9000

ISO/TS 16949

Juran advocates a never-ending spiral of activities that includes all of the following EXCEPT:

Product development

Workers

Inspection and testing

Purchasing

To increase the awareness and involvement of people, employee training should cover all of the following EXCEPT:

The vision for the future of the organization

The organization's policies and objectives

**Organizations's
intangible resources such as intellectual property ??**

The initiation and implementation of improvement activities

Which of the following is a follow up activity relating to internal audit?

Collection of quality cost data

Questionnaires and surveys

**Verification
of implementation**

Utilization of technologies

All of the following are the consideration of the organizations when analyzing how all the processes interact to make sure that all processes operate as an efficient network EXCEPT:

Ensuring inputs, activities and outputs are clearly defined and controlled

Managing risks and opportunities

**Monitoring
inputs and outputs to confirm that, group processes are unrelated**

Establishing data analysis that facilitates continual improvement across all processes

Which of the following is a valid statement about

design and development outputs?

It identifies and rectifies against potential conformance

It is used where excess in product will only be apparent in use

It insignifies and conceals the significant features of the processes

It
meets the input requirements for design and development

Against which of the following scales, the Capability Maturity Model can be used to assess an organization?

Three process maturity levels

Four process maturity levels

Five
process maturity levels

Six process maturity levels

In the levels of organizational maturity, the dysfunctional system precedes which of the following systems?

Awakening System

Developing System

Maturing System

World Class System

Which of the following is TRUE for lower costs of quality?

Doing the wrong thing in the wrong way

Doing the wrong thing in the right way

Doing the right thing in the wrong way

Doing
the right thing in the right way

Which of the following defines the boundaries of the business in which the organization operates?

Vision

Mission

Plan

Objectives

What are the core competencies of leadership?

Activities that get the opportunity to meet many people with similar interests

Activities and processes needed to meet customers' minimum requirements

Activities that gives you an opportunity to learn and experience new things

Activities

that underpin competitive advantage and are difficult for competitors to copy

Which of the following is NOT an organizational weakness?

Obsolete facilities

Obsolete processes

Weak management team

Higher

quality competitors ??

Which of the following base all actions, activities, and decisions on what is most likely within an ethical framework to ensure successful performance in the marketplace?

Operation Management

**Strategic
Management**

Financial Management

Network Management

Total quality relates to which of the following which enhances an organization's ability to gain a sustainable competitive advantage in the marketplace?

Operation Management

**Strategic
Management**

Financial Management

Network Management

Which of the following is a communication tool used to tell a story of how value is created for the organization?

Injective map

**Strategy
map ???**

Surjective map

Bijjective map

Under which perspective of the balance scorecard, would you classify inventory management measurement?

**The
customer perspective ??**

The business process perspective

The financial perspective

The learning & growth perspective

Under which perspective of the balance scorecard, would you classify corporate goals measurement?

The customer perspective

The
business process perspective ??

The financial perspective

The learning & growth perspective

Under which perspective of the balance scorecard, would you classify operating cost management measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Under which perspective of the balance scorecard, would you classify average time to process orders measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

Under which perspective of the balance scorecard, would you classify number of strategic skills learned measurement?

The customer perspective

The business process perspective

The financial perspective

The learning & growth perspective

All of the following are the examples of product concerns from customers and end-users EXCEPT:

Conformance

Transportation

Availability

Delivery

Brainstorming refers to which of the following?

Mess Finding

Fact Finding

Problem Finding

Idea
Finding

The Pareto Principle is a technique used by quality managers to determine the quality control problems concerning a particular service or manufacturing process to be corrected. Which of the following statements best represents the problems employed by this principle?

In order to minimize financial losses, all problems should be corrected

The majority of defects are caused by a small percentage of the identifiable problems.

In order to achieve zero defects, all quality control problems should be corrected

Generally, 80% of the quality control problems are justifiable for correction via cost-benefit analysis.

Which of the following helps you distinguish between normal and unusual variation in a process?

Control Chart

Cause and effect diagram

Pareto Chart

Scatter Diagram

The observation in the 'middle' of sorted data is called:

Mode

Median

Range

Variance

Which of the following is the contributor to materials as a cause in free-throwing cause-and-effect diagram?

Basketball

Games

Video camera

Touched rim

Which of the following is the contributor to equipments as a cause in free-throwing cause-and-effect diagram?

Basketball

Games

**Video
camera**

Touched rim

In which of the following teams, the team members communicate by computers, take turns as leaders, and jump in and out as necessary?

**Virtual
teams**

Quality circles

Management teams

Natural work teams

Which of the following is performed in the process of benchmarking?

To collect related and structured activities that produce a specific product

To plan and control the performance of any type of activity

To
compare external data collected to the internal data collected

To create goods and services that meet needs and wants of customers

Which of the following is NOT the responsibility of organization regarding Competence, Awareness and Training of employees?

To
implement the country's legal requirements for efficient transportation of people

To make training available to employees or take other actions to satisfy competency needs

To determine the necessary competence for employees performing work affecting product quality

To ensure that employees are aware of the relevance and importance of their activities

You know how to paint, but your reactions are not automatic---- you have to work at it. This statement refers to which of the following stages of learning?

Unconscious incompetence

Conscious incompetence

Conscious
competence ??

Unconscious competence

Mouser Electronics determines, collects, and analyzes data to demonstrate the continuing suitability and effectiveness of quality management system. What will be the contents of data used for such analysis ? 3

What are the responsibilities of the owner of a critical success factor? 3

What are the advantages and uses of a fish bone diagram? Marks=2+1)

The policy of ABC Machine Works Ltd. is to provide products and services that meet the requirements of our customers in accordance with our quality policy. How the management and employees of this company are committed to achieving this policy? 5

Compare and contrast run charts and control charts. 5

Explain the following in the light of seven-step method for problem solving: Marks=5+5)

- Collect and Analyze Data

- Identify the Root Cause

Differentiate between control limits and specification limits. 10

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Re: MGMT 510 TQM All Quiz

by **Admin** on Sat Jan 23, 2010 12:59 pm

MIDTERM

EXAMINATION FALL 2007 Marks: 40

MGMT510 - TOTAL QUALITY MANAGEMENT (Session

Which of the following is the odd one out?

Peter Drucker

Philip Crosby

W. Edwards Deming

Kaoro Ishikawa

One of the most effective means of implementing quality initiatives for executive management is to:

Establish
quality goals tied to organizational performance

Conduct meetings on quality and demonstrate support for initiatives

Make public announcements explaining the company's quality goals

Hire a quality consultant to develop a total quality plan and lead its implementation

ISO 9000 certification:

Aims to promote top management's commitment to TQM

Certifies products and their manufacturing processes

Does not apply to service industries

Is seen as a customer requirement

According to Crosby,

most companies spend _____ percent of their sales dollars on quality costs.

15 to 20

20 to 25

25 to 30

None of the given options

The control of quality involves the following processes except:

Assess actual
quality performance

Compare
performance with goals

Act on
differences between performance and goals

None of the given options

Methods of continuous improvement include _____.

Six Sigma

Lean Production

Kaizen

**All
of the given options**

Shareholder is an example of _____.

Internal
customer

External
customer

Investor customer

Social customer

_____ concepts are referred to as statistical
quality control (SQC).

Shewhart's

Deming's

Juran's

Crosby's

_____ is the judge of quality.

Customer

Supplier

Producer

None of the given options

_____ is not an example of organization.

Factory

Mosque

Hospital

**None
of the given options**

Deming cycle was developed to link the production of a

product with consumer needs.

True

False

Avoidable costs are related to preventing defects.

True

False

Total quality requires a set of guiding principles.

True

False
??

Crosby is best known for his advocacy of zero defects management.

True

False

The Six Sigma is a new filed separate from the total quality philosophy.

True

False

How would you establish a system to measure customer satisfaction?

Question No: 17
(Marks: 5)

Which of the quality gurus, in your view, have most significantly influenced the quality

movement since World War II?

Question No: 18
(Marks: 5)

Discuss the various types of quality costs. What kind of costs should a firm be more

concerned with and why?

Question No: 19
(Marks: 10)

Discuss the absolutes of quality in context to the organization you are working in or are

familiar with.

Being
a quality control manager you are supposed to visit a restaurant of your area,
how would you measure its quality

Deming's 14/7 as a guideline

**Discuss the importance of the 3C concept in PCAP's
implementation of Total Quality Management**

You are planning to open an internet café and have secured the necessary

capital. Your aim is to attract both regular customers and passing trade. Discuss

the key implications of this for the management of the business.

Question

Why do you think that companies in the United States were slow to adopt the

quality management principles Japanese companies had used to gain market share

worldwide?

Being the manager of XYZ, the ISO 9000:2000 certified company, what type of

problems you encountered while implementing ISO 9000:2000, and what steps you

took to handle such problems?

Admin
Administrator

Number of posts: 8058

Age: 36

Location: Sharqi Abadi Mustafabad Kasur

Job/hobbies: Do more for other

Mode (i.e. cool, angry etc): Serious

Warning:



Rate by Admin:



Reputation: 22

Registration date: 2007-10-23



Re: MGMT 510 TQM All Quiz

by **Admin** on Sat Jan 23, 2010 1:00 pm

FINAL TERM EXAMINATION FALL 2007 MGT510 - TOTAL QUALITY
MANAGEMENT (Session -2)

Marks: 60 Time: 150min

Exam Date: Thursday, March 06, 2008

Which of the following statements best describe the
acronym SWOT?

Specific, Witness, Opportunities, and Time

Single, Weaknesses, Occasion, and Tactics

Strengths,
Weaknesses, Opportunities, and Threats

Strategic, Weaknesses, Objectives, and Tactics

Which of the following is NOT a law of organizational change?

Involve
everyone affected by change in making it ??

Be prepared to listen and observe

Trust people with dignity and respect

Understand the history behind the current culture

The most effective leaders can be found in what part of an organization?

Top-level
management

Mid-level management

Lowest paid wage earner

All of the given options

Which of the following character traits promote successful teamwork?

Honesty

Controlling

Accountability

Bossy

Which of the following statements outlines the relationship factor?

ISO 9000 and total quality are not in competition

ISO 9000 and total quality are not interchangeable

ISO 9000 is compatible with total quality

**All
of the given options**

What is the only tool of the seven tools that is not based on statistics?

Pareto Chart

**Fishbone
Diagram**

Scatter Diagram

Histogram

What portion of the QFD house explanation is where customer requirements are converted into manufacturing terms?

The wall

The roof

The middle

The bottom

The most common inhibitor of SPC is:

Inadequate training

Lack of resources

Failure to have processes under control

Low production rates

Little Q regards the customer as:

The person who buys the product

All people involved, internal, and external

The decision-maker

Having continually changing needs

Which of the following behavior best describes the concept of benchmarking?

Unethical

Illegal

Industrial espionage

None
of the given options ??

Which of the following statements is considered true concerning the affects of global integration?

It has helped fuel economic growth throughout the industrialized world

It can be the Achilles heel of economic growth

Downturns in one country can now have a ripple effect that quickly spreads to other

countries

All
of the given options

Which of the following statements is not true about Total Quality Management (TQM)?

TQM is focused on improving product and customer service quality

TQM is expensive to initiate

TQM requires top management support

TQM
requires ISO 9000 certification

Which of the following may be a dimension of quality?

Performance

Conformance

Reliability

Durability

According to Juran trilogy, quality control involves which of the following process(es).

Assessment of actual quality performance

Comparison of performance with goals

Action on differences between performance and goals

**All
of the given options**

is
best known for his zero defects management.

Crosby

Deming

Juran

None of the given options

The need to improve an organization's financial condition correlated directly with the process of making and measuring quality improvements.

True

False

SWOT Analysis is the only step needed in the strategic planning process.

True

False

The concept of quality has often been defined, from a transcendent view, as innate excellence.

True

False

Quality materials produced by Juran have been translated into 12 different languages.

True

False

Philosophy helps us to understand people, interactions between people and circumstances.

True

False

According to view quality means meeting or exceeding customer expectations.

Poorly trained operator is an example of .

Six Sigma was developed by to improve its business processes.

ISO 9001 and ISO 9004 are known as .

Scatter diagram is used to measure between two variables.

Question No: 26 (
Marks: 5)

Explain the total quality philosophy of training.

Question No: 27 (
Marks: 5)

Explain the purpose of a Pareto Chart. Give an example of when it should be used.

Question No: 28 (
Marks: 5)

Why do some quality initiatives fail?

Question No: 29 (Marks: 10)

Defend the statement that the operator of the process should be the owner and data plotter of the control chart, as opposed to a person from quality assurance or engineer.

Question No: 30 (Marks: 10)

Explain the following statement “Successful implementation of empowerment requires

change in the corporate culture.”

Question # 6 of 15 (Start time: 06:10:39 PM)

Total Marks:

Which of the following is NOT a Juran's step to quality improvement?

Select correct option:



Provide a healthy ambiance



Organize to meet the goals that have been set



Implement projects aimed at solving problems



Set goals for improvement

Question # 7 of 15 (Start time: 06:11:53 PM)

Total Marks: 1

If workers are given rewards based on meeting quotas, what sort of improvements will it cause within an organization?

Select correct option:

- ☐ Long-term improvements
- ☒ Short-term improvements
- ☐ Both long-term and short-term improvements
- ☐ No improvement

Question # 8 of 15 (Start time: 06:13:18 PM)

Total Marks: 1

What was the orientation approach during 1950s?

Select correct option:

- ☒ Builds in quality
- ☐ Inspects in quality
- ☐ Controls in quality
- ☐ Manages in quality

Question # 11 of 15 (Start time: 12:56:43 PM)

Total Marks:

When Juran taught quality principles to the Japanese?

Select correct option:

- ☐ 1980s
- ☐ 1970s
- ☐ 1960s
- ☒ 1950s

Question # 15 of 15 (Start time: 12:59:47 PM)

Total Marks: 1

All of the following is the consideration of the organizations to ensure how all processes operate as an efficient network EXCEPT:

Select correct option:

- ☐ Ensuring inputs, activities and outputs are clearly defined and controlled
- ☐ Ensuring that the sequence and interaction of processes are designed to achieve the desired results
- ☒ Monitoring inputs and outputs to confirm that group processes are unrelated
- ☐ Establishing data analysis that facilitates continual improvement across all processes

The customer value strategy may be addressed in which of the following topics?

Select correct option:

- ☐ Projects ^
v
 - ☐ Products ^
v
 - ☒ Services ^
v
 - ☐ Measurement ^
v
-